T2K Neutrino and Anti-Neutrino 3-Flavour Joint Analysis of Run 1-8 (1.4734×10^{21}-POT ν 7.558×10^{20}-POT $\bar{\nu}$) data sets

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Abstract

We report the results of the 3-flavour, 5 sample $\nu/\bar{\nu}$ joint analysis, performed by the *VALOR* group, on the Run 1-8 dataset, using data taken in both neutrino and antineutrino beam mode. Two sets of results are reported; section 5 presents results before and section 6 results after the inclusion of a new systematic parameter and smearing procedure to address biases associated with fake data studies involving binding energy. The goal of this analysis is to search for CP violation by setting limits on δ_{CP} and measure the oscillation parameters $\sin^2\theta_{13}$, $\sin^2\theta_{23}$, Δm^2_{32} (Δm^2_{13}). Fits are performed under both mass hierarchy hypotheses to show which is preferred by data.

The Run 1-8 dataset corresponds to an integrated J-PARC neutrino beam exposure of 1.4734×10^{21} POT in neutrino mode and 7.558×10^{20} POT in antineutrino mode.

This analysis predicts 268.4 ± 11.8 (syst) for neutrino μ-like, 64.3 ± 2.4 (syst) for antineutrino μ-like, 73.5 ± 5.3 (syst) for neutrino e-like, and 7.9 ± 0.6 (syst) for antineutrino e-like single-ring 0π events in Super-K and 6.9 ± 1.5 (syst) ν_e CC1π^+ events, given the oscillation parameters of the Asimov data set A in 2.1.1. We observe respectively 240 for neutrino μ-like, 68 for antineutrino μ-like,74 for neutrino e-like, 7 for antineutrino e-like and 15 for ν_e CC1π^+-like.

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1. Introduction

In this technical note, we report the results of a 3-flavour $\nu/\bar{\nu}$ joint analysis performed by the *VALOR* group, on the combined Run 1-8 dataset, using data taken in both neutrino-enhanced forward horn current (FHC) mode (1.4734×10^{21} POT), and antineutrino-enhanced reverse horn current (RHC) mode (7.558×10^{20} POT).

This analysis updates the previous analyses based on the additional data gathered at Super-K, a new cross-section model and a change in the Super-K reconstruction method.

Section 2 describes the Super-K samples, section 3 describes the systematic errors considered and their effects on the extrapolated single-ring event spectra, whilst the sensitivity of the various fits is considered in section 4. Fit results are presented in two sections; preliminary results can be found in section 5, with the final results, following the inclusion of a new systematic parameter and smearing procedure to address biases seen in fake data studies involving binding energy, presented in section 6.

In this analysis we move from the APFit reconstruction algorithm to the fiTQun algorithm [17]. Selection cuts are revisited and a significant improvement in statistics is achieved by modifying the fiducial volume cuts to allow for a larger interaction mass [16]. This analysis includes new cross-section model parameters. A 2p-2h shape parameter is added to the 2p2h normalisation parameters already present in the previous analysis. Furthermore, the Random Phase Approximation (RPA) is parameterised by a Bernstein polynomial (BeRPA) [15]. Each of these new parameters will be described in section 3.

Confidence intervals for δ_{CP} are produced using the Feldman-Cousins method (see section 1.1.1), while confidence intervals of the other oscillation parameters are set using the constant $\Delta\chi^2$ method. Sensitivity studies are shown in section 4. The results of the measurement using the Run 1-8 dataset are given in section 5 of this note.

Finally, a summary is given in section 7.

1.1. Analysis procedure

This analysis builds on our previous analysis in [12]. In this analysis we fit Super-K data using a hybrid frequentist-Bayesian method [70], which consists of marginalising instead of profiling the nuisance parameters. A description of the method can be found in section 4.2 of [10].

The analysis uses a binned likelihood-ratio method [66] where the estimation of the parameters of interest, $\vec{\theta}$, is obtained by comparing the observed and predicted Super-K spectra for 1-ring events. For the two one-ring μ-like samples the binning of data events is in one-dimensional reconstructed neutrino energy space, whilst the three one-ring e-like samples bin data events in two-dimensional reconstructed neutrino energy and lepton angle space.

The 2D binning scheme used for the e-like samples is to improve the separation of the different reaction / flavour components, in particular $\bar{\nu}_\mu \to \bar{\nu}_e$ and $\nu_\mu \to \nu_e$, fundamental in measuring δ_{CP}. The ν_e intrinsic beam and NC components can also be better separated from signal.

μ-like templates have 84 true energy bins and 73 reconstructed energy bins. e-like templates have 84 true energy bins, 25 reconstructed energy bins and 15 reconstructed lepton angle bins.

For all templates the 84 true energy bins of constant width are the following:

- 6 50-MeV bins from 0-0.3 GeV,
- 28 25-MeV bins from 0.3-1 GeV,
- 40 50-MeV bins from 1-3 GeV,
- 5 100-MeV bin from 3-3.5 GeV,
- 1 bin from 3.5-4 GeV,
- 1 bin from 4-5 GeV,
- 1 bin from 5-7 GeV,
- 1 bin from 7-10 GeV and
- 1 bin from 10-30 GeV

For μ-like samples the 73 reconstructed energy bins are the following:

- 60 50-MeV bins from 0-3 GeV,
- 4 250-MeV bins from 3-4 GeV,
- 4 500-MeV bins from 4-6 GeV,
- 4 1000-MeV bins from 6-10 GeV and
- 1 bin from 10-30 GeV.

For e-like samples the 25 reconstructed energy bins with constant width are the following:

- 25 50-MeV bins from 0-1.25 GeV.

The θ binning used for e-like samples is

- 14 10° bins from 0° to 140°,
- 1 bin for the range $140° - 180°$.

The method of calculation of these predicted spectra is presented in Appendix A. This analysis uses flux tuning based on NA61/SHINE 2009 data [23][60]. It also uses a model-based extrapolation from ND280 to Super-K, which is implemented via a fit of flux model and cross-section model parameters to ND280 measurements of the muon momentum-angle distributions of $\bar{\nu}_\mu$ and ν_μ charged current interaction candidates for different event topologies with the neutrino interaction vertex either in FGD1 or FGD2 (BANFF fit) [36].

The log-likelihood function used in the analysis is

$$-2\ln\lambda(\vec{\theta},\mathbf{a}) = 2\cdot\sum_{i=0}^{N-1}\left(n_i^{obs}\cdot\ln(n_i^{obs}/n_i^{exp}) + (n_i^{exp}-n_i^{obs})\right) \quad (1)$$

where n_i^{obs} is the observed number of events in the ith bin and $n_i^{exp} = n_i^{exp}(\vec{\theta};\mathbf{a})$ is the corresponding expected number of events, N is the number of reconstructed energy and lepton angle bins and $\mathbf{a}$ is a vector of systematic parameters.

The total likelihood is then given by the sum of $-2\ln\lambda(\vec{\theta},\mathbf{a})$ from all of the Super-K samples

$$-2\ln\lambda(\vec{\theta},\mathbf{a}) = \sum_{s=1}^{N_{samples}} -2\ln\lambda_s(\vec{\theta},\mathbf{a}) \quad (2)$$

The postfit errors of covariance matrices which result from the BANFF fit and Super-K fits are used as prior uncertainties. The parameters are thrown using the Cholesky method, which allows throws according the parameters' multivariate distribution and existing correlations. Where systematic parameters have physical boundaries, any toy experiment where such a parameter is generated beyond its physical limits is rejected. Furthermore, values of systematic parameters that give a negative predicted number of events in any reconstructed energy bin in any interaction mode are not allowed. If a systematic throw gives a negative number of events, that toy experiment is rejected. All rejected toys are regenerated.

A single set of oscillation parameters in the 3-flavour framework is used to describe both neutrino and antineutrino data. All 6 oscillation parameters float in the fit, with nuisance oscillation parameters marginalised while the parameters of interest are minimized with respect to the likelihood distribution. Flat priors are used for $\sin^2\theta_{23}$, Δm^2_{32} and δ_{CP} in all fits, whereas $\sin^2\theta_{13}$ uses a flat prior when not applying constraints from reactor experiments and a gaussian prior when using the measurement found in reactor experiments. $\sin^2\theta_{12}$ and Δm^2_{21} uses gaussian priors in all fits due to the limited sensitivity of T2K to the solar parameters. Fits are performed for each mass hierarchy and both with and without the reactor constraint applied.

Oscillation probabilities include constant-density matter effects, with details of our oscillation probability calculations, estimates of their numerical accuracy and comparisons with calculations used by the alternative T2K oscillation analyses are given in Appendices A and B of Ref. [4].

1.1.1. Feldman-Cousins

In the gaussian approximation the constant $\Delta\chi^2$ method can be used. However, the oscillation probability is not linear in oscillation parameters and so in some cases the gaussian approximation is not reliable and the Feldman-Cousins method [69] must be used. In this method a different critical value for each point of the $\vec{\theta}$ grid, $\Delta\chi^2(\vec{\theta})_{crit}$, is calculated. Since the most important result of this analysis is the search for CP violation, this method is performed only for the simultaneous fit of δ_{CP} and mass hierarchy. In particular, critical values are calculated for 9 evenly spaced true values of δ_{CP} in the range $[-\pi,+\pi]$ for both normal and inverted mass hierarchy, and linearly interpolated

between these points.

In the previous analysis the Feldman-Cousins implementation used the disappearance likelihood surface from the best-fit of the data to generate fake data sets during the procedure. However, this approach causes the critical values to become smaller in the vicinity of the best-fit for δ_{CP} because the generated fake data sets are closer to the real data. The new method [90] instead uses the data best-fit for disappearance parameters, along with the PDG 2016 best-fit values for the remaining oscillation parameters (table 1), to perform additional Asimov fits in normal and inverted hierarchy. It is then the likelihood surfaces from these fits that are used to generate the fake data sets during the procedure. The procedure consists of the following steps:

1. At least 1×10^4 fake data sets are produced assuming the oscillation hypothesis of the grid point $\vec{\theta}$ and taking into account both the statistical and systematic uncertainties (i.e. nuisance parameters, including non-fit oscillation parameters, are marginalised);
2. For each fake data set:
   (a) $\chi^2(\vec{\theta}_{fix})$ is obtained by computing $\chi^2(\vec{\theta})$ with respect to the marginalisation toys produced for the data fit, with the oscillation parameters $\vec{\theta}$ fixed to the values corresponding to the grid point.
   (b) $\chi^2(\vec{\theta}_{bf})$ is obtained by minimizing $\chi^2(\vec{\theta})$ with respect to the marginalisation toys produced for the data fit, leaving the oscillation parameters $\vec{\theta}$ free.
   (c) The test statistic is calculated as

$$\Delta\chi^2 = \chi^2(\vec{\theta}_{fix}) - \chi^2(\vec{\theta}_{bf}) \tag{3}$$

3. This results in a distribution of $\Delta\chi^2(\vec{\theta})$, $f(\Delta\chi^2)$.
4. Several critical values for X% CL (e.g. 68%, 90%...), $\Delta\chi^2(\vec{\theta})_{crit}$, can be defined as:

$$\Delta\chi^2_{crit} : \int_0^{\Delta\chi^2_{crit}} f(\Delta\chi^2) d(\Delta\chi^2) = X\% \tag{4}$$

   where $f(\Delta\chi^2)$ is the p.d.f. of $\Delta\chi^2$ and $d(\Delta\chi^2)$ is the differential of the integral.
5. This procedure is repeated to produce critical values for all the points of the $\vec{\theta}$ grid.

Once a critical value, $\Delta\chi^2(\vec{\theta})_{crit}$, is computed for each oscillation hypothesis in the $\vec{\theta}$ grid, the confidence intervals are set by the condition

$$\Delta\chi^2(\vec{\theta}) > \Delta\chi^2_{crit} \tag{5}$$

with those values of $\Delta\chi^2(\vec{\theta})$ being excluded at the relevant confidence interval.

2. Super-K samples used in the analysis

This analysis is comprised of five samples based on the fiTQun selection algorithm, which has improved reconstruction capabilities and can therefore reduce the background components of the selected samples. The five samples are:

- neutrino mode single μ-like ring sample
- antineutrino mode single μ-like ring sample
- neutrino mode single e-like ring sample
- antineutrino mode single e-like ring sample
- neutrino mode single e-like ring sample with one decay electron (referred to as ν_e CC1π^+)

The Super-K MC templates contain events from the ν_μ, $\bar{\nu}_\mu$, ν_e, $\bar{\nu}_e$, oscillated ν_e and oscillated $\bar{\nu}_e$ samples passing the Super-K selection cuts. Details of the selection cuts can be found in [42].

2.1. Predicted event rates

Predicted event rates for this analysis are determined for the case of two different sets of oscillation parameters based on previous best fit results from T2K and NOvA. These parameter sets are referred to as Asimov A and B, with their definitions and associated event rates given in the following sections.

2.1.1. Definition of Asimov A and B

Table 1 shows the definitions used for the Asimov data sets. Asimov A represents parameter values close to the T2K best fit for Run 1-7, whilst Asimov B modifies these parameters such that CP is conserved and changes the $\sin^2\theta_{23}$ octant.

Parameter(s)	Asimov A	Asimov B
$\sin^2\theta_{23}$	0.528	0.450
$\sin^2\theta_{13}$ reactors	0.0219	0.0219
$\sin^2\theta_{12}$	0.304	0.304
$\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{31}\rvert$ (IH)	2.509×10^{-3} eV^2/c^4	2.509×10^{-3} eV^2/c^4
Δm^2_{21}	7.53×10^{-5} eV^2/c^4	7.53×10^{-5} eV^2/c^4
δ_{CP}	-1.601	0
Mass Hierarchy	Normal	Normal

Table 1: Values of oscillation parameters used to compute the event rates, systematic effects and sensitivity studies. Each set of oscillation parameters correspond to a different Asimov data set, which is the MC expected distribution in a certain oscillation hypothesis. In the Asimov data set A the nominal values of $\sin^2 2\theta_{13}$, $\sin^2 2\theta_{12}$ and Δm^2_{21} are from [81], while all the other oscillation parameter values corresponds to the most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [20]. The mass hierarchy is not marginalised but fixed to either NH or IH.

2.1.2. Event rates

Table 2 shows the expected event rates in Super-K for the Asimov A data set, but with δ_{CP} varied at $-\pi/2$, 0, $\pi/2$ and π.

	$\delta_{CP} = -\pi/2$	$\delta_{CP} = 0$	$\delta_{CP} = \pi/2$	$\delta_{CP} = \pi$
FHC e-like sample	73.47354	61.43388	49.90582	61.94533
FHC μ-like sample	268.40351	268.05327	268.37281	268.81441
FHC ν_e CC1π^+-like sample	6.92243	6.00922	4.86656	5.77983
RHC e-like sample	7.91206	9.02311	10.02989	8.91886
RHC μ-like sample	64.27306	64.12292	64.27392	64.43654

Table 2: Predicted total number of events in each sample obtained after applying the BANFF weight for a total exposure of 1.4734×10^{21} POT. The Asimov data set A in 1 is used, but with varying δ_{CP}.

Tables 3 to 7 present the expected event rates in Super-K corresponding to the Asimov A data set. Event rates in other oscillation scenarios can be found in Appendix B. In each case the normal hierarchy is assumed and all rates are generated for an integrated exposure corresponding to Run 1-8.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	175.62883	0.00413	11.03473	0.00019	0.02245	0.00015	186.69049
CC1pi	27.69898	0.00199	2.61604	0.00009	0.02406	0.00007	30.34123
CCcoherent	0.28837	0.00000	0.09218	0.00000	0.00000	0.00000	0.38055
2p2h	35.57284	0.00151	1.41213	0.00008	0.02551	0.00004	37.01211
CCother	5.46647	0.00086	0.40615	0.00008	0.00011	0.00001	5.87368
NC1pi0	0.61495	0.01841	0.02120	0.00143	N/A	N/A	0.65599
NC1piPM	4.90783	0.09799	0.17531	0.00975	N/A	N/A	5.19089
NCcoherent	0.00000	0.00000	0.00043	0.00003	N/A	N/A	0.00047
NCother	2.05619	0.07378	0.13055	0.00912	N/A	N/A	2.26963
Total	252.23446	0.19867	15.88872	0.02078	0.07214	0.00027	268.41504

Table 3: Predicted number of events in the neutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 1.4734×10^{21} POT. The Asimov data set A in 1 is used.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	15.29349	0.00085	29.88622	0.00093	0.00094	0.00137	45.18379
CC1pi	3.95485	0.00033	4.53265	0.00025	0.00060	0.00061	8.48929
CCcoherent	0.04705	0.00000	0.17567	0.00005	0.00000	0.00006	0.22283
2p2h	4.47453	0.00026	2.84291	0.00027	0.00044	0.00021	7.31861
CCother	0.94199	0.00012	0.60584	0.00000	0.00000	0.00000	1.54795
NC1pi0	0.06000	0.00222	0.06147	0.00169	N/A	N/A	0.12538
NC1piPM	0.35950	0.01527	0.46009	0.01235	N/A	N/A	0.84721
NCcoherent	0.00000	0.00000	0.00201	0.00000	N/A	N/A	0.00201
NCother	0.31580	0.01640	0.20064	0.00789	N/A	N/A	0.54073
Total	25.44720	0.03544	38.76749	0.02342	0.00198	0.00226	64.27780

Table 4: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 7.558×10^{20} POT. The Asimov data set A in 1 is used.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.16115	6.30178	0.00634	0.25566	46.25561	0.27811	53.25866
CC1pi	0.04703	0.93168	0.00280	0.06750	4.49505	0.04745	5.59150
CCcoherent	0.00011	0.00830	0.00018	0.00397	0.03445	0.00327	0.05028
2p2h	0.03033	1.59694	0.00117	0.04275	8.68112	0.03893	10.39125
CCother	0.01329	0.12803	0.00034	0.00942	0.08850	0.00326	0.24284
NC1pi0	1.70203	0.03706	0.06012	0.00356	N/A	N/A	1.80278
NC1piPM	0.16854	0.00472	0.01020	0.00057	N/A	N/A	0.18404
NCcoherent	0.52817	0.00715	0.04794	0.00294	N/A	N/A	0.58619
NCother	0.31767	0.01695	0.02120	0.00128	N/A	N/A	0.35711
NC1gamma	0.94222	0.01687	0.04983	0.00233	N/A	N/A	1.01125
Total	3.91054	9.04948	0.20014	0.38998	59.55473	0.37103	73.47589

Table 5: Predicted number of events in the neutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 1.4734×10^{21} POT. The Asimov data set A in 1 is used.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.01288	0.50911	0.02388	0.84721	1.02586	2.85198	5.27090
CC1pi	0.00763	0.10111	0.00733	0.14431	0.14337	0.33485	0.73858
CCcoherent	0.00002	0.00099	0.00042	0.01066	0.00114	0.03130	0.04455
2p2h	0.00295	0.14379	0.00219	0.11301	0.23285	0.33042	0.82521
CCother	0.00390	0.02116	0.00120	0.01154	0.01042	0.00668	0.05491
NC1pi0	0.13849	0.00541	0.17659	0.00468	N/A	N/A	0.32516
NC1piPM	0.02475	0.00127	0.02001	0.00080	N/A	N/A	0.04683
NCcoherent	0.04776	0.00221	0.20260	0.00317	N/A	N/A	0.25574
NCother	0.05664	0.00221	0.02672	0.00097	N/A	N/A	0.08654
NC1gamma	0.07943	0.00439	0.17490	0.00383	N/A	N/A	0.26255
Total	0.37445	0.79164	0.63583	1.14019	1.41363	3.55523	7.91097

Table 6: Predicted number of events in the antineutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 1.4734×10^{21} POT. The Asimov data set A in 1 is used.

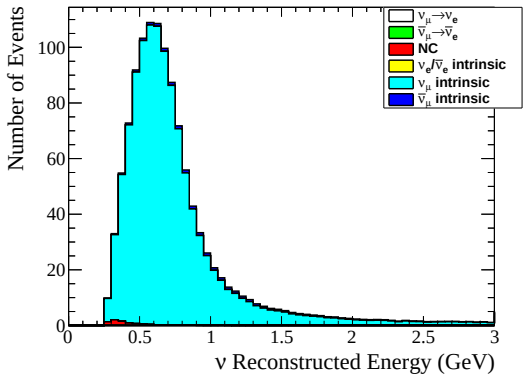
	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.02857	0.02442	0.00117	0.00174	0.21123	0.00265	0.26979
CC1pi	0.07582	0.76065	0.00257	0.00383	4.85325	0.00227	5.69839
CCcoherent	0.00027	0.01765	0.00008	0.00007	0.11121	0.00011	0.12939
2p2h	0.01012	0.02195	0.00049	0.00095	0.16017	0.00105	0.19473
CCother	0.03770	0.11225	0.00116	0.00279	0.12085	0.00150	0.27625
NC1pi0	0.02049	0.00076	0.00083	0.00008	N/A	N/A	0.02216
NC1piPM	0.06474	0.00165	0.00382	0.00033	N/A	N/A	0.07053
NCother	0.20584	0.00830	0.01304	0.00088	N/A	N/A	0.22807
NC1gamma	0.02844	0.00000	0.00067	0.00007	N/A	N/A	0.02918
Total	0.47199	0.94764	0.02382	0.01075	5.45671	0.00758	6.91850

Table 7: Predicted number of events in the neutrino mode ν_e CC1π^+-like sample obtained after applying the BANFF weight for a total exposure of 1.4734×10^{21} POT. The Asimov data set A in 1 is used.

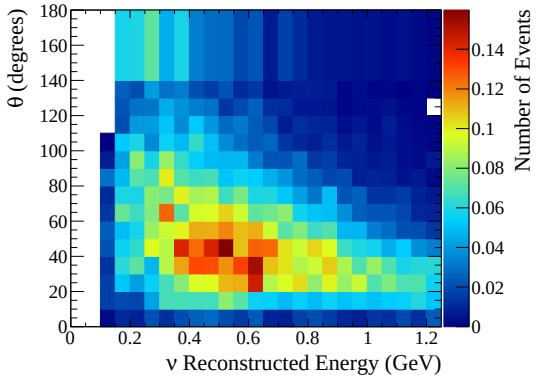
2.1.3. Predicted and observed spectra

In this section, we present the expected spectra in Super-K for Asimov A and unoscillated scenarios, assuming normal hierarchy. All plots are generated for an integrated exposure corresponding to Run 1-8.

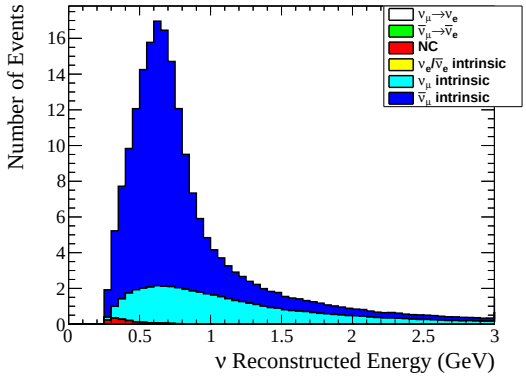
Predicted single ring Super-K spectra are shown in figs. 1 and 2. By comparing neutrino and antineutrino mode e-like plots it is clear that θ vs E_{reco} distributions are differently populated by $\nu_\mu \to \nu_e$ and $\bar{\nu}_\mu \to \bar{\nu}_e$ events.



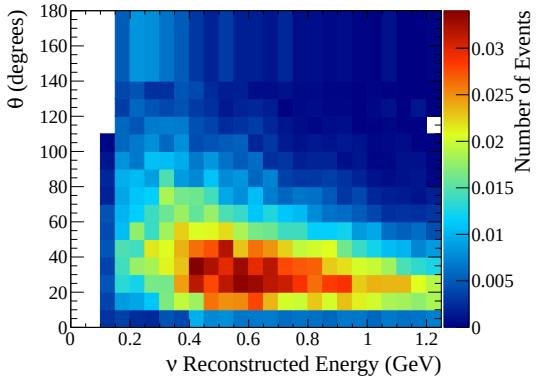
(a) Muon-like, neutrino mode



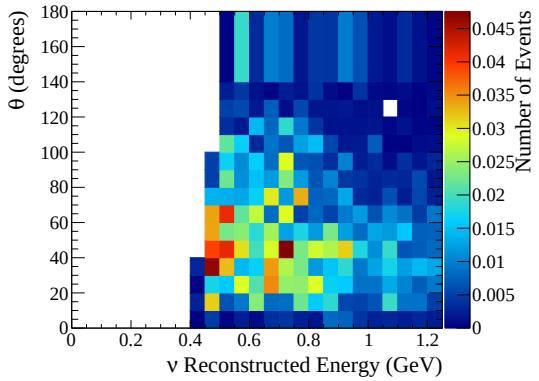
(b) Electron-like, neutrino mode



(c) Muon-like, antineutrino mode

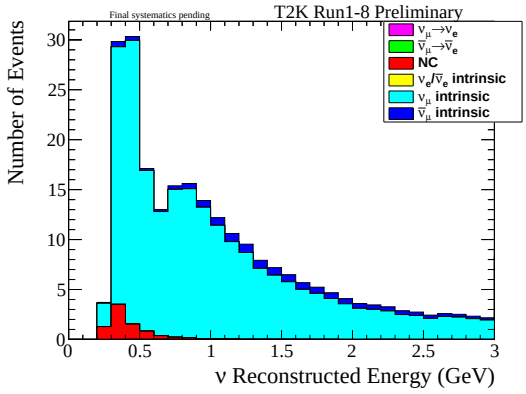


(d) Electron-like antineutrino mode

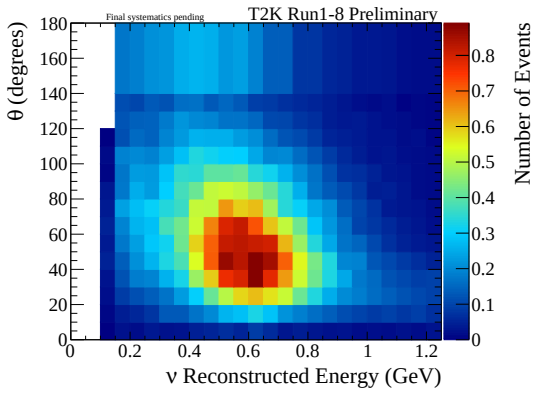


(e) ν_e CC1π^+-like, neutrino mode

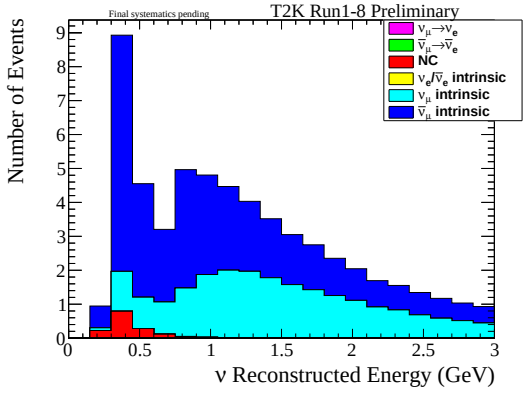
Figure 1: Predicted unoscillated spectra. μ-like distributions are a function of the reconstructed neutrino energy, while the e-like, including ν_e CC1π^+, distributions are functions of both the reconstructed neutrino energy and the reconstructed angle between the outgoing lepton and the neutrino direction. The distributions correspond to the statistics collected in the full Run 1-8 data set. The spectra are generated with the systematic parameters described in section 3.



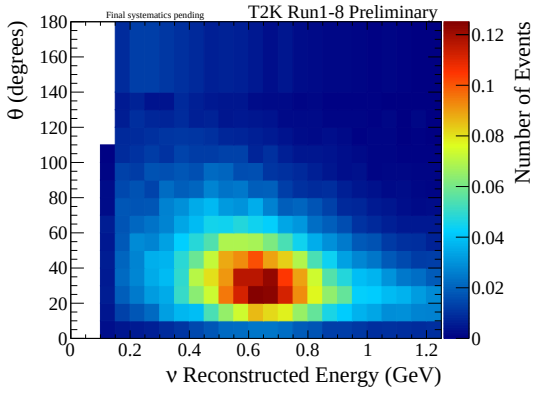
(a) Muon-like, neutrino mode



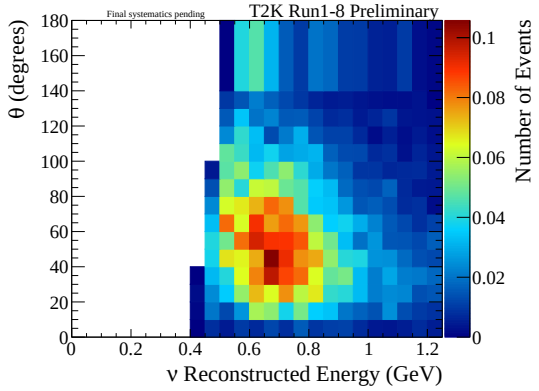
(b) Electron-like, neutrino mode



(c) Muon-like, antineutrino mode



(d) Electron-like antineutrino mode



(e) ν_e CC1π^+-like, neutrino mode

Figure 2: Predicted Asimov A spectra. μ-like distributions are a function of the reconstructed neutrino energy, while the e-like, including ν_e CC1π^+, distributions are functions of both the reconstructed neutrino energy and the reconstructed angle between the outgoing lepton and the neutrino direction. The distributions correspond to the statistics collected in the full Run 1-8 data set. The spectra are generated with the systematic parameters described in section 3 and the oscillation parameters corresponding to the Asimov data set A shown in table 1.

3. Systematic effects in the prediction of the single μ-like ring event energy spectrum

This analysis considers 110 flux, cross section and Super-K detector systematic parameters plus 7 oscillation parameters, which can be treated as either nuisance ps or as parameters of interest, depending on the analysis that is performed.

3.1. Flux and cross-section parameters (73 parameters)

In this analysis, all the flux and cross-section systematic parameters are included in the ND280 data (BANFF) fit. However, since ND280 is not sensitive to the electron neutrino cross section and the NC1γ parameters they remain unconstrained. The parameters fit are 11 ν_μ, 5 $\bar{\nu}_\mu$, 7 ν_e and 2 $\bar{\nu}_e$ neutrino mode (FHC) flux parameters, 5 ν_μ, 11 $\bar{\nu}_\mu$, 2 ν_e and 7 $\bar{\nu}_e$ antineutrino mode (RHC) flux parameters and 23 cross-section parameters. Best-fit values of these parameters and a covariance matrix giving their correlated uncertainties are obtained by the BANFF group from a fit of the $\bar{\nu}_\mu$ and ν_μ ND280 samples as described in Ref [36]. A list of the BANFF parameters, their best-fit values and their prefit and postfit errors can be found in tables 8 to 10.

The best-fit values are used to reweight the templates as described in Appendix A, and the postfit errors in the covariance matrix are used as prior uncertainties in this analysis. All the cross section parameters set to 1 correspond to the NEUT nominal value, except "CC other shape" and "2p2h shape O", for which the NEUT nominal value is 0. For all the normalisation parameters, 1 is the nominal pre-BANFF fit value.

The effects of these correlated uncertainties (see fig. 3) are evaluated using the Cholesky method, that allows throws of the systematic parameters according to their multivariate distribution taking into account existing correlations.

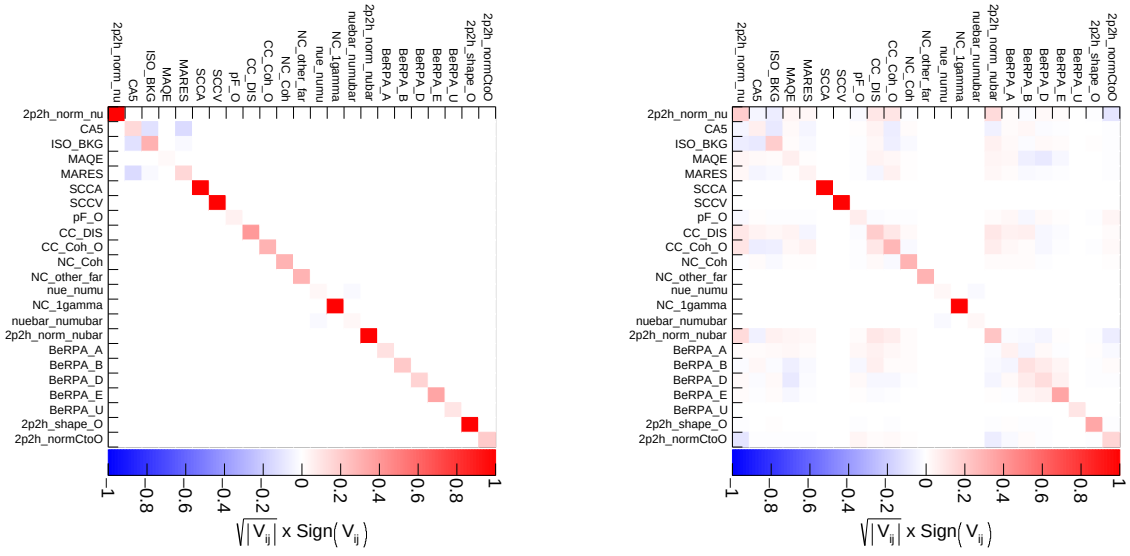


Figure 3: The pre-BANFF (left) and post-BANFF (right) cross-section covariance matrix, with the values shown as the sign of the element times the square root of the absolute value of the element.

Index	Parameter	Description	Best fit	1σ pre/postfit fractional error
0	$f^{banff}_{0;t,r}$	FHC ν_μ flux normalisation, E = 0.0 - 0.4 GeV	1.026	0.099 / 0.057
1	$f^{banff}_{1;t,r}$	FHC ν_μ flux normalisation, E = 0.4 - 0.5 GeV	1.050	0.103 / 0.052
2	$f^{banff}_{2;t,r}$	FHC ν_μ flux normalisation, E = 0.5 - 0.6 GeV	1.032	0.096 / 0.045
3	$f^{banff}_{3;t,r}$	FHC ν_μ flux normalisation, E = 0.6 - 0.7 GeV	0.991	0.087 / 0.042
4	$f^{banff}_{4;t,r}$	FHC ν_μ flux normalisation, E = 0.7 - 1.0 GeV	0.943	0.113 / 0.054
5	$f^{banff}_{5;t,r}$	FHC ν_μ flux normalisation, E = 1.0 - 1.5 GeV	0.963	0.092 / 0.049
6	$f^{banff}_{6;t,r}$	FHC ν_μ flux normalisation, E = 1.5 - 2.5 GeV	1.030	0.070 / 0.042
7	$f^{banff}_{7;t,r}$	FHC ν_μ flux normalisation, E = 2.5 - 3.5 GeV	1.047	0.074 / 0.044
8	$f^{banff}_{8;t,r}$	FHC ν_μ flux normalisation, E = 3.5 - 5.0 GeV	1.038	0.087 / 0.041
9	$f^{banff}_{9;t,r}$	FHC ν_μ flux normalisation, E = 5.0 - 7.0 GeV	1.003	0.098 / 0.040
10	$f^{banff}_{10;t,r}$	FHC ν_μ flux normalisation, E = 7.0 - 30.0 GeV	0.991	0.114 / 0.048
11	$f^{banff}_{11;t,r}$	FHC $\bar{\nu_\mu}$ flux normalisation, E = 0.0 - 0.7 GeV	0.989	0.103 / 0.074
12	$f^{banff}_{12;t,r}$	FHC $\bar{\nu_\mu}$ flux normalisation, E = 0.7 - 1.0 GeV	0.986	0.079 / 0.048
13	$f^{banff}_{13;t,r}$	FHC $\bar{\nu_\mu}$ flux normalisation, E = 1.0 - 1.5 GeV	0.996	0.084 / 0.057
14	$f^{banff}_{14;t,r}$	FHC $\bar{\nu_\mu}$ flux normalisation, E = 1.5 - 2.5 GeV	1.042	0.086 / 0.062
15	$f^{banff}_{15;t,r}$	FHC $\bar{\nu_\mu}$ flux normalisation, E = 2.5 - 30.0 GeV	1.104	0.086 / 0.065
16	$f^{banff}_{16;t,r}$	FHC ν_e flux normalisation, E = 0.0 - 0.5 GeV	1.030	0.090 / 0.046
17	$f^{banff}_{17;t,r}$	FHC ν_e flux normalisation, E = 0.5 - 0.7 GeV	1.031	0.090 / 0.042
18	$f^{banff}_{18;t,r}$	FHC ν_e flux normalisation, E = 0.7 - 0.8 GeV	1.029	0.086 / 0.041
19	$f^{banff}_{19;t,r}$	FHC ν_e flux normalisation, E = 0.8 - 1.5 GeV	1.017	0.081 / 0.039
20	$f^{banff}_{20;t,r}$	FHC ν_e flux normalisation, E = 1.5 - 2.5 GeV	1.034	0.079 / 0.040
21	$f^{banff}_{21;t,r}$	FHC ν_e flux normalisation, E = 2.5 - 4.0 GeV	1.036	0.084 / 0.041
22	$f^{banff}_{22;t,r}$	FHC ν_e flux normalisation, E = 4.0 - 30.0 GeV	1.050	0.094 / 0.059
23	$f^{banff}_{23;t,r}$	FHC $\bar{\nu_e}$ flux normalisation, E = 0.0 - 2.5 GeV	1.050	0.074 / 0.054
24	$f^{banff}_{24;t,r}$	FHC $\bar{\nu_e}$ flux normalisation, E = 2.5 - 30.0 GeV	1.093	0.128 / 0.114

Table 8: Summary of neutrino mode flux systematics included in the VALOR joint fit analysis. A description of how the BANFF fit constrained these systematics can be found in [36].

Many of the cross-section parameters are the same as those used in the previous analysis [12]. However, the binding energy for ^{16}O is no longer included in this analysis and a number of new parameters have been included, see section 3.1.1. Five of these new parameters (2p2h, BeRPA A, B, D and E) are implemented using cubic splines, details for which can be found in [12], while one of the new parameters (BeRPA U) is fixed in the analysis.

3.1.1. Neutrino-nucleus interaction uncertainties

Overall normalisation uncertainties for 2p2h interactions were included in previous analyses, but this analysis includes a 2p2h shape uncertainty (using best-fit value and errors from the BANFF fit) to describe the uncertainty in the cross-section strength between Delta-like, non-Delta-like and interference terms of the 2p2h model, determined by redistributing the cross-section strength of

Index	Parameter	Description	Best fit	1σ pre/postfit fractional error
25	$f^{banff}_{0;t,r}RHC$	RHC ν_μ flux normalisation, E = 0.0 - 0.7 GeV	0.993	0.094 / 0.066
26	$f^{banff}_{1;t,r}RHC$	RHC ν_μ flux normalisation, E = 0.7 - 1.0 GeV	1.003	0.079 / 0.049
27	$f^{banff}_{2;t,r}RHC$	RHC ν_μ flux normalisation, E = 1.0 - 1.5 GeV	1.018	0.077 / 0.045
28	$f^{banff}_{3;t,r}RHC$	RHC ν_μ flux normalisation, E = 1.5 - 2.5 GeV	1.061	0.081 / 0.049
29	$f^{banff}_{4;t,r}RHC$	RHC ν_μ flux normalisation, E = 2.5 - 30.0 GeV	1.051	0.080 / 0.043
30	$f^{banff}_{5;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.0 - 0.4 GeV	1.012	0.104 / 0.065
31	$f^{banff}_{6;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.4 - 0.5 GeV	1.028	0.102 / 0.052
32	$f^{banff}_{7;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.5 - 0.6 GeV	1.008	0.096 / 0.045
33	$f^{banff}_{8;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.6 - 0.7 GeV	0.988	0.085 / 0.041
34	$f^{banff}_{9;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.7 - 1.0 GeV	0.986	0.125 / 0.052
35	$f^{banff}_{10;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 1.0 - 1.5 GeV	0.997	0.105 / 0.047
36	$f^{banff}_{11;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 1.5 - 2.5 GeV	1.036	0.080 / 0.042
37	$f^{banff}_{12;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 2.5 - 3.5 GeV	1.063	0.074 / 0.046
38	$f^{banff}_{13;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 3.5 - 5.0 GeV	1.069	0.094 / 0.063
39	$f^{banff}_{14;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 5.0 - 7.0 GeV	1.044	0.093 / 0.056
40	$f^{banff}_{15;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 7.0 - 30.0 GeV	1.007	0.130 / 0.093
41	$f^{banff}_{16;t,r}RHC$	RHC ν_e flux normalisation, E = 0.0 - 2.5 GeV	1.050	0.069 / 0.047
42	$f^{banff}_{17;t,r}RHC$	RHC ν_e flux normalisation, E = 2.5 - 30.0 GeV	1.048	0.085 / 0.065
43	$f^{banff}_{18;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 0.0 - 0.5 GeV	1.018	0.095 / 0.051
44	$f^{banff}_{19;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 0.5 - 0.7 GeV	1.018	0.091 / 0.043
45	$f^{banff}_{20;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 0.7 - 0.8 GeV	1.016	0.091 / 0.044
46	$f^{banff}_{21;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 0.8 - 1.5 GeV	1.018	0.084 / 0.040
47	$f^{banff}_{22;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 1.5 - 2.5 GeV	1.043	0.080 / 0.051
48	$f^{banff}_{23;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 2.5 - 4.0 GeV	1.046	0.089 / 0.064
49	$f^{banff}_{24;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 4.0 - 30.0 GeV	1.083	0.156 / 0.133

Table 9: Summary of antineutrino mode flux systematics included in the VALOR joint fit analysis. A description of how the BANFF fit constrained these systematics can be found in [36].

these terms subject to (approximately) maintaining the total 2p2h event rate (see Section 2 of [14]). In terms of energy transfer (q0) and three-momentum transfer (q3) kinematics, the distribution of events in this space for neutrino and antineutrino interactions on oxygen are shown in figs. 4 and 5 respectively (see Section 2 of [14]).

This analysis also introduces 5 BeRPA parameters (A, B, D, E and U) to describe the collective effects of interactions between neutrinos and nucleons inside nuclei. These effects modify the Q^2 dependence of the neutrino-nucleus cross-section and in BeRPA are described by a cubic polynomial transitioning to an exponential above a Q^2 threshold, as described in [15]. Neither the 2p2h shape parameter nor the BeRPA parameters apply to interactions on free protons.

Index	Parameter	Description	Best fit	1σ pre/postfit fractional error
50	$f^{banff}_{Norm_{2p2h}}$	Two particle two hole normalisation for ^{16}O	1.426	1.000 / 0.182
51	$f^{banff}_{C_5^A}$	C_5^A nucleon to Δ transition axial form factor	0.977	0.149 / 0.059
52	$f^{banff}_{Bg_{RES}}$	Scale of isospin 1/2 nonresonant background	0.979	0.308 / 0.190
53	$f^{banff}_{M_A^{QE}}$	CCQE axial-mass scaling factor	0.903	0.025 / 0.057
54	$f^{banff}_{M_A^{RES}}$	Resonance-production axial-mass scaling factor	0.822	0.158 / 0.045
55	f^{banff}_{SCCA}	BANFF; Second current class axial	1.000	- / -
56	f^{banff}_{SCCV}	BANFF; Second current class vector	1.000	- / -
57	$f^{banff}_{p_f}$	Fermi momentum for ^{16}O	0.916	0.058 / 0.072
58	$f^{banff}_{Shape_{CCoth}}$	CC other shape	0.519	0.400 / 0.187
59	$f^{banff}_{Norm_{CCcoh}}$	CC coherent for ^{16}O normalisation	0.906	0.300 / 0.275
60	$f^{banff}_{Norm_{NCcoh}}$	NC coherent normalisation	0.939	0.300 / 0.297
61	$f^{banff}_{Norm_{NCother}}$	NC other normalisation	1.000	- / 0.300
62	$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$	CC ν_e normalisation	1.000	- / 0.028
63	$f^{banff}_{Norm_{NC1\gamma}}$	NC 1γ normalisation	1.000	- / 1.000
64	$f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$	CC $\bar{\nu}_e$ normalisation	1.000	- / 0.028
65	$f^{banff}_{Norm_{2p2hBar}}$	Antineutrino two particle two hole normalisation for ^{16}O	0.552	1.000 / 0.201
66	$f^{banff}_{Shape_{BeRPA_A}}$	Bernstein Polynomial coefficient A	0.663	0.118 / 0.056
67	$f^{banff}_{Shape_{BeRPA_B}}$	Bernstein Polynomial coefficient B	1.647	0.210 / 0.117
68	$f^{banff}_{Shape_{BeRPA_D}}$	Bernstein Polynomial coefficient D	0.988	0.170 / 0.128
69	$f^{banff}_{Shape_{BeRPA_E}}$	Bernstein Polynomial coefficient E	0.876	0.352 / 0.355
70	$f^{banff}_{Shape_{BeRPA_U}}$	Bernstein Polynomial coefficient U	1.200	0.100 / 0.100
71	$f^{banff}_{Shape_{2p2h\nu}}$	Neutrino two particle two hole ^{16}O shape	0.997	3.000 / 0.338
72	$f^{banff}_{Norm_{2p2hCtoO}}$	Two particle two hole ^{12}C to ^{16}O normalisation	0.940	0.200 / 0.161

Table 10: Summary of cross section systematics included in the VALOR joint fit analysis. Parameters with no prefit error were unconstrained. A description of how the BANFF fit constrained these systematics can be found in [36].

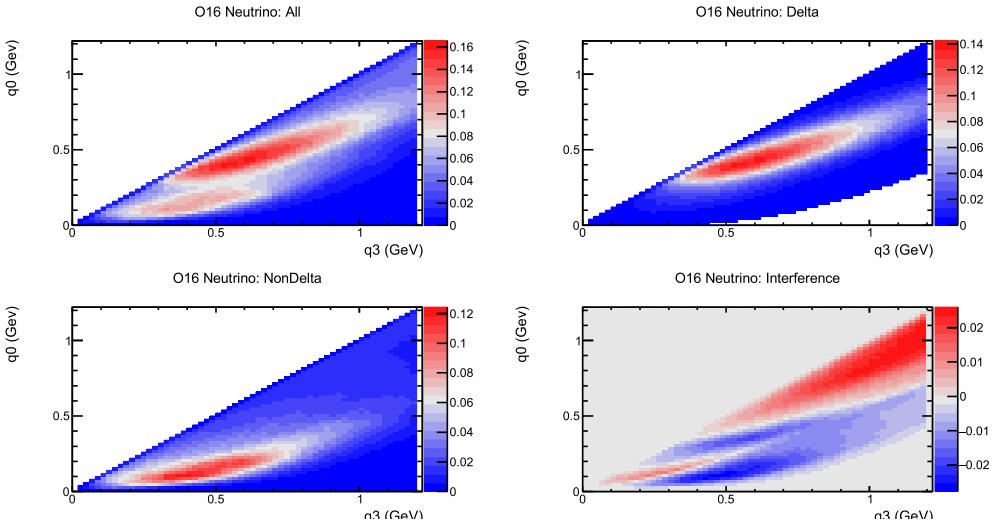


Figure 4: Distribution of neutrino events for 2p2h and each 2p2h term in q0-q3 space [14].

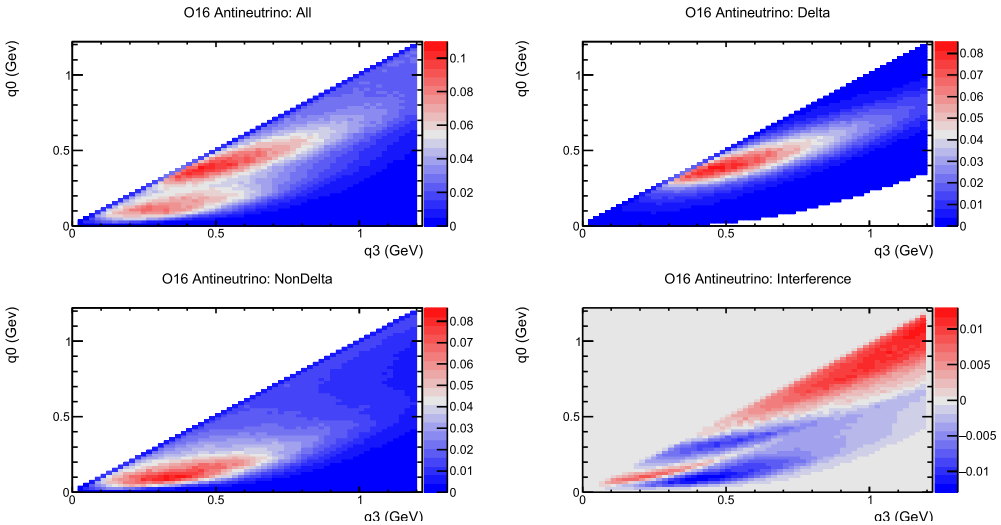


Figure 5: Distribution of antineutrino events for 2p2h and each 2p2h term in q0-q3 space [14].

3.2. Super-K efficiencies & effects of intranuclear and secondary re-interactions (44 parameters)

The Super-K detector uncertainties include the efficiencies of the fiducial volume and reduction chain, and the OD, ring-counting, PID, momentum and decay-electron cuts. These uncertainties have been evaluated by comparisons between SK-IV atmospheric data and atmospheric neutrino Monte Carlo. Ring counting errors are evaluated using ν_μ CCQE, ν_μ CCnQE, NC, and ν_e CC-enriched control samples from atmospheric neutrinos. The NC PID error is evaluated using a NC-enriched control sample, obtained by not using the PID information during the selection. All the Monte Carlo control samples are fitted simultaneously to the data with a χ^2 function using the selection efficiencies as fit parameters [37] [38].

Intranuclear final-state interactions (FSI) have significant effects in the T2K energy range and result in event topologies, for scattering from nuclear targets, which are drastically different from those created in neutrino scattering from free nucleons. Uncertainties due to FSI were estimated by simultaneously varying NEUT parameters that scale interaction probabilities as described in section 2 of [26]. Uncertainties due to secondary pion interactions (SI) were also evaluated by allowing variations in the interaction probabilities. Details of this are given in [29]. Since the same model is used for FSI and SI, it is possible to evaluate the uncertainties in both FSI and SI simultaneously. Details of this procedure are described in section 8 of [25].

A list of the Super-K parameters and their errors can be found in table 11. We consider the Super-K detector efficiency and FSI+SI+PN systematics together, adding their covariance matrices linearly and evaluating the effects of these uncertainties using the Cholesky method. Details of all parameters can be found in [12].

3.3. Super-K energy scale (1 parameter)

The systematic parameter $f^{SK}_{E;r}$ is included to estimate the effects of uncertainty in the Super-K reconstructed energy scale. This uncertainty is estimated to be 2.4% [39]; its effects are calculated

Index	Parameter	Description	1σ fractional error
0	$f^{SK+FSI}_{0;t,r}$	SKDet + FSI/SI 0; Ereco range 0.00 - 0.40 GeV; numu/numubar CCQE (1Rmu)	0.009
1	$f^{SK+FSI}_{1;t,r}$	SKDet + FSI/SI 1; Ereco range 0.40 - 1.10 GeV; numu/numubar CCQE (1Rmu)	0.007
2	$f^{SK+FSI}_{2;t,r}$	SKDet + FSI/SI 2; Ereco range 1.10 - 30.00 GeV; numu/numubar CCQE (1Rmu)	0.007
3	$f^{SK+FSI}_{3;t,r}$	SKDet + FSI/SI 3; Ereco range 0.00 - 30.00 GeV; numu/numubar CCnQE (1Rmu)	0.169
4	$f^{SK+FSI}_{4;t,r}$	SKDet + FSI/SI 4; Ereco range 0.00 - 30.00 GeV; nue/nuebar/signue CC (1Rmu)	1.005
5	$f^{SK+FSI}_{5;t,r}$	SKDet + FSI/SI 5; Ereco range 0.00 - 30.00 GeV; all NC (1Rmu)	0.658
6	$f^{SK+FSI}_{6;t,r}$	SKDet + FSI/SI 6; Ereco range 0.00 - 0.35 GeV; oscillated nue CC (1Re)	0.181
7	$f^{SK+FSI}_{7;t,r}$	SKDet + FSI/SI 7; Ereco range 0.35 - 0.80 GeV; oscillated nue CC (1Re)	0.036
8	$f^{SK+FSI}_{8;t,r}$	SKDet + FSI/SI 8; Ereco range 0.80 - 1.25 GeV; oscillated nue CC (1Re)	0.042
9	$f^{SK+FSI}_{9;t,r}$	SKDet + FSI/SI 9; Ereco range 0.00 - 0.35 GeV; numu/numubar CC (1Re)	0.300
10	$f^{SK+FSI}_{10;t,r}$	SKDet + FSI/SI 10; Ereco range 0.35 - 0.80 GeV; numu/numubar CC (1Re)	0.321
11	$f^{SK+FSI}_{11;t,r}$	SKDet + FSI/SI 11; Ereco range 0.80 - 1.25 GeV; numu/numubar CC (1Re)	0.393
12	$f^{SK+FSI}_{12;t,r}$	SKDet + FSI/SI 12; Ereco range 0.00 - 0.35 GeV; nue/nuebar CC (1Re)	0.135
13	$f^{SK+FSI}_{13;t,r}$	SKDet + FSI/SI 13; Ereco range 0.35 - 0.80 GeV; nue/nuebar CC (1Re)	0.070
14	$f^{SK+FSI}_{14;t,r}$	SKDet + FSI/SI 14; Ereco range 0.80 - 1.25 GeV; nue/nuebar CC (1Re)	0.077
15	$f^{SK+FSI}_{15;t,r}$	SKDet + FSI/SI 15; Ereco range 0.00 - 0.35 GeV; all NC (1Re)	0.194
16	$f^{SK+FSI}_{16;t,r}$	SKDet + FSI/SI 16; Ereco range 0.35 - 0.80 GeV; all NC (1Re)	0.181
17	$f^{SK+FSI}_{17;t,r}$	SKDet + FSI/SI 17; Ereco range 0.80 - 1.25 GeV; all NC (1Re)	0.472
18	$f^{SK+FSI}_{0;t,r}RHC$	SKDet + FSI/SI 0; Ereco range 0.00 - 0.40 GeV; numu/numubar CCQE (1Rmu); RHC	0.009
19	$f^{SK+FSI}_{1;t,r}RHC$	SKDet + FSI/SI 1; Ereco range 0.40 - 1.10 GeV; numu/numubar CCQE (1Rmu); RHC	0.007
20	$f^{SK+FSI}_{2;t,r}RHC$	SKDet + FSI/SI 2; Ereco range 1.10 - 30.00 GeV; numu/numubar CCQE (1Rmu); RHC	0.007
21	$f^{SK+FSI}_{3;t,r}RHC$	SKDet + FSI/SI 3; Ereco range 0.00 - 30.00 GeV; numu/numubar CCnQE (1Rmu); RHC	0.130
22	$f^{SK+FSI}_{4;t,r}RHC$	SKDet + FSI/SI 4; Ereco range 0.00 - 30.00 GeV; nue/nuebar/signue CC (1Rmu); RHC	1.005
23	$f^{SK+FSI}_{5;t,r}RHC$	SKDet + FSI/SI 5; Ereco range 0.00 - 30.00 GeV; all NC (1Rmu); RHC	0.657
24	$f^{SK+FSI}_{6;t,r}RHC$	SKDet + FSI/SI 6; Ereco range 0.00 - 0.35 GeV; oscillated nue CC (1Re); RHC	0.108
25	$f^{SK+FSI}_{7;t,r}RHC$	SKDet + FSI/SI 7; Ereco range 0.35 - 0.80 GeV; oscillated nue CC (1Re); RHC	0.036
26	$f^{SK+FSI}_{8;t,r}RHC$	SKDet + FSI/SI 8; Ereco range 0.80 - 1.25 GeV; oscillated nue CC (1Re); RHC	0.057
27	$f^{SK+FSI}_{9;t,r}RHC$	SKDet + FSI/SI 9; Ereco range 0.00 - 0.35 GeV; numu/numubar CC (1Re); RHC	0.349
28	$f^{SK+FSI}_{10;t,r}RHC$	SKDet + FSI/SI 10; Ereco range 0.35 - 0.80 GeV; numu/numubar CC (1Re); RHC	0.342
29	$f^{SK+FSI}_{11;t,r}RHC$	SKDet + FSI/SI 11; Ereco range 0.80 - 1.25 GeV; numu/numubar CC (1Re); RHC	0.417
30	$f^{SK+FSI}_{12;t,r}RHC$	SKDet + FSI/SI 12; Ereco range 0.00 - 0.35 GeV; nue/nuebar CC (1Re); RHC	0.085
31	$f^{SK+FSI}_{13;t,r}RHC$	SKDet + FSI/SI 13; Ereco range 0.35 - 0.80 GeV; nue/nuebar CC (1Re); RHC	0.056
32	$f^{SK+FSI}_{14;t,r}RHC$	SKDet + FSI/SI 14; Ereco range 0.80 - 1.25 GeV; nue/nuebar CC (1Re); RHC	0.078
33	$f^{SK+FSI}_{15;t,r}RHC$	SKDet + FSI/SI 15; Ereco range 0.00 - 0.35 GeV; all NC (1Re); RHC	0.214
34	$f^{SK+FSI}_{16;t,r}RHC$	SKDet + FSI/SI 16; Ereco range 0.35 - 0.80 GeV; all NC (1Re); RHC	0.191
35	$f^{SK+FSI}_{17;t,r}RHC$	SKDet + FSI/SI 17; Ereco range 0.80 - 1.25 GeV; all NC (1Re); RHC	0.465
36	$f^{SK+FSI}_{0;t,r}MultiRing$	SKDet + FSI/SI 0; Ereco range 0.30 - 0.80 GeV; oscillated nue CC (MultiRe); MultiRing	0.249
37	$f^{SK+FSI}_{1;t,r}MultiRing$	SKDet + FSI/SI 1; Ereco range 0.80 - 1.25 GeV; oscillated nue CC (MultiRe); MultiRing	0.215
38	$f^{SK+FSI}_{2;t,r}MultiRing$	SKDet + FSI/SI 2; Ereco range 0.30 - 0.80 GeV; numu/numubar CC (MultiRe); MultiRing	0.507
39	$f^{SK+FSI}_{3;t,r}MultiRing$	SKDet + FSI/SI 3; Ereco range 0.80 - 1.25 GeV; numu/numubar CC (MultiRe); MultiRing	0.236
40	$f^{SK+FSI}_{4;t,r}MultiRing$	SKDet + FSI/SI 4; Ereco range 0.30 - 0.80 GeV; nue/nuebar CC (MultiRe); MultiRing	0.244
41	$f^{SK+FSI}_{5;t,r}MultiRing$	SKDet + FSI/SI 5; Ereco range 0.80 - 1.25 GeV; nue/nuebar CC (MultiRe); MultiRing	0.235
42	$f^{SK+FSI}_{6;t,r}MultiRing$	SKDet + FSI/SI 6; Ereco range 0.30 - 0.80 GeV; all NC (MultiRe); MultiRing	0.983
43	$f^{SK+FSI}_{7;t,r}MultiRing$	SKDet + FSI/SI 7; Ereco range 0.80 - 1.25 GeV; all NC (MultiRe); MultiRing	0.524
44	$f^{SK}_{E;r}$	SK energy scale	0.024

Table 11: Summary of SK detector + FSI + SI + PN systematics included in the VALOR joint fit analysis.

by scaling the bin edges of the MC templates and, assuming uniform distribution of events within the bins, calculating the number of events gained from/lost to other bins (including bins that are not immediately adjacent). This systematic is applied after all other systematics have been applied, because it affects reconstructed rather than true energy, does not commute with spline-based systematics, which are generated without consideration of energy scale variation and also does not commute with the other Super-K systematics, which are determined from data without any energy scale variation.

Parameter(s)	Prior	Range
$\sin^2\theta_{23}$	uniform	$[0.3; 0.7]$
$\sin^2 2\theta_{13}$ ($\sin^2\theta_{13}$) reactors	gaussian	0.0857 ± 0.0046
$\sin^2 2\theta_{13}$ ($\sin^2\theta_{13}$) T2K only	uniform	$[0; 0.4]$
$\sin^2 2\theta_{12}$	gaussian	0.846 ± 0.021
$\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{13}\rvert$ (IH)	uniform	$[2.3; 2.7] \times 10^{-3}$ eV^2/c^4
Δm^2_{21}	gaussian	$(7.53 \pm 0.18) \times 10^{-5}$ eV^2/c^4
δ_{CP}	uniform	$[-\pi; +\pi]$
Mass Hierarchy	fixed	NH or IH

Table 12: Treatment of the oscillation parameters in the $\nu/\bar{\nu}$ joint analysis. All the gaussian priors are from [81]. The parameter $\sin^2 2\theta_{13}$ can be constrained with the measurement of reactor experiments using the gaussian prior, otherwise a uniform prior is used and the measurement relies only on the T2K data. The nominal values of $\sin^2 2\theta_{13}$, $\sin^2 2\theta_{12}$ and Δm^2_{21} are from [81], while all the other oscillation parameter values corresponds to the most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [20]. The mass hierarchy is not marginalised but fixed to either NH or IH.

3.4. Oscillation parameters and mass hierarchy (7 parameters)

The 3-flavour oscillation probability is used. All the nuisance oscillation parameters, i.e. those parameters which are not shown in a given contour, are marginalised except the mass hierarchy which is fixed to either normal or inverted hierarchy.

For $\sin^2\theta_{13}$ two different priors can be used, depending on whether the reactor constraint is applied, denoted as "reactors", when applied and "T2K only" when not applied.

3.5. Systematic error on Super-K predictions

The effect of each category of systematic parameter on the expected event rate for Super-K sample is given in tables 43 to 47 (details of per-sample event rates for 1σ and 3σ variations of each independent systematic parameter can be found in Appendix C). The mean and RMS of 10000 throws of the respective systematics (with correlations taken into account) is computed for each sample and in the case of pre-BANFF and post-BANFF errors. When not measuring the effect of oscillation parameters variations, the oscillation parameters are fixed at Asimov A values (see section 2.1.1).

For the measurement of δ_{CP} it is important to understand the effect of the systematic uncertainties on the ratio of the number of events between neutrino and antineutrino mode samples. If correlations do not reproduce the physics correctly, a fake asymmetry between neutrino and antineutrino oscillations can be seen. Furthermore, if correlations exist but are not taken into account, the measurement of δ_{CP} could be biased. The true value of δ_{CP} determines whether correlations or anti-correlations between systematic parameters have the greatest effect in hiding any oscillation asymmetry. In particular, if true $\delta_{CP} = \pm\pi/2$ then anti-correlation in the e-like samples could hide any oscillation asymmetry. Alternatively, in the case that true $\delta_{CP} \sim 0$ it is uncorrelated parameters that reduce the sensitivity to δ_{CP}. The uncertainties on the ratios between neutrino and antineutrino mode events due to the systematic parameters are shown in table 13. The most critical systematic parameters are σ_{ν_e} and $\sigma_{\bar{\nu}_e}$ which are applied respectively to ν_e and $\bar{\nu}_e$ events.

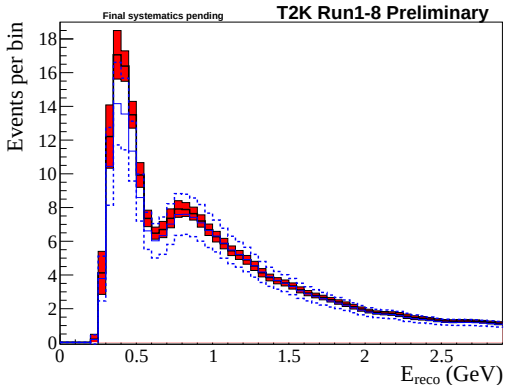
Their fractional error from the BANFF covariance matrix is 2.83% (see table 10) and they are partially anti-correlated with non-diagonal elements in the covariance matrix of -0.0004.

It can be seen in tables 43 to 47 (Appendix E) that the reduction in the error on flux and constrained cross-section systematics achieved through the BANFF fit reduces the fractional error due to all systematics from $\sim 14\%$ to $\sim 4\%$ and $\sim 11\%$ to $\sim 3\%$ in the neutrino- and antineutrino-mode μ-like samples respectively, from $\sim 16\%$ to $\sim 7\%$ and $\sim 13\%$ to $\sim 8\%$ in the neutrino- and antineutrino-mode e-like samples respectively, and from $\sim 25\%$ to $\sim 22\%$ in the ν_e CC1π^+-like sample. A summary of the effects of the systematic errors is given in table 13.

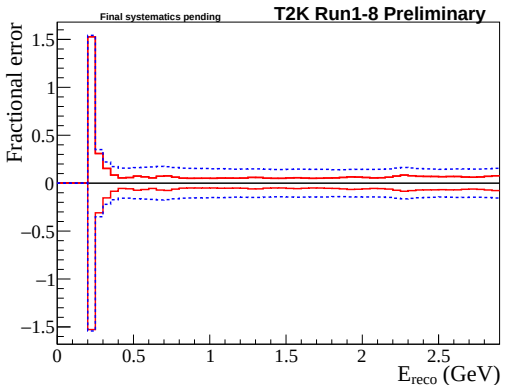
	1-Ring μ		**1-Ring e**			
Error source	FHC	RHC	FHC	RHC	FHC 1 d.e.	FHC/RHC
SK Detector	1.86	1.51	3.03	4.22	16.69	1.60
SK FSI+SI+PN	2.20	1.98	3.01	2.31	11.43	1.57
Flux + Xsec constrained	3.22	2.72	3.22	2.88	4.05	2.50
$\sigma(\nu_e)/\sigma(\bar{\nu}_e)$	0.00	0.00	2.63	1.46	2.62	3.03
NC1γ	0.00	0.00	1.08	2.59	0.33	1.49
NC Other	0.25	0.25	0.14	0.33	0.98	0.18
Osc	0.04	0.03	3.86	3.60	3.78	0.79
All Systematics	4.40	3.76	6.10	6.51	20.94	4.77
All with osc	4.40	3.76	7.27	7.44	21.24	4.85

Table 13: Percentage error on event rate by error source and sample. Final column is the percentage error on the ratio of FHC/RHC events in the one-ring e sample.

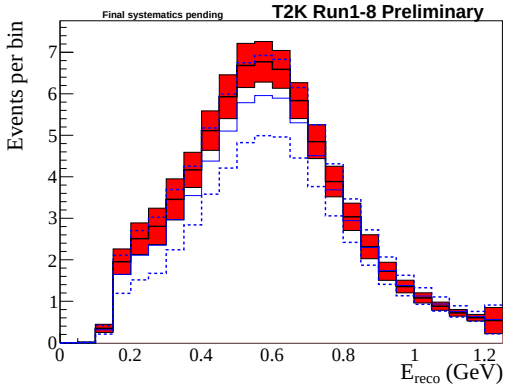
The error envelopes corresponding to tables 43 to 47 are shown in figs. 6 to 21. These error envelopes are determined by performing 1×10^4 correlated throws of the systematic parameter group under consideration and plotting the mean and Gaussian 1σ error resulting.



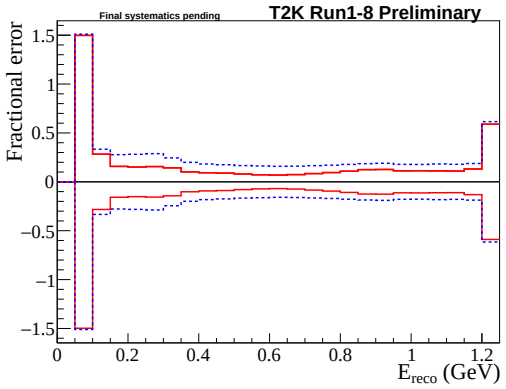
(a) Muon-like, neutrino mode



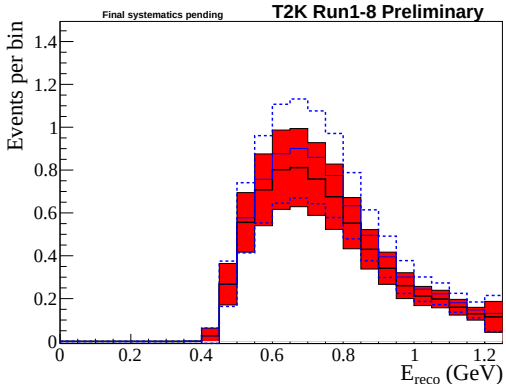
(b) Muon-like, neutrino mode



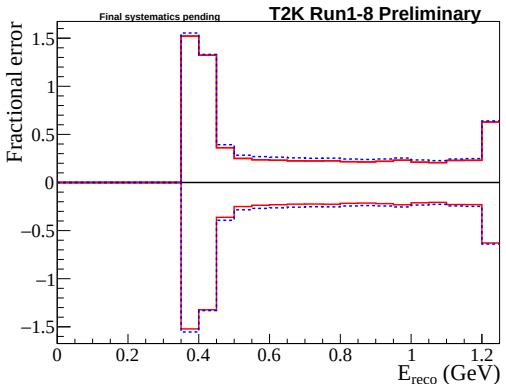
(c) Electron-like, neutrino mode



(d) Electron-like, neutrino mode

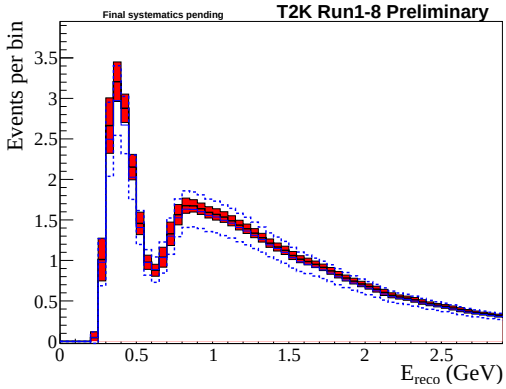


(e) ν_e CC1π^+-like, neutrino mode

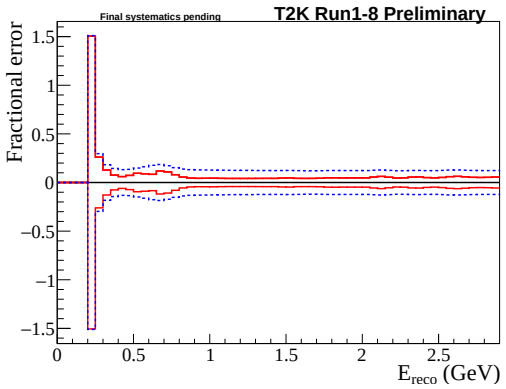


(f) ν_e CC1π^+-like, neutrino mode

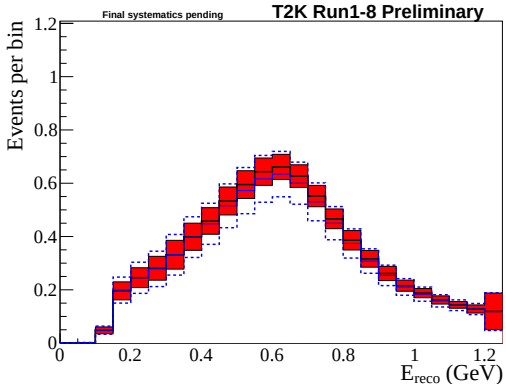
Figure 6: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum for all systematic parameters for pre-BANFF (blue) and post-BANFF (red) errors. 1×10^4 toys are generated for the Asimov data set A with randomized systematic parameters with correlations taken into account. Since δ_{CP}, $\sin^2\theta_{23}$, Δm^2_{32} and the mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.



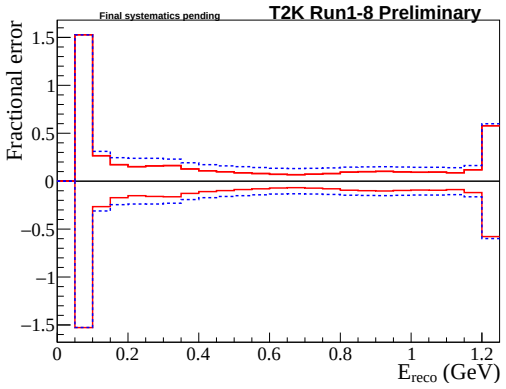
(a) Muon-like, antineutrino mode



(b) Muon-like, antineutrino mode

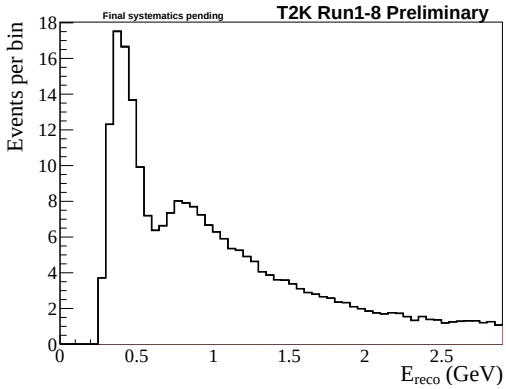


(c) Electron-like antineutrino mode

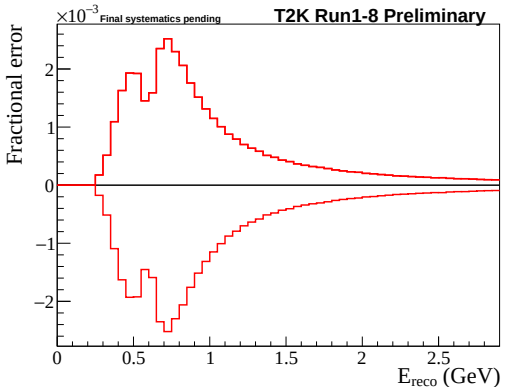


(d) Electron-like antineutrino mode

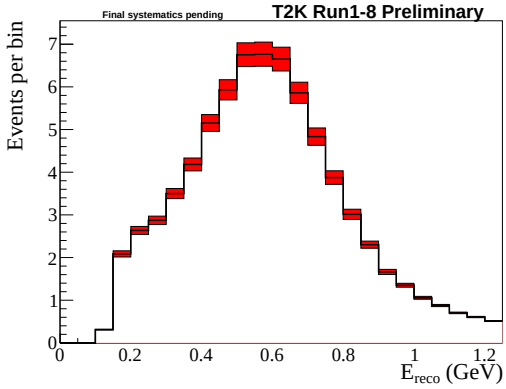
Figure 7: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum for all systematic parameters for pre-BANFF (blue) and post-BANFF (red) errors. 1×10^4 toys are generated for the Asimov data set A with randomized systematic parameters with correlations taken into account. Since δ_{CP}, $\sin^2\theta_{23}$, Δm^2_{32} and the mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.



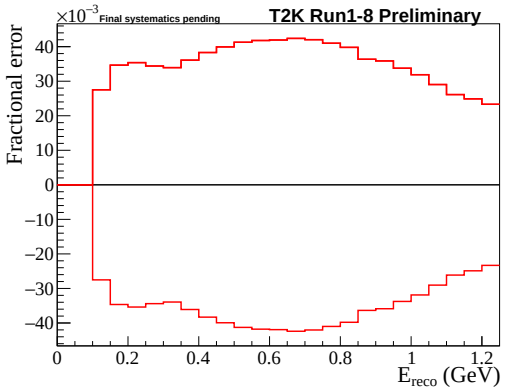
(a) Muon-like, neutrino mode



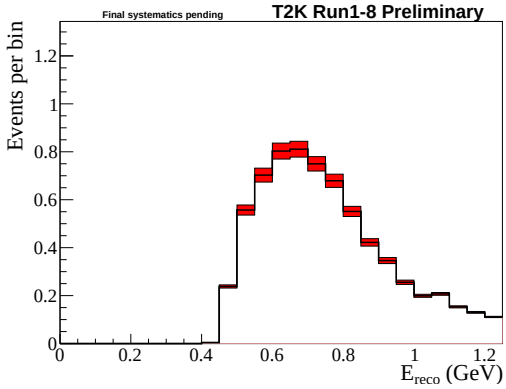
(b) Muon-like, neutrino mode



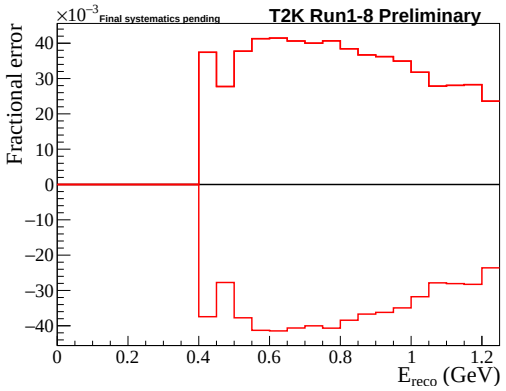
(c) Electron-like, neutrino mode



(d) Electron-like, neutrino mode

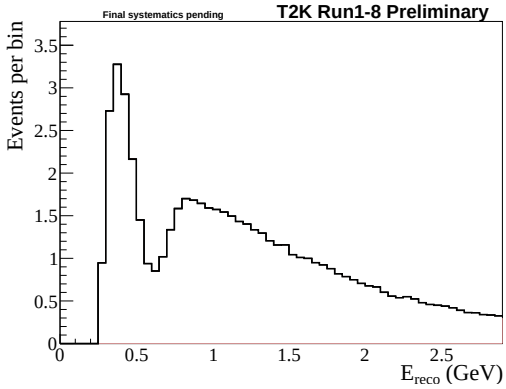


(e) ν_e CC1π^+-like, neutrino mode

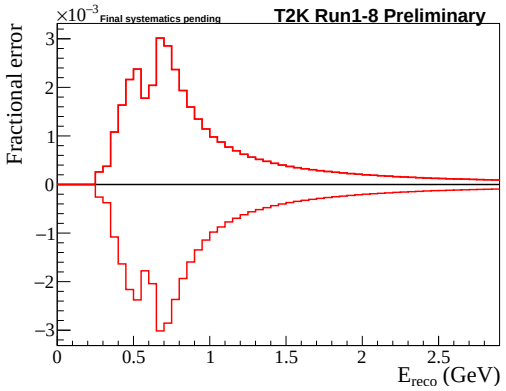


(f) ν_e CC1π^+-like, neutrino mode

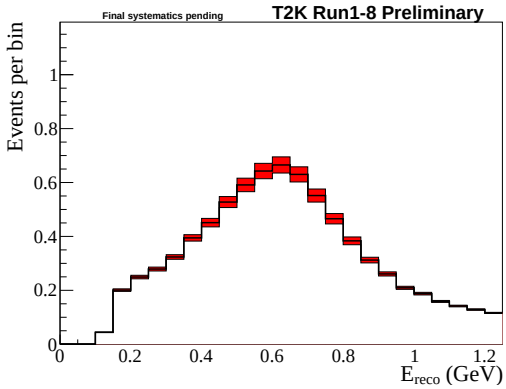
Figure 8: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum for all oscillation nuisance parameters. 1×10^4 toys are generated for the Asimov data set A with randomized systematic parameters with correlations taken into account. Since δ_{CP}, $\sin^2\theta_{23}$, Δm^2_{32} and the mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.



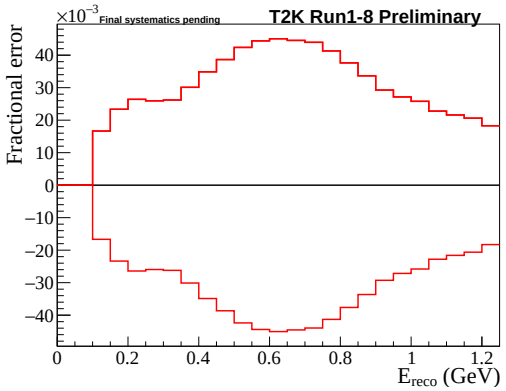
(a) Muon-like, antineutrino mode



(b) Muon-like, antineutrino mode

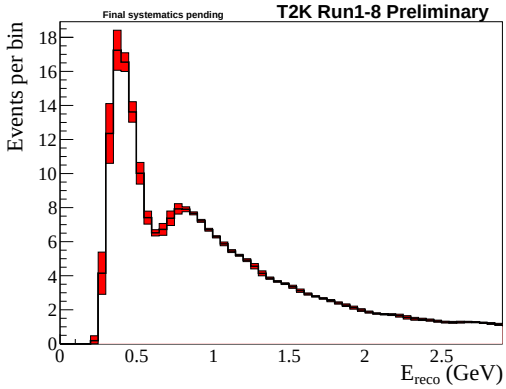


(c) Electron-like antineutrino mode

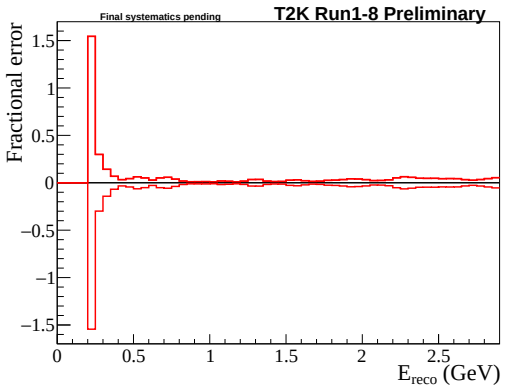


(d) Electron-like antineutrino mode

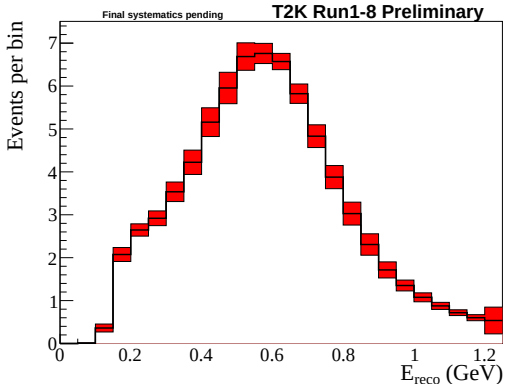
Figure 9: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum for all oscillation nuisance parameters. 1×10^4 toys are generated for the Asimov data set A with randomized systematic parameters with correlations taken into account. Since δ_{CP}, $\sin^2\theta_{23}$, Δm^2_{32} and the mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.



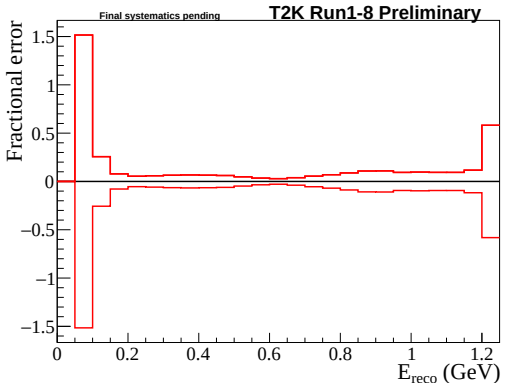
(a) Muon-like, neutrino mode



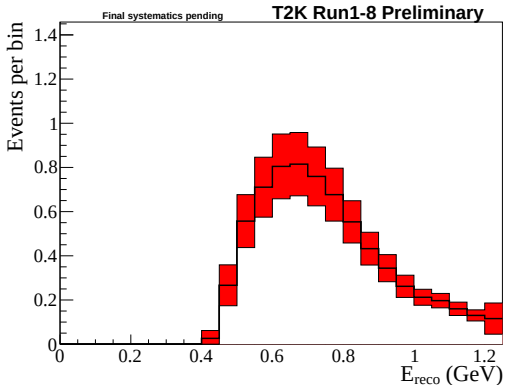
(b) Muon-like, neutrino mode



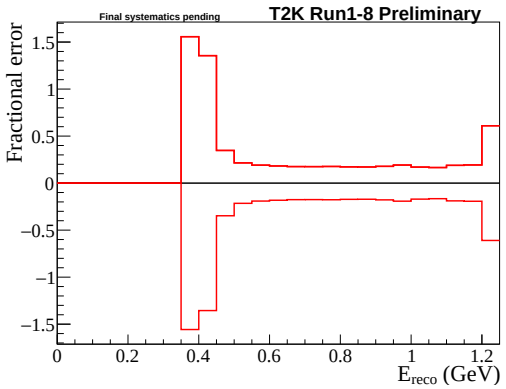
(c) Electron-like, neutrino mode



(d) Electron-like, neutrino mode

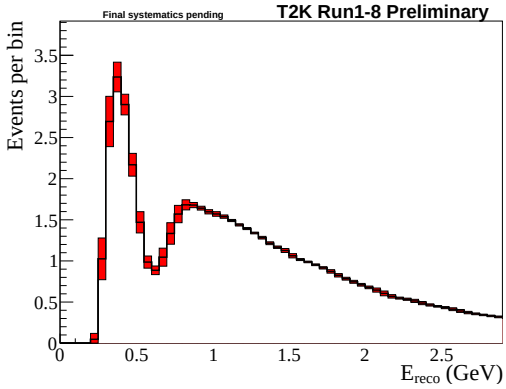


(e) ν_e CC1π^+-like, neutrino mode

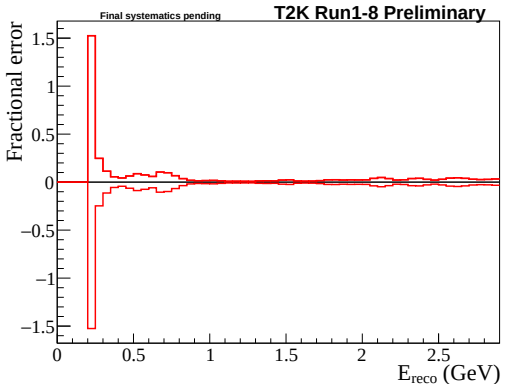


(f) ν_e CC1π^+-like, neutrino mode

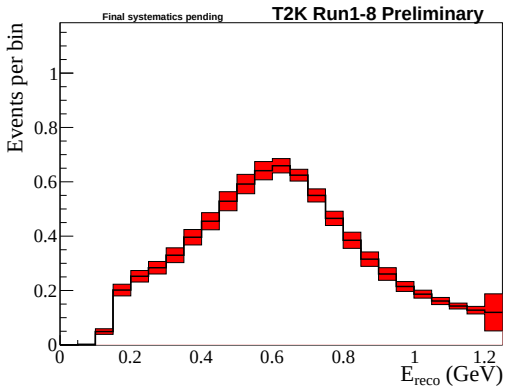
Figure 10: Error envelopes (left) and fractional error (right) for the reconstructed neutrino energy spectrum for Super-K detector systematic parameters. 1×10^4 toys are generated for the Asimov data set A with randomized systematic parameters with correlations taken into account.



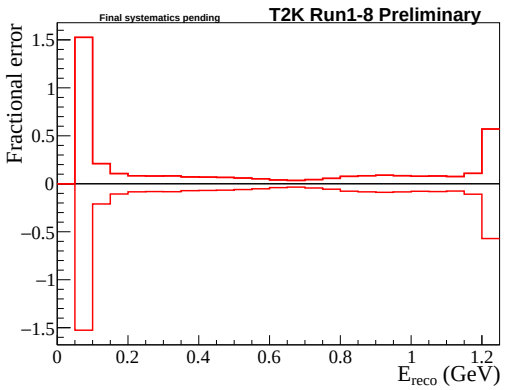
(a) Muon-like, antineutrino mode



(b) Muon-like, antineutrino mode

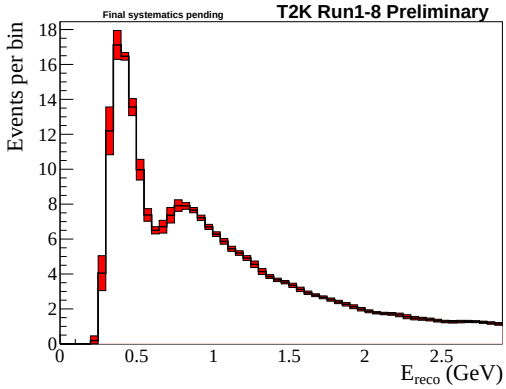


(c) Electron-like antineutrino mode

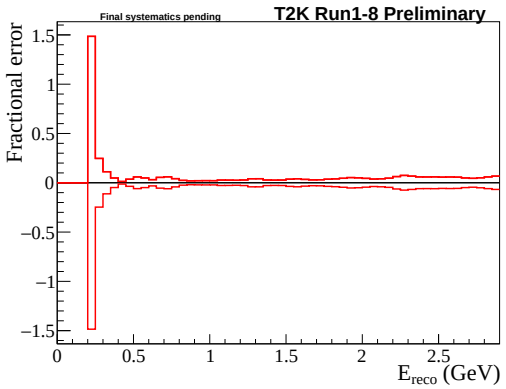


(d) Electron-like antineutrino mode

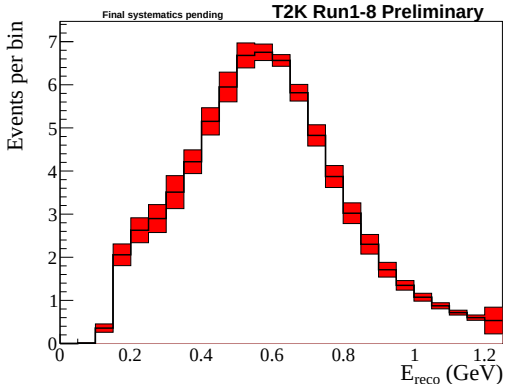
Figure 11: Error envelopes (left) and fractional error (right) for the reconstructed neutrino energy spectrum for Super-K detector systematic parameters. 1×10^4 toys are generated for the Asimov data set A with randomized systematic parameters with correlations taken into account.



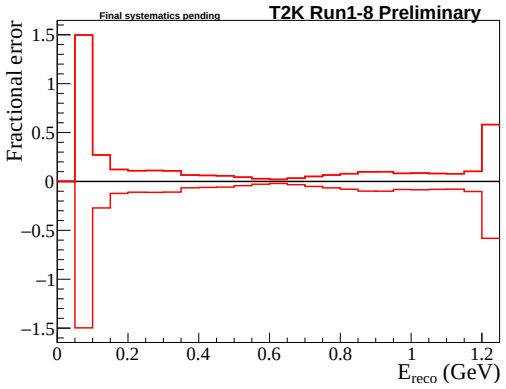
(a) Muon-like, neutrino mode



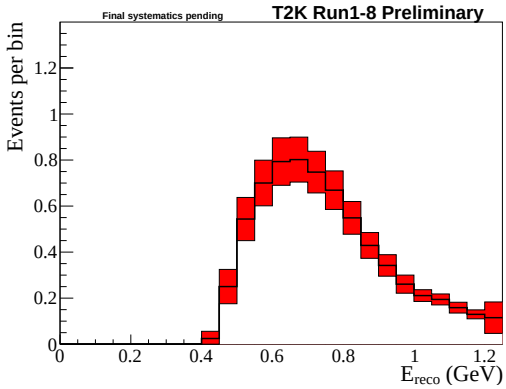
(b) Muon-like, neutrino mode



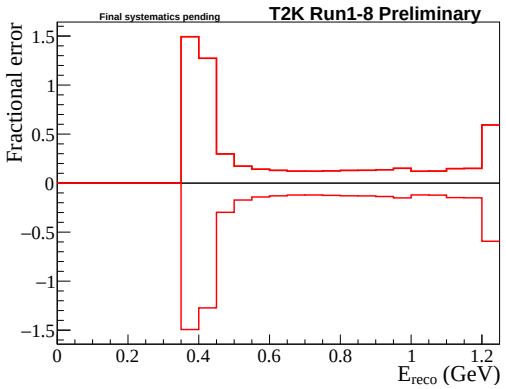
(c) Electron-like, neutrino mode



(d) Electron-like, neutrino mode

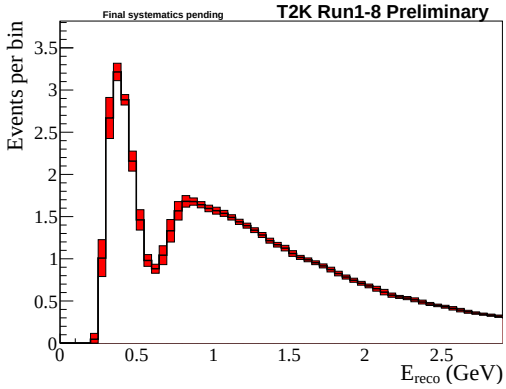


(e) ν_e CC1π^+-like, neutrino mode

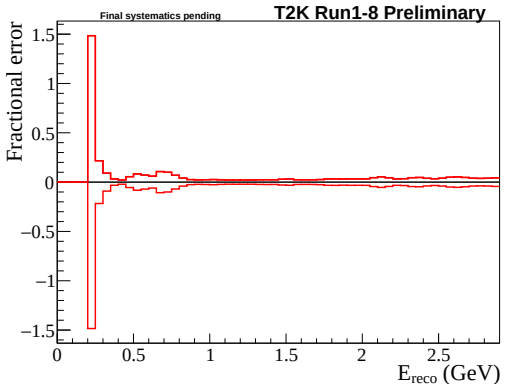


(f) ν_e CC1π^+-like, neutrino mode

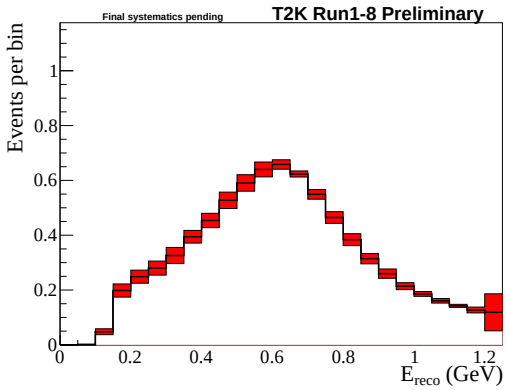
Figure 12: Error envelopes (left) and fractional error (right) for the reconstructed neutrino energy spectrum for Super-K FSI+SI+PN systematic parameters. 1×10^4 toys are generated for the Asimov data set A with randomized systematic parameters with correlations taken into account.



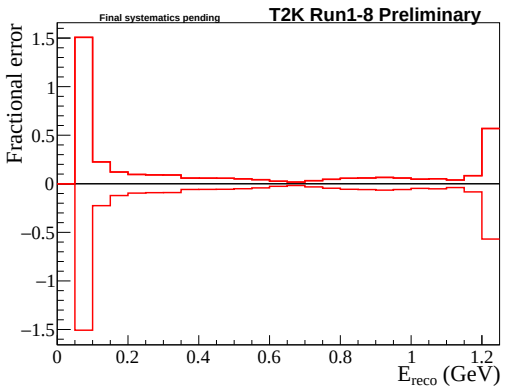
(a) Muon-like, antineutrino mode



(b) Muon-like, antineutrino mode

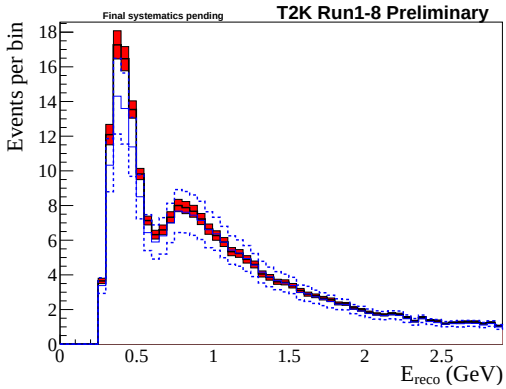


(c) Electron-like antineutrino mode

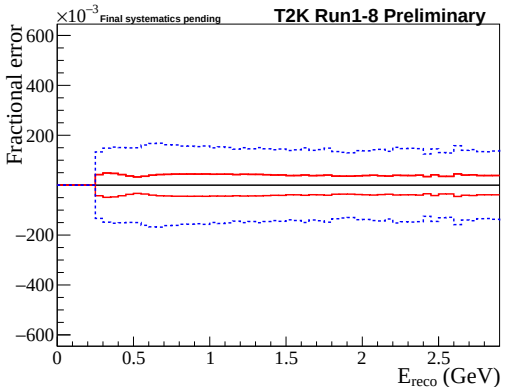


(d) Electron-like antineutrino mode

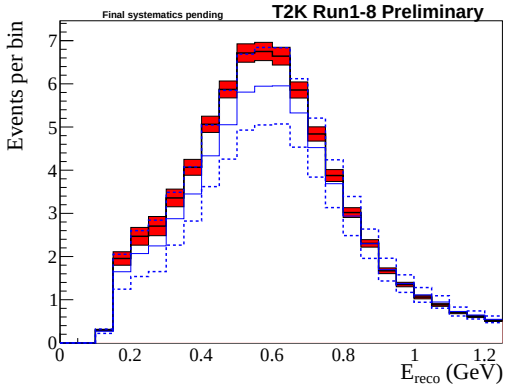
Figure 13: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum for Super-K FSI+SI+PN systematic parameters. 1×10^4 toys are generated for the Asimov data set A with randomized systematic parameters with correlations taken into account.



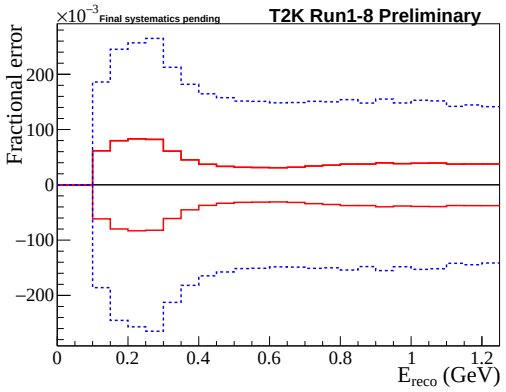
(a) Muon-like, neutrino mode



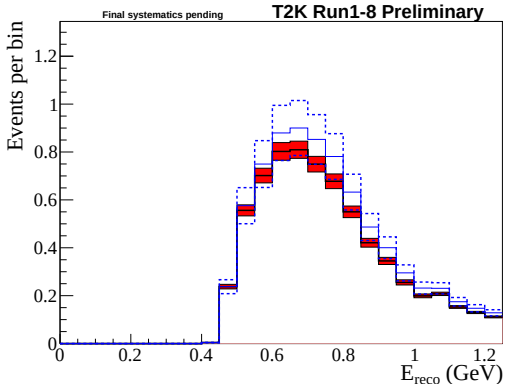
(b) Muon-like, neutrino mode



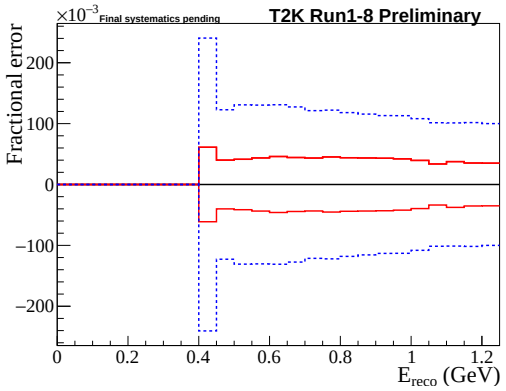
(c) Electron-like, neutrino mode



(d) Electron-like, neutrino mode

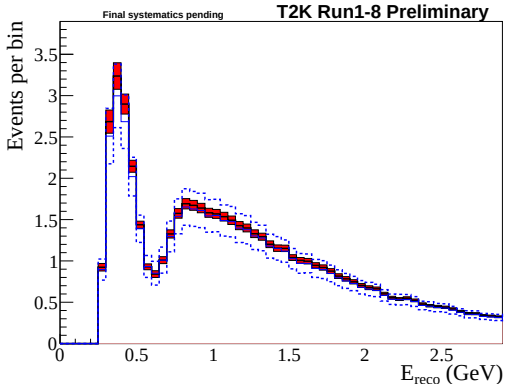


(e) ν_e CC1π^+-like, neutrino mode

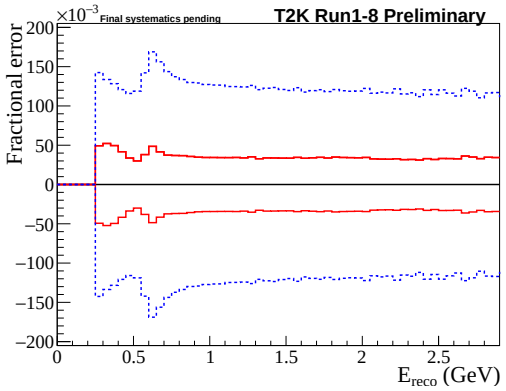


(f) ν_e CC1π^+-like, neutrino mode

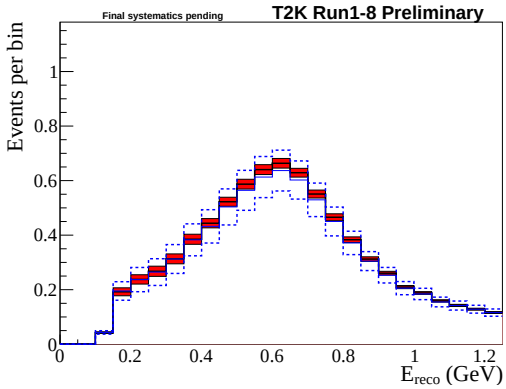
Figure 14: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum for the flux and constrained cross-section systematic parameters for pre-BANFF (blue) and post-BANFF (red) errors. 1×10^4 toys are generated for the Asimov data set A with randomized systematic parameters with correlations taken into account.



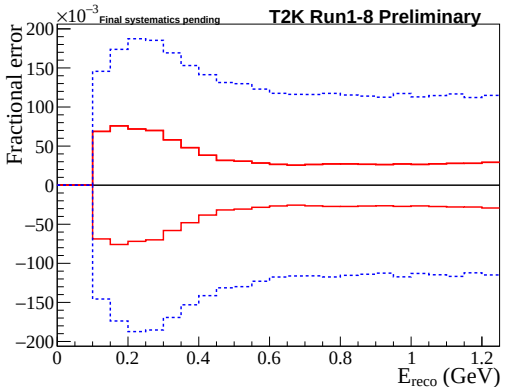
(a) Muon-like, antineutrino mode



(b) Muon-like, antineutrino mode

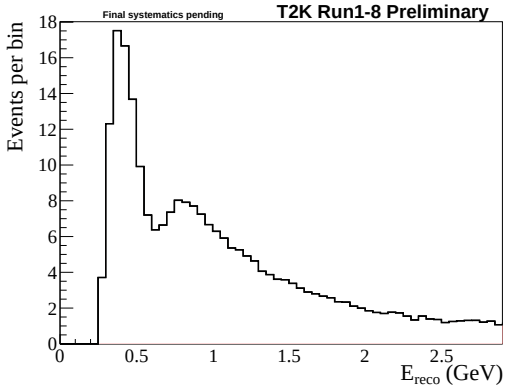


(c) Electron-like antineutrino mode

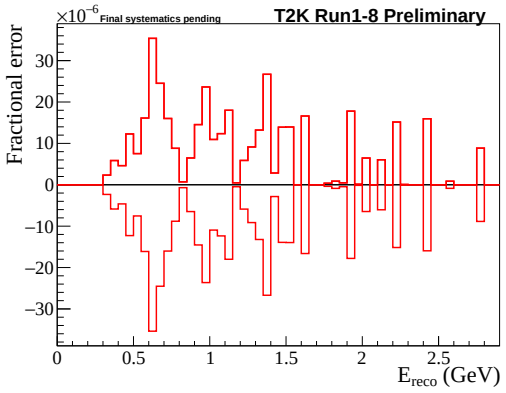


(d) Electron-like antineutrino mode

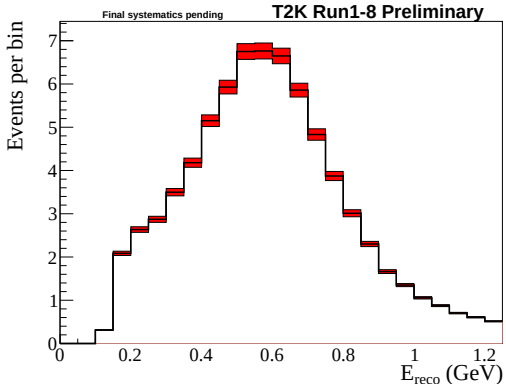
Figure 15: Error envelopes (left) and fractional error (right) for the reconstructed neutrino energy spectrum for the flux and constrained cross-section systematic parameters for pre-BANFF (blue) and post-BANFF (red) errors. 1×10^4 toys are generated for the Asimov data set A with randomized systematic parameters with correlations taken into account.



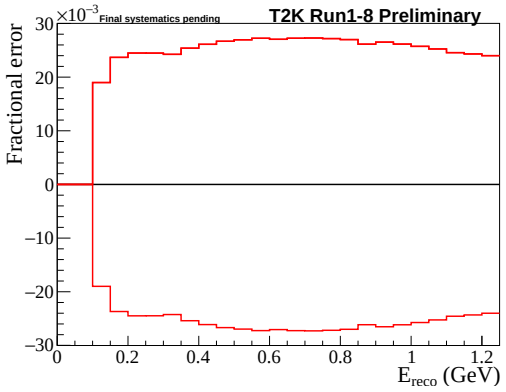
(a) Muon-like, neutrino mode



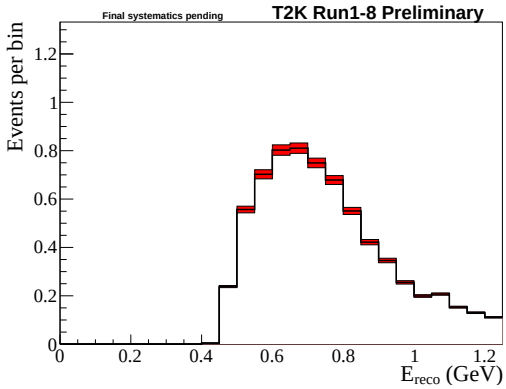
(b) Muon-like, neutrino mode



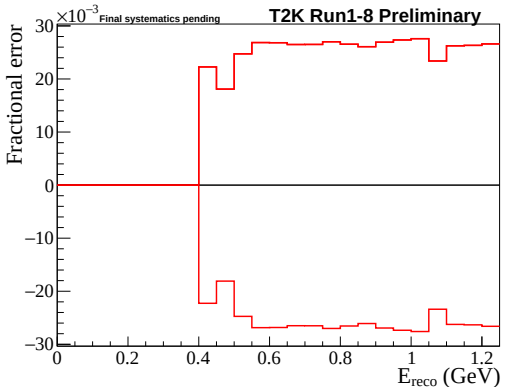
(c) Electron-like, neutrino mode



(d) Electron-like, neutrino mode

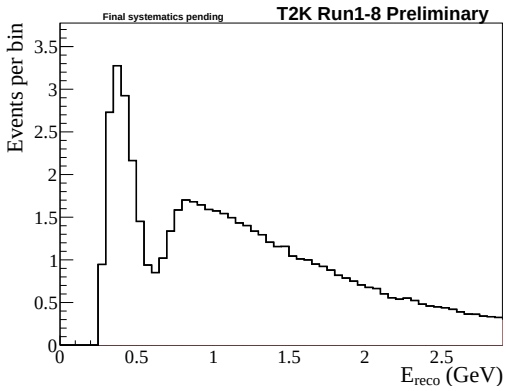


(e) ν_e CC1π^+-like, neutrino mode

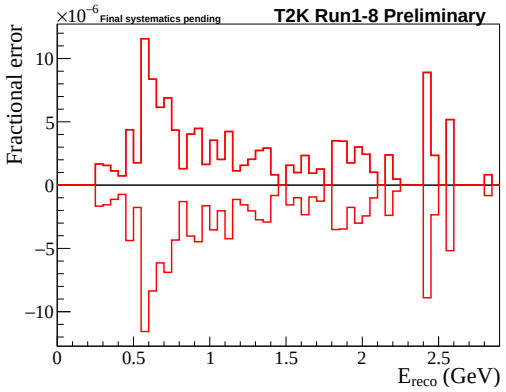


(f) ν_e CC1π^+-like, neutrino mode

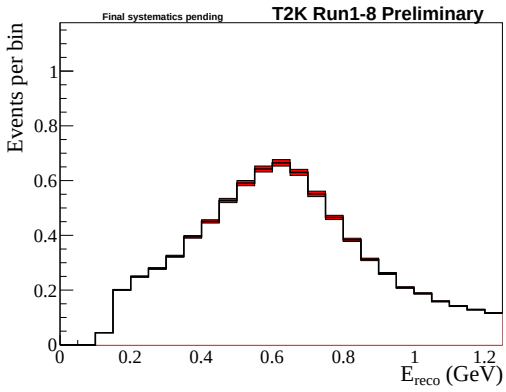
Figure 16: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum for the unconstrained electron neutrino cross-section systematic parameters. 1×10^4 toys are generated for the Asimov data set A with randomized systematic parameters with correlations taken into account.



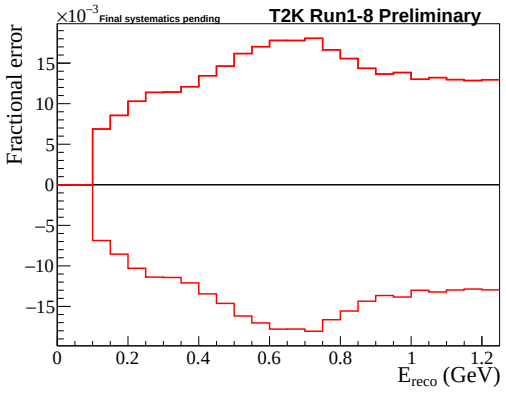
(a) Muon-like, antineutrino mode



(b) Muon-like, antineutrino mode

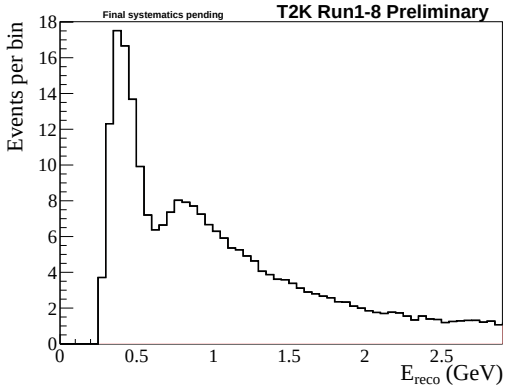


(c) Electron-like antineutrino mode

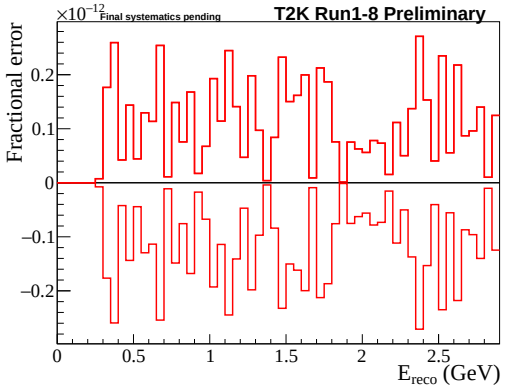


(d) Electron-like antineutrino mode

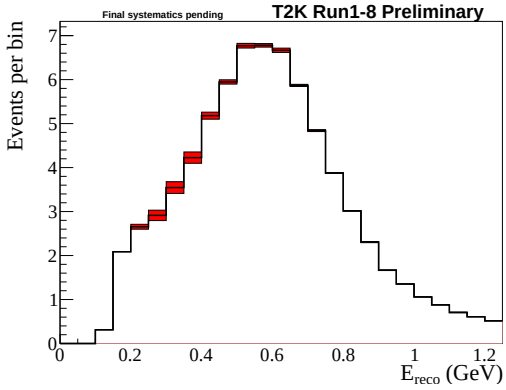
Figure 17: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum for the unconstrained electron neutrino cross-section systematic parameters. 1×10^4 toys are generated for the Asimov data set A with randomized systematic parameters with correlations taken into account.



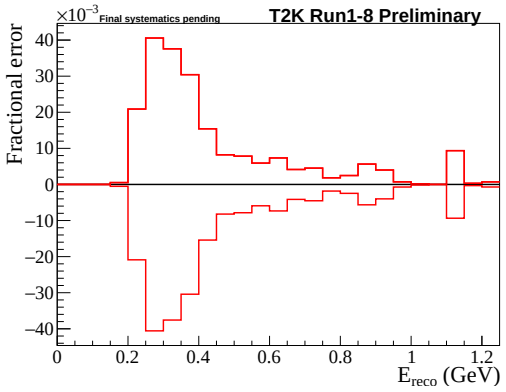
(a) Muon-like, neutrino mode



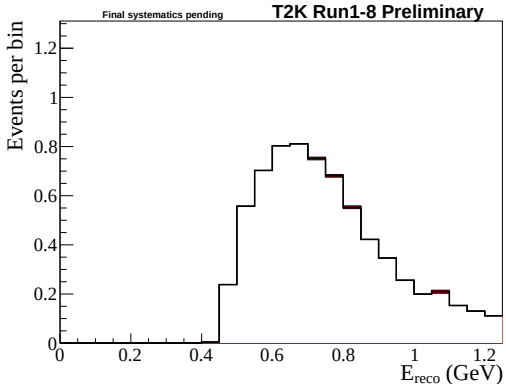
(b) Muon-like, neutrino mode



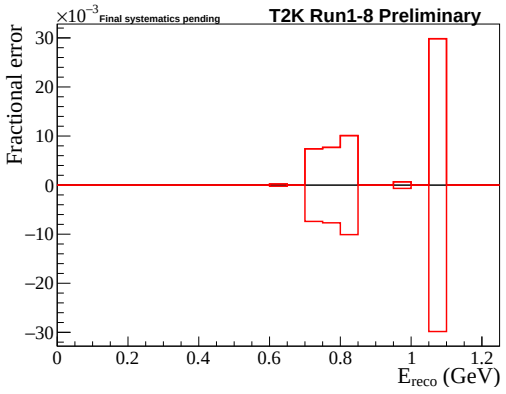
(c) Electron-like, neutrino mode



(d) Electron-like, neutrino mode

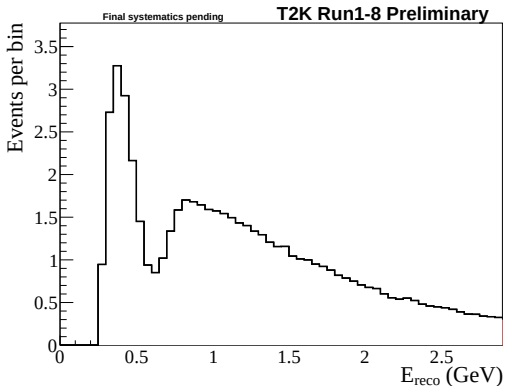


(e) ν_e CC1π^+-like, neutrino mode

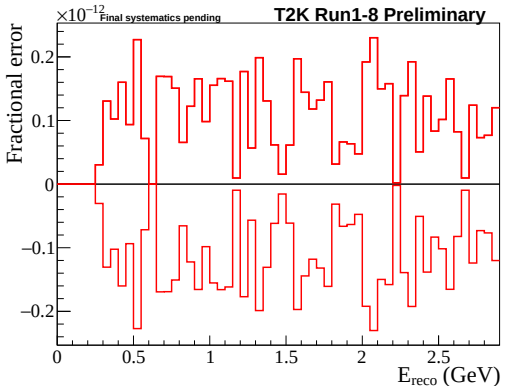


(f) ν_e CC1π^+-like, neutrino mode

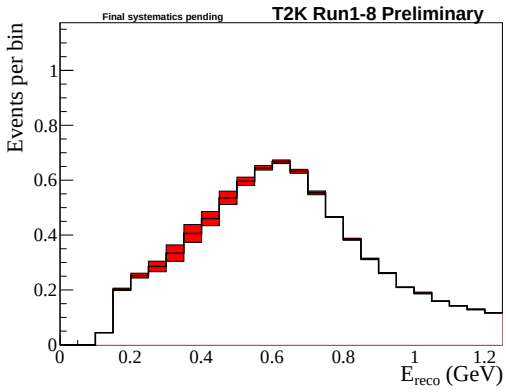
Figure 18: Error envelopes (left) and fractional error (right) for the reconstructed neutrino energy spectrum for the unconstrained NC1γ systematic parameters. 1×10^4 toys are generated for the Asimov data set A with randomized systematic parameters with correlations taken into account.



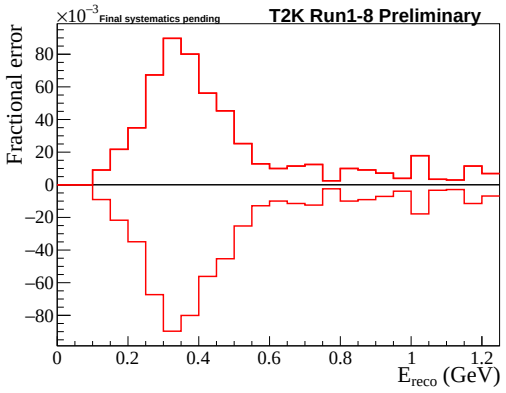
(a) Muon-like, antineutrino mode



(b) Muon-like, antineutrino mode

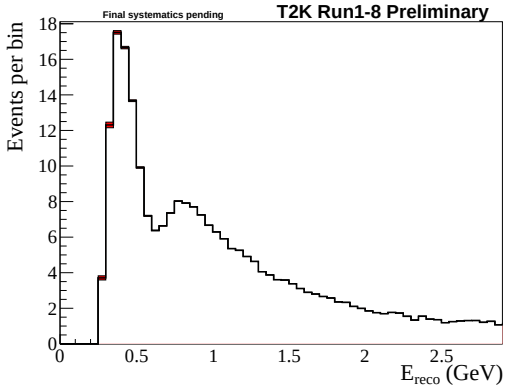


(c) Electron-like antineutrino mode

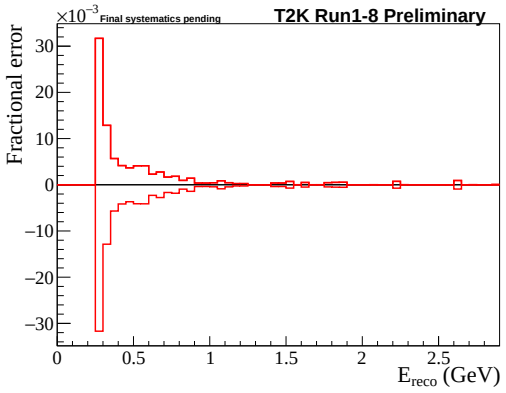


(d) Electron-like antineutrino mode

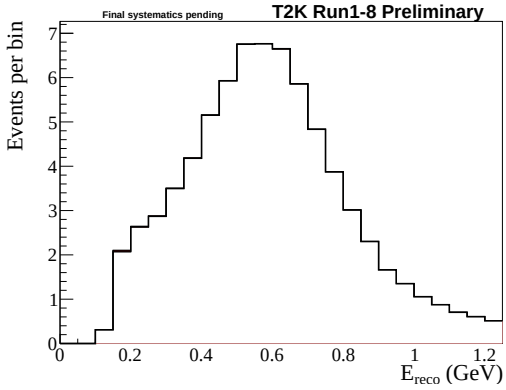
Figure 19: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum for the unconstrained NC1γ systematic parameters. 1×10^4 toys are generated for the Asimov data set A with randomized systematic parameters with correlations taken into account.



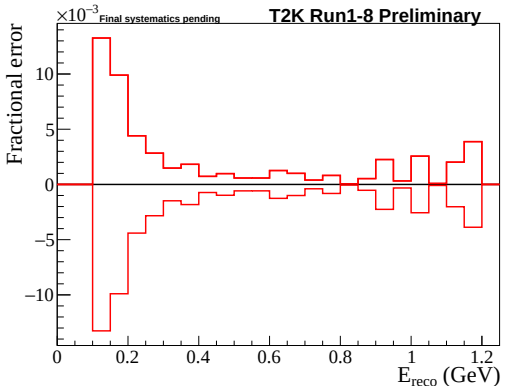
(a) Muon-like, neutrino mode



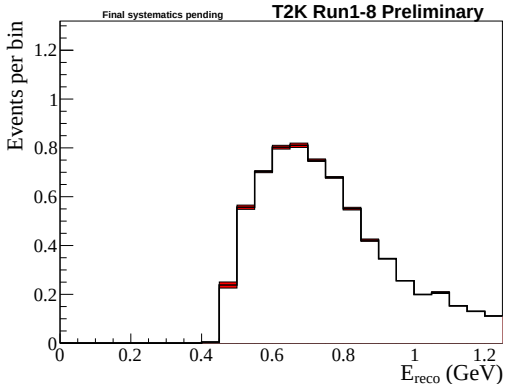
(b) Muon-like, neutrino mode



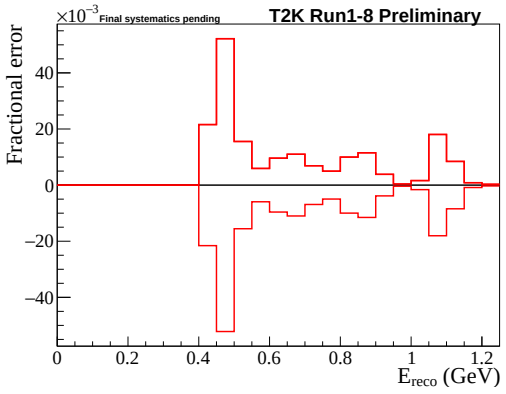
(c) Electron-like, neutrino mode



(d) Electron-like, neutrino mode

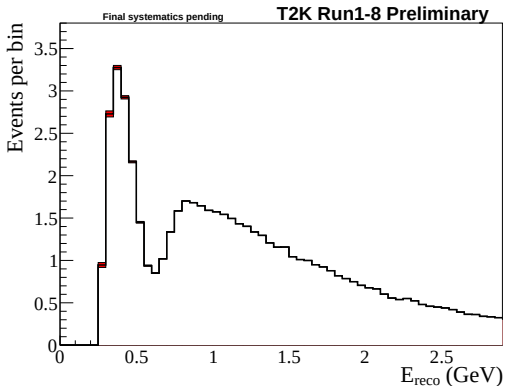


(e) ν_e CC1π^+-like, neutrino mode

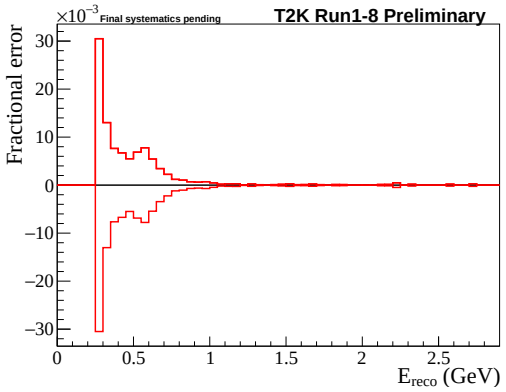


(f) ν_e CC1π^+-like, neutrino mode

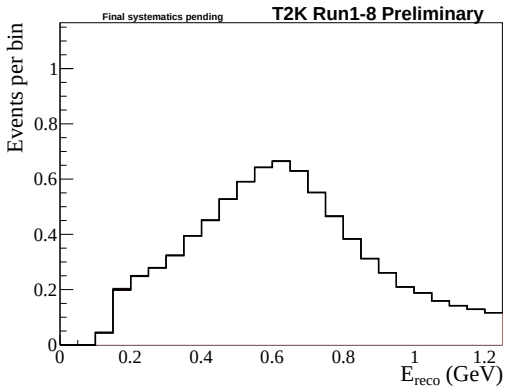
Figure 20: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum for the unconstrained NCother systematic parameters. 1×10^4 toys are generated for the Asimov data set A with randomized systematic parameters with correlations taken into account.



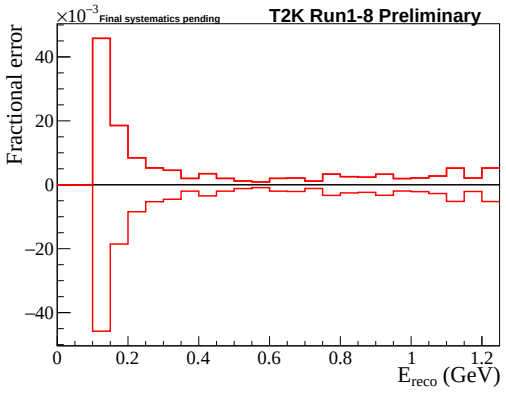
(a) Muon-like, antineutrino mode



(b) Muon-like, antineutrino mode



(c) Electron-like antineutrino mode



(d) Electron-like antineutrino mode

Figure 21: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum for the unconstrained NCother systematic parameters. 1×10^4 toys are generated for the Asimov data set A with randomized systematic parameters with correlations taken into account.

4. Sensitivity studies

To determine the sensitivity of the analysis a number of fits were performed using Asimov data set A, the contours for which indicate the ability of the analysis to measure the oscillation parameters in the absence of statistical fluctuations and under the assumption that their true values correspond to the Asimov A values. Fits of single parameters are performed for each of $\sin^2\theta_{13}$, δ_{CP}, $\sin^2\theta_{23}$ and Δm^2_{32}, with two-dimensional fits being performed for δ_{CP} vs $\sin^2\theta_{13}$ and Δm^2 vs $\sin^2\theta_{23}$. Fits are performed for each of normal and inverted hierarchy, and also with and without the reactor constraint on $\sin^2\theta_{13}$.

4.1. Asimov data sets

The definitions of the Asimov data sets given in section 2.1.1 are repeated here for reference.

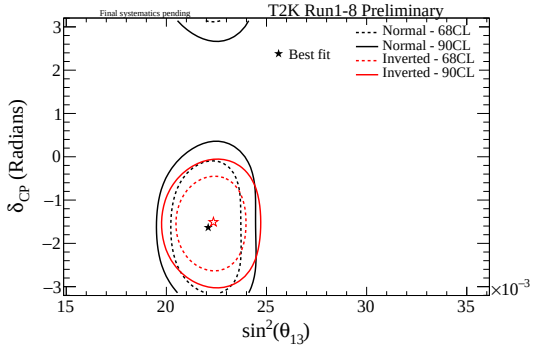
Parameter(s)	Asimov A	Asimov B
$\sin^2\theta_{23}$	0.528	0.450
$\sin^2\theta_{13}$ reactors	0.0219	0.0219
$\sin^2\theta_{12}$	0.304	0.304
$\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{31}\rvert$ (IH)	2.509×10^{-3} eV^2/c^4	2.509×10^{-3} eV^2/c^4
Δm^2_{21}	7.53×10^{-5} eV^2/c^4	7.53×10^{-5} eV^2/c^4
δ_{CP}	-1.601	0
Mass Hierarchy	Normal	Normal

Table 14: Values of oscillation parameters used to compute the event rates, systematic effects and sensitivity studies. Each set of oscillation parameters correspond to a different Asimov data set, which is the MC expected distribution in a certain oscillation hypothesis. In the Asimov data set A the nominal values of $\sin^2 2\theta_{13}$, $\sin^2 2\theta_{12}$ and Δm^2_{21} are from [81], while all the other oscillation parameter values corresponds to the most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [20]. The mass hierarchy is not marginalised but fixed to either NH or IH.

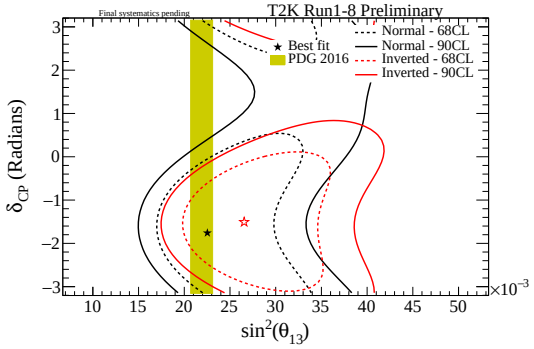
4.2. Results for Asimov sets A and B

4.2.1. Results for δ_{CP} vs $\sin^2\theta_{13}$

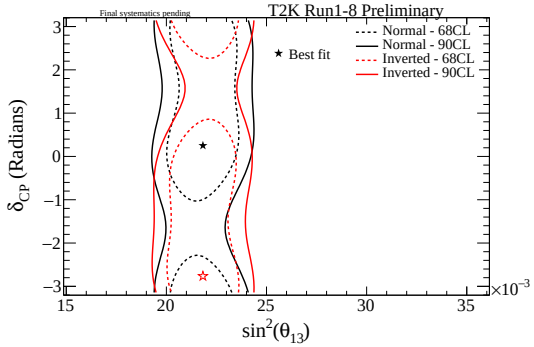
In fig. 22 the sensitivity to δ_{CP} vs $\sin^2\theta_{13}$ with reactor constraint and T2K only is shown for the Asimov data sets A and B. Note that in the Asimov B case it can be seen that the best-fit point is shifted away from the true δ_{CP} value of zero towards $-\pi$. In addition to best-fit points being moved from the true value due to marginalisation of oscillation and systematic parameters there are additional contributing effects. There is a lack of sensitivity to distinguish $\delta_{CP} = 0$ and $\delta_{CP} = \pi$ due to the spectra for $\delta_{CP} = 0$ in inverted hierarchy being similar to the spectra for $\delta_{CP} = \pi$ in normal hierarchy and vice versa (see fig. 74 in Appendix H), leading to similar $\Delta\chi^2$ values for the two points. This can be seen in fig. 24(c) and fig. 24(d), with the difference in $\Delta\chi^2_{true}$ and $\Delta\chi^2_{best-fit}$ being only ~ 0.2. Furthermore, degeneracy between δ_{CP} and mass hierarchy can cause $\delta_{CP} = 0$ to behave more like $\delta_{CP} = \pi/2$ in inverted hierarchy, and more like $\delta_{CP} = -\pi/2$ in normal hierarchy, shifting the best-fit for inverted hierarchy away from zero to a value between $-\pi$ and $-\pi/2$.



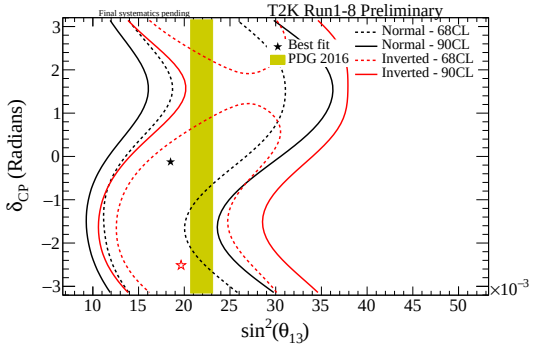
(a) Asimov A δ_{CP} vs $\sin^2\theta_{13}$ with reactor constraint



(b) Asimov A δ_{CP} vs $\sin^2\theta_{13}$ T2K only



(c) Asimov B δ_{CP} vs $\sin^2\theta_{13}$ with reactor constraint

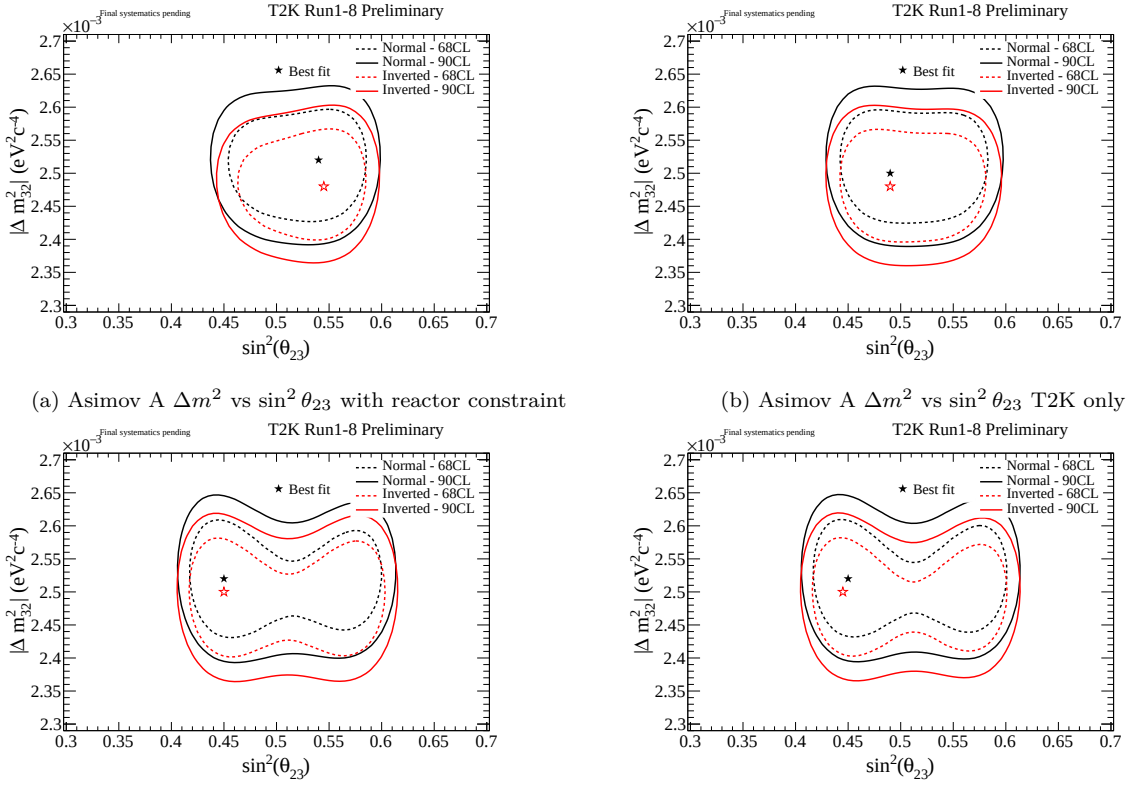


(d) Asimov B δ_{CP} vs $\sin^2\theta_{13}$ T2K only

Figure 22: Contours at 68% and 90% CL for δ_{CP} vs $\sin^2\theta_{13}$ with and without reactor constraint for Asimov data sets A and B. Normal and inverted hierarchy contours are independent. All 5 samples were used to produce these fits.

4.2.2. Results for Δm^2 vs $\sin^2\theta_{23}$

The sensitivity to Δm^2 vs $\sin^2\theta_{23}$ with reactor constraint and T2K only is shown for the Asimov data sets A and B in fig. 23.



(a) Asimov A Δm^2 vs $\sin^2\theta_{23}$ with reactor constraint

(b) Asimov A Δm^2 vs $\sin^2\theta_{23}$ T2K only

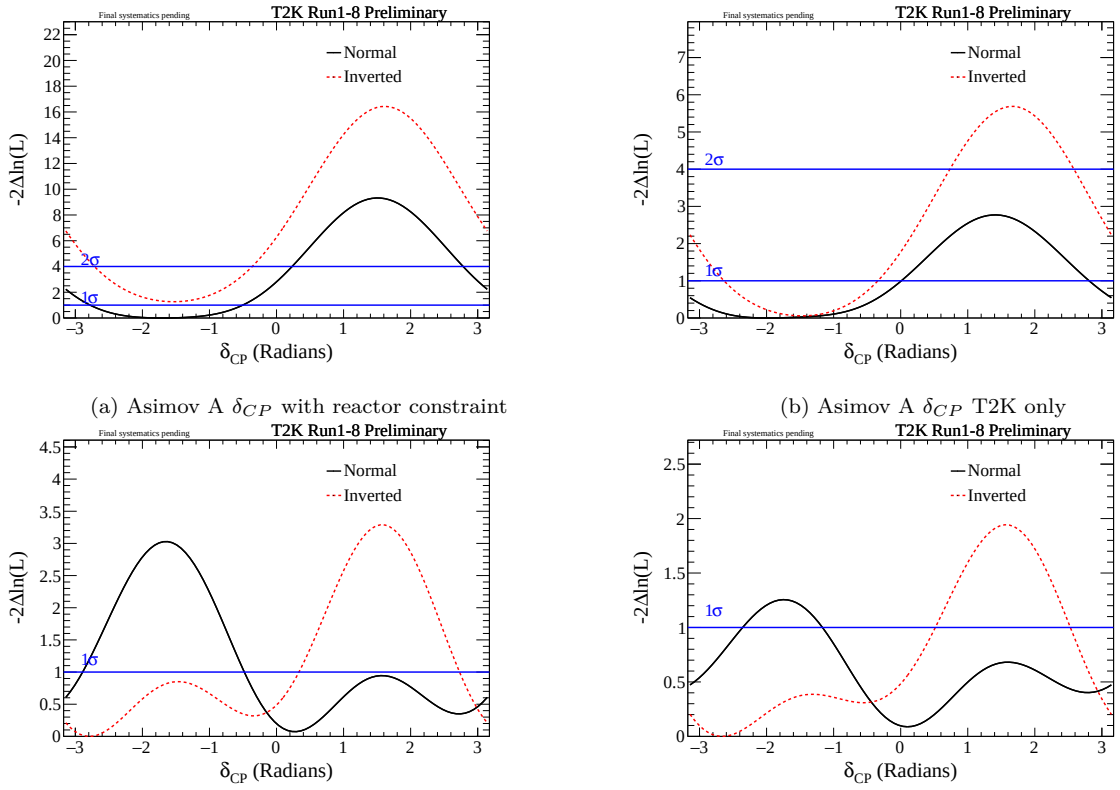
(c) Asimov B Δm^2 vs $\sin^2\theta_{23}$ with reactor constraint

(d) Asimov B Δm^2 vs $\sin^2\theta_{23}$ T2K only

Figure 23: Contours at 68% and 90% CL for Δm^2 vs $\sin^2\theta_{23}$ with and without reactor constraint for Asimov data sets A and B. Normal and inverted hierarchy contours are independent. All 5 samples were used to produce these fits.

4.2.3. Results for δ_{CP}

The sensitivity to δ_{CP} with reactor constraint and T2K only is shown for the Asimov data sets A and B in fig. 24.



(a) Asimov A δ_{CP} with reactor constraint

(b) Asimov A δ_{CP} T2K only

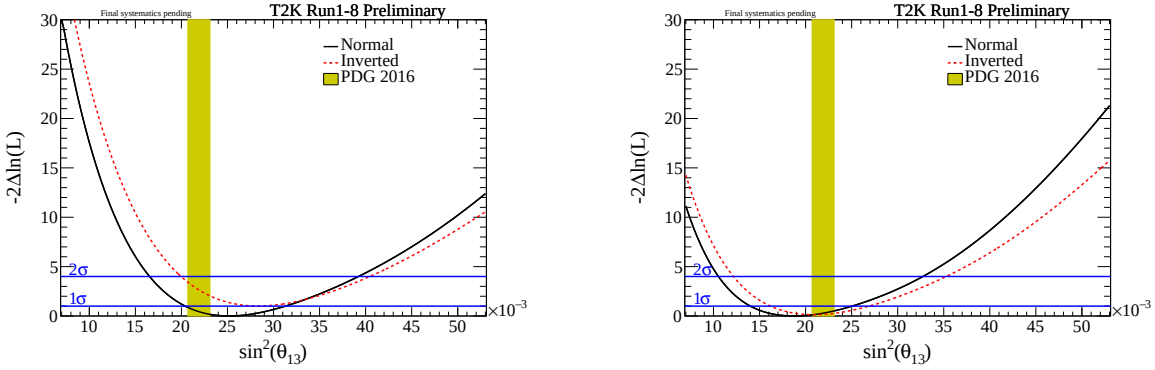
(c) Asimov B δ_{CP} with reactor constraint

(d) Asimov B δ_{CP} T2K only

Figure 24: The expected $\Delta\chi^2$ distribution as a function of δ_{CP} with and without reactor constraint for Asimov data sets A and B. The mass hierarchy is minimized, with normal and inverted hierarchy $\Delta\chi^2$ distributions shifted to the same global best-fit χ^2 value, which is taken to be the minimum between normal and inverted hierarchy. All 5 samples were used to produce these fits.

4.2.4. *Results for* $\sin^2\theta_{13}$

The sensitivity to $\sin^2\theta_{13}$ (without reactor constraint) is shown for the Asimov data sets A and B in fig. 25.



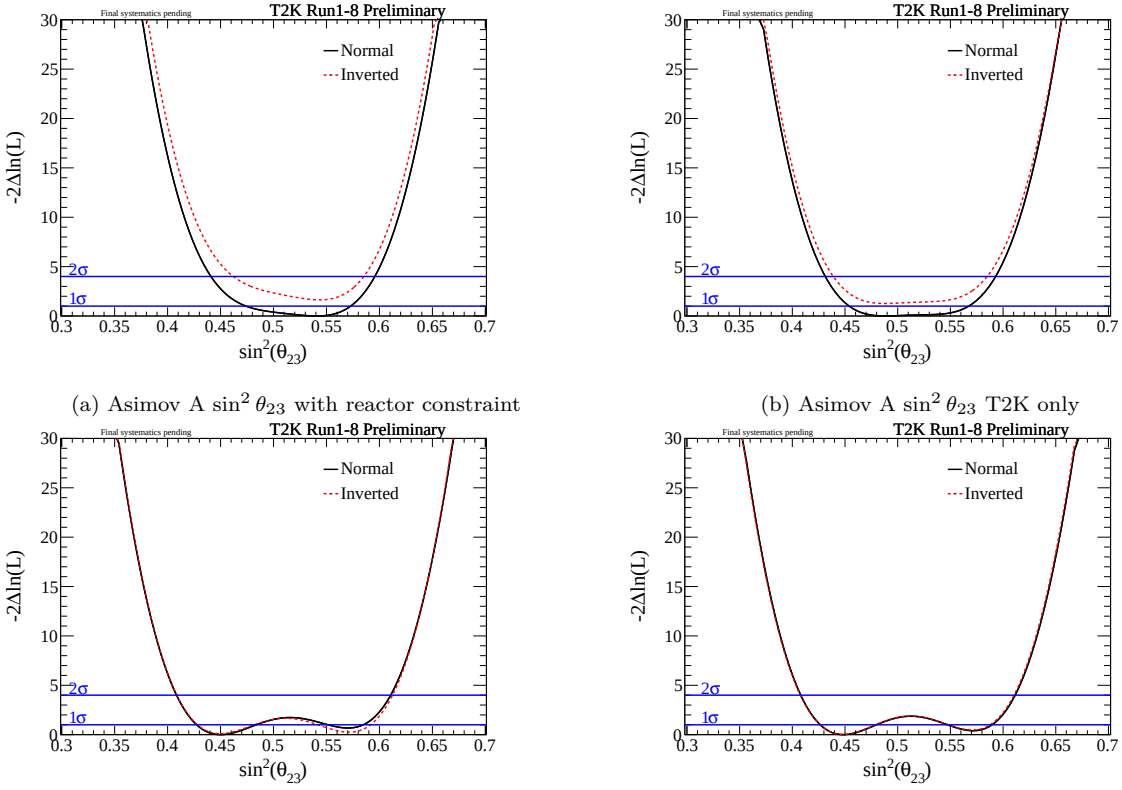
(a) Asimov A $\sin^2\theta_{13}$ T2K only

(b) Asimov B $\sin^2\theta_{13}$ T2K only

Figure 25: The expected $\Delta\chi^2$ distribution as a function of $\sin^2\theta_{13}$ without reactor constraint for Asimov data sets A and B. The mass hierarchy is minimized, with normal and inverted hierarchy $\Delta\chi^2$ distributions shifted to the same global best-fit χ^2 value, which is taken to be the minimum between normal and inverted hierarchy. All 5 samples were used to produce these fits.

4.2.5. *Results for* $\sin^2\theta_{23}$

The sensitivity to $\sin^2\theta_{23}$ with reactor constraint and T2K only is shown for the Asimov data sets A and B in fig. 26.



(a) Asimov A $\sin^2\theta_{23}$ with reactor constraint

(b) Asimov A $\sin^2\theta_{23}$ T2K only

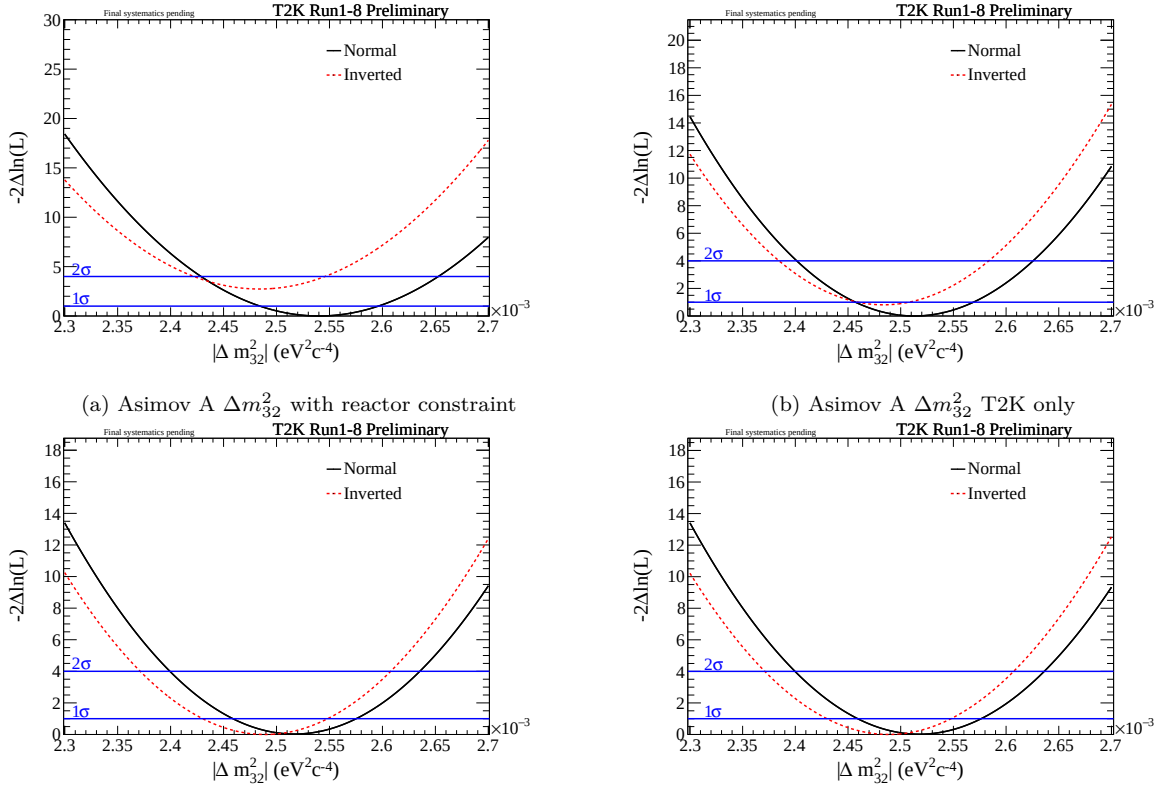
(c) Asimov B $\sin^2\theta_{23}$ with reactor constraint

(d) Asimov B $\sin^2\theta_{23}$ T2K only

Figure 26: The expected $\Delta\chi^2$ distribution as a function of $\sin^2\theta_{23}$ with and without reactor constraint for Asimov data sets A and B. The mass hierarchy is minimized, with normal and inverted hierarchy $\Delta\chi^2$ distributions shifted to the same global best-fit χ^2 value, which is taken to be the minimum between normal and inverted hierarchy. All 5 samples were used to produce these fits.

4.2.6. *Results for* Δm^2_{32}

The sensitivity to Δm^2_{32} with reactor constraint and T2K only is shown for the Asimov data sets A and B in fig. 27.



(a) Asimov A Δm^2_{32} with reactor constraint

(b) Asimov A Δm^2_{32} T2K only

(c) Asimov B Δm^2_{32} with reactor constraint

(d) Asimov B Δm^2_{32} T2K only

Figure 27: The expected $\Delta\chi^2$ distribution as a function of Δm^2_{32} with and without reactor constraint for Asimov data sets A and B. The mass hierarchy is minimized, with normal and inverted hierarchy $\Delta\chi^2$ distributions shifted to the same global best-fit χ^2 value, which is taken to be the minimum between normal and inverted hierarchy. All 5 samples were used to produce these fits.

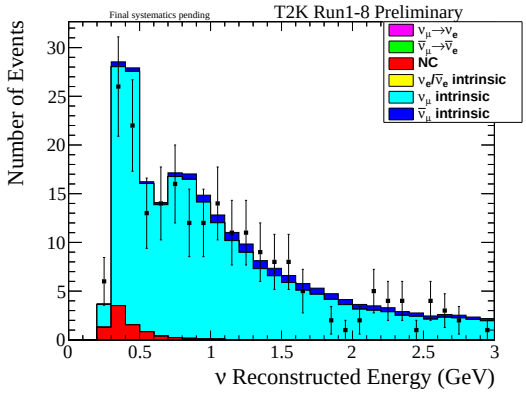
5. Results of the $\nu\bar{\nu}$ joint analysis with the Run 1-8 dataset

The results of the Run 1-8 data set fit, for an exposure of 1.4734×10^{21}POT in neutrino mode and of 7.558×10^{20}POT in antineutrino mode, are shown in this chapter. In fig. 28 the predicted spectra of μ-like and e-like samples are compared to the observed data. In table 15 the expected and observed number of events in Run 1-8 data set are shown. The predicted number of events are shown for the oscillation parameters of Asimov data sets A and B. The comparison of the confidence intervals obtained by the fit of the Run 1-8 data set among all the oscillation analysis groups is shown in [88].

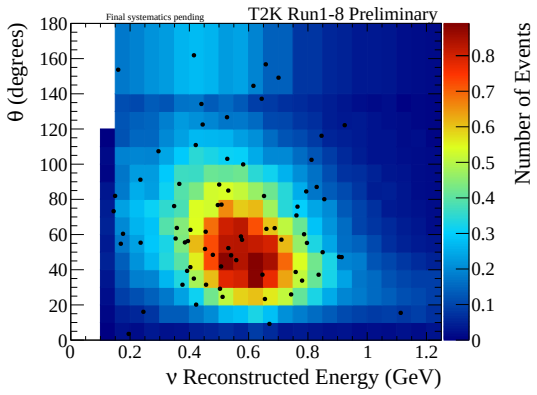
Beam mode	Sample	Exp. Asimov A	Exp. Asimov A ($\delta_{CP} = 0$)	Exp. Asimov B	Exp. Not Osc	Observed
ν	μ-like	268.415	268.053	280.994	1209.918	240
ν	e-like	73.476	61.434	54.684	15.356	74
$\bar{\nu}$	μ-like	64.278	64.123	66.010	211.548	68
$\bar{\nu}$	e-like	7.911	9.023	8.187	3.243	7
ν	ν_e CC1π^+-like	6.919	6.009	5.389	2.543	15

Table 15: The observed, expected and best-fit number of events in Run 1-8 data set for an exposure of 1.4734×10^{21}POT in neutrino mode and of 7.558×10^{20}POT in antineutrino mode are shown for each selected sample. The prediction is produced using the BANFF tuning and the oscillation parameters shown in table 1.

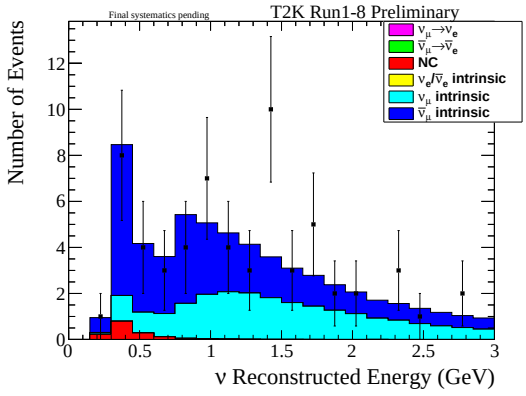
The predicted and observed spectra are shown for each of the samples in fig. 28. The e-like samples are broken down into E_{rec} and θ projections in fig. 29.



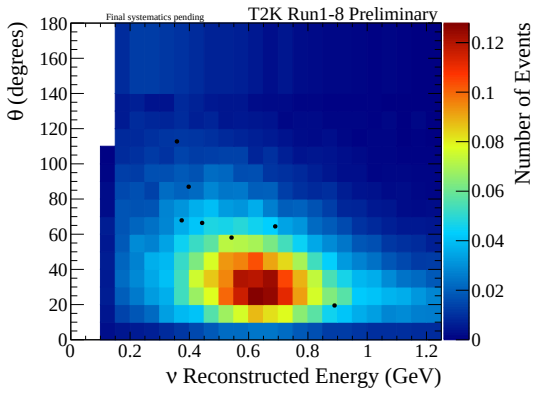
(a) Muon-like, neutrino mode



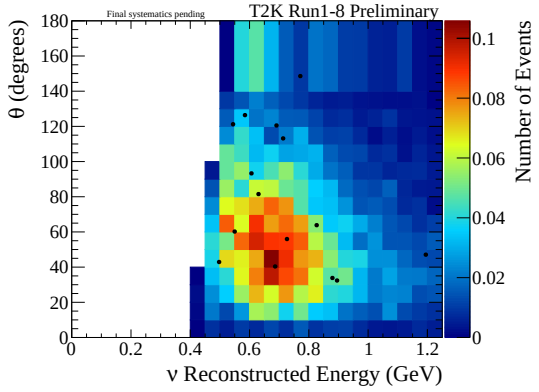
(b) Electron-like, neutrino mode



(c) Muon-like, antineutrino mode

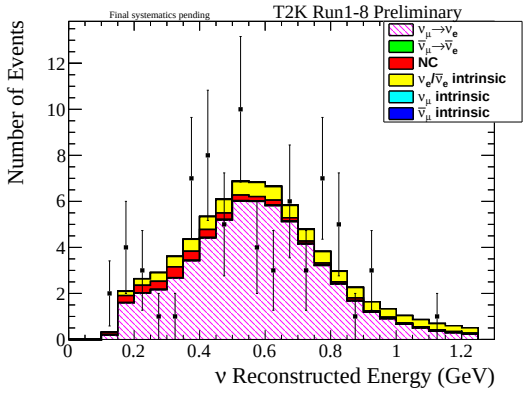


(d) Electron-like antineutrino mode

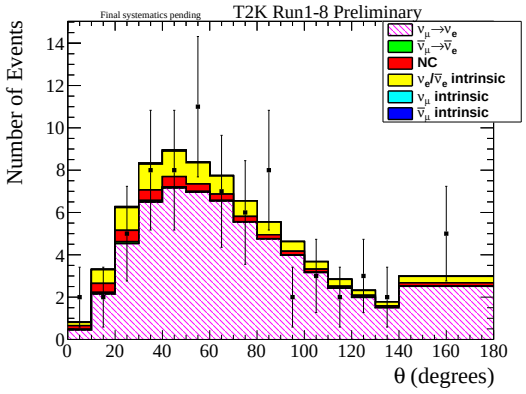


(e) ν_e CC1π^+-like, neutrino mode

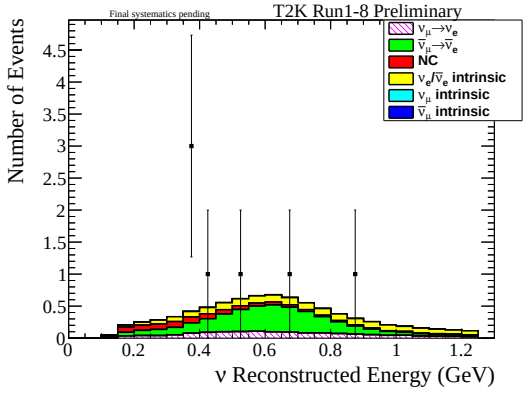
Figure 28: Predicted spectra and observed events (points). μ-like distributions are a function of the reconstructed neutrino energy, while the e-like, including ν_e CC1π^+, distributions are functions of both the reconstructed neutrino energy and the reconstructed angle between the outgoing lepton and the neutrino direction. The distributions correspond to the statistics collected in the full Run 1-8 data set. The spectra are generated with the systematic parameters described in section 3 and the oscillation parameters corresponding to the best-fit values from the data fit (solar parameters at PDG 2016).



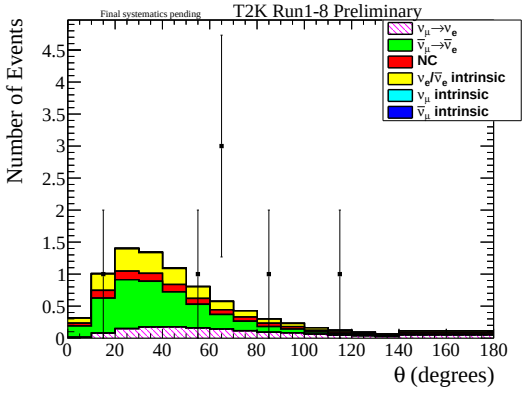
(a) Electron-like, neutrino mode



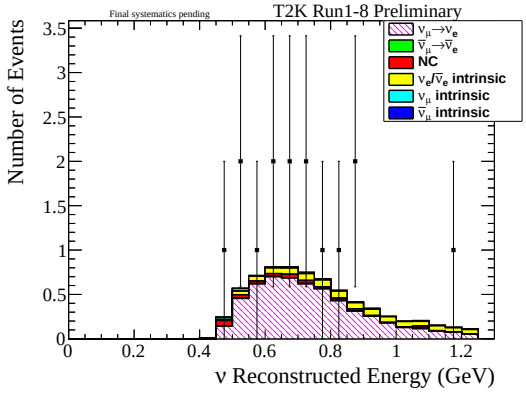
(b) Electron-like, neutrino mode



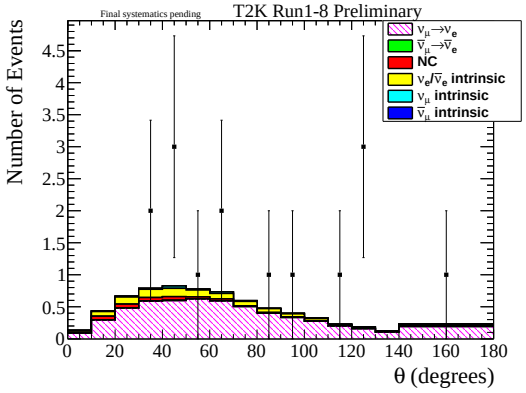
(c) Electron-like antineutrino mode



(d) Electron-like antineutrino mode



(e) ν_e CC1π^+-like, neutrino mode



(f) ν_e CC1π^+-like, neutrino mode

Figure 29: Predicted spectra and observed events (points). The e-like, including ν_e CC1π^+, distributions are functions of both the reconstructed neutrino energy and the reconstructed angle between the outgoing lepton and the neutrino direction, with the projections in each variable shown here. The distributions correspond to the statistics collected in the full Run 1-8 data set. The spectra are generated with the systematic parameters described in section 3 and the oscillation parameters corresponding to the best-fit values from the data fit (solar parameters at PDG 2016).

Parameter	Reactor	Best-fit (NH)	$\pm1\sigma$ (NH)	Best-fit (IH)	$\pm1\sigma$ (IH)
δ_{CP}	Yes	-1.833	[-2.476,-1.151]	-1.374	[-1.947,-0.886]
δ_{CP}	No	-2.083	[-2.972,-1.151]	-1.157	[-1.965,-0.403]
$\sin^2\theta_{13}$	No	0.0277	[0.0230,0.0331]	0.0331	[0.0265,0.0378]
$\sin^2\theta_{23}$	Yes	0.530	[0.493,0.561]	0.530	[0.496,0.559]
Δm^2_{32} or Δm^2_{13} (10^{-3} eV^2/c^4)	Yes	2.462	[2.407,2.519]	2.436	[2.383,2.491]

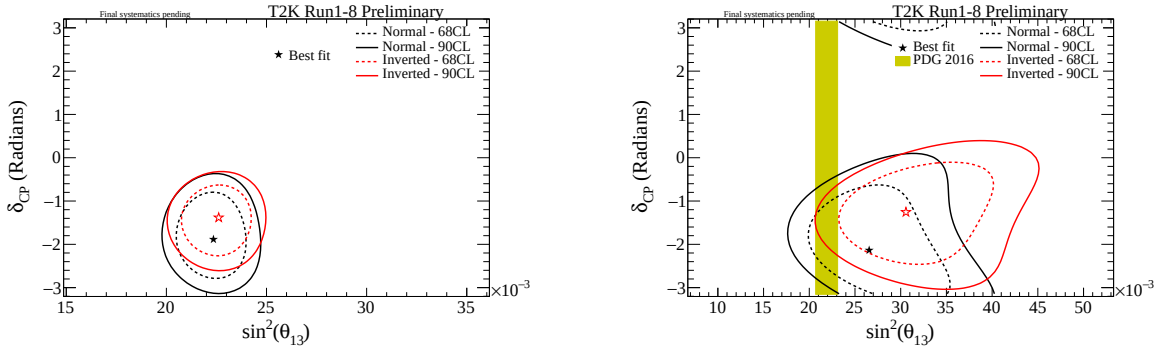
Table 16: The Run 1-8 measured oscillation parameter best-fit and the $\pm1\sigma$ intervals are shown for normal and inverted hierarchies with respect to the hierarchy best-fit. The $\pm1\sigma$ interval is obtained by using the constant $\Delta\chi^2$ method. For each parameter normal hierarchy is the global best-fit.

The best-fit values and 1σ ranges for each oscillation parameter are shown in table 16 for normal hierarchy (which is preferred in all data fits). The 1σ ranges are extracted for each oscillation parameter from the 1-dimensional $\Delta\chi^2$ distributions shown in figs. 32 to 35.

The 1-dimensional contours are plotted with respect to the global minimum from normal and inverted hierarchies, while the 2-dimensional confidence intervals are shown drawing the normal and inverted hierarchy contours independently, i.e. each contour is produced with respect to the global minimum under that hierarchy hypothesis.

5.1. *Results for* δ_{CP} *vs* $\sin^2\theta_{13}$

In fig. 30 the result for δ_{CP} vs $\sin^2\theta_{13}$ with reactor constraint and T2K only is shown.



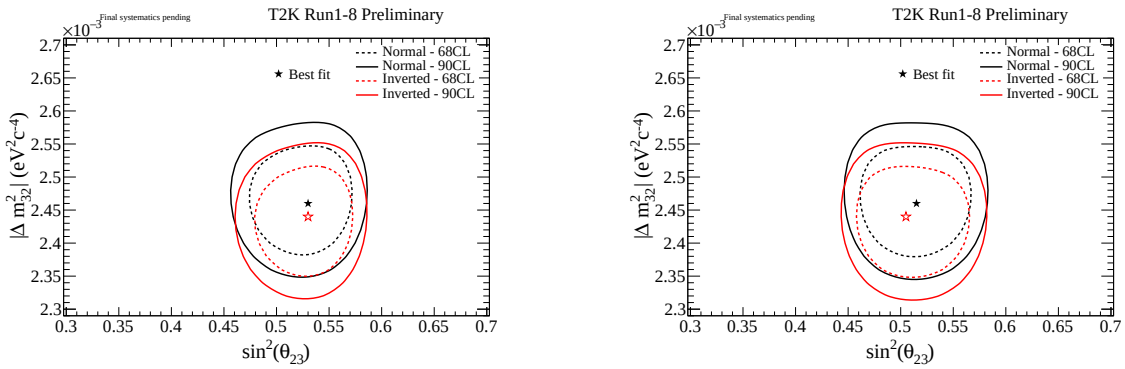
(a) δ_{CP} vs $\sin^2\theta_{13}$ with reactor constraint

(b) δ_{CP} vs $\sin^2\theta_{13}$ T2K only

Figure 30: Contours at 68% and 90% CL for δ_{CP} vs $\sin^2\theta_{13}$ with and without reactor constraint. Normal and inverted hierarchy contours are independent. All 5 samples were used to produce these fits.

5.2. *Results for* Δm^2 *vs* $\sin^2\theta_{23}$

The result for Δm^2 vs $\sin^2\theta_{23}$ with reactor constraint and T2K only is shown in fig. 31.



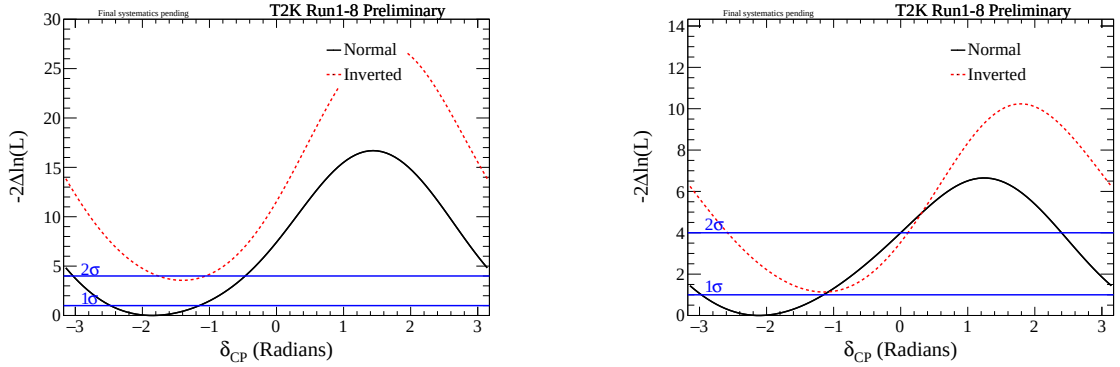
(a) Δm^2 vs $\sin^2\theta_{23}$ with reactor constraint

(b) Δm^2 vs $\sin^2\theta_{23}$ T2K only

Figure 31: Contours at 68% and 90% CL for Δm^2 vs $\sin^2\theta_{23}$ with and without reactor constraint. Normal and inverted hierarchy contours are independent. All 5 samples were used to produce these fits.

5.3. *Results for* δ_{CP}

The result for δ_{CP} with reactor constraint and T2K only is shown in fig. 32.



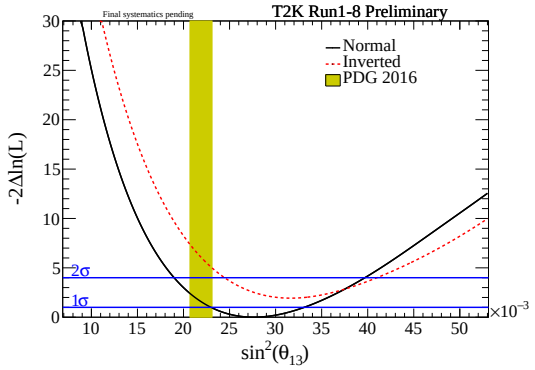
(a) δ_{CP} with reactor constraint

(b) δ_{CP} T2K only

Figure 32: The expected $\Delta\chi^2$ distribution as a function of δ_{CP} with and without reactor constraint. The mass hierarchy is minimized, with normal and inverted hierarchy $\Delta\chi^2$ distributions shifted to the same global best-fit χ^2 value, which is taken to be the minimum between normal and inverted hierarchy. All 5 samples were used to produce these fits.

5.4. *Results for* $\sin^2\theta_{13}$

The result for $\sin^2\theta_{13}$ (without reactor constraint) is shown in fig. 33.

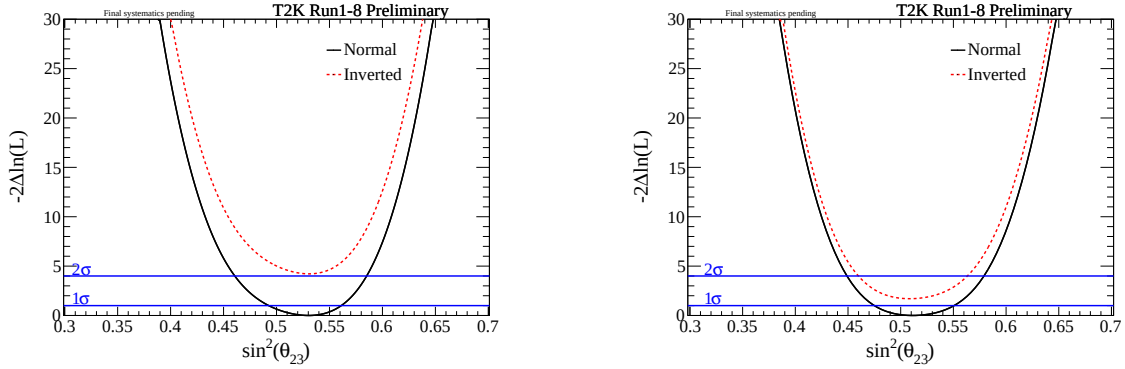


(a) $\sin^2\theta_{13}$ T2K only

Figure 33: The expected $\Delta\chi^2$ distribution as a function of $\sin^2\theta_{13}$ without reactor constraint. The mass hierarchy is minimized, with normal and inverted hierarchy $\Delta\chi^2$ distributions shifted to the same global best-fit χ^2 value, which is taken to be the minimum between normal and inverted hierarchy. All 5 samples were used to produce these fits.

5.5. *Results for* $\sin^2\theta_{23}$

The result for $\sin^2\theta_{23}$ with reactor constraint and T2K only is shown in fig. 34.



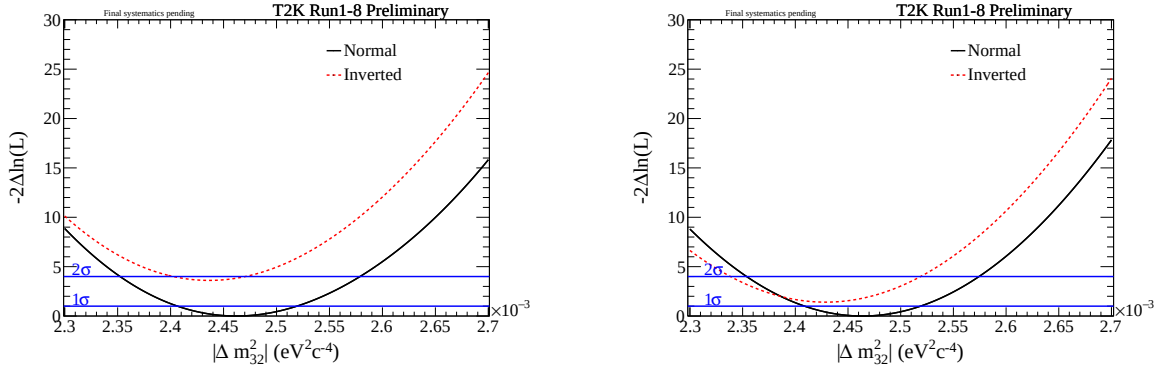
(a) $\sin^2\theta_{23}$ with reactor constraint

(b) $\sin^2\theta_{23}$ T2K only

Figure 34: The expected $\Delta\chi^2$ distribution as a function of $\sin^2\theta_{23}$ with and without reactor constraint. The mass hierarchy is minimized, with normal and inverted hierarchy $\Delta\chi^2$ distributions shifted to the same global best-fit χ^2 value, which is taken to be the minimum between normal and inverted hierarchy. All 5 samples were used to produce these fits.

5.6. *Results for* Δm^2_{32}

The result for Δm^2_{32} with reactor constraint and T2K only is shown in fig. 35.



(a) Δm^2_{32} with reactor constraint

(b) Δm^2_{32} T2K only

Figure 35: The expected $\Delta\chi^2$ distribution as a function of Δm^2_{32} with and without reactor constraint. The mass hierarchy is minimized, with normal and inverted hierarchy $\Delta\chi^2$ distributions shifted to the same global best-fit χ^2 value, which is taken to be the minimum between normal and inverted hierarchy. All 5 samples were used to produce these fits.

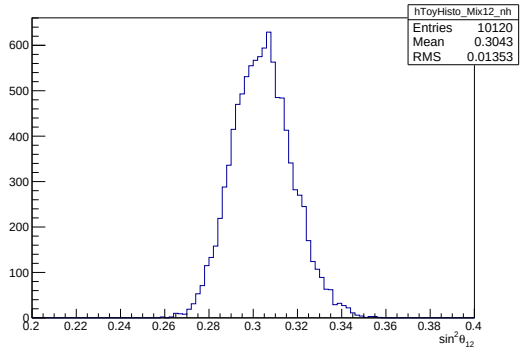
5.7. *Feldman-Cousins simultaneous fit of* δ_{CP} *and mass hierarchy*

This section presents the results of the Feldman-Cousins fit of δ_{CP} and mass hierarchy. Throws of oscillation parameters are made according to PDG 2016 best-fit values and errors for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, whilst $\sin^2\theta_{23}$ and Δm^2_{32} are thrown according to the likelihood distribution produced by an Asimov fit using the best-fit from data for these parameters, this is an updated

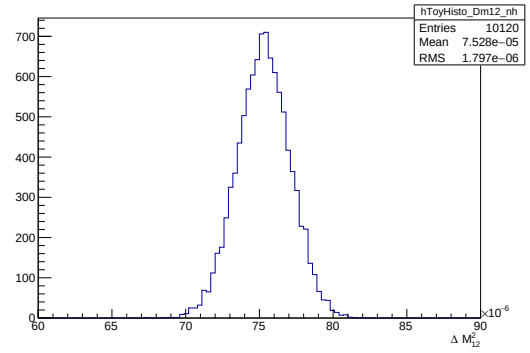
method from the previous analysis and the proposal is given in [90]. δ_{CP} is fixed at one of 9 evenly specified values on the range $[-\pi, \pi]$.

5.7.1. Input oscillation parameters

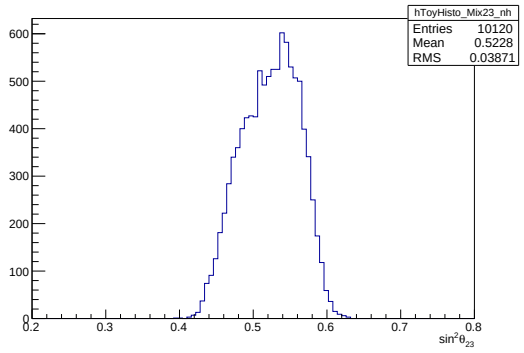
The distributions of oscillation parameters (for normal hierarchy) are shown in fig. 36. Note that whilst the distributions of the atmospheric parameters are shown independently they are generated jointly from the likelihood distribution shown in fig. 37.



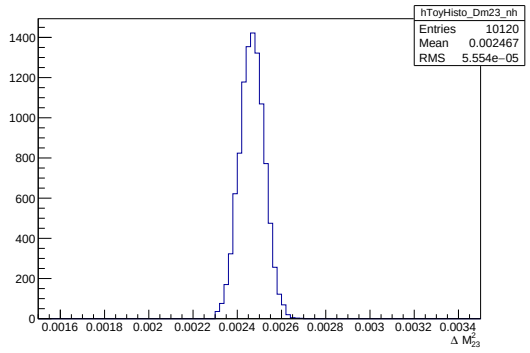
(a) $\sin^2\theta_{12}$



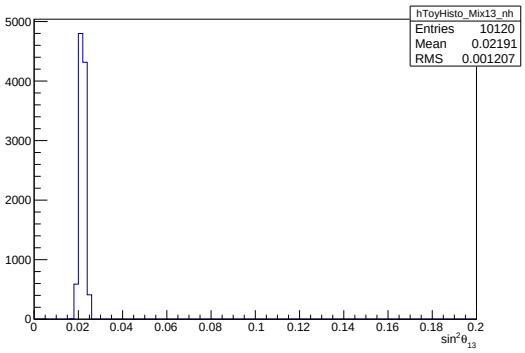
(b) Δm^2_{21}



(c) $\sin^2\theta_{23}$

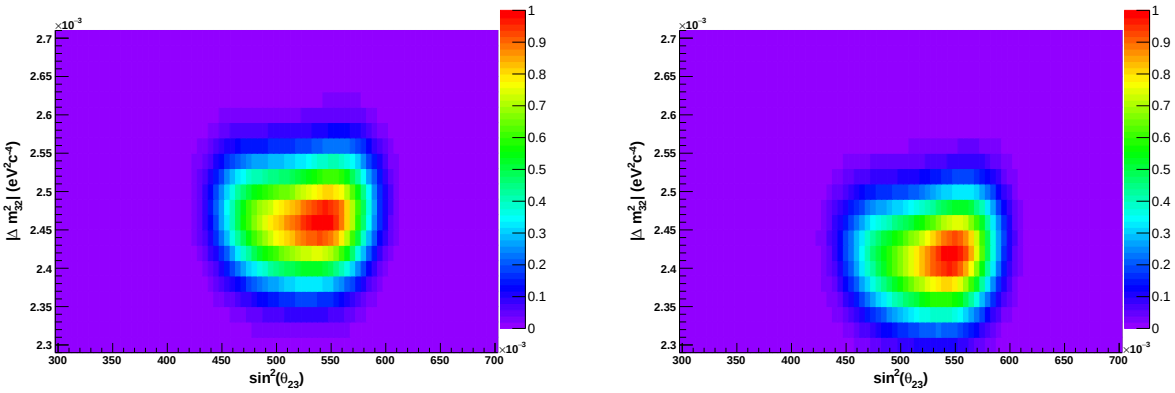


(d) Δm^2_{32}



(e) $\sin^2\theta_{13}$

Figure 36: Distribution of oscillation parameter throws for Feldman-Cousins fits.



(a) Normal hierarchy

(b) Inverted hierarchy

Figure 37: Likelihood surfaces for throws of the disappearance parameter.

5.7.2. Feldman-Cousins critical values and confidence intervals

In fig. 38 the confidence intervals obtained with the 'Feldman-Cousins' method are shown for δ_{CP} and mass hierarchy with reactor constraint for the Run 1-8 data set. The critical values at 90%, 2σ and 3σ CL are computed for both the mass hierarchies at the following true values of δ_{CP}; $-\pi$, $-3\pi/4$, $-\pi/2$, $-\pi/4$, 0, $+\pi/4$, $+\pi/2$, $+3\pi/4$, and $+\pi$. Inverted hierarchy is almost excluded at 2σ, except for a small interval around $\delta_{CP} = -\pi/2$. Both $\delta_{CP} = 0$ and $\delta_{CP} = \pi$ are excluded at 2σ. In table 17 the 2σ CL intervals for δ_{CP} versus mass hierarchy, obtained with the Feldman-Cousins method, are shown.

Parameter	Reactor	CL	Normal hierarchy	Inverted hierarchy
δ_{CP}	Yes	90%	$[-2.805, -0.830]$	-
δ_{CP}	Yes	2σ	$[-2.981, -0.600]$	$[-1.531, -1.184]$

Table 17: The confidence intervals at 90% CL and 2σ CL obtained with the Feldman-Cousins method for the Run 1-8 data set is shown for the measurement of δ_{CP} versus mass hierarchy.

5.8. Discussion of the Run 1-8 data fit results

The ν_e CC1π^+data set shows an excess in the number of observed appearance events, observing 15 events compared to an Asimov A expectation of 7. Unlike in our previous analysis, an excess is not observed in the neutrino mode one ring e-like sample, rather in this analysis we see a deficit in the number of observed events in neutrino mode one ring μ-like sample, observing 240 against an Asimov A expectation of 268.

The value of δ_{CP} favoured by the joint fit analysis, when the reactor constraint is applied, is -1.833 and corresponds to normal hierarchy. If the reactor constraint is not applied, the best-fit value of δ_{CP} is -2.083 and $\delta_{CP} = 0$ is still excluded at 2σ CL.

When considering the constant $\Delta\chi^2$ method, we see that the constraint on δ_{CP} is stronger than expected. To investigate the source of this difference a number of fits were undertaken using hybrids of data and MC. In particular fig. 39 shows the expected sensitivity from the Asimov A data set

(dashed), the data fit (solid black) and then a number of additional fits where one of the data samples was replaced by the corresponding Asimov A sample (e.g. Data x1Rmu indicates a fit of 4 data samples, with the one-ring μ-like sample being replaced by the Asimov A prediction). From this one can see that the largest contribution to the stronger constraint comes from the upward fluctuation observed in the ν_e CC1π^+ sample, but contributions are also evident from the FHC one-ring μ-like sample, where a deficit in events is seen, and from the two one-ring e-like samples. This results in the exclusion of CP-conservation at more than 2σ with the reactor constraint applied and the inverted hierarchy is disfavoured at 90%CL. The effect is also apparent in the appearance contour.

A stronger than expected constraint is also evident in the disappearance contour, with greater exclusion in $\sin^2\theta_{23}$. Similarly to the δ_{CP} hybrid fits above, fig. 40 shows the expected sensitivity from the Asimov A data set (red), the data fit (black) and then a number of additional fits where one of the data samples was replaced by the corresponding Asimov A sample. From this one can see that the stronger constraint can be explained by the deficit in the observed number of events in the one-ring μ-like sample.

The value of $\sin^2\theta_{13}$ favoured by the analysis is 0.0277 and corresponds to normal hierarchy, with the 1D contours in agreement with the PDG 2016 reactor constraint (0.0219 ± 0.0012) at $\sim 1.2\sigma$ (see table 16).

The value of $\sin^2\theta_{23}$ favoured by the analysis is 0.530 (0.510) with (without) the reactor constraint and corresponds to normal hierarchy, with the 1D contours in agreement with and without the reactor constraint at 1σ, favouring maximal mixing as seen in our previous analyses.

The value of Δm^2_{32} favoured by the analysis is 2.462 $\times 10^{-3}$ eV^2/c^4 (2.466 $\times 10^{-3}$ eV^2/c^4) with (without) the reactor constraint and corresponds to normal hierarchy, with the 1D contours in agreement with and without the reactor constraint at 1σ.

5.9. Comparison of δ_{CP} versus mass hierarchy result with sensitivity

In order to provide a better understanding of the results of the Run 1-8 data fit, the result has been compared with the expected sensitivity obtained from a large number (10k) of toy experiments for different true value of δ_{CP} (currently only $\delta_{CP} = -\pi/2$ is included) and the mass hierarchy.

5.9.1. Expected sensitivity method

This section outlines the procedure to produce the expected sensitivity comparison plots. The example given is for true normal hierarchy and $\delta_{CP} = -\pi/2$.

- 1×10^4 fake data sets are generated following the Feldman-Cousins method for a true value of δ_{CP} and mass hierarchy, in this case $\delta_{CP} = -\pi/2$ and normal hierarchy.
- A χ^2 distribution is computed with respect to the marginalisation toys as a function of δ_{CP} and mass hierarchy.
- For each fake data set, the minimum value of χ^2 for each bin of δ_{CP}, $\chi^2_{bin,min}$ is found with respect to both normal and inverted hierarchy.

- The test statistic is then computed as $\Delta\chi^2_{bin} = \chi^2_{bin} - \chi^2_{bin,min}$, which is slightly different to the Feldman-Cousins method, where $\Delta\chi^2_{FC} = \chi^2_{true} - \chi^2_{best-fit}$. The distributions of $\Delta\chi^2_{bin}$ and $\Delta\chi^2_{FC}$ coincide if the bin is $\delta_{CP} = -\pi/2$ and normal hierarchy, i.e. the true point used to generate the toy experiments.

- This results in a $\Delta\chi^2$ distribution from which we can draw conclusions about the expected $\Delta\chi^2$ for each value of δ_{CP} given a hypothesised true value and hierarchy.

- The spread of the $\Delta\chi^2$ distribution for each point of δ_{CP} and mass hierarchy is obtained as the range of the $\Delta\chi^2_{bin}$ distribution that contains respectively the 68.27% and 95.45% of the toy experiments on its left side. The test statistic $\Delta\chi^2_{bin}$ is assumed ot be 1-sided because by construction $\chi^2_{bin,min} \leq \chi^2_{bin}$.

- An alternative to the one-sided distribution is also produced, for which values of $\Delta\chi^2$ of each bin of δ_{CP} are themselves binned (100 evenly-spaced bins on the range $[\Delta\chi^2_{bin,min}, \Delta\chi^2_{bin,max}]$) and then toys are added to the expected sensitivity bands starting from the bin of $\Delta\chi^2$ that contains the most toys, until 68.27% and 95.45% of the toy experiments have been included, thereby producing two-sided distributions.

- A plot is also produced to show the fraction of toy experiments which exclude each value of δ_{CP} to greater than 2σ significance in the Feldman-Cousins corrected critical values along with the probabilities to exclude $\delta_{CP} = 0$ and $\delta_{CP} = \pi$ for a given true value of δ_{CP}.

5.9.2. $\delta_{CP} = -\pi/2$ - *normal hierarchy*

In fig. 41 the Run 1-8 data result for δ_{CP} versus mass hierarchy is compared to the 1σ and 2σ uncertainties on the $\Delta\chi^2$ distributions corresponding to $\delta_{CP} = -\pi/2$ and (a) normal hierarchy and (b) inverted hierarchy. We see that the data contour is contained within the 2σ band for inverted hierarchy, whilst the data contour falls just outside the 2σ band for normal hierarchy in the region around $\delta_{CP} = 0$. This suggests that the data constraint is consistent with the PMNS matrix at around the 2σ level.

We also see an interesting feature near $\delta_{CP} = -\pi/2$ where the $\Delta\chi^2$ values have a wider distribution. This is due to the boundary at $\delta_{CP} = -\pi/2$ which causes a wider distribution when the parameter δ_{CP} would like to be pushed beyond the boundary in the fit.

Finally from fig. 41(c) we see that $\delta_{CP} = 0$ and $\delta_{CP} = \pi$ are excluded at 90% confidence level in around 40% (see table 18) of the toy experiments and excluded at 2σ confidence level in between 25% and 30% of the toy experiments, with $\delta_{CP} = \pi$ excluded in slightly fewer toy experiments in each case.

5.9.3. $\delta_{CP} = -1.833$ - *normal hierarchy*

In fig. 42 the Run 1-8 data result for δ_{CP} versus mass hierarchy is compared to the 1σ and 2σ uncertainties on the $\Delta\chi^2$ distributions corresponding to $\delta_{CP} = -1.833$ and (a) normal hierarchy

δ_{CP}	Hierarchy	90%	2σ
0	NH	0.430	0.300
π	NH	0.383	0.250
0	IH	0.760	0.656
π	IH	0.769	0.667

Table 18: The fraction of toy experiments for which $\delta_{CP} = 0, \pi$ and normal and inverted hierarchy are excluded at 90% and 2σ CL is shown. 10k toy experiments are used.

and (b) inverted hierarchy. We see that the data contour is consistent with the PMNS matrix at around the 2σ level.

5.9.4. Sample-by-sample expected sensitivity

An extension to the two-sided method described above was applied in order to assess the effect of each sample on the δ_{CP} contour obtained. There key difference to the two-sided method described in section 5.9.1 is that the generation of the 1×10^4 fake data sets is undertaken for a single sample of interest, with the remaining 4 samples being fixed at the observed spectra.

Figures 43 to 44

It can be seen from fig. 43 that for normal hierarchy the observed contour resides within the 1σ band of expected sensitivities for 4 of the samples and thus while three of these contours are contributing to a stronger constraint than expected, the additional strength of the constraint in these samples is by no means surprising.

The contribution from the ν_e CC1π^+sample is rather extreme, as might be expected given the excess events observed in this sample, with the observed contour falling outside the 2σ band of expected sensitivities for this sample. Nonetheless, with all 5 samples considered together, as in fig. 41(a), the observed constraint resides approximately within the 2σ expected sensitivity.

For the case of inverted hierarchy (fig. 44) a similar scenario is observed, the principal difference being that the FHC one-ring μ-like sample has an observed contour contained with the 2σ band of expected sensitivities, though the full 5 sample contour remains within the 2σ expected sensitivity (fig. 41).

5.10. Comparison of $\sin^2\theta_{23}$ versus mass hierarchy result with sensitivity

This section provides a comparison of the Run 1-8 data fit with the expected sensitivity for $\sin^2\theta_{23}$.

The procedure used is the one outline in section 5.9.1 for the two-sided distribution, with the appropriate change in oscillation parameter. Furthermore, whereas the expected sensitivity method for δ_{CP} throws the atmospheric parameters from a 2D likelihood distribution from on an Asimov fake data set, whose oscillation parameter values are those from the disappearance data fit, in this case the Δm^2_{32} parameter is thrown from the 1D projection of this likelihood distribution.

5.10.1. $\sin^2\theta_{23} = 0.530$ - *normal hierarchy*

In fig. 45 the Run 1-8 data result for $\sin^2\theta_{23}$ versus mass hierarchy is compared to the 1σ and 2σ uncertainties on the $\Delta\chi^2$ distributions corresponding to $\sin^2\theta_{23} = 0.530$ for (a) normal hierarchy and (b) inverted hierarchy.

5.11. Comparison of Δm^2_{32} *versus mass hierarchy result with sensitivity*

This section provides a comparison of the Run 1-8 data fit with the expected sensitivity for Δm^2_{32}.

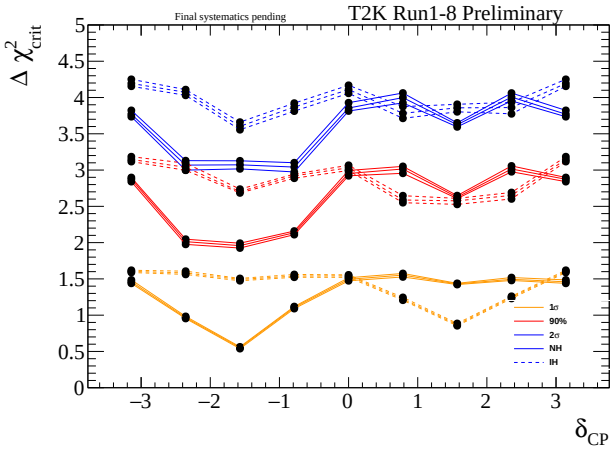
The procedure used is the one outline in section 5.9.1 for the two-sided distribution, with the appropriate change in oscillation parameter. Furthermore, whereas the expected sensitivity method for δ_{CP} throws the atmospheric parameters from a 2D likelihood distribution from on an Asimov fake data set, whose oscillation parameter values are those from the disappearance data fit, in this case the $\sin^2\theta_{23}$ parameter is thrown from the 1D projection of this likelihood distribution.

5.11.1. $\Delta m^2_{32} = -2.462 \times 10^{-3} eV^2\ c^{-4}$ - *normal hierarchy*

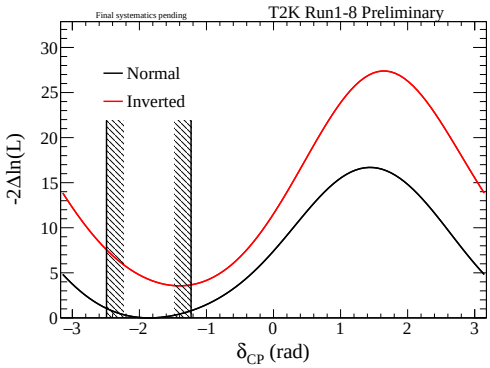
In fig. 46 the Run 1-8 data result for Δm^2_{32} versus mass hierarchy is compared to the 1σ and 2σ uncertainties on the $\Delta\chi^2$ distributions corresponding to $\Delta m^2_{32} = -2.462 \times 10^{-3}\mathrm{eV}^2\ \mathrm{c}^{-4}$ for (a) normal hierarchy and (b) inverted hierarchy.

5.12. Comparison of e-like event rates with expectation

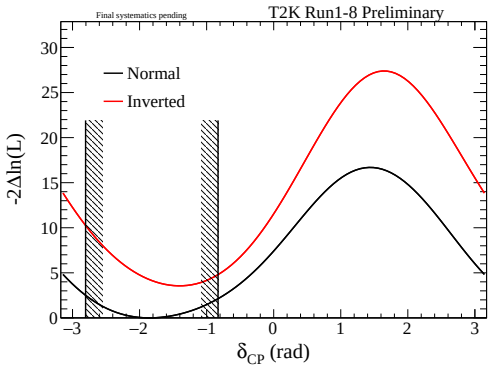
It is valuable to see how different values of δ_{CP}, $\sin^2\theta_{23}$ and hierarchy affect the predicted event rates. Figure 47 shows the predicted $\bar{\nu}_e$ event rate vs ν_e event rate for best-fit values of oscillation parameters where δ_{CP} is varied between CP conserving and maximally CP violating values, and $\sin^2\theta_{23}$ is varied around its best-fit, for both hierarchies. Event rates for a given value of $\sin^2\theta_{23}$ and mass hierarchy are linearly interpolated between those computed for 9 evenly-spaced values of δ_{CP} from $-\pi$ to $+\pi$ to indicate the behaviour produced by varying δ_{CP}. The observed event rates with 1σ errors are also shown. Figure 47(a) shows e-like event rates where the FHC sample includes the ν_e CC1π^+sample, while fig. 47(b) shows e-like event rates where the FHC sample does not include the ν_e CC1π^+sample.



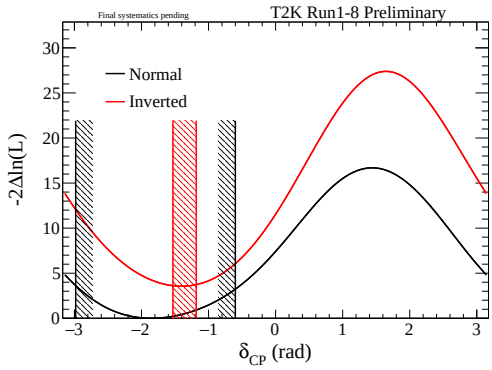
(a) Critical $\Delta\chi^2$ values



(b) 1σ CL exclusion region



(c) 90% CL exclusion region



(d) 2σ CL exclusion region

Figure 38: $\Delta\chi^2$ critical values and confidence intervals for the measured $\Delta\chi^2$ distributions for Run 1-8. Critical values obtained with the Feldman-Cousins method for Run 1-8 for 9 evenly spaced values on the range $[-\pi, \pi]$. Critical values are shown for 1σ, 2σ and 90% CL for normal (solid lines) and inverted (dashed lines) hierarchies. At least 1×10^4 toy experiments are performed for each point. The three bands of lines show the $\pm 1\sigma$ uncertainty on the critical values. Also shown are the measured $\Delta\chi^2$ distributions shifted with respect to the same global minium and the 1σ, 90% and 2σ exclusion regions for both mass hierarchies.

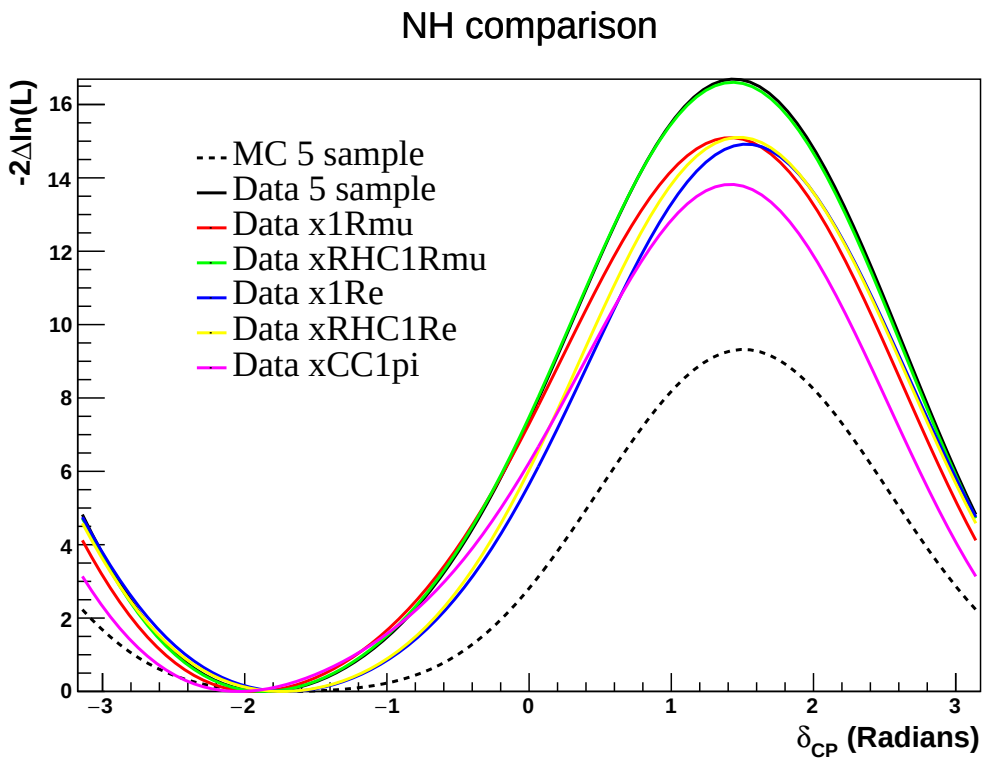


Figure 39: The expected $\Delta\chi^2$ distribution as a function of δ_{CP} with reactor constraint for MC, data and data/MC hybrid data sets.

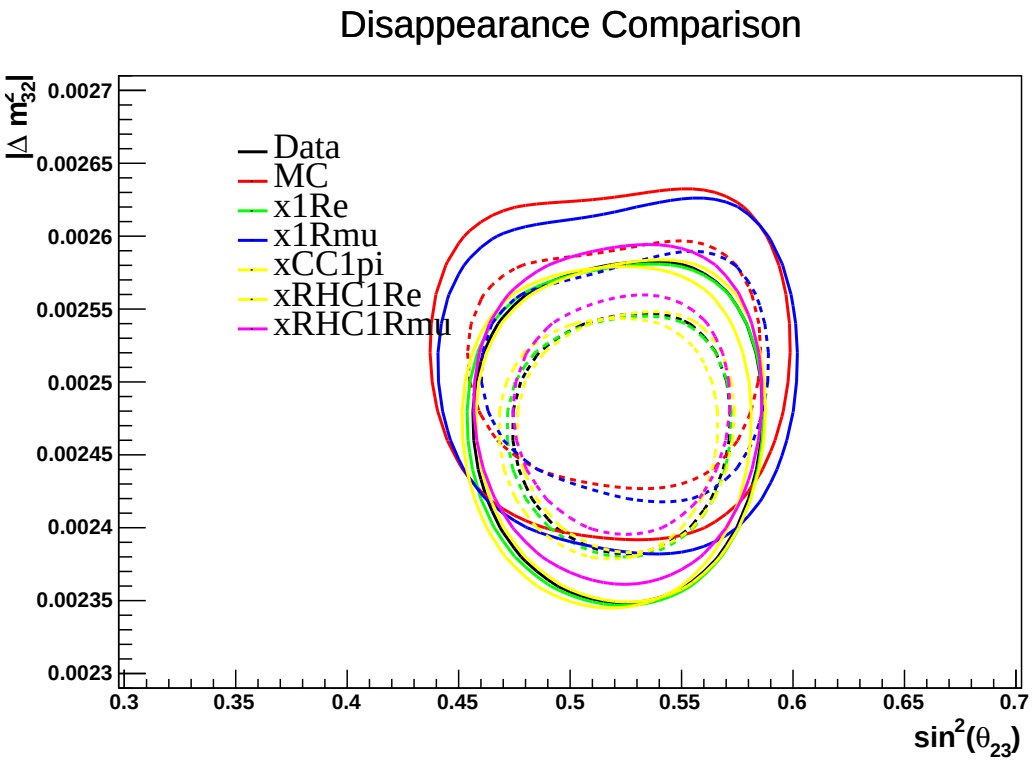
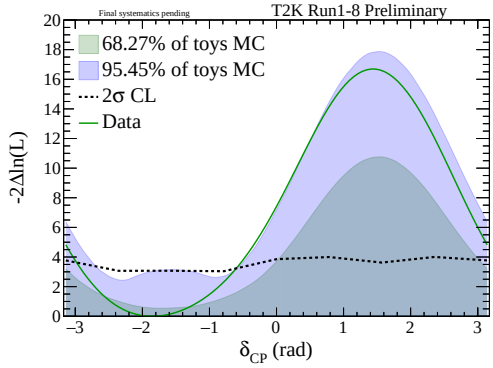
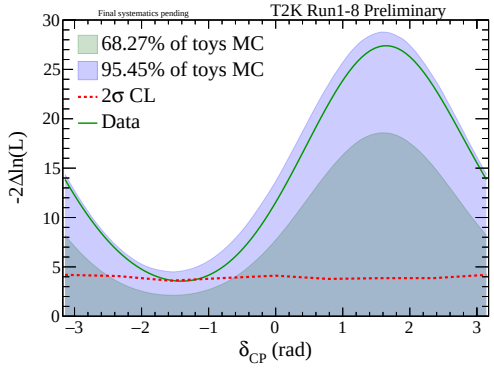


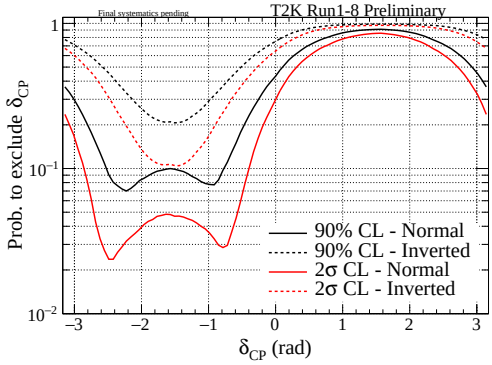
Figure 40: The expected disappearance contours with reactor constraint for normal hierarchy for MC, data and data/MC hybrid data sets.



(a) Normal hierarchy

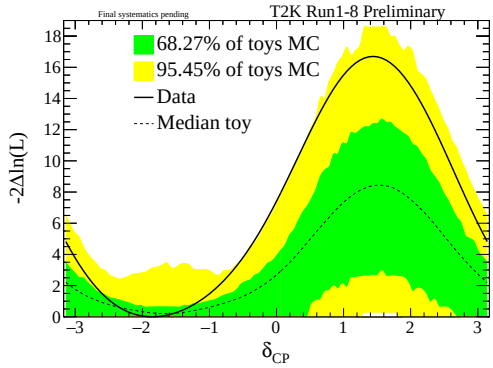


(b) Inverted hierarchy

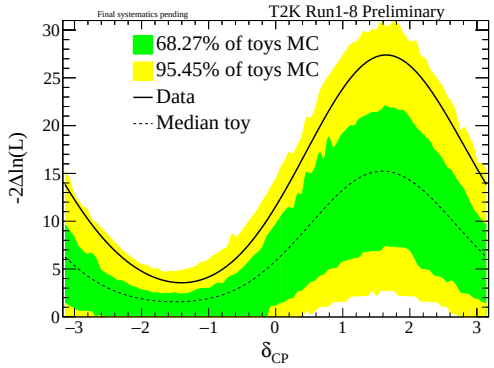


(c) Data $\Delta\chi^2$ probablility

Figure 41: The distribution of $\Delta\chi^2$ vs δ_{CP} obtained with 1×10^4 toy experiments generated with $\delta_{CP} = -\pi/2$ and normal hierarchy is shown. The $\Delta\chi^2$ distributions obtained by fixing the mass hierarchy to normal (left) and inverted (right) are shown, along with the $\Delta\chi^2$ values corresponding to 68.27% and 95.45% of the toy experiments and the average $\Delta\chi^2$ values from all experiments. Also shown are the fractions of toys which exclude each value of δ_{CP} to 90% and 2σ using the Feldman-Cousins corrected critical values (bottom).

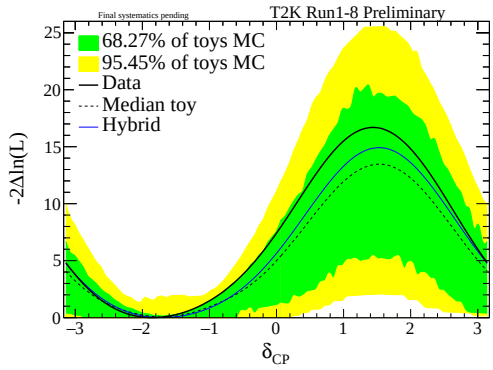


(a) Normal hierarchy

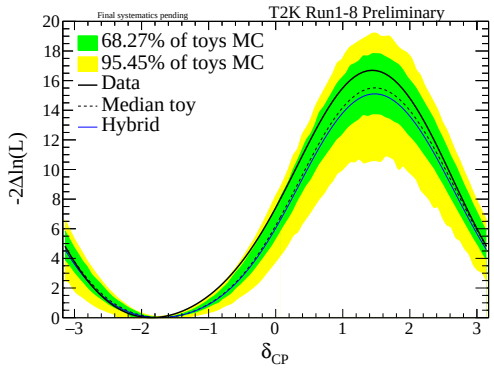


(b) Inverted hierarchy

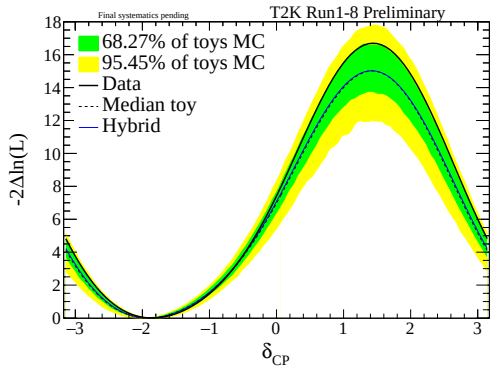
Figure 42: The distribution of $\Delta\chi^2$ vs δ_{CP} obtained with 1×10^4 toy experiments generated with $\delta_{CP} = -1,833$ and normal hierarchy is shown. The $\Delta\chi^2$ distributions obtained by fixing the mass hierarchy to normal (left) and inverted (right) are shown, along with the $\Delta\chi^2$ values corresponding to 68.27% and 95.45% of the toy experiments, the median toy and the data contour.



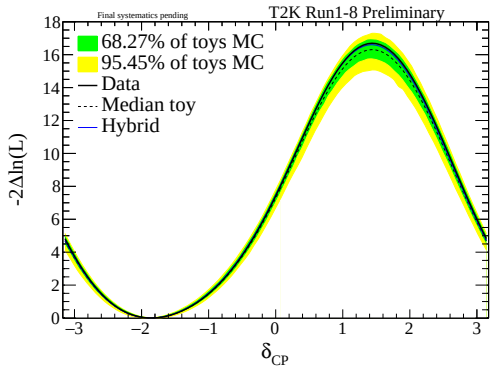
(a) Electron-like, neutrino mode



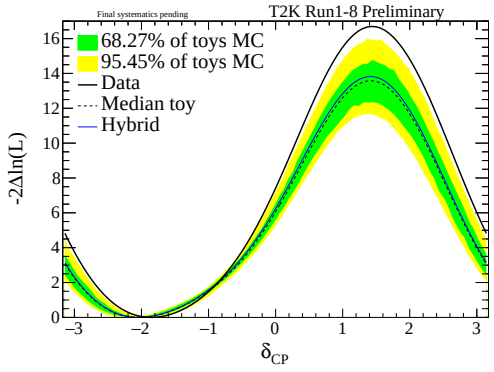
(b) Electron-like, antineutrino mode



(c) Muon-like, neutrino mode

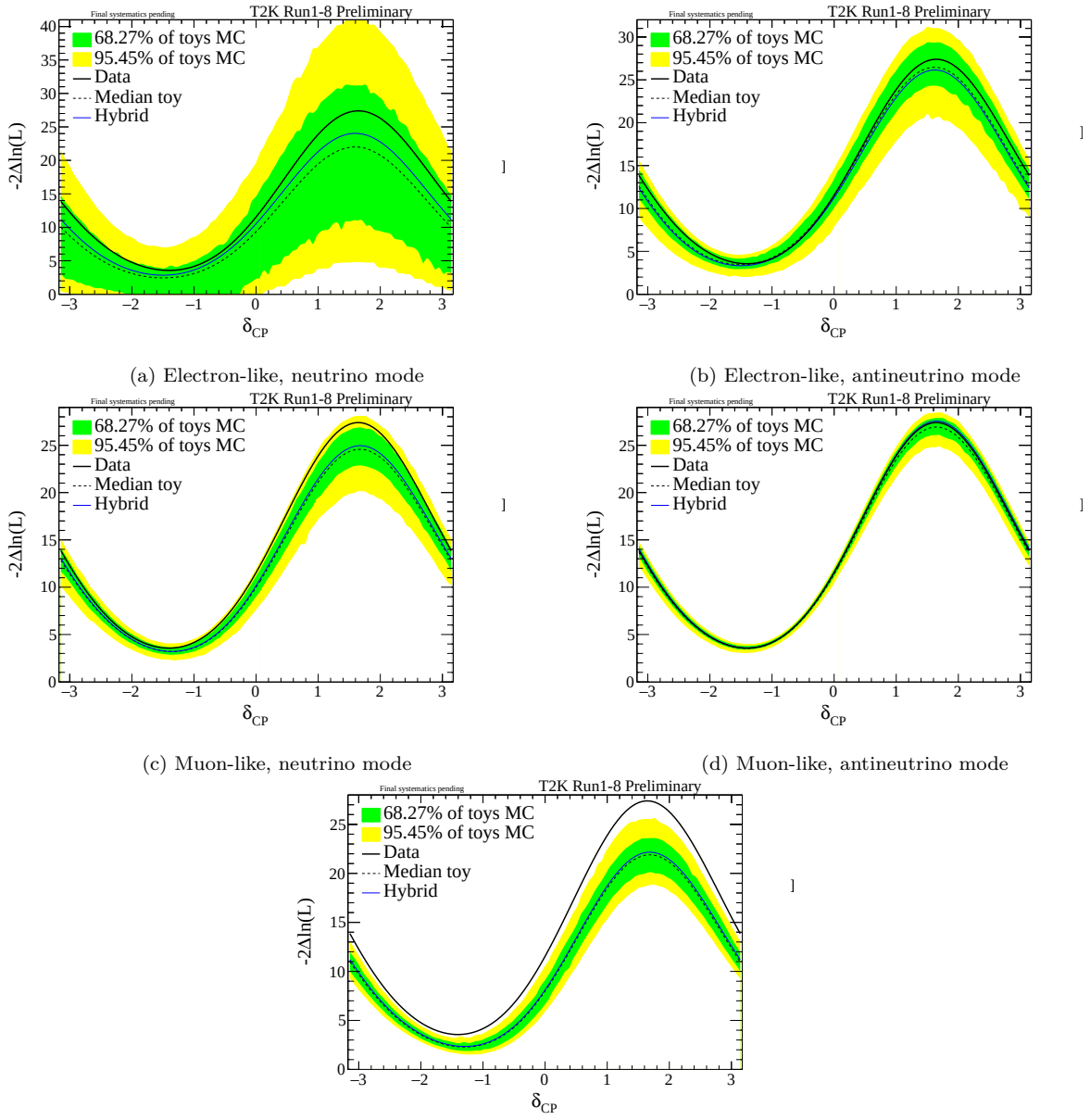


(d) Muon-like, antineutrino mode



(e) ν_e CC1π^+-like, neutrino mode

Figure 43: The distribution of $\Delta\chi^2$ vs δ_{CP} obtained with 1×10^4 toy experiments generated with $\delta_{CP} = -1.833$ and normal hierarchy is shown. The $\Delta\chi^2$ distribution obtained from the full 5-sample fit, obtained by fixing the mass hierarchy to normal, is shown, along with the $\Delta\chi^2$ values corresponding to 68.27% and 95.45% of the toy experiments for the indicated sample, with all other samples fixed at their observed spectra.



(a) Electron-like, neutrino mode

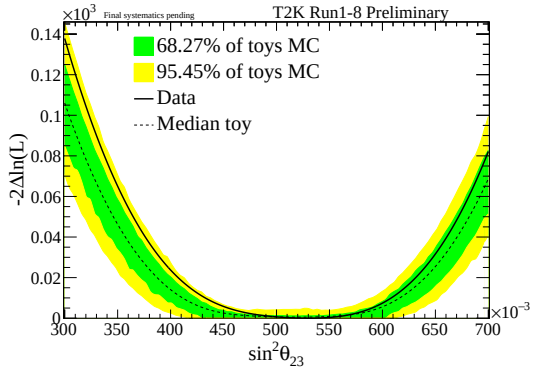
(b) Electron-like, antineutrino mode

(c) Muon-like, neutrino mode

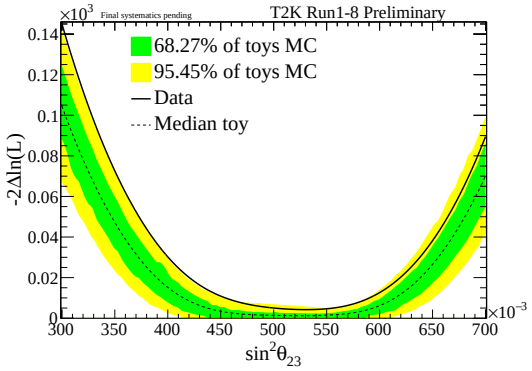
(d) Muon-like, antineutrino mode

(e) ν_e CC1π^+-like, neutrino mode

Figure 44: The distribution of $\Delta\chi^2$ vs δ_{CP} obtained with 1×10^4 toy experiments generated with $\delta_{CP} = -1.833$ and normal hierarchy is shown. The $\Delta\chi^2$ distribution obtained from the full 5-sample fit, obtained by fixing the mass hierarchy to inverted, is shown, along with the $\Delta\chi^2$ values corresponding to 68.27% and 95.45% of the toy experiments for the indicated sample, with all other samples fixed at their observed spectra.

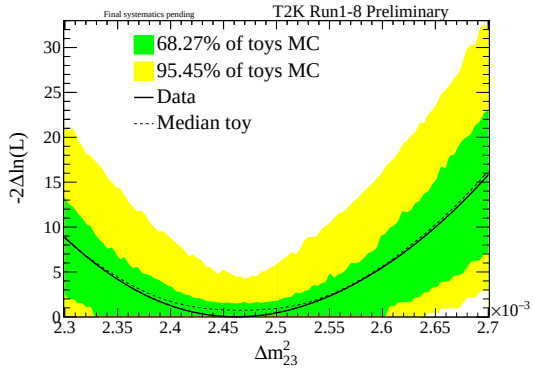


(a) Normal hierarchy

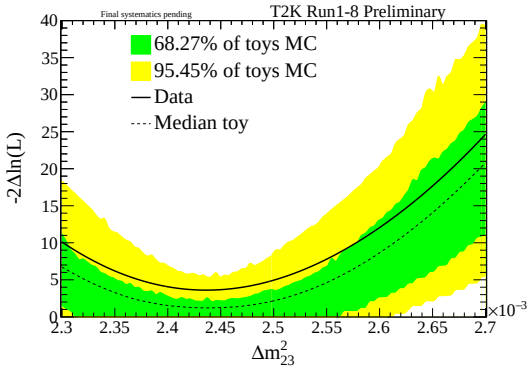


(b) Inverted hierarchy

Figure 45: The distribution of $\Delta\chi^2$ vs $\sin^2\theta_{23}$ obtained with 1×10^4 toy experiments generated with $\sin^2\theta_{23} = 0.530$ and normal hierarchy is shown. The $\Delta\chi^2$ distributions obtained by fixing the mass hierarchy to normal (left) and inverted (right) are shown, along with the $\Delta\chi^2$ values corresponding to 68.27% and 95.45% of the toy experiments, the median toy and the data contour.



(a) Normal hierarchy



(b) Inverted hierarchy

Figure 46: The distribution of $\Delta\chi^2$ vs Δm^2_{32} obtained with 1×10^4 toy experiments generated with $\Delta m^2_{32} = -2.462 \times 10^{-3}\text{eV}^2\ \text{c}^{-4}$ and normal hierarchy is shown. The $\Delta\chi^2$ distributions obtained by fixing the mass hierarchy to normal (left) and inverted (right) are shown, along with the $\Delta\chi^2$ values corresponding to 68.27% and 95.45% of the toy experiments, the median toy and the data contour.

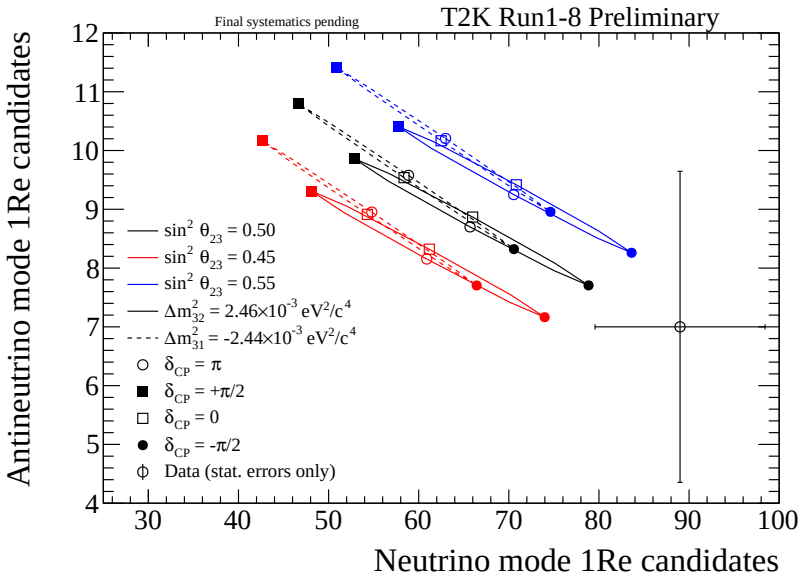


Figure 47: This plot shows the candidate RHC one-ring e-like event rate vs the candidate FHC one-ring e-like event rate for a variety of different oscillation parameter values. Predictions are generated for best-fit values of oscillation parameters for different values of δ_{CP}, $\sin^2\theta_{23}$ and mass hierarchy. In particular the FHC and RHC event rates are plotted for the CP conserving and maximally CP violating values of δ_{CP}, with $\sin^2\theta_{23}$ set at the best-fit value and $\pm 1\sigma$ values in both normal and inverted hierarchy. The observed rates with statistical error shown are also included.

6. Updated results with E_b parameter and Δm^2 smearing

6.1. Findings from the 2017 fake data studies

As part of the analysis procedure, a variety of fake data fits were performed to assess the robustness of the analysis to changes in the cross-section model [89]. The conclusion of these fake data studies was that an additional source of uncertainty was necessary to address biases seen in $\sin^2\theta_{23}$ and Δm^2 (Δm^2_{32} for normal hierarchy or Δm^2_{13} for inverted hierarchy) in the E_b fake data studies. Furthermore, the inclusion of the additional systematic parameter was insufficient to address the bias in Δm^2 and therefore an additional smearing of the likelihood surface for this parameter was added.

6.2. Systematic parameter update and smearing procedure

The additional E_b systematic parameter is implemented as a response function, based on weights determined by the ratio of near-to-far predicted fake data event rates and event rates seen in the fake data set. The ratio was calculated for 3 points; -0.5σ, 0σ and $+1\sigma$, with a further 3 linearly extrapolated points at -1.5σ, $+3\sigma$ and $+5\sigma$ included to allow the response function to better conform to bi-linearity throughout the allowed $[-1.5\sigma, +5\sigma]$ range. Negative weights that can result from the extrapolation are set to zero.

Post-fit, the Δm^2 contour is inflated via a smearing procedure. In particular, the likelihood surface resulting from the fit is scaled by a factor of 10 and resampled using Catmull-Rom cubic interpolation, before each source bin in Δm^2 (for a particular value of $\sin^2\theta_{23}$ in the 2D fit) is smeared over the remaining Δm^2 bins, with the contribution weighted by a Gaussian centred on the source bin, integrated over the target bin interval.

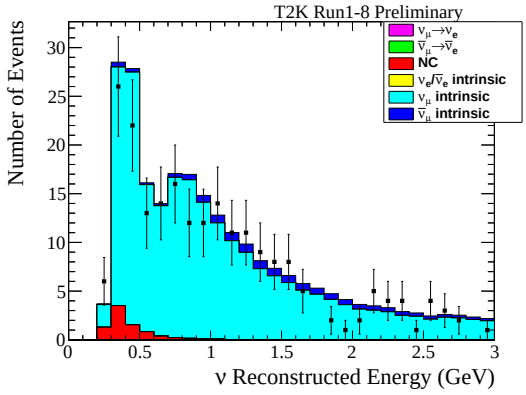
6.3. Results

The results of the Run 1-8 data set fit, for an exposure of 1.4734×10^{21}POT in neutrino mode and of 7.558×10^{20}POT in antineutrino mode, with the E_b parameter and smearing applied, are shown in this chapter. In fig. 48 the best-fit predicted spectra of μ-like and e-like samples are compared to the observed data. In table 19 the expected and observed number of events in Run 1-8 data set are shown.

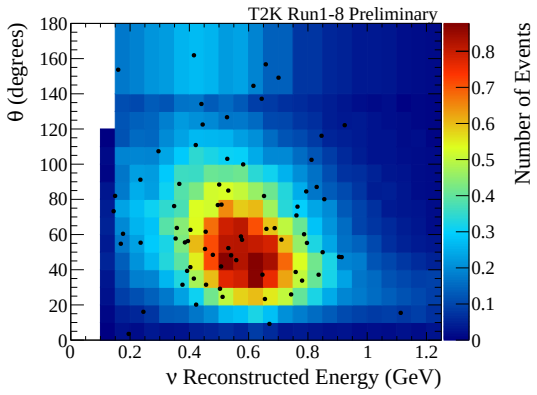
Beam mode	Sample	Exp. Best-fit	Observed
ν	μ-like	270.792	240
ν	e-like	73.099	74
$\bar{\nu}$	μ-like	64.997	68
$\bar{\nu}$	e-like	7.947	7
ν	ν_e CC1π^+-like	6.812	15

Table 19: The observed and best-fit expected number of events in Run 1-8 data set for an exposure of 1.4734×10^{21}POT in neutrino mode and of 7.558×10^{20}POT in antineutrino mode are shown for each selected sample. The prediction is produced using the BANFF tuning.

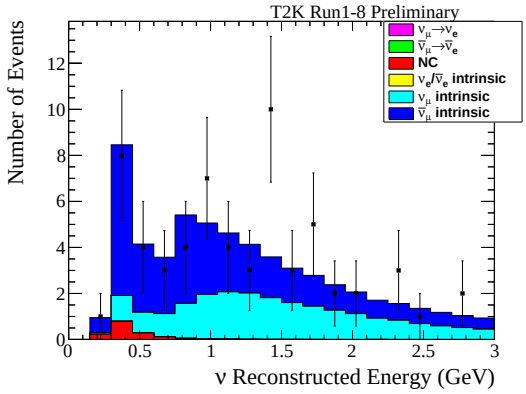
The predicted and observed spectra are shown for each of the samples in fig. 48. The e-like samples are broken down into E_{rec} and θ projections in fig. 49.



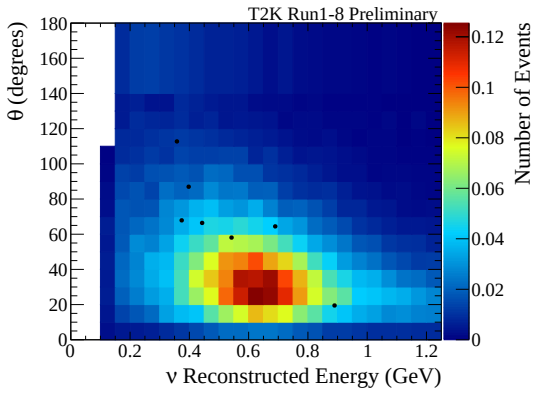
(a) Muon-like, neutrino mode



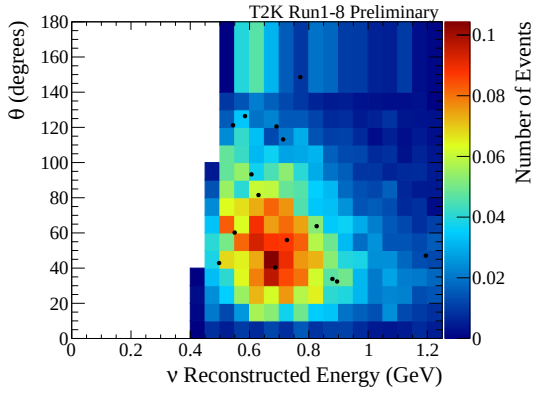
(b) Electron-like, neutrino mode



(c) Muon-like, antineutrino mode

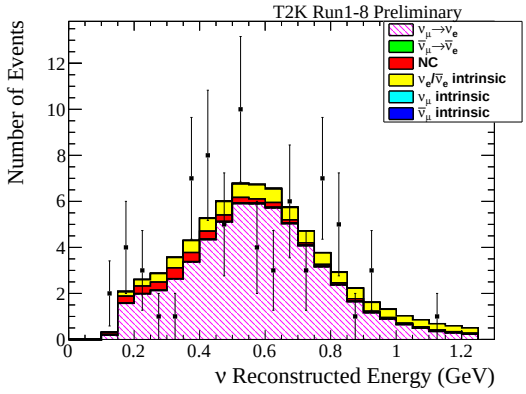


(d) Electron-like antineutrino mode

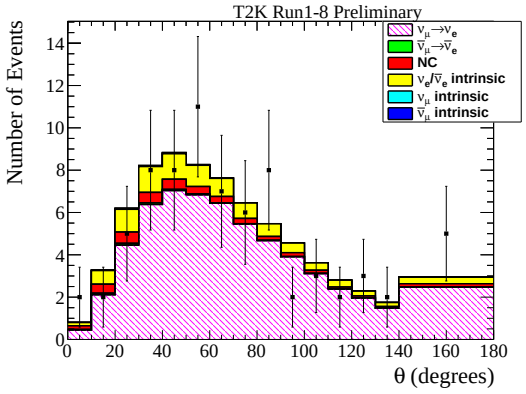


(e) ν_e CC1π^+-like, neutrino mode

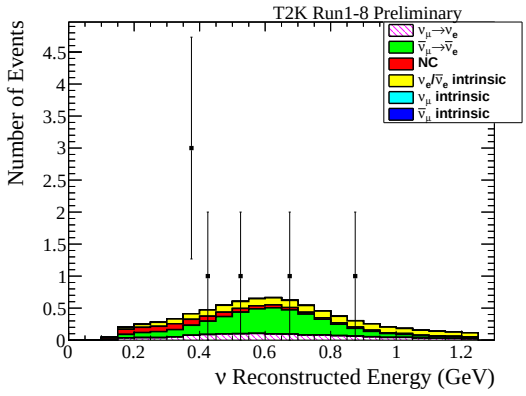
Figure 48: Predicted spectra and observed events (points). μ-like distributions are a function of the reconstructed neutrino energy, while the e-like, including ν_e CC1π^+, distributions are functions of both the reconstructed neutrino energy and the reconstructed angle between the outgoing lepton and the neutrino direction. The distributions correspond to the statistics collected in the full Run 1-8 data set. The spectra are generated with the systematic parameters described in section 3 and the oscillation parameters corresponding to the best-fit values from the data fit (solar parameters at PDG 2016).



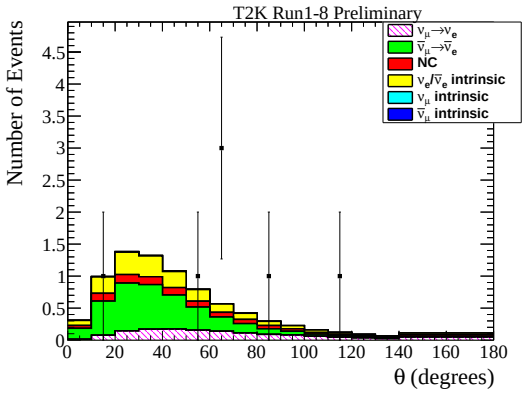
(a) Electron-like, neutrino mode



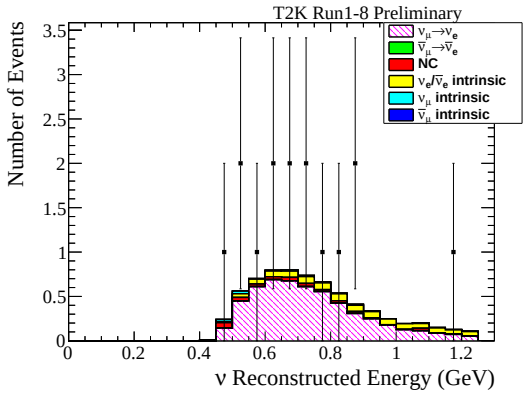
(b) Electron-like, neutrino mode



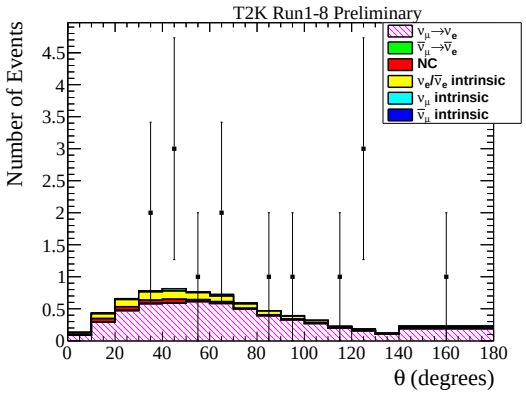
(c) Electron-like antineutrino mode



(d) Electron-like antineutrino mode



(e) ν_e CC1π^+-like, neutrino mode



(f) ν_e CC1π^+-like, neutrino mode

Figure 49: Predicted spectra and observed events (points). The e-like, including ν_e CC1π^+, distributions are functions of both the reconstructed neutrino energy and the reconstructed angle between the outgoing lepton and the neutrino direction, with the projections in each variable shown here. The distributions correspond to the statistics collected in the full Run 1-8 data set. The spectra are generated with the systematic parameters described in section 3 and the oscillation parameters corresponding to the best-fit values from the data fit (solar parameters at PDG 2016).

Parameter	Reactor	Best-fit (NH)	$\pm1\sigma$ (NH)	Best-fit (IH)	$\pm1\sigma$ (IH)
δ_{CP}	Yes	-1.872	[-2.480,-1.137]	-1.428	[-1.961,-0.880]
δ_{CP}	No	-2.126	[-3.048,-1.129]	-1.174	[-1.988,-0.440]
$\sin^2\theta_{13}$	No	0.0279	[0.0231,0.0343]	0.0316	[0.0264,0.0380]
$\sin^2\theta_{23}$	Yes	0.526	[0.490,0.558]	0.530	[0.496,0.560]
Δm^2_{32} or Δm^2_{13} (10^{-3} eV^2/c^4)	Yes	2.463	[2.393,2.534]	2.432	[2.361,2.502]

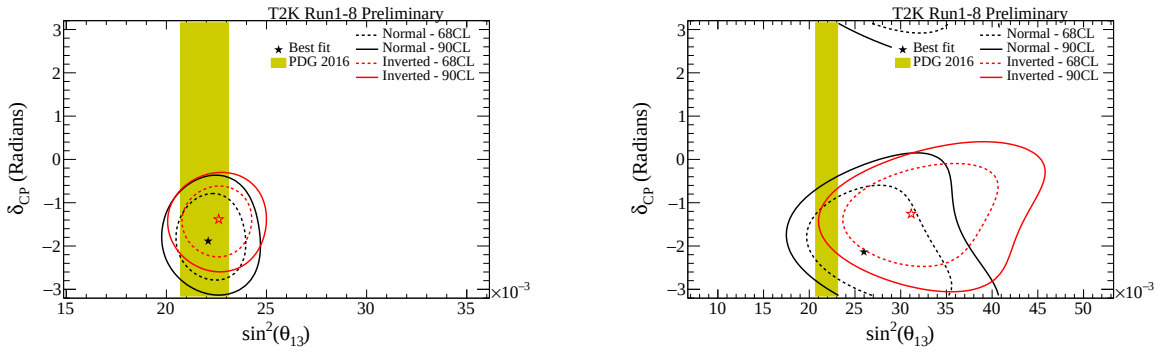
Table 20: The Run 1-8 measured oscillation parameter best-fit and the $\pm1\sigma$ intervals are shown for normal and inverted hierarchies with respect to the hierarchy best-fit. The $\pm1\sigma$ interval is obtained by using the constant $\Delta\chi^2$ method. For each parameter normal hierarchy is the global best-fit.

The best-fit values and 1σ ranges for each oscillation parameter are shown in table 20 for normal hierarchy (which is preferred in all data fits). The 1σ ranges are extracted for each oscillation parameter from the 1-dimensional $\Delta\chi^2$ distributions shown in figs. 52 to 55.

The 1-dimensional contours are plotted with respect to the global minimum from normal and inverted hierarchies, while the 2-dimensional confidence intervals are shown drawing the normal and inverted hierarchy contours independently, i.e. each contour is produced with respect to the global minimum under that hierarchy hypothesis.

6.3.1. *Results for δ_{CP} vs $\sin^2\theta_{13}$*

In fig. 50 the result for δ_{CP} vs $\sin^2\theta_{13}$ with reactor constraint and T2K only is shown.



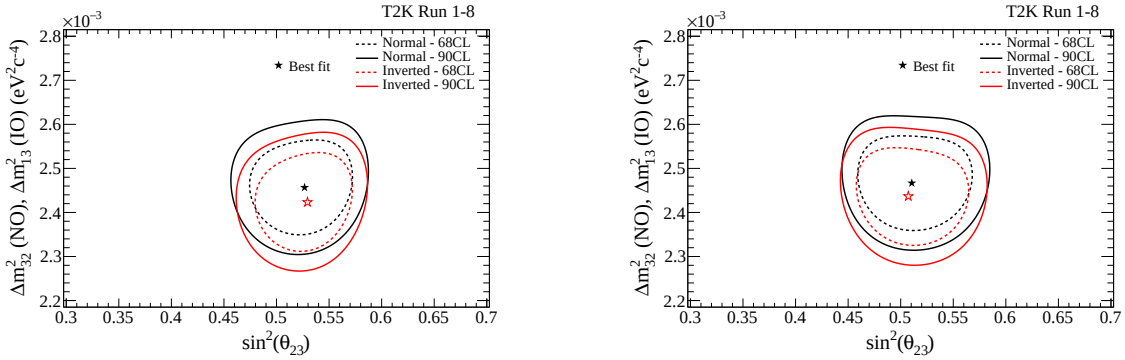
(a) δ_{CP} vs $\sin^2\theta_{13}$ with reactor constraint

(b) δ_{CP} vs $\sin^2\theta_{13}$ T2K only

Figure 50: Contours at 68% and 90% CL for δ_{CP} vs $\sin^2\theta_{13}$ with and without reactor constraint. Normal and inverted hierarchy contours are independent. All 5 samples were used to produce these fits.

6.3.2. *Results for Δm^2 vs $\sin^2\theta_{23}$*

The result for Δm^2 vs $\sin^2\theta_{23}$ with reactor constraint and T2K only is shown in fig. 51.



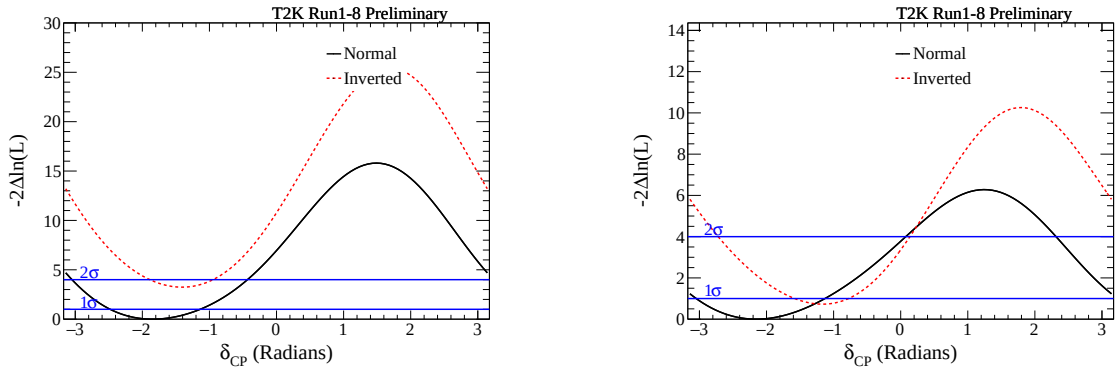
(a) Δm^2 vs $\sin^2\theta_{23}$ with reactor constraint

(b) Δm^2 vs $\sin^2\theta_{23}$ T2K only

Figure 51: Contours at 68% and 90% CL for Δm^2 vs $\sin^2\theta_{23}$ with and without reactor constraint. Normal and inverted hierarchy contours are independent. All 5 samples were used to produce these fits.

6.3.3. Results for δ_{CP}

The result for δ_{CP} with reactor constraint and T2K only is shown in fig. 52.



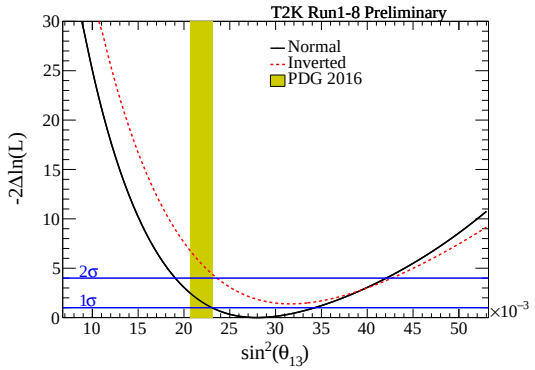
(a) δ_{CP} with reactor constraint

(b) δ_{CP} T2K only

Figure 52: The expected $\Delta\chi^2$ distribution as a function of δ_{CP} with and without reactor constraint. The mass hierarchy is minimized, with normal and inverted hierarchy $\Delta\chi^2$ distributions shifted to the same global best-fit χ^2 value, which is taken to be the minimum between normal and inverted hierarchy. All 5 samples were used to produce these fits.

6.3.4. Results for $\sin^2\theta_{13}$

The result for $\sin^2\theta_{13}$ (without reactor constraint) is shown in fig. 53.

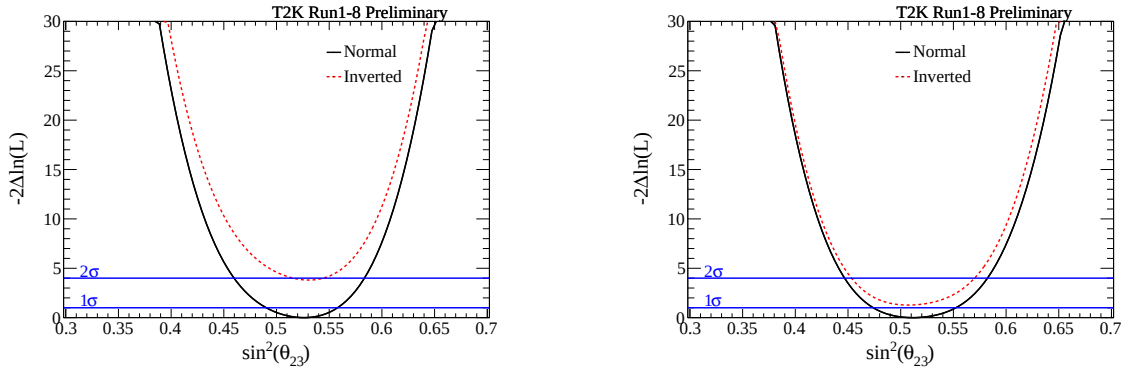


(a) $\sin^2\theta_{13}$ T2K only

Figure 53: The expected $\Delta\chi^2$ distribution as a function of $\sin^2\theta_{13}$ without reactor constraint. The mass hierarchy is minimized, with normal and inverted hierarchy $\Delta\chi^2$ distributions shifted to the same global best-fit χ^2 value, which is taken to be the minimum between normal and inverted hierarchy. All 5 samples were used to produce these fits.

6.3.5. *Results for* $\sin^2\theta_{23}$

The result for $\sin^2\theta_{23}$ with reactor constraint and T2K only is shown in fig. 54.



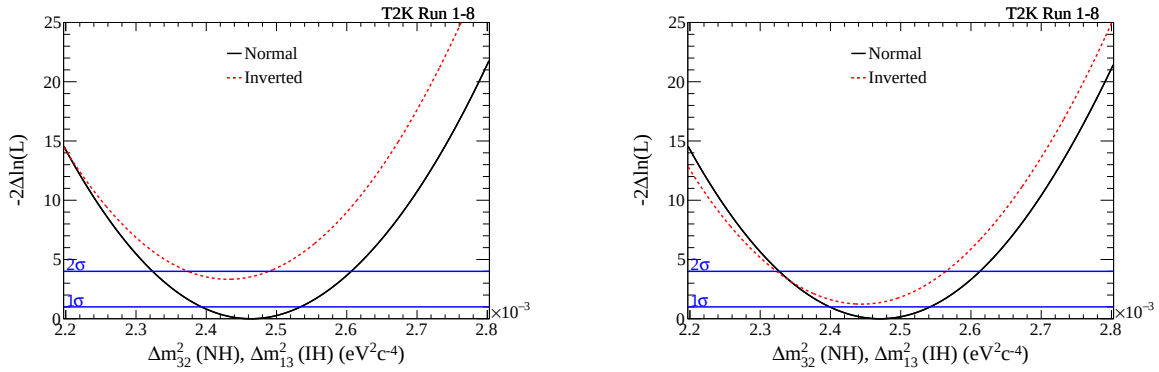
(a) $\sin^2\theta_{23}$ with reactor constraint

(b) $\sin^2\theta_{23}$ T2K only

Figure 54: The expected $\Delta\chi^2$ distribution as a function of $\sin^2\theta_{23}$ with and without reactor constraint. The mass hierarchy is minimized, with normal and inverted hierarchy $\Delta\chi^2$ distributions shifted to the same global best-fit χ^2 value, which is taken to be the minimum between normal and inverted hierarchy. All 5 samples were used to produce these fits.

6.3.6. *Results for* Δm^2

The result for Δm^2 with reactor constraint and T2K only is shown in fig. 55.



(a) Δm^2 with reactor constraint

(b) Δm^2 T2K only

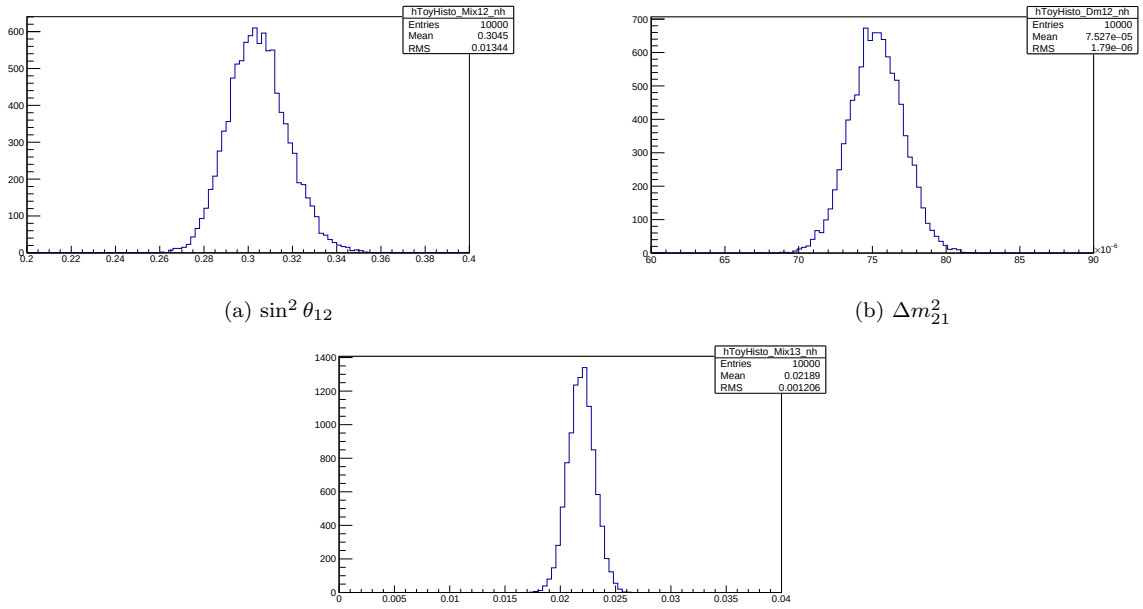
Figure 55: The expected $\Delta\chi^2$ distribution as a function of Δm^2 with and without reactor constraint. The mass hierarchy is minimized, with normal and inverted hierarchy $\Delta\chi^2$ distributions shifted to the same global best-fit χ^2 value, which is taken to be the minimum between normal and inverted hierarchy. All 5 samples were used to produce these fits.

6.4. *Feldman-Cousins simultaneous fit of* δ_{CP} *and mass hierarchy*

This section presents the results of the Feldman-Cousins fit of δ_{CP} and mass hierarchy. Throws of oscillation parameters are made according to PDG 2016 best-fit values and errors for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, whilst $\sin^2\theta_{23}$ and Δm^2 are thrown according to the smeared likelihood distribution produced by an Asimov fit using the best-fit from data for these parameters. δ_{CP} is fixed at one of 9 evenly specified values on the range $[-\pi, \pi]$.

6.4.1. Input oscillation parameters

The distributions of oscillation parameters (for normal hierarchy) are shown in fig. 56. The distributions of the atmospheric parameters are shown in fig. 57.

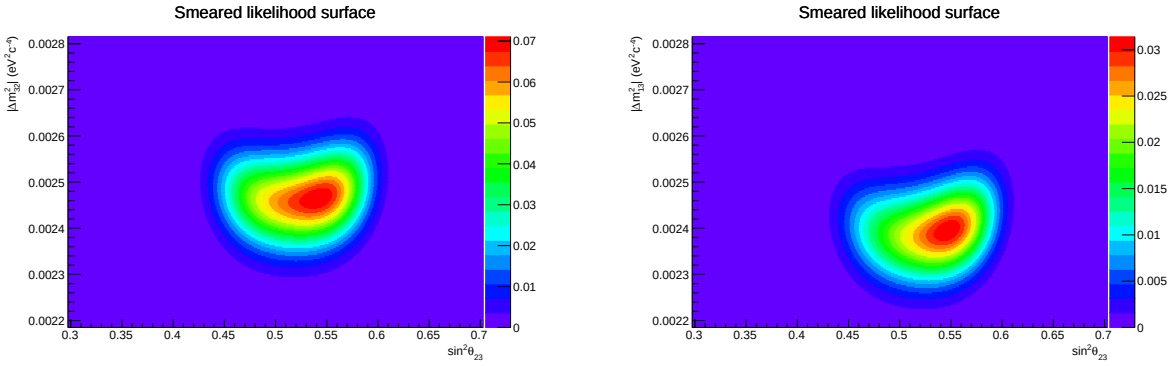


(a) $\sin^2\theta_{12}$

(b) Δm^2_{21}

(c) $\sin^2\theta_{13}$

Figure 56: Distribution of oscillation parameter throws for Feldman-Cousins fits.



(a) Normal hierarchy

(b) Inverted hierarchy

Figure 57: Likelihood surfaces for throws of the disappearance parameter.

6.4.2. Feldman-Cousins critical values and confidence intervals

In fig. 58 the confidence intervals obtained with the 'Feldman-Cousins' method are shown for δ_{CP} and mass hierarchy with reactor constraint for the Run 1-8 data set. The critical values at 1σ, 90% and 2σ CL are computed for both the mass hierarchies at the following true values of δ_{CP}; $-\pi$, $-3\pi/4$, $-\pi/2$, $-\pi/4$, 0, $+\pi/4$, $+\pi/2$, $+3\pi/4$, and $+\pi$. Inverted hierarchy is almost excluded at 2σ, except for a small interval around $\delta_{CP} = -\pi/2$. Both $\delta_{CP} = 0$ and $\delta_{CP} = \pi$ are excluded at 2σ.

In table 21 the 2σ CL intervals for δ_{CP} versus mass hierarchy, obtained with the Feldman-Cousins method, are shown. Comparison to p-theta critical values are shown in Appendix J.

Parameter	Reactor	CL	Normal hierarchy	Inverted hierarchy
δ_{CP}	Yes	1σ	$[-2.510, -1.262]$	-
δ_{CP}	Yes	90%	$[-2.815, -0.847]$	-
δ_{CP}	Yes	2σ	$[-2.992, -0.592]$	$[-1.808, -1.005]$

Table 21: The confidence intervals at 1σ, 90% CL and 2σ CL obtained with the Feldman-Cousins method for the Run 1-8 data set is shown for the measurement of δ_{CP} versus mass hierarchy.

6.4.3. $\delta_{CP} = -\pi/2$ - *normal hierarchy*

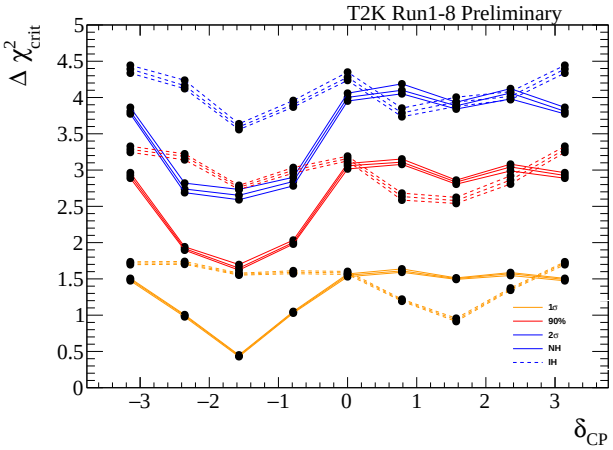
In fig. 59 the Run 1-8 data result for δ_{CP} versus mass hierarchy is compared to 68.27% and 95.45% of toys from a $\Delta\chi^2$ distributioncorresponding to $\delta_{CP} = -\pi/2$ and (a) normal hierarchy and (b) inverted hierarchy. We see that more than 4.45% (2σ level) of toys provide a stronger constraint than the data contour for both normal and inverted hierarchy, consistent with the PMNS matrix.

We also see an interesting feature near $\delta_{CP} = -\pi/2$ where the $\Delta\chi^2$ values have a wider distribution. This is due to the boundary at $\delta_{CP} = -\pi/2$ which causes a wider distribution when the parameter δ_{CP} would like to be pushed beyond the boundary in the fit.

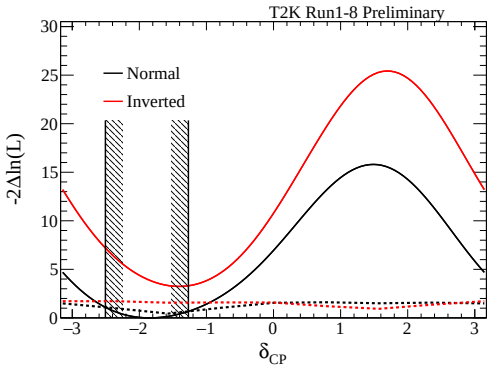
Finally from fig. 59(c) we see that $\delta_{CP} = 0$ and $\delta_{CP} = \pi$ are excluded at 90% confidence level at between 40% and 50% (see table 22) of the toy experiments and excluded at 2σ confidence level in between 25% and 35% of the toy experiments, with $\delta_{CP} = \pi$ excluded in slightly fewer toy experiments in each case.

δ_{CP}	Hierarchy	90%	2σ
0	NH	0.474	0.332
π	NH	0.414	0.279
0	IH	0.803	0.708
π	IH	0.796	0.693

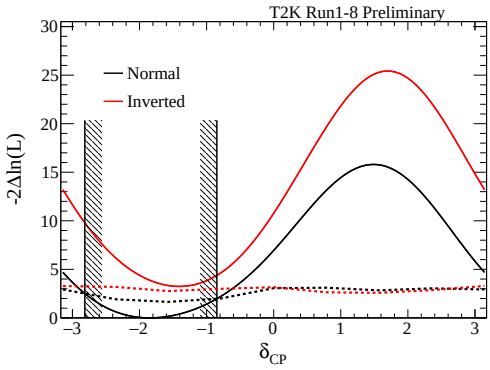
Table 22: The fraction of toy experiments for which $\delta_{CP} = 0, \pi$ and normal and inverted hierarchy are excluded at 90% and 2σ CL is shown. 10k toy experiments are used.



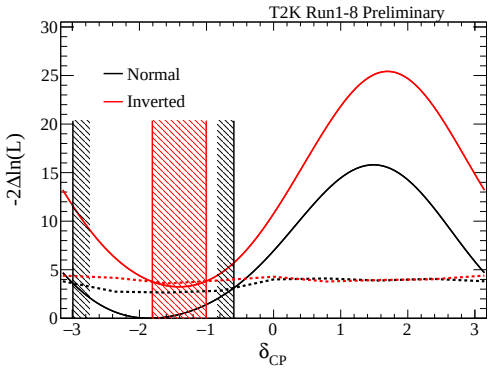
(a) Critical $\Delta\chi^2$ values



(b) 1σ CL exclusion region

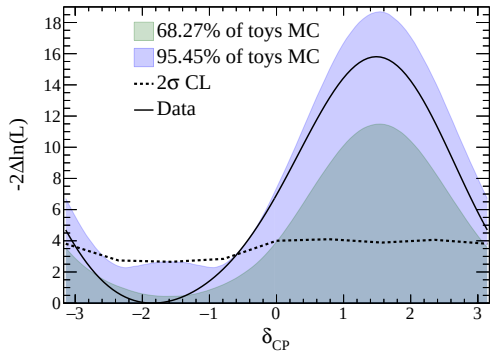


(c) 90% CL exclusion region

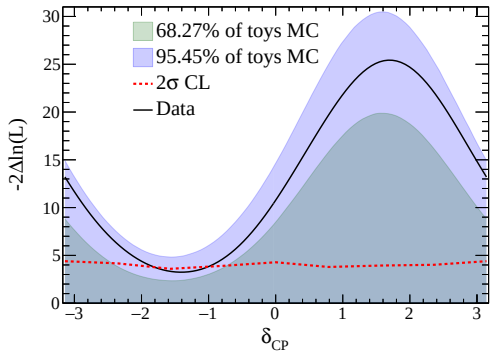


(d) 2σ CL exclusion region

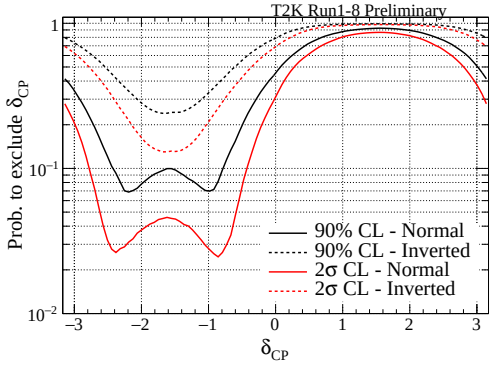
Figure 58: $\Delta\chi^2$ critical values and confidence intervals for the measured $\Delta\chi^2$ distributions for Run 1-8. Critical values obtained with the Feldman-Cousins method for Run 1-8 for 9 evenly spaced values on the range $[-\pi, \pi]$. Critical values are shown for 1σ, 2σ and 90% CL for normal (solid lines) and inverted (dashed lines) hierarchies. At least 1×10^4 toy experiments are performed for each point. The three bands of lines show the $\pm 1\sigma$ uncertainty on the critical values. Also shown are the measured $\Delta\chi^2$ distributions shifted with respect to the same global minium and the 1σ, 90% and 2σ exclusion regions for both mass hierarchies.



(a) Normal hierarchy



(b) Inverted hierarchy



(c) Data $\Delta\chi^2$ probability

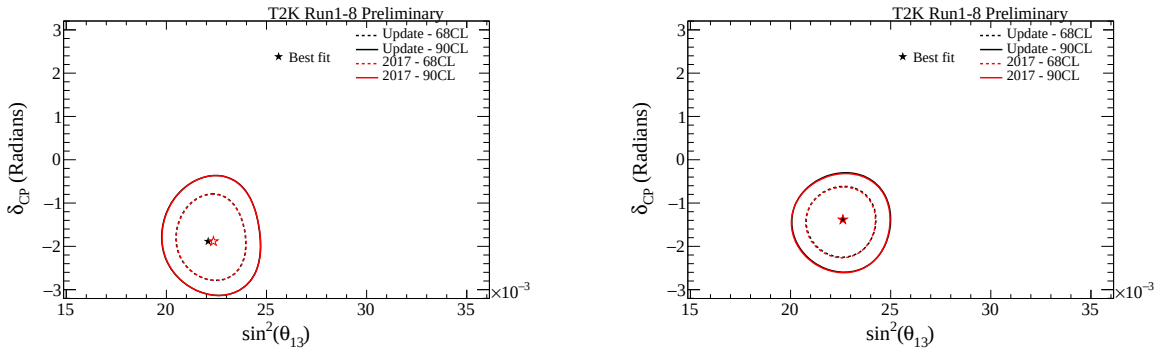
Figure 59: The distribution of $\Delta\chi^2$ vs δ_{CP} obtained with 1×10^4 toy experiments generated with $\delta_{CP} = -\pi/2$ and normal hierarchy is shown. The $\Delta\chi^2$ distributions obtained by fixing the mass hierarchy to normal (left) and inverted (right) are shown, along with the $\Delta\chi^2$ values corresponding to 68.27% and 95.45% of the toy experiments and the average $\Delta\chi^2$ values from all experiments. Also shown are the fractions of toys which exclude each value of δ_{CP} to 90% and 2σ using the Feldman-Cousins corrected critical values (bottom).

6.5. *Comparison to 2017 results*

This section presents comparisons of the updated fits with the 2017 fits described in section 5 for the case of reactor fits (except in the case of $\sin^2\theta_{13}$, which does not apply the reactor constraint).

6.5.1. *Results for δ_{CP} vs $\sin^2\theta_{13}$*

In fig. 60 the result for δ_{CP} vs $\sin^2\theta_{13}$ with reactor constraint.



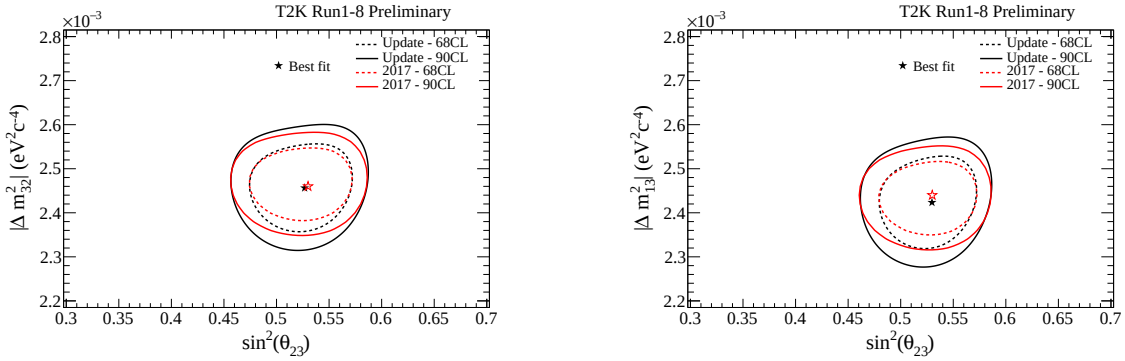
(a) δ_{CP} vs $\sin^2\theta_{13}$ NH

(b) δ_{CP} vs $\sin^2\theta_{13}$ IH

Figure 60: Contours at 68% and 90% CL for δ_{CP} vs $\sin^2\theta_{13}$ with reactor constraint. Normal and inverted hierarchy contours are independent. All 5 samples were used to produce these fits.

6.5.2. *Results for Δm^2 vs $\sin^2\theta_{23}$*

The result for Δm^2 vs $\sin^2\theta_{23}$ with reactor constraint is shown in fig. 61.



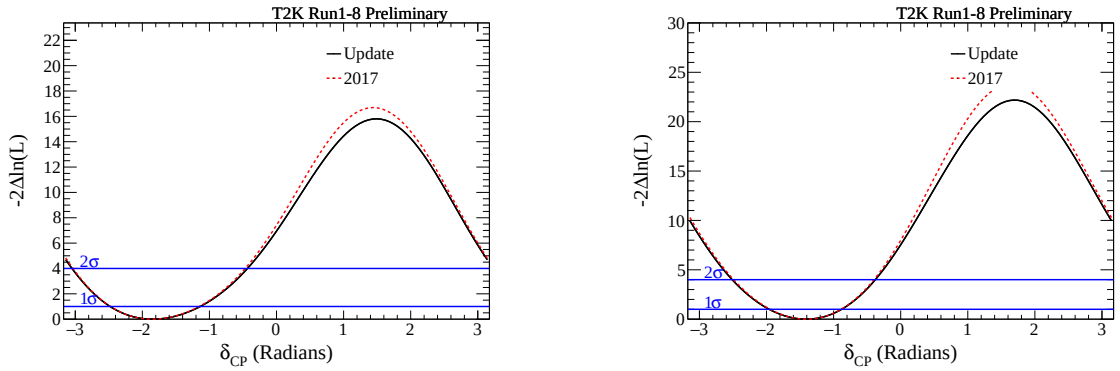
(a) Δm^2 vs $\sin^2\theta_{23}$ NH

(b) Δm^2 vs $\sin^2\theta_{23}$ IH

Figure 61: Contours at 68% and 90% CL for Δm^2 vs $\sin^2\theta_{23}$ with reactor constraint. Normal and inverted hierarchy contours are independent. All 5 samples were used to produce these fits.

6.5.3. *Results for* δ_{CP}

The result for δ_{CP} with reactor constraint is shown in fig. 62.

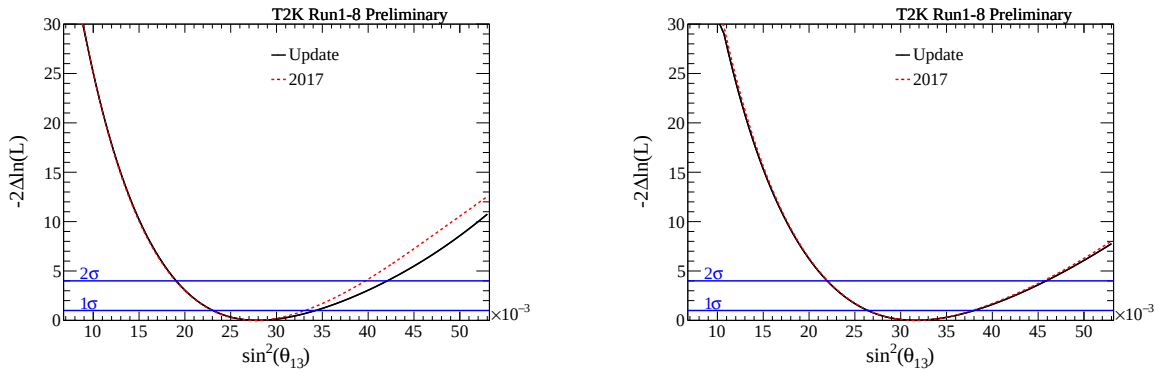


(a) δ_{CP} NH (b) δ_{CP} IH

Figure 62: The expected $\Delta\chi^2$ distribution as a function of δ_{CP} with reactor constraint. The mass hierarchy is minimized, with normal and inverted hierarchy $\Delta\chi^2$ distributions shifted to the same global best-fit χ^2 value, which is taken to be the minimum between normal and inverted hierarchy. All 5 samples were used to produce these fits.

6.5.4. *Results for* $\sin^2\theta_{13}$

The result for $\sin^2\theta_{13}$ (without reactor constraint) is shown in fig. 63.

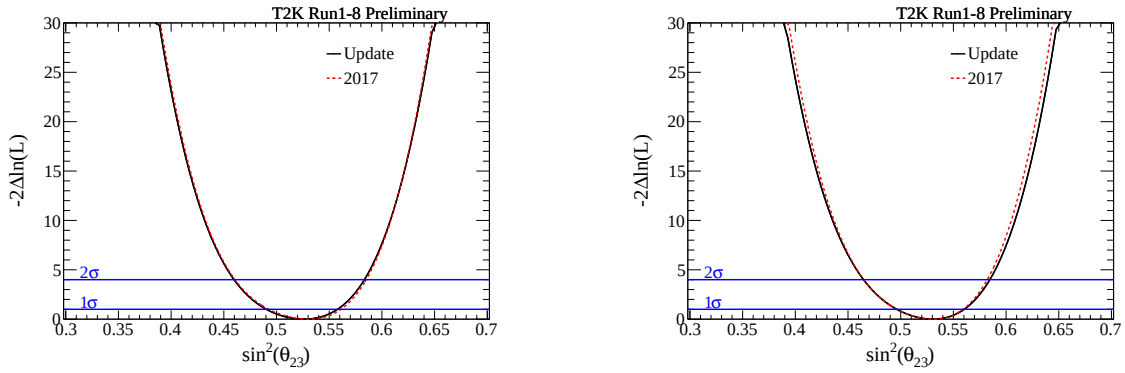


(a) $\sin^2\theta_{13}$ NH (b) $\sin^2\theta_{13}$ IH

Figure 63: The expected $\Delta\chi^2$ distribution as a function of $\sin^2\theta_{13}$ without reactor constraint. The mass hierarchy is minimized, with normal and inverted hierarchy $\Delta\chi^2$ distributions shifted to the same global best-fit χ^2 value, which is taken to be the minimum between normal and inverted hierarchy. All 5 samples were used to produce these fits.

6.5.5. *Results for* $\sin^2\theta_{23}$

The result for $\sin^2\theta_{23}$ with reactor constraint is shown in fig. 64.

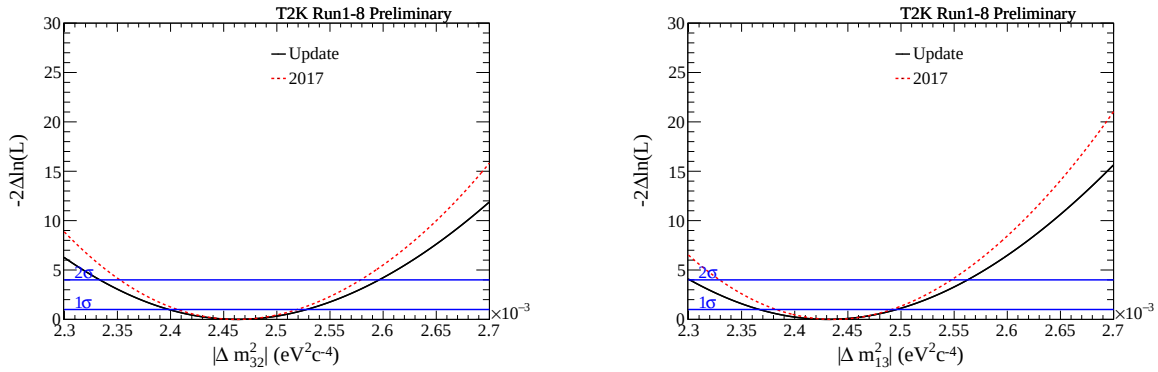


(a) $\sin^2\theta_{23}$ NH (b) $\sin^2\theta_{23}$ IH

Figure 64: The expected $\Delta\chi^2$ distribution as a function of $\sin^2\theta_{23}$ with reactor constraint. The mass hierarchy is minimized, with normal and inverted hierarchy $\Delta\chi^2$ distributions shifted to the same global best-fit χ^2 value, which is taken to be the minimum between normal and inverted hierarchy. All 5 samples were used to produce these fits.

6.5.6. *Results for* Δm^2

The result for Δm^2 with reactor constraint is shown in fig. 65.



(a) Δm^2 NH (b) Δm^2 IH

Figure 65: The expected $\Delta\chi^2$ distribution as a function of Δm^2 with reactor constraint. The mass hierarchy is minimized, with normal and inverted hierarchy $\Delta\chi^2$ distributions shifted to the same global best-fit χ^2 value, which is taken to be the minimum between normal and inverted hierarchy. All 5 samples were used to produce these fits.

6.6. *Discussion of the Run 1-8 data fit results*

It can be seen from section section 6.5 that the majority of fits show little change between the 2017 and updated analyses for fits of $\sin^2\theta_{13}$, $\sin^2\theta_{23}$ and δ_{CP}. The best-fit values show changes of no more than a single bin in the fit parameter; this being most evident in the normal hierarchy δ_{CP} vs $\sin^2\theta_{13}$ fit, where the binning is coarser. The contours for the δ_{CP} vs $\sin^2\theta_{13}$ fit are almost identical. A moderate inflation of the T2K-only $\sin^2\theta_{13}$ contour in normal hierarchy can be seen. A possible explanation of the asymmetry in the inflation of this contour is that as the

value of $\sin^2\theta_{13}$ increases a larger combination of values of other oscillation parameters can produce equivalent spectra; the addition of the Eb parameter then provides additional freedom in this respect and thereby weakens the fit in this region. The effect diminishes for smaller values of $\sin^2\theta_{13}$ as there is less of an appearance signal for Eb to act upon. Modest changes in the δ_{CP} contours are evident, with slightly less extreme values of $\Delta\chi^2$ around the peak near $\delta_{CP} = \pi/2$, but this does not alter the rejection of CP conservation at 2σ, as demonstrated in section 6.4.2.

Differences are evident in the fits of the atmospheric parameters, particularly Δm^2. The addition of the Eb parameter and Δm^2 smearing has inflated the contours, most notably in the Δm^2 parameter, as expected. The best-fit point is little changed in normal hierarchy (again being most evident in the coarser-binned 2D fit), and a small shift in Δm^2 is observed in both 2D and 1D fits in inverted hierarchy. The 1D fits of $\sin^2\theta_{23}$ show slight inflation of the contours, with the inflation clearly evident in the 1D fits of Δm^2.

7. Summary

In this technical note, we have presented the results of the 3-flavour $\bar{\nu}/\nu$ joint analysis performed by the *VALOR* group, on the combined Run 1-8 data set. The goal of the analysis was to search for CP violation in the leptonic sector and determine limits on δ_{CP} as well as measure all the other oscillation parameters to which T2K is sensitive: $\sin^2\theta_{13}$, $\sin^2\theta_{23}$, Δm^2_{32} (Δm^2_{13}) and partially mass hierarchy, to which T2K has little sensitivity.

This analysis predicts 268.4 ± 11.8 (syst) for neutrino μ-like, 64.3 ± 2.4 (syst) for antineutrino μ-like, 73.5 ± 5.3 (syst) for neutrino e-like, 7.9 ± 0.6 (syst) for antineutrino e-like and 6.9 ± 1.5 (syst) for neutrino ν_e CC1π^+ events in Super K and observes respectively 240 for neutrino μ-like, 68 for antineutrino μ-like, 74 for neutrino e-like, 7 for antineutrino e-like and 15 for neutrino ν_e CC1π^+.

The $\bar{\nu}/\nu$ analysis was performed in a framework of 3-flavour oscillations including matter effects in constant-density matter. The observed reconstructed energy spectrum of single μ-like ring events and reconstructed energy versus lepton angle of single e-like ring events in addition to e-like events with an additional decay electron were used to compute the confidence intervals. All 110 systematic parameters as well as the oscillation parameters which were not parameters of interest considered in this analysis were marginalised.

From the fit of the Run 1-8 data we can see that the favoured value of δ_{CP} is -1.872, close to $-\pi/2$. The Feldman-Cousins method is used to construct confidence intervals with neither over or under coverage from which the significance to exclude the hypothesis of CP conservation is greater than 2 standard deviations.

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Appendices

A. Fitter details

A.1. Spectra prediction methodology

In this analysis a comparison of the reconstructed energy (E_r) and reconstructed lepton angle (θ) spectrum of single ring events, including the ν_e CC1π^+-like sample, observed at Super-K with the predicted spectrum is used to build the test statistic to quote confidence intervals for neutrino oscillation parameters. For μ-like samples the templates are built as a function of the reconstructed neutrino energy, as in previous analyses, and a single θ bin is used. The predicted number $N_{SK;r,\theta}$ of single ring events in the r-th and θ-th bin is computed for each selected sample as follows:

$$N_{SK;r,\theta} = \sum_m \sum_t \sum_{r\prime} P_{m;t} \cdot T_{r;r\prime;f^{SK}_{E;r}} \cdot S_{m;t;r\prime,\theta;\vec{f}} \cdot N^{MC}_{SK;m;r\prime,\theta;t} \tag{6}$$

In 6, $N^{MC}_{SK;m;r\prime,\theta;t}$ is the input Super-K Monte Carlo (MC) template containing the number of events in the single ring MC sample with true reaction mode m in the true energy bin t, the reconstructed energy bin $r\prime$ and reconstructed lepton angle bin θ. $S_{m;t;r\prime,\theta;\vec{f}}$ is an overall, multiplicative, systematic error factor depending on a vector of nuisance (systematic) parameters $\vec{f}$, which is function of the reaction mode m, the true energy bin t, the reconstructed energy bin $r\prime$, and the reconstructed lepton angle bin θ. $T_{r;r\prime;f^{SK}_{E;r}}$ is a transfer function describing the migration of events between the reconstructed energy bins r and $r\prime$ due to uncertainty in the Super-K reconstructed energy scale, expressed here in terms of the nuisance parameter $f^{SK}_{E;r}$. Finally, $P_{m;t}$ is the 3-flavour oscillation probability applied to the true energy bin t of the Super-K MC template which corresponds to mode m; application of the term $P_{m;t}$ is discussed in A.3.

A.2. Construction of the nominal Super-K Monte Carlo templates

The nominal Super-K single ring and ν_e CC1π^+ MC analysis templates $N^{MC}_{SK;m;r\prime,\theta;t}$ are constructed by applying the single e-like ring selection cuts, the ν_e CC1π^+ selection cuts, or the single μ-like ring selection cuts to the official Super-K MC samples [42]. The calculation of the normalisation of each MC sample is described in section A.2.1. The list of MC templates used in this analysis is shown in section A.2.2, while the true and reconstructed energy as well as the reconstructed lepton angle binning of the templates are discussed in section A.2.3.

A.2.1. Normalisation of Monte Carlo samples

The normalisation (integrated exposure in terms of POT) of each event sample is calculated from the number of events with a MC truth interaction vertex within the 22.5 kt fiducial volume.

$$N = \int dSdIdE \cdot \frac{d^3\Phi_{SK}}{dSdIdE_\nu} \cdot \sigma_{H_2O} \cdot \frac{N_A}{A} \cdot \rho \cdot L \tag{7}$$

where $d^3\Phi_{SK}/dSdIdE$ is the number of flux particles for the given neutrino species per neutrino energy bin dE_ν, per unit area dS and per POT, σ_{H_2O} is the total interaction cross section on water

for the given neutrino species, I is the beam exposure in terms of POT, N_A is Avogadro's number, A is the mass number for water, ρ is the water density and L is the neutrino path length in the water volume.

The Super-K MC samples are normalised to the Run 1-8 POT. Assuming that all the MC files are used, each sample is first normalised to 10^{21} POT by weighting it with the ratio of the number of events per 22.5 kt per 10^{21} POT to the number of generated events in the true fiducial volume in all the available MC. Then each sample is normalised to 1.4734×10^{21} POT for neutrino and 7.558×10^{20} for antineutrino mode, by weighting the result respectively by 1.4734 and 0.7558 (the Run 1-8 POT in units of 10^{21} POT).

A.2.2. List of Monte Carlo templates

For each Super-K MC sample, a number of different MC templates is constructed corresponding to different true reaction modes. The template number depends on the type of oscillation analysis and the specific systematic parameters considered in the analysis. 50 are used for the current analysis, these are the same templates as in our 2016 analysis [12]. In particular

- $\bar{\nu}_\mu$ CCQE,
- $\bar{\nu}_\mu$ CC1π,
- $\bar{\nu}_\mu$ CC coherent,
- $\bar{\nu}_\mu$ CC 2p-2h,
- $\bar{\nu}_\mu$ CC other,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC1π^0,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC1$\pi^\pm$,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC coherent,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC other,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC 1γ,
- ν_μ CCQE,
- ν_μ CC1π,
- ν_μ CC coherent,
- ν_μ CC 2p-2h,
- ν_μ CC other,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC1π^0,

- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC1$\pi^\pm$,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC coherent,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC other,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC 1γ,
- $\bar{\nu}_e$ CCQE,
- $\bar{\nu}_e$ CC1π,
- $\bar{\nu}_e$ CC coherent,
- $\bar{\nu}_e$ CC 2p-2h,
- $\bar{\nu}_e$ CC other,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC1π^0,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC1$\pi^\pm$,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC coherent,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC other,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC 1γ,
- ν_e CCQE,
- ν_e CC1π,
- ν_e CC coherent,
- ν_e CC 2p-2h,
- ν_e CC other,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC1π^0,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC1$\pi^\pm$,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC coherent,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC other,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC 1γ,
- oscillated $\nu_\mu \rightarrow \nu_e$ CCQE,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC1π,

- oscillated $\nu_\mu \to \nu_e$ CC coherent,
- oscillated $\nu_\mu \to \nu_e$ CC 2p-2h,
- oscillated $\nu_\mu \to \nu_e$ CC other,
- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CCQE,
- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC1π,
- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC coherent,
- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC 2p-2h,
- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC other.

A.2.3. Binning of Monte Carlo templates

The binning scheme for the current analysis remains unchanged from the previous analysis [12]. For μ-like samples the templates are built as a function of the reconstructed neutrino energy, having 84 true energy bins and 73 reconstructed energy bins. e-like templates have 84 true energy bins, 25 reconstructed energy bins and 15 reconstructed lepton angle bins.

For all templates the 84 true energy bins of constant width are:

- 6 50-MeV bins from 0-0.3 GeV,
- 28 25-MeV bins from 0.3-1 GeV,
- 40 50-MeV bins from 1-3 GeV,
- 5 100-MeV bin from 3-3.5 GeV,
- 1 bin from 3.5-4 GeV,
- 1 bin from 4-5 GeV,
- 1 bin from 5-7 GeV,
- 1 bin from 7-10 GeV and
- 1 bin from 10-30 GeV

For μ-like samples the 73 reconstructed energy bins are:

- 60 50-MeV bins from 0-3 GeV,
- 4 250-MeV bins from 3-4 GeV,
- 4 500-MeV bins from 4-6 GeV,
- 4 1000-MeV bins from 6-10 GeV and

- 1 bin from 10-30 GeV.

For e-like samples the 25 reconstructed energy bins with constant width are the following:

- 25 50-MeV bins from 0-1.25 GeV.

The θ binning used for e-like samples is

- 14 10° bins from 0° to 140°,
- 1 bin for the range $140° - 180°$.

For μ-like samples a single θ bin, that covers the whole range 0° - 180°, is used. The predicted e-like 2-dimensional spectra were studied in detail in [11] for each flavour component and were compared with the 1-dimensional reconstructed energy spectra.

A study was performed to decide the optimum binning scheme for reconstructing neutrino energy, balancing the needs for accuracy and speed. Details are given in Appendix I of Ref. [4].

A.3. Applying neutrino oscillations

The MC templates are constructed from the unoscillated MC samples and must be weighted with the appropriate oscillation probability. Oscillations are applied as a function of true energy to the MC templates for all CC interactions.

The MC templates calculated from the $\bar{\nu}_\mu$ MC sample are weighted with $\mathrm{P}(\bar{\nu}_\mu \to \bar{\nu}_\mu)$, with this survival probability being calculated using the oscillation parameters, namely $\sin^2\theta_{23}$, $\sin^2\theta_{12}$, $\sin^2\theta_{13}$, $|\Delta m^2_{32}|$ (normal hierarchy) or $|\Delta m^2_{31}|$ (inverted hierarchy), Δm^2_{21} and δ_{CP}. The $\bar{\nu}_e$ templates are weighted with $\mathrm{P}(\bar{\nu}_e \to \bar{\nu}_e)$, while the oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ templates are weighted with $\mathrm{P}(\bar{\nu}_\mu \to \bar{\nu}_e)$ (the sample was generated assuming 100% of $\bar{\nu}_\mu$ transform to $\bar{\nu}_e$). All these oscillation probabilities are computed in a 3-flavour framework including matter effects in constant-density matter (assuming an Earth crust density of 2.6 g/cm^3). Custom oscillation probability calculation code was developed within the *VALOR* analysis framework. Details and estimates of the numerical accuracy of this code are first presented in [3], and is also shown in Appendix A of [4]. Comparisons between the *VALOR* oscillation probability calculation code and Prob3++ [85], which is typically used by the alternative T2K oscillation analyses, are shown in Appendix B of [4].

In the standard 3-flavour oscillation framework, oscillations of $\bar{\nu}_e$ and $\bar{\nu}_\mu$ can yield $\bar{\nu}_\tau$. In this analysis, we neglect contributions from ν_τ-CC and $\bar{\nu}_\tau$-CC as their energy threshold is around 3.5 GeV and their effect is negligible[1]. Accordingly, this analysis uses no ν_τ-CC and $\bar{\nu}_\tau$-CC MC templates (and, in fact, no ν_τ and $\bar{\nu}_\tau$ Super-K MC samples are available).

It should be emphasized here that the ν_μ NC MC templates for a mode m contain the NC MC templates for the mixture of ν_e+ν_μ+ν_τ resulting from 3-flavour ν_μ oscillations for that mode m.

[1] A back-of-the-envelope calculation can be found in the replies to comments and revision history document of [4] that gives $\ll 1\%$ ν_τ contamination in the 1-ring μ-like Super-K sample. A Super-K MC study is required to determine whether the Super-K efficiency assumptions made there are valid.

The same applies to the $\bar{\nu}_\mu$, ν_e and $\bar{\nu}_e$ NC MC templates. These NC MC templates are unchanged under standard 3-flavour oscillations.

Also it should be noted that there are no explicit NC MC templates made from the oscillated $\bar{\nu}_e$ and ν_e samples. If they were used, the oscillated $\bar{\nu}_e$ (i.e. $\bar{\nu}_e$ coming from oscillations of $\bar{\nu}_\mu$) would be double counted since they are already included in the $\bar{\nu}_\mu$ NC MC templates. The same applies to oscillated ν_e.

B. Event rates

Tables 23-27 present unoscillated rates.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	953.15505	0.00429	23.18824	0.00020	0.00000	0.00000	976.34778
CC1pi	76.40088	0.00211	4.12922	0.00010	0.00000	0.00000	80.53230
CCcoherent	0.75678	0.00000	0.17387	0.00000	0.00000	0.00000	0.93065
2p2h	134.02475	0.00161	2.51636	0.00008	0.00000	0.00000	136.54280
CCother	6.94520	0.00090	0.50167	0.00008	0.00000	0.00000	7.44785
NC1pi0	0.61495	0.01841	0.02120	0.00143	N/A	N/A	0.65599
NC1piPM	4.90783	0.09799	0.17531	0.00975	N/A	N/A	5.19089
NCcoherent	0.00000	0.00000	0.00043	0.00003	N/A	N/A	0.00047
NCother	2.05619	0.07378	0.13055	0.00912	N/A	N/A	2.26963
Total	1178.86162	0.19909	30.83686	0.02080	0.00000	0.00000	1209.91837

Table 23: Predicted number of events in the neutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 1.4734×10^{21} POT. Oscillations are not applied.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	36.21445	0.00089	133.92633	0.00096	0.00000	0.00000	170.14262
CC1pi	6.51012	0.00035	11.57229	0.00027	0.00000	0.00000	18.08303
CCcoherent	0.06956	0.00000	0.75838	0.00005	0.00000	0.00000	0.82799
2p2h	8.28538	0.00028	10.73613	0.00028	0.00000	0.00000	19.02208
CCother	1.18677	0.00012	0.77049	0.00000	0.00000	0.00000	1.95738
NC1pi0	0.06000	0.00222	0.06147	0.00169	N/A	N/A	0.12538
NC1piPM	0.35950	0.01527	0.46009	0.01235	N/A	N/A	0.84721
NCcoherent	0.00000	0.00000	0.00201	0.00000	N/A	N/A	0.00201
NCother	0.31580	0.01640	0.20064	0.00789	N/A	N/A	0.54073
Total	53.00159	0.03552	158.48783	0.02349	0.00000	0.00000	211.54843

Table 24: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 7.558×10^{20}POT. Oscillations are not applied.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.99622	6.79206	0.02082	0.27180	0.00000	0.00000	8.08090
CC1pi	0.18090	0.99514	0.00514	0.07082	0.00000	0.00000	1.25199
CCcoherent	0.00076	0.00885	0.00031	0.00419	0.00000	0.00000	0.01412
2p2h	0.13667	1.71219	0.00236	0.04511	0.00000	0.00000	1.89633
CCother	0.02827	0.13255	0.00057	0.00971	0.00000	0.00000	0.17111
NC1pi0	1.70203	0.03706	0.06012	0.00356	N/A	N/A	1.80278
NC1piPM	0.16854	0.00472	0.01020	0.00057	N/A	N/A	0.18404
NCcoherent	0.52817	0.00715	0.04794	0.00294	N/A	N/A	0.58619
NCother	0.31767	0.01695	0.02120	0.00128	N/A	N/A	0.35711
NC1gamma	0.94222	0.01687	0.04983	0.00233	N/A	N/A	1.01125
Total	5.00145	9.72355	0.21850	0.41231	0.00000	0.00000	15.35581

Table 25: Predicted number of events in the neutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 1.4734×10^{21} POT. Oscillations are not applied.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.04096	0.54758	0.13906	0.90195	0.00000	0.00000	1.62955
CC1pi	0.01660	0.10715	0.01787	0.15254	0.00000	0.00000	0.29415
CCcoherent	0.00006	0.00105	0.00143	0.01133	0.00000	0.00000	0.01388
2p2h	0.00652	0.15324	0.00694	0.11994	0.00000	0.00000	0.28664
CCother	0.00568	0.02184	0.00232	0.01193	0.00000	0.00000	0.04177
NC1pi0	0.13849	0.00541	0.17659	0.00468	N/A	N/A	0.32516
NC1piPM	0.02475	0.00127	0.02001	0.00080	N/A	N/A	0.04683
NCcoherent	0.04776	0.00221	0.20260	0.00317	N/A	N/A	0.25574
NCother	0.05664	0.00221	0.02672	0.00097	N/A	N/A	0.08654
NC1gamma	0.07943	0.00439	0.17490	0.00383	N/A	N/A	0.26255
Total	0.41688	0.84635	0.76842	1.21115	0.00000	0.00000	3.24280

Table 26: Predicted number of events in the antineutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 7.558×10^{20} POT. Oscillations are not applied.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.34542	0.02646	0.00615	0.00187	0.00000	0.00000	0.37990
CC1pi	0.54696	0.82041	0.00710	0.00403	0.00000	0.00000	1.37850
CCcoherent	0.00364	0.01905	0.00021	0.00008	0.00000	0.00000	0.02298
2p2h	0.08323	0.02355	0.00176	0.00100	0.00000	0.00000	0.10953
CCother	0.17904	0.11704	0.00287	0.00289	0.00000	0.00000	0.30184
NC1pi0	0.02049	0.00076	0.00083	0.00008	N/A	N/A	0.02216
NC1piPM	0.06474	0.00165	0.00382	0.00033	N/A	N/A	0.07053
NCother	0.20584	0.00830	0.01304	0.00088	N/A	N/A	0.22807
NC1gamma	0.02844	0.00000	0.00067	0.00007	N/A	N/A	0.02918
Total	1.47780	1.01722	0.03645	0.01122	0.00000	0.00000	2.54269

Table 27: Predicted number of events in the neutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 1.4734×10^{21} POT. Oscillations are not applied.

1094 Tables 28-32 present oscillated rates corresponding to the Asimov B data set.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	186.41514	0.00413	11.15862	0.00019	0.01514	0.00018	197.59340
CC1pi	28.20533	0.00199	2.62809	0.00009	0.01651	0.00009	30.85209
CCcoherent	0.29385	0.00000	0.09293	0.00000	0.00000	0.00001	0.38678
2p2h	36.71916	0.00151	1.42161	0.00008	0.01693	0.00005	38.15934
CCother	5.47728	0.00086	0.40680	0.00008	0.00011	0.00001	5.88513
NC1pi0	0.61495	0.01841	0.02120	0.00143	N/A	N/A	0.65599
NC1piPM	4.90783	0.09799	0.17531	0.00975	N/A	N/A	5.19089
NCcoherent	0.00000	0.00000	0.00043	0.00003	N/A	N/A	0.00047
NCother	2.05619	0.07378	0.13055	0.00912	N/A	N/A	2.26963
Total	264.68973	0.19867	16.03554	0.02078	0.04868	0.00032	280.99373

Table 28: Predicted number of events in the neutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 1.4734×10^{21} POT. The Asimov data set B in 1 is used.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	15.53910	0.00085	31.15342	0.00093	0.00068	0.00181	46.69679
CC1pi	3.97662	0.00033	4.60026	0.00025	0.00043	0.00073	8.57862
CCcoherent	0.04724	0.00000	0.18223	0.00005	0.00000	0.00007	0.22959
2p2h	4.50940	0.00026	2.92853	0.00027	0.00030	0.00025	7.43900
CCother	0.94370	0.00012	0.60700	0.00000	0.00000	0.00000	1.55082
NC1pi0	0.06000	0.00222	0.06147	0.00169	N/A	N/A	0.12538
NC1piPM	0.35950	0.01527	0.46009	0.01235	N/A	N/A	0.84721
NCcoherent	0.00000	0.00000	0.00201	0.00000	N/A	N/A	0.00201
NCother	0.31580	0.01640	0.20064	0.00789	N/A	N/A	0.54073
Total	25.75135	0.03544	40.19565	0.02342	0.00142	0.00286	66.01015

Table 29: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 7.558×10^{20} POT. The Asimov data set B in 1 is used.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.17174	6.30178	0.00647	0.25566	31.20630	0.33860	38.28056
CC1pi	0.04847	0.93168	0.00282	0.06750	3.28336	0.05561	4.38943
CCcoherent	0.00012	0.00830	0.00018	0.00397	0.02515	0.00382	0.04154
2p2h	0.03143	1.59694	0.00118	0.04275	6.08289	0.04642	7.80162
CCother	0.01346	0.12803	0.00034	0.00942	0.07482	0.00374	0.22981
NC1pi0	1.70203	0.03706	0.06012	0.00356	N/A	N/A	1.80278
NC1piPM	0.16854	0.00472	0.01020	0.00057	N/A	N/A	0.18404
NCcoherent	0.52817	0.00715	0.04794	0.00294	N/A	N/A	0.58619
NCother	0.31767	0.01695	0.02120	0.00128	N/A	N/A	0.35711
NC1gamma	0.94222	0.01687	0.04983	0.00233	N/A	N/A	1.01125
Total	3.92386	9.04948	0.20030	0.38998	40.67252	0.44819	54.68432

Table 30: Predicted number of events in the neutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 1.4734×10^{21} POT. The Asimov data set B in 1 is used.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.01318	0.50911	0.02513	0.84721	0.72476	3.38779	5.50718
CC1pi	0.00771	0.10111	0.00742	0.14431	0.11344	0.39683	0.77081
CCcoherent	0.00002	0.00099	0.00043	0.01066	0.00090	0.03680	0.04981
2p2h	0.00298	0.14379	0.00224	0.11301	0.17444	0.39087	0.82733
CCother	0.00392	0.02116	0.00121	0.01154	0.00920	0.00777	0.05480
NC1pi0	0.13849	0.00541	0.17659	0.00468	N/A	N/A	0.32516
NC1piPM	0.02475	0.00127	0.02001	0.00080	N/A	N/A	0.04683
NCcoherent	0.04776	0.00221	0.20260	0.00317	N/A	N/A	0.25574
NCother	0.05664	0.00221	0.02672	0.00097	N/A	N/A	0.08654
NC1gamma	0.07943	0.00439	0.17490	0.00383	N/A	N/A	0.26255
Total	0.37487	0.79164	0.63724	1.14019	1.02275	4.22006	8.18675

Table 31: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 7.558×10^{20} POT. The Asimov data set B in 1 is used.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.03298	0.02442	0.00123	0.00174	0.14531	0.00339	0.20908
CC1pi	0.08149	0.76065	0.00261	0.00383	3.47649	0.00267	4.32774
CCcoherent	0.00031	0.01765	0.00008	0.00007	0.07633	0.00013	0.09457
2p2h	0.01100	0.02195	0.00050	0.00095	0.11660	0.00125	0.15225
CCother	0.03947	0.11225	0.00117	0.00279	0.09788	0.00174	0.25531
NC1pi0	0.02049	0.00076	0.00083	0.00008	N/A	N/A	0.02216
NC1piPM	0.06474	0.00165	0.00382	0.00033	N/A	N/A	0.07053
NCother	0.20584	0.00830	0.01304	0.00088	N/A	N/A	0.22807
NC1gamma	0.02844	0.00000	0.00067	0.00007	N/A	N/A	0.02918
Total	0.48477	0.94764	0.02396	0.01075	3.91261	0.00917	5.38890

Table 32: Predicted number of events in the neutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 1.4734×10^{21} POT. The Asimov data set B in 1 is used.

C. Systematic variations

Tables table 33 through table 42 show the effect of systematic variations on the event rates for pre-BANFF and post-BANFF errors.

Parameter	1 σ value	BANFF tuned -3σ		BANFF tuned -1σ		BANFF tuned +1σ		BANFF tuned +3σ	
All syst		61.9761	−74.2947(%)	143.921	−40.307 (%)	385.63	59.945(%)	900.64	273.551(%)
BANFF 0, SK numu flux, 0.0 - 0.4 GeV	0.098 732 3	234.448	−2.760 12	238.884	−0.920 04	243.321	0.920 04	247.757	2.760 12
BANFF 1, SK numu flux, 0.4 - 0.5 GeV	0.103 488	234.367	−2.793 41	238.857	−0.931 14	243.347	0.931 14	247.837	2.793 41
BANFF 2, SK numu flux, 0.5 - 0.6 GeV	0.096 444	239.435	−0.691 534	240.547	−0.230 51	241.658	0.230 51	242.77	0.691 534
BANFF 3, SK numu flux, 0.6 - 0.7 GeV	0.086 695 8	239.741	−0.564 733	240.648	−0.188 24	241.556	0.188 24	242.464	0.564 733
BANFF 4, SK numu flux, 0.7 - 1.0 GeV	0.113 052	224.426	−6.916 84	235.543	−2.3056	246.661	2.3056	257.779	6.916 84
BANFF 5, SK numu flux, 1.0 - 1.5 GeV	0.091 747 6	228.616	−5.178 78	236.94	−1.7263	245.264	1.7263	253.588	5.178 78
BANFF 6, SK numu flux, 1.5 - 2.5 GeV	0.070 174 2	233.261	−3.252 25	238.489	−1.0841	243.716	1.0841	248.944	3.252 25
BANFF 7, SK numu flux, 2.5 - 3.5 GeV	0.073 680 5	237.002	−1.700 83	239.735	−0.566 94	242.469	0.566 94	245.203	1.700 83
BANFF 8, SK numu flux, 3.5 - 5.0 GeV	0.087 373 3	236.859	−1.760 11	239.688	−0.5867	242.517	0.5867	245.346	1.760 11
BANFF 9, SK numu flux, 5.0 - 7.0 GeV	0.097 936 9	239.83	−0.527 644	240.678	−0.175 88	241.526	0.175 88	242.374	0.527 644
BANFF 10, SK numu flux, 7.0 - 30.0 GeV	0.114 36	240.982	−0.049 815 7	241.062	−0.016 605	241.142	0.016 605	241.222	0.049 815 7
BANFF 11, SK numubar flux, 0.0 - 0.7 GeV	0.102 576	240.821	−0.116 474	241.009	−0.038 825	241.196	0.038 825	241.383	0.116 474
BANFF 12, SK numubar flux, 0.7 - 1.0 GeV	0.078 533 1	240.849	−0.105 028	241.018	−0.035 009	241.187	0.035 009	241.356	0.105 028
BANFF 13, SK numubar flux, 1.0 - 1.5 GeV	0.084 453 6	240.163	−0.389 64	240.789	−0.129 88	241.415	0.129 88	242.042	0.389 64
BANFF 14, SK numubar flux, 1.5 - 2.5 GeV	0.085 568 3	239.779	−0.548 919	240.661	−0.182 97	241.543	0.182 97	242.426	0.548 919
BANFF 15, SK numubar flux, 2.5 - 30.0 GeV	0.086 428 4	240.158	−0.391 625	240.788	−0.130 54	241.417	0.130 54	242.046	0.391 625
BANFF 16, SK nue flux, 0.0 - 0.5 GeV	0.089 698 3	241.101	−0.000 449 436	241.102	−0.000 149 81	241.103	0.000 149 81	241.103	0.000 449 436
BANFF 17, SK nue flux, 0.5 - 0.7 GeV	0.089 952	241.1	−0.001 126 08	241.101	−0.000 375 36	241.103	0.000 375 36	241.105	0.001 126 08
BANFF 18, SK nue flux, 0.7 - 0.8 GeV	0.085 964 8	241.1	−0.000 823 573	241.102	−0.000 274 52	241.103	0.000 274 52	241.104	0.000 823 573
BANFF 19, SK nue flux, 0.8 - 1.5 GeV	0.080 918 2	241.088	−0.005 849 91	241.098	−0.001 95	241.107	0.001 95	241.116	0.005 849 91
BANFF 20, SK nue flux, 1.5 - 2.5 GeV	0.078 971 9	241.085	−0.007 055 64	241.097	−0.002 351 9	241.108	0.002 351 9	241.119	0.007 055 64
BANFF 21, SK nue flux, 2.5 - 4.0 GeV	0.083 85	241.09	−0.005 217 14	241.098	−0.001 739	241.106	0.001 739	241.115	0.005 217 14
BANFF 22, SK nue flux, 4.0 - 30.0 GeV	0.093 894 3	241.096	−0.002 758 5	241.1	−0.000 919 5	241.104	0.000 919 5	241.109	0.002 758 5
BANFF 23, SK nuebar flux, 0.0 - 2.5 GeV	0.074 027 9	241.1	−0.000 885 825	241.102	−0.000 295 28	241.103	0.000 295 28	241.104	0.000 885 825
BANFF 24, SK nuebar flux, 2.5 - 30.0 GeV	0.128 416	241.097	−0.002 092 35	241.101	−0.000 697 45	241.104	0.000 697 45	241.107	0.002 092 35
BANFF; Norm; 2p2h	1	214.649	−10.972	214.649	−10.972	267.556	10.972	320.463	32.9159
BANFF; CA5 RES	0.148 515	223.261	−7.399 78	234.198	−2.8636	248.963	3.2605	267.557	10.9725
BANFF; Norm; BgRES Isospin 1/2	0.307 692	237.768	−1.3831	238.756	−0.973 28	244.684	1.4855	255.552	5.993 38
BANFF, Ma QE	0.025	231.751	−3.878 74	238.038	−1.2711	244.113	1.2487	249.969	3.677 67
BANFF, Ma RES	0.157 895	223.458	−7.318 19	235.185	−2.4544	246.811	2.3678	257.26	6.701 57
BANFF; Fermi Momentum	0.057 777 8	246.226	2.125 31	243.832	1.1323	238.262	−1.1782	232.278	−3.659 93
BANFF; Shape; CC Oth	0.4	238.696	−0.998 015	240.292	−0.336 17	241.913	0.336 17	243.534	1.008 52
BANFF; Norm, CC Coh	0.3	240.713	−0.161 511	240.972	−0.053 837	241.232	0.053 837	241.492	0.161 511
BANFF; Norm, NC Coh	0.3	241.102	−0.000 187 994	241.102	-6.2665×10^{-5}	241.102	6.2665×10^{-5}	241.103	0.000 187 994
BANFF; Norm, NC Oth	0.3	239.074	−0.841 32	240.426	−0.280 44	241.778	0.280 44	243.131	0.841 32
BANFF; Norm, ν_e To ν_μ	0.028 284 3	241.097	−0.002 326 35	241.1	−0.000 775 45	241.104	0.000 775 45	241.108	0.002 326 35
BANFF; Norm, $\bar{\nu}_e$ To $\bar{\nu}_\mu$	0.028 284 3	241.102	$-2.252\,73 \times 10^{-5}$	241.102	-7.5091×10^{-6}	241.102	7.5091×10^{-6}	241.102	$2.252\,73 \times 10^{-5}$
BANFF; Norm; 2p2hBar	1	239.969	−0.470 064	239.969	−0.470 06	242.236	0.470 06	244.502	1.410 19
BANFF; BeRPA A	0.118	207.044	−14.1262	229.749	−4.7088	252.455	4.7088	275.161	14.1263
BANFF; BeRPA B	0.21	208.023	−13.72	230.076	−4.5734	252.129	4.5734	274.182	13.7201
BANFF; BeRPA D	0.1695	216.789	−10.0843	232.998	−3.3614	249.207	3.3615	265.416	10.0843
BANFF; BeRPA E	0.352	240.035	−0.442 852	240.672	−0.178 53	241.537	0.180 37	242.416	0.545 028
BANFF; Shape; 2p2h	3	242.411	0.542 684	241.756	0.271 34	240.37	−0.303 51	239.639	−0.607 028
BANFF; Norm; 2p2h C to O	0.2	224.55	−6.865 22	235.585	−2.2884	246.62	2.2884	257.654	6.865 22
SKDet + FSI/SI 0; E_{reco}(0.00 - 0.40)GeV; $\nu_\mu/\bar{\nu}_\mu$ CCQE ($1R_\mu$)	0.008 785 93	240.569	−0.221 14	240.925	−0.073 713	241.28	0.073 713	241.635	0.221 14
SKDet + FSI/SI 1; E_{reco}(0.40 - 1.10)GeV; $\nu_\mu/\bar{\nu}_\mu$ CCQE ($1R_\mu$)	0.007 403 28	239.211	−0.784 536	240.472	−0.261 51	241.733	0.261 51	242.994	0.784 536
SKDet + FSI/SI 2; E_{reco}(1.10 - 30.00)GeV; $\nu_\mu/\bar{\nu}_\mu$ CCQE ($1R_\mu$)	0.007 204 4	239.287	−0.752 812	240.497	−0.250 94	241.707	0.250 94	242.917	0.752 812
SKDet + FSI/SI 3; E_{reco}(0.00 - 30.00)GeV; $\nu_\mu/\bar{\nu}_\mu$ CCnQE ($1R_\mu$)	0.169 035	219.654	−8.895 93	233.953	−2.9653	248.252	2.9653	262.551	8.895 93
SKDet + FSI/SI 4; E_{reco}(0.00 - 30.00)GeV; $\nu_e/\bar{\nu}_e$/sigν_e CC ($1R_\mu$)	1.005 48	240.901	−0.083 500 7	241.035	−0.027 834	241.169	0.027 834	241.304	0.083 500 7
SKDet + FSI/SI 5; E_{reco}(0.00 - 30.00)GeV; all NC ($1R_\mu$)	0.658 101	222.61	−7.669 98	234.938	−2.5567	247.266	2.5567	259.595	7.669 98
SK energy scale	0.024	241.102	$-2.357\,65 \times 10^{-14}$	241.102	-3.5365×10^{-14}	241.102	-1.1788×10^{-14}	241.102	4.7153×10^{-14}

Table 33: $\pm1\sigma$ and $\pm3\sigma$ effect of the systematic parameters on the total number of μ-like Super-K events with Run 1-8 POT for neutrino mode pre-BANFF Asimov data set A. Only the systematic parameters which affect this sample are shown.

Parameter	1 σ value	BANFF tuned -3σ		BANFF tuned -1σ		BANFF tuned +1σ		BANFF tuned +3σ	
All syst		19.3721	−66.6249(%)	37.6935	−35.06 (%)	88.523	52.511(%)	201.123	246.503(%)
BANFF 0, SK numu flux, 0.0 - 0.7 GeV; RHC	0.093 682 2	57.6767	−0.632 032	57.9213	−0.210 68	58.1658	0.210 68	58.4104	0.632 032
BANFF 1, SK numu flux, 0.7 - 1.0 GeV; RHC	0.079 343	57.5584	−0.835 793	57.8819	−0.2786	58.2053	0.2786	58.5287	0.835 793
BANFF 2, SK numu flux, 1.0 - 1.5 GeV; RHC	0.076 727 1	56.6231	−2.447 23	57.5701	−0.815 74	58.5171	0.815 74	59.464	2.447 23
BANFF 3, SK numu flux, 1.5 - 2.5 GeV; RHC	0.080 558 7	56.1916	−3.190 67	57.4262	−1.0636	58.6609	1.0636	59.8955	3.190 67
BANFF 4, SK numu flux, 2.5 - 30.0 GeV; RHC	0.080 289 6	56.7027	−2.310 12	57.5966	−0.770 04	58.4905	0.770 04	59.3844	2.310 12
BANFF 5, SK numubar flux, 0.0 - 0.4 GeV; RHC	0.104 476	56.5378	−2.594 24	57.5416	−0.864 75	58.5455	0.864 75	59.5494	2.594 24
BANFF 6, SK numubar flux, 0.4 - 0.5 GeV; RHC	0.101 533	57.067	−1.682 51	57.718	−0.560 84	58.3691	0.560 84	59.0202	1.682 51
BANFF 7, SK numubar flux, 0.5 - 0.6 GeV; RHC	0.096 167 3	57.8301	−0.367 714	57.9724	−0.122 57	58.1147	0.122 57	58.257	0.367 714
BANFF 8, SK numubar flux, 0.6 - 0.7 GeV; RHC	0.084 636 6	57.8844	−0.274 248	57.9905	−0.091 416	58.0966	0.091 416	58.2027	0.274 248
BANFF 9, SK numubar flux, 0.7 - 1.0 GeV; RHC	0.125 091	55.6825	−4.067 81	57.2565	−1.3559	58.8306	1.3559	60.4047	4.067 81
BANFF 10, SK numubar flux, 1.0 - 1.5 GeV; RHC	0.105 291	55.8659	−3.751 76	57.3177	−1.2506	58.7695	1.2506	60.2212	3.751 76
BANFF 11, SK numubar flux, 1.5 - 2.5 GeV; RHC	0.079 987	56.4133	−2.808 66	57.5001	−0.936 22	58.587	0.936 22	59.6738	2.808 66
BANFF 12, SK numubar flux, 2.5 - 3.5 GeV; RHC	0.073 938 4	57.3425	−1.207 82	57.8099	−0.402 61	58.2773	0.402 61	58.7446	1.207 82
BANFF 13, SK numubar flux, 3.5 - 5.0 GeV; RHC	0.093 992 3	57.4113	−1.089 34	57.8328	−0.363 11	58.2543	0.363 11	58.6759	1.089 34
BANFF 14, SK numubar flux, 5.0 - 7.0 GeV; RHC	0.092 512 6	57.9201	−0.212 685	58.0024	−0.070 895	58.0847	0.070 895	58.167	0.212 685
BANFF 15, SK numubar flux, 7.0 - 30.0 GeV; RHC	0.130 312	58.032	−0.019 895 6	58.0397	−0.006 631 9	58.0474	0.006 631 9	58.0551	0.019 895 6
BANFF 16, SK nue flux, 0.0 - 2.5 GeV; RHC	0.068 881 3	58.0397	−0.006 697 58	58.0423	−0.002 232 5	58.0449	0.002 232 5	58.0475	0.006 697 58
BANFF 17, SK nue flux, 2.5 - 30.0 GeV; RHC	0.084 944 6	58.0384	−0.008 949 23	58.0418	−0.002 983 1	58.0453	0.002 983 1	58.0488	0.008 949 23
BANFF 18, SK nuebar flux, 0.0 - 0.5 GeV; RHC	0.094 695 4	58.0435	−0.000 168 308	58.0435	-5.6103×10^{-5}	58.0436	5.6103×10^{-5}	58.0437	0.000 168 308
BANFF 19, SK nuebar flux, 0.5 - 0.7 GeV; RHC	0.091 039 3	58.0432	−0.000 576 142	58.0435	−0.000 192 05	58.0437	0.000 192 05	58.0439	0.000 576 142
BANFF 20, SK nuebar flux, 0.7 - 0.8 GeV; RHC	0.091 012 4	58.0434	−0.000 288 631	58.0435	-9.621×10^{-5}	58.0436	9.621×10^{-5}	58.0437	0.000 288 631
BANFF 21, SK nuebar flux, 0.8 - 1.5 GeV; RHC	0.083 856 5	58.042	−0.002 779 29	58.043	−0.000 926 43	58.0441	0.000 926 43	58.0452	0.002 779 29
BANFF 22, SK nuebar flux, 1.5 - 2.5 GeV; RHC	0.079 577 6	58.042	−0.002 617 59	58.0431	−0.000 872 53	58.0441	0.000 872 53	58.0451	0.002 617 59
BANFF 23, SK nuebar flux, 2.5 - 4.0 GeV; RHC	0.089 007 5	58.0416	−0.003 421 41	58.0429	−0.001 140 5	58.0442	0.001 140 5	58.0456	0.003 421 41
BANFF 24, SK nuebar flux, 4.0 - 30.0 GeV; RHC	0.155 813	58.0419	−0.002 941 6	58.043	−0.000 980 53	58.0441	0.000 980 53	58.0453	0.002 941 6
BANFF; Norm; 2p2h	1	55.0471	−5.162 51	55.0471	−5.1625	61.0401	5.1625	67.0331	15.4875
BANFF; CA5 RES	0.148 515	54.0272	−6.919 63	56.3167	−2.9751	60.1584	3.6436	65.5523	12.9363
BANFF; Norm; BgRES Isospin 1/2	0.307 692	57.1647	−1.514 08	57.4252	−1.0654	58.9874	1.6261	61.8515	6.560 39
BANFF, Ma QE	0.025	56.3858	−2.856 11	57.4906	−0.952 62	58.5957	0.951 22	59.6946	2.844 45
BANFF, Ma RES	0.157 895	54.3275	−6.402 21	56.6448	−2.4099	59.5013	2.5114	62.3804	7.471 77
BANFF; Fermi Momentum	0.057 777 8	59.1725	1.944 97	58.637	1.0224	57.4477	−1.0265	56.2466	−3.095 93
BANFF; Shape; CC Oth	0.4	57.4048	−1.100 53	57.8291	−0.369 55	58.2581	0.369 55	58.6871	1.108 64
BANFF; Norm, CC Coh	0.3	57.8175	−0.389 455	57.9682	−0.129 82	58.1189	0.129 82	58.2696	0.389 455
BANFF; Norm, NC Coh	0.3	58.0416	−0.003 404 46	58.0429	−0.001 134 8	58.0442	0.001 134 8	58.0455	0.003 404 46
BANFF; Norm, NC Oth	0.3	57.5743	−0.808 423	57.8872	−0.269 47	58.2	0.269 47	58.5128	0.808 423
BANFF; Norm, ν_e To ν_μ	0.028 284 3	58.0433	−0.000 419 022	58.0435	−0.000 139 67	58.0436	0.000 139 67	58.0438	0.000 419 022
BANFF; Norm, $\bar{\nu}_e$ To $\bar{\nu}_\mu$	0.028 284 3	58.0433	−0.000 467 005	58.0435	−0.000 155 67	58.0437	0.000 155 67	58.0438	0.000 467 005
BANFF; Norm; 2p2hBar	1	55.5746	−4.253 65	55.5746	−4.2536	60.5125	4.2536	65.4505	12.7609
BANFF; BeRPA A	0.118	51.1456	−11.8841	55.7443	−3.9613	60.3428	3.9613	64.9414	11.8839
BANFF; BeRPA B	0.21	52.501	−9.549 05	56.196	−3.183	59.8911	3.183	63.5862	9.549 09
BANFF; BeRPA D	0.1695	54.0755	−6.836 36	56.7209	−2.2788	59.3663	2.2788	62.0118	6.836 64
BANFF; BeRPA E	0.352	57.8798	−0.282 06	57.9772	−0.114 38	58.111	0.116 18	58.2481	0.352 391
BANFF; Shape; 2p2h	3	58.1578	0.196 836	58.1007	0.098 417	57.9753	−0.117 56	57.9071	−0.235 113
BANFF; Norm; 2p2h C to O	0.2	54.7643	−5.649 69	56.9505	−1.8832	59.1367	1.8832	61.3228	5.649 69
SKDet + FSI/SI 0; E_{reco}(0.00 - 0.40)GeV; $\nu_\mu/\bar{\nu}_\mu$ CCQE ($1R_\mu$); RHC	0.008 512 49	57.9186	−0.215 368	58.0019	−0.071 789	58.0852	0.071 789	58.1686	0.215 368
SKDet + FSI/SI 1; E_{reco}(0.40 - 1.10)GeV; $\nu_\mu/\bar{\nu}_\mu$ CCQE ($1R_\mu$); RHC	0.007 050 99	57.7186	−0.559 823	57.9353	−0.186 61	58.1519	0.186 61	58.3685	0.559 823
SKDet + FSI/SI 2; E_{reco}(1.10 - 30.00)GeV; $\nu_\mu/\bar{\nu}_\mu$ CCQE ($1R_\mu$); RHC	0.006 784	57.5365	−0.873 521	57.8746	−0.291 17	58.2126	0.291 17	58.5506	0.873 521
SKDet + FSI/SI 3; E_{reco}(0.00 - 30.00)GeV; $\nu_\mu/\bar{\nu}_\mu$ CCnQE ($1R_\mu$); RHC	0.129 646	53.7137	−7.4596	56.6003	−2.4865	59.4868	2.4865	62.3734	7.4596
SKDet + FSI/SI 4; E_{reco}(0.00 - 30.00)GeV; $\nu_e/\bar{\nu}_e$/sigν_e CC ($1R_\mu$); RHC	1.004 54	58.0253	−0.031 468	58.0375	−0.010 489	58.0497	0.010 489	58.0618	0.031 468
SKDet + FSI/SI 5; E_{reco}(0.00 - 30.00)GeV; all NC ($1R_\mu$); RHC	0.657 251	54.6208	−5.896 82	56.9027	−1.9656	59.1845	1.9656	61.4663	5.896 82
SK energy scale	0.024	58.0436	$-2.448\,31 \times 10^{-14}$	58.0436	-8.5691×10^{-14}	58.0436	-1.2242×10^{-13}	58.0436	$-6.120\,77 \times 10^{-14}$

Table 34: $\pm1\sigma$ and $\pm3\sigma$ effect of the systematic parameters on the total number of μ-like Super-K events with Run 1-8 POT for antineutrino mode pre-BANFF Asimov data set A. Only the systematic parameters which affect this sample are shown.

Parameter	1 σ value	BANFF tuned -3σ		BANFF tuned -1σ		BANFF tuned +1σ		BANFF tuned +3σ	
All syst		13.4912	−79.0516(%)	36.0858	−43.968 (%)	109.17	69.507(%)	276.4	329.178(%)
BANFF 0, SK numu flux, 0.0 - 0.4 GeV	0.098 732 3	64.0102	−0.608 829	64.2716	−0.202 94	64.533	0.202 94	64.7944	0.608 829
BANFF 1, SK numu flux, 0.4 - 0.5 GeV	0.103 488	62.682	−2.671 26	63.8289	−0.890 42	64.9758	0.890 42	66.1227	2.671 26
BANFF 2, SK numu flux, 0.5 - 0.6 GeV	0.096 444	60.6675	−5.799 23	63.1574	−1.9331	65.6473	1.9331	68.1372	5.799 23
BANFF 3, SK numu flux, 0.6 - 0.7 GeV	0.086 695 8	60.3974	−6.218 64	63.0673	−2.0729	65.7373	2.0729	68.4073	6.218 64
BANFF 4, SK numu flux, 0.7 - 1.0 GeV	0.113 052	58.8035	−8.693 53	62.536	−2.8978	66.2686	2.8978	70.0011	8.693 53
BANFF 5, SK numu flux, 1.0 - 1.5 GeV	0.091 747 6	63.687	−1.110 72	64.1639	−0.370 24	64.6408	0.370 24	65.1176	1.110 72
BANFF 6, SK numu flux, 1.5 - 2.5 GeV	0.070 174 2	64.2733	−0.200 259	64.3593	−0.066 753	64.4453	0.066 753	64.5313	0.200 259
BANFF 7, SK numu flux, 2.5 - 3.5 GeV	0.073 680 5	64.3235	−0.122 345	64.376	−0.040 782	64.4286	0.040 782	64.4811	0.122 345
BANFF 8, SK numu flux, 3.5 - 5.0 GeV	0.087 373 3	64.2921	−0.171 13	64.3656	−0.057 043	64.439	0.057 043	64.5125	0.171 13
BANFF 9, SK numu flux, 5.0 - 7.0 GeV	0.097 936 9	64.3488	−0.083 04	64.3845	−0.027 68	64.4201	0.027 68	64.4558	0.083 04
BANFF 10, SK numu flux, 7.0 - 30.0 GeV	0.114 36	64.3732	−0.045 278 9	64.3926	−0.015 093	64.412	0.015 093	64.4315	0.045 278 9
BANFF 11, SK numubar flux, 0.0 - 0.7 GeV	0.102 576	64.3597	−0.066 141 9	64.3881	−0.022 047	64.4165	0.022 047	64.4449	0.066 141 9
BANFF 12, SK numubar flux, 0.7 - 1.0 GeV	0.078 533 1	64.3627	−0.061 523 9	64.3891	−0.020 508	64.4155	0.020 508	64.4419	0.061 523 9
BANFF 13, SK numubar flux, 1.0 - 1.5 GeV	0.084 453 6	64.3682	−0.053 012 6	64.3909	−0.017 671	64.4137	0.017 671	64.4365	0.053 012 6
BANFF 14, SK numubar flux, 1.5 - 2.5 GeV	0.085 568 3	64.3811	−0.033 001 7	64.3952	−0.011 001	64.4094	0.011 001	64.4236	0.033 001 7
BANFF 15, SK numubar flux, 2.5 - 30.0 GeV	0.086 428 4	64.3817	−0.031 934 7	64.3955	−0.010 645	64.4092	0.010 645	64.4229	0.031 934 7
BANFF 16, SK nue flux, 0.0 - 0.5 GeV	0.089 698 3	64.0426	−0.558 585	64.2824	−0.1862	64.5222	0.1862	64.7621	0.558 585
BANFF 17, SK nue flux, 0.5 - 0.7 GeV	0.089 952	63.9068	−0.769 331	64.2372	−0.256 44	64.5675	0.256 44	64.8978	0.769 331
BANFF 18, SK nue flux, 0.7 - 0.8 GeV	0.085 964 8	64.1548	−0.384 367	64.3198	−0.128 12	64.4848	0.128 12	64.6499	0.384 367
BANFF 19, SK nue flux, 0.8 - 1.5 GeV	0.080 918 2	63.6486	−1.170 31	64.1511	−0.3901	64.6535	0.3901	65.156	1.170 31
BANFF 20, SK nue flux, 1.5 - 2.5 GeV	0.078 971 9	64.3446	−0.089 644 1	64.3831	−0.029 881	64.4216	0.029 881	64.46	0.089 644 1
BANFF 21, SK nue flux, 2.5 - 4.0 GeV	0.083 85	64.3936	−0.013 484 5	64.3994	−0.004 494 8	64.4052	0.004 494 8	64.411	0.013 484 5
BANFF 22, SK nue flux, 4.0 - 30.0 GeV	0.093 894 3	64.3987	−0.005 670 38	64.4011	−0.001 890 1	64.4035	0.001 890 1	64.406	0.005 670 38
BANFF 23, SK nuebar flux, 0.0 - 2.5 GeV	0.074 027 9	64.3297	−0.112 81	64.3781	−0.037 603	64.4265	0.037 603	64.475	0.112 81
BANFF 24, SK nuebar flux, 2.5 - 30.0 GeV	0.128 416	64.3995	−0.004 383 1	64.4014	−0.001 461	64.4033	0.001 461	64.4051	0.004 383 1
BANFF; Norm; 2p2h	1	57.5264	−10.6765	57.5264	−10.677	71.2782	10.677	85.0301	32.0295
BANFF; CA5 RES	0.148 515	60.5772	−5.939 36	62.946	−2.2612	66.0398	2.5426	69.8586	8.472 18
BANFF; Norm; BgRES Isospin 1/2	0.307 692	63.9148	−0.756 971	64.0593	−0.532 67	64.9259	0.813 02	66.5148	3.280 13
BANFF, Ma QE	0.025	61.7753	−4.078 99	63.5516	−1.3209	65.2283	1.2825	66.8074	3.734 46
BANFF, Ma RES	0.157 895	61.1503	−5.0496	63.4044	−1.5494	65.3124	1.4132	66.9029	3.882 76
BANFF; Fermi Momentum	0.057 777 8	65.765	2.115 87	65.128	1.1268	63.6315	−1.1968	61.9681	−3.779 65
BANFF; Shape; CC Oth	0.4	64.2479	−0.239 744	64.3445	−0.089 736	64.4601	0.089 736	64.5757	0.269 208
BANFF; Norm, CC Coh	0.3	64.3496	−0.081 866 2	64.3847	−0.027 289	64.4199	0.027 289	64.455	0.081 866 2
BANFF; Norm, NC Coh	0.3	63.8321	−0.885 326	64.2123	−0.295 11	64.5924	0.295 11	64.9725	0.885 326
BANFF; Norm, NC Oth	0.3	64.0843	−0.493 823	64.2963	−0.164 61	64.5083	0.164 61	64.7203	0.493 823
BANFF; Norm, ν_e To ν_μ	0.028 284 3	59.4949	−7.619 97	62.7665	−2.54	66.0381	2.54	69.3097	7.619 97
BANFF; Norm; NC 1 γ	1	62.3329	−3.213 31	62.3329	−3.2133	66.4718	3.2133	70.6107	9.639 94
BANFF; Norm, $\bar{\nu}_e$ To $\bar{\nu}_\mu$	0.028 284 3	64.347	−0.085 852 9	64.3839	−0.028 618	64.4207	0.028 618	64.4576	0.085 852 9
BANFF; Norm; 2p2hBar	1	64.3424	−0.093 002 2	64.3424	−0.093 002	64.4622	0.093 002	64.582	0.279 007
BANFF; BeRPA A	0.118	55.2091	−14.2747	61.3379	−4.7582	67.4667	4.7582	73.5956	14.2747
BANFF; BeRPA B	0.21	53.464	−16.9843	60.7562	−5.6614	68.0484	5.6614	75.3406	16.9843
BANFF; BeRPA D	0.1695	58.0806	−9.815 99	62.2951	−3.272	66.5096	3.272	70.724	9.815 97
BANFF; BeRPA E	0.352	64.0489	−0.548 743	64.2615	−0.2187	64.5432	0.218 73	64.8249	0.656 212
BANFF; Shape; 2p2h	3	63.9869	−0.644 983	64.1946	−0.322 49	64.603	0.311 56	64.8036	0.623 127
BANFF; Norm; 2p2h C to O	0.2	60.2408	−6.4617	63.0152	−2.1539	65.7895	2.1539	68.5638	6.4617
SKDet + FSI/SI 6; E_{reco}(0.00 - 0.35)GeV; Osc. ν_e CC ($1R_e$)	0.181 271	61.2702	−4.863 41	63.3583	−1.6211	65.4464	1.6211	67.5345	4.863 41
SKDet + FSI/SI 7; E_{reco}(0.35 - 0.80)GeV; Osc. ν_e CC ($1R_e$)	0.036 083	60.4597	−6.1219	63.0881	−2.0406	65.7165	2.0406	68.345	6.1219
SKDet + FSI/SI 8; E_{reco}(0.80 - 1.25)GeV; Osc. ν_e CC ($1R_e$)	0.041 663 6	63.3311	−1.663 28	64.0452	−0.554 43	64.7594	0.554 43	65.4735	1.663 28
SKDet + FSI/SI 9; E_{reco}(0.00 - 0.35)GeV; $\nu_\mu/\bar{\nu}_\mu$ CC ($1R_e$)	0.300 431	64.3482	−0.083 951 1	64.3843	−0.027 984	64.4203	0.027 984	64.4564	0.083 951 1
SKDet + FSI/SI 10; E_{reco}(0.35 - 0.80)GeV; $\nu_\mu/\bar{\nu}_\mu$ CC ($1R_e$)	0.321 346	64.2457	−0.243 223	64.3501	−0.081 074	64.4545	0.081 074	64.559	0.243 223
SKDet + FSI/SI 11; E_{reco}(0.80 - 1.25)GeV; $\nu_\mu/\bar{\nu}_\mu$ CC ($1R_e$)	0.392 969	64.3731	−0.045 282 3	64.3926	−0.015 094	64.412	0.015 094	64.4315	0.045 282 3
SKDet + FSI/SI 12; E_{reco}(0.00 - 0.35)GeV; $\nu_e/\bar{\nu}_e$ CC ($1R_e$)	0.135 399	64.0289	−0.579 849	64.2778	−0.193 28	64.5268	0.193 28	64.7757	0.579 849
SKDet + FSI/SI 13; E_{reco}(0.35 - 0.80)GeV; $\nu_e/\bar{\nu}_e$ CC ($1R_e$)	0.069 892 8	63.5422	−1.335 58	64.1156	−0.445 19	64.689	0.445 19	65.2625	1.335 58
SKDet + FSI/SI 14; E_{reco}(0.80 - 1.25)GeV; $\nu_e/\bar{\nu}_e$ CC ($1R_e$)	0.077 136 1	63.7745	−0.974 818	64.193	−0.324 94	64.6116	0.324 94	65.0301	0.974 818
SKDet + FSI/SI 15; E_{reco}(0.00 - 0.35)GeV; all NC ($1R_e$)	0.194 453	63.1388	−1.961 83	63.9812	−0.653 94	64.8235	0.653 94	65.6658	1.961 83
SKDet + FSI/SI 16; E_{reco}(0.35 - 0.80)GeV; all NC ($1R_e$)	0.181 015	62.8486	−2.412 51	63.8844	−0.804 17	64.9202	0.804 17	65.956	2.412 51
SKDet + FSI/SI 17; E_{reco}(0.80 - 1.25)GeV; all NC ($1R_e$)	0.472 078	63.4933	−1.4115	64.0993	−0.4705	64.7053	0.4705	65.3113	1.4115
SK energy scale	0.024	64.4023	0	64.4023	-2.2066×10^{-14}	64.4023	0	64.4023	0

Table 35: $\pm 1\sigma$ and $\pm 3\sigma$ effect of the systematic parameters on the total number of e-like Super-K events with Run 1-8 POT for neutrino mode pre-BANFF Asimov data set A. Only the systematic parameters which affect this sample are shown.

Parameter	1 σ value	BANFF tuned -3σ		BANFF tuned -1σ		BANFF tuned +1σ		BANFF tuned +3σ	
All syst		1.979 92	−73.1627(%)	4.194 57	−43.144 (%)	12.51	69.576(%)	32.2025	336.497(%)
BANFF 0, SK numu flux, 0.0 - 0.7 GeV; RHC	0.093 682 2	7.219 18	−2.145 75	7.324 71	−0.715 25	7.430 25	0.715 25	7.535 78	2.145 75
BANFF 1, SK numu flux, 0.7 - 1.0 GeV; RHC	0.079 343	7.256 37	−1.641 56	7.337 11	−0.547 19	7.417 85	0.547 19	7.498 59	1.641 56
BANFF 2, SK numu flux, 1.0 - 1.5 GeV; RHC	0.076 727 1	7.304 03	−0.995 621	7.353	−0.331 87	7.401 96	0.331 87	7.450 93	0.995 621
BANFF 3, SK numu flux, 1.5 - 2.5 GeV; RHC	0.080 558 7	7.341 56	−0.486 92	7.365 51	−0.162 31	7.389 46	0.162 31	7.4134	0.486 92
BANFF 4, SK numu flux, 2.5 - 30.0 GeV; RHC	0.080 289 6	7.337 62	−0.540 283	7.364 19	−0.180 09	7.390 77	0.180 09	7.417 34	0.540 283
BANFF 5, SK numubar flux, 0.0 - 0.4 GeV; RHC	0.104 476	7.353 38	−0.326 617	7.369 45	−0.108 87	7.385 51	0.108 87	7.401 58	0.326 617
BANFF 6, SK numubar flux, 0.4 - 0.5 GeV; RHC	0.101 533	7.290 96	−1.172 75	7.348 64	−0.390 92	7.406 32	0.390 92	7.464	1.172 75
BANFF 7, SK numubar flux, 0.5 - 0.6 GeV; RHC	0.096 167 3	7.156 96	−2.989 11	7.303 97	−0.996 37	7.450 99	0.996 37	7.598	2.989 11
BANFF 8, SK numubar flux, 0.6 - 0.7 GeV; RHC	0.084 636 6	7.110 41	−3.620 06	7.288 46	−1.2067	7.4665	1.2067	7.644 55	3.620 06
BANFF 9, SK numubar flux, 0.7 - 1.0 GeV; RHC	0.125 091	6.877 86	−6.772 27	7.210 94	−2.2574	7.544 02	2.2574	7.8771	6.772 27
BANFF 10, SK numubar flux, 1.0 - 1.5 GeV; RHC	0.105 291	7.285 18	−1.251 17	7.346 71	−0.417 06	7.408 25	0.417 06	7.469 79	1.251 17
BANFF 11, SK numubar flux, 1.5 - 2.5 GeV; RHC	0.079 987	7.353 22	−0.328 861	7.369 39	−0.109 62	7.385 57	0.109 62	7.401 74	0.328 861
BANFF 12, SK numubar flux, 2.5 - 3.5 GeV; RHC	0.073 938 4	7.3661	−0.154 206	7.373 69	−0.051 402	7.381 27	0.051 402	7.388 86	0.154 206
BANFF 13, SK numubar flux, 3.5 - 5.0 GeV; RHC	0.093 992 3	7.364 21	−0.179 93	7.373 06	−0.059 977	7.381 91	0.059 977	7.390 76	0.179 93
BANFF 14, SK numubar flux, 5.0 - 7.0 GeV; RHC	0.092 512 6	7.373 16	−0.058 575 7	7.376 04	−0.019 525	7.378 92	0.019 525	7.3818	0.058 575 7
BANFF 15, SK numubar flux, 7.0 - 30.0 GeV; RHC	0.130 312	7.374 81	−0.036 139	7.376 59	−0.012 046	7.378 37	0.012 046	7.380 15	0.036 139
BANFF 16, SK nue flux, 0.0 - 2.5 GeV; RHC	0.068 881 3	7.243 97	−1.809 73	7.332 98	−0.603 24	7.421 99	0.603 24	7.510 99	1.809 73
BANFF 17, SK nue flux, 2.5 - 30.0 GeV; RHC	0.084 944 6	7.375 09	−0.032 427 5	7.376 68	−0.010 809	7.378 28	0.010 809	7.379 87	0.032 427 5
BANFF 18, SK nuebar flux, 0.0 - 0.5 GeV; RHC	0.094 695 4	7.325 35	−0.706 645	7.3601	−0.235 55	7.394 86	0.235 55	7.429 61	0.706 645
BANFF 19, SK nuebar flux, 0.5 - 0.7 GeV; RHC	0.091 039 3	7.317 31	−0.815 59	7.357 42	−0.271 86	7.397 54	0.271 86	7.437 65	0.815 59
BANFF 20, SK nuebar flux, 0.7 - 0.8 GeV; RHC	0.091 012 4	7.344 44	−0.447 891	7.366 47	−0.1493	7.3885	0.1493	7.410 52	0.447 891
BANFF 21, SK nuebar flux, 0.8 - 1.5 GeV; RHC	0.083 856 5	7.268 36	−1.4791	7.341 11	−0.493 03	7.413 85	0.493 03	7.4866	1.4791
BANFF 22, SK nuebar flux, 1.5 - 2.5 GeV; RHC	0.079 577 6	7.370 79	−0.090 695 5	7.375 25	−0.030 232	7.379 71	0.030 232	7.384 17	0.090 695 5
BANFF 23, SK nuebar flux, 2.5 - 4.0 GeV; RHC	0.089 007 5	7.376 11	−0.018 546 3	7.377 02	−0.006 182 1	7.377 94	0.006 182 1	7.378 85	0.018 546 3
BANFF 24, SK nuebar flux, 4.0 - 30.0 GeV; RHC	0.155 813	7.376 35	−0.015 332 2	7.3771	−0.005 110 7	7.377 86	0.005 110 7	7.378 61	0.015 332 2
BANFF; Norm; 2p2h	1	7.132 28	−3.323 58	7.132 28	−3.3236	7.622 68	3.3236	8.113 07	9.970 75
BANFF; CA5 RES	0.148 515	6.848 93	−7.164 38	7.1284	−3.3762	7.699 46	4.3644	8.562 12	16.0575
BANFF; Norm; BgRES Isospin 1/2	0.307 692	7.286 15	−1.237 96	7.313 21	−0.871 15	7.475 58	1.3297	7.773 24	5.364 46
BANFF, Ma QE	0.025	7.198 64	−2.424 09	7.318 05	−0.805 61	7.436 54	0.800 56	7.553 11	2.380 56
BANFF, Ma RES	0.157 895	6.986 82	−5.295 35	7.221 81	−2.1101	7.540 13	2.2047	7.857 05	6.500 38
BANFF; Fermi Momentum	0.057 777 8	7.530 15	2.069 35	7.458 34	1.0961	7.295 45	−1.1119	7.127 52	−3.3882
BANFF; Shape; CC Oth	0.4	7.344 09	−0.452 601	7.365 67	−0.160 04	7.389 29	0.160 04	7.4129	0.480 121
BANFF; Norm, CC Coh	0.3	7.3311	−0.628 631	7.362 02	−0.209 54	7.392 94	0.209 54	7.423 86	0.628 631
BANFF; Norm, NC Coh	0.3	7.131 97	−3.327 78	7.295 65	−1.1093	7.459 32	1.1093	7.622 99	3.327 78
BANFF; Norm, NC Oth	0.3	7.302 47	−1.016 77	7.352 48	−0.338 92	7.402 48	0.338 92	7.452 49	1.016 77
BANFF; Norm, ν_e To ν_μ	0.028 284 3	7.221 29	−2.117 13	7.325 42	−0.705 71	7.429 54	0.705 71	7.533 67	2.117 13
BANFF; Norm; NC 1 γ	1	6.853 36	−7.1043	6.853 36	−7.1043	7.9016	7.1043	8.949 84	21.3129
BANFF; Norm, $\bar{\nu}_e$ To $\bar{\nu}_\mu$	0.028 284 3	7.0285	−4.730 33	7.261 15	−1.5768	7.493 81	1.5768	7.726 46	4.730 33
BANFF; Norm; 2p2hBar	1	7.054 87	−4.372 86	7.054 87	−4.3729	7.700 09	4.3729	8.3453	13.1186
BANFF; BeRPA A	0.118	6.480 99	−12.1517	7.078 65	−4.0506	7.676 31	4.0506	8.273 97	12.1517
BANFF; BeRPA B	0.21	6.6711	−9.574 82	7.142 02	−3.1916	7.612 94	3.1916	8.083 86	9.574 79
BANFF; BeRPA D	0.1695	7.055 99	−4.357 67	7.270 32	−1.4526	7.484 64	1.4526	7.698 97	4.357 65
BANFF; BeRPA E	0.352	7.359 57	−0.242 812	7.370 34	−0.096 784	7.384 62	0.096 802	7.398 91	0.290 453
BANFF; Shape; 2p2h	3	7.337 46	−0.542 431	7.357 47	−0.271 22	7.409 66	0.436 17	7.441 84	0.872 345
BANFF; Norm; 2p2h C to O	0.2	7.0368	−4.617 87	7.263 92	−1.5393	7.491 04	1.5393	7.718 16	4.617 87
SKDet + FSI/SI 6; E_{reco}(0.00 - 0.35)GeV; Osc. ν_e CC ($1R_e$); RHC	0.108 165	7.263 51	−1.544 86	7.339 49	−0.514 95	7.415 47	0.514 95	7.491 45	1.544 86
SKDet + FSI/SI 7; E_{reco}(0.35 - 0.80)GeV; Osc. ν_e CC ($1R_e$); RHC	0.036 303 9	7.0458	−4.4959	7.266 92	−1.4986	7.488 04	1.4986	7.709 17	4.4959
SKDet + FSI/SI 8; E_{reco}(0.80 - 1.25)GeV; Osc. ν_e CC ($1R_e$); RHC	0.056 731 5	7.216 19	−2.186 26	7.323 72	−0.728 75	7.431 24	0.728 75	7.538 77	2.186 26
SKDet + FSI/SI 9; E_{reco}(0.00 - 0.35)GeV; $\nu_\mu/\bar{\nu}_\mu$ CC ($1R_e$); RHC	0.348 644	7.367 59	−0.134 016	7.374 19	−0.044 672	7.380 78	0.044 672	7.387 37	0.134 016
SKDet + FSI/SI 10; E_{reco}(0.35 - 0.80)GeV; $\nu_\mu/\bar{\nu}_\mu$ CC ($1R_e$); RHC	0.341 647	7.340 54	−0.500 721	7.365 17	−0.166 91	7.389 79	0.166 91	7.414 42	0.500 721
SKDet + FSI/SI 11; E_{reco}(0.80 - 1.25)GeV; $\nu_\mu/\bar{\nu}_\mu$ CC ($1R_e$); RHC	0.417 164	7.360 76	−0.226 635	7.371 91	−0.075 545	7.383 05	0.075 545	7.3942	0.226 635
SKDet + FSI/SI 12; E_{reco}(0.00 - 0.35)GeV; $\nu_e/\bar{\nu}_e$ CC ($1R_e$); RHC	0.084 556 3	7.336 87	−0.550 513	7.363 94	−0.1835	7.391 02	0.1835	7.418 09	0.550 513
SKDet + FSI/SI 13; E_{reco}(0.35 - 0.80)GeV; $\nu_e/\bar{\nu}_e$ CC ($1R_e$); RHC	0.056 069 7	7.2436	−1.814 67	7.332 86	−0.604 89	7.422 11	0.604 89	7.511 36	1.814 67
SKDet + FSI/SI 14; E_{reco}(0.80 - 1.25)GeV; $\nu_e/\bar{\nu}_e$ CC ($1R_e$); RHC	0.078 024 1	7.2246	−2.072 23	7.326 52	−0.690 74	7.428 44	0.690 74	7.530 36	2.072 23
SKDet + FSI/SI 15; E_{reco}(0.00 - 0.35)GeV; all NC ($1R_e$); RHC	0.214 066	7.064 69	−4.239 82	7.273 22	−1.4133	7.481 74	1.4133	7.690 27	4.239 82
SKDet + FSI/SI 16; E_{reco}(0.35 - 0.80)GeV; all NC ($1R_e$); RHC	0.190 575	6.973 98	−5.469 32	7.242 98	−1.8231	7.511 98	1.8231	7.780 98	5.469 32
SKDet + FSI/SI 17; E_{reco}(0.80 - 1.25)GeV; all NC ($1R_e$); RHC	0.464 86	7.137 17	−3.2574	7.297 38	−1.0858	7.457 59	1.0858	7.617 79	3.2574
SK energy scale	0.024	7.377 48	0	7.377 48	-1.2039×10^{-14}	7.377 48	-2.4078×10^{-14}	7.377 48	0

Table 36: $\pm 1\sigma$ and $\pm 3\sigma$ effect of the systematic parameters on the total number of e-like Super-K events with Run 1-8 POT for antineutrino mode pre-BANFF Asimov data set A. Only the systematic parameters which affect this sample are shown.

Parameter	1 σ value	BANFF tuned -3σ		BANFF tuned -1σ		BANFF tuned +1σ		BANFF tuned +3σ	
All syst		0.424 293	−94.5073(%)	3.6093	−53.276 (%)	14.919	93.127(%)	45.8854	494.009(%)
BANFF 0, SK numu flux, 0.0 - 0.4 GeV	0.098 732 3	7.721 82	−0.037 355 8	7.723 75	−0.012 452	7.725 67	0.012 452	7.727 59	0.037 355 8
BANFF 1, SK numu flux, 0.4 - 0.5 GeV	0.103 488	7.633 42	−1.181 75	7.694 28	−0.393 92	7.755 14	0.393 92	7.815 99	1.181 75
BANFF 2, SK numu flux, 0.5 - 0.6 GeV	0.096 444	7.359 53	−4.727 45	7.602 98	−1.5758	7.846 43	1.5758	8.089 89	4.727 45
BANFF 3, SK numu flux, 0.6 - 0.7 GeV	0.086 695 8	7.204 13	−6.739 09	7.551 18	−2.2464	7.898 23	2.2464	8.245 28	6.739 09
BANFF 4, SK numu flux, 0.7 - 1.0 GeV	0.113 052	6.907 02	−10.5853	7.452 15	−3.5284	7.997 27	3.5284	8.542 39	10.5853
BANFF 5, SK numu flux, 1.0 - 1.5 GeV	0.091 747 6	7.623 55	−1.309 51	7.690 99	−0.4365	7.758 43	0.4365	7.825 86	1.309 51
BANFF 6, SK numu flux, 1.5 - 2.5 GeV	0.070 174 2	7.702 59	−0.286 339	7.717 33	−0.095 446	7.732 08	0.095 446	7.746 83	0.286 339
BANFF 7, SK numu flux, 2.5 - 3.5 GeV	0.073 680 5	7.710 56	−0.183 134	7.719 99	−0.061 045	7.729 42	0.061 045	7.738 85	0.183 134
BANFF 8, SK numu flux, 3.5 - 5.0 GeV	0.087 373 3	7.6978	−0.348 282	7.715 74	−0.116 09	7.733 68	0.116 09	7.751 61	0.348 282
BANFF 9, SK numu flux, 5.0 - 7.0 GeV	0.097 936 9	7.713 17	−0.149 381	7.720 86	−0.049 794	7.728 55	0.049 794	7.736 25	0.149 381
BANFF 10, SK numu flux, 7.0 - 30.0 GeV	0.114 36	7.7233	−0.018 241	7.724 24	−0.006 080 3	7.725 18	0.006 080 3	7.726 12	0.018 241
BANFF 11, SK numubar flux, 0.0 - 0.7 GeV	0.102 576	7.723 85	−0.011 103 3	7.724 42	−0.003 701 1	7.724 99	0.003 701 1	7.725 56	0.011 103 3
BANFF 12, SK numubar flux, 0.7 - 1.0 GeV	0.078 533 1	7.723 78	−0.012 024	7.7244	−0.004 008	7.725 02	0.004 008	7.725 64	0.012 024
BANFF 13, SK numubar flux, 1.0 - 1.5 GeV	0.084 453 6	7.723 26	−0.018 690 7	7.724 23	−0.006 230 2	7.725 19	0.006 230 2	7.726 15	0.018 690 7
BANFF 14, SK numubar flux, 1.5 - 2.5 GeV	0.085 568 3	7.722 55	−0.027 962 6	7.723 99	−0.009 320 9	7.725 43	0.009 320 9	7.726 87	0.027 962 6
BANFF 15, SK numubar flux, 2.5 - 30.0 GeV	0.086 428 4	7.7221	−0.033 754 7	7.723 84	−0.011 252	7.725 58	0.011 252	7.727 31	0.033 754 7
BANFF 16, SK nue flux, 0.0 - 0.5 GeV	0.089 698 3	7.714 48	−0.132 379	7.7213	−0.044 126	7.728 12	0.044 126	7.734 93	0.132 379
BANFF 17, SK nue flux, 0.5 - 0.7 GeV	0.089 952	7.668 23	−0.731 188	7.705 88	−0.243 73	7.743 53	0.243 73	7.781 19	0.731 188
BANFF 18, SK nue flux, 0.7 - 0.8 GeV	0.085 964 8	7.6858	−0.503 612	7.711 74	−0.167 87	7.737 67	0.167 87	7.763 61	0.503 612
BANFF 19, SK nue flux, 0.8 - 1.5 GeV	0.080 918 2	7.585 96	−1.796 18	7.678 46	−0.598 73	7.770 96	0.598 73	7.863 46	1.796 18
BANFF 20, SK nue flux, 1.5 - 2.5 GeV	0.078 971 9	7.713 81	−0.141 038	7.721 08	−0.047 013	7.728 34	0.047 013	7.7356	0.141 038
BANFF 21, SK nue flux, 2.5 - 4.0 GeV	0.083 85	7.723 24	−0.018 965 5	7.724 22	−0.006 321 8	7.7252	0.006 321 8	7.726 17	0.018 965 5
BANFF 22, SK nue flux, 4.0 - 30.0 GeV	0.093 894 3	7.724 12	−0.007 605 52	7.724 51	−0.002 535 2	7.7249	0.002 535 2	7.725 29	0.007 605 52
BANFF 23, SK nuebar flux, 0.0 - 2.5 GeV	0.074 027 9	7.722 91	−0.023 243	7.724 11	−0.007 747 7	7.725 31	0.007 747 7	7.7265	0.023 243
BANFF 24, SK nuebar flux, 2.5 - 30.0 GeV	0.128 416	7.724 28	−0.005 480 76	7.724 57	−0.001 826 9	7.724 85	0.001 826 9	7.725 13	0.005 480 76
BANFF; Norm; 2p2h	1	7.621 98	−1.329 85	7.621 98	−1.3298	7.827 43	1.3298	8.032 89	3.989 54
BANFF; CA5 RES	0.148 515	4.948 98	−35.9331	6.684 39	−13.467	8.880 11	14.957	11.5362	49.3409
BANFF; Norm; BgRES Isospin 1/2	0.307 692	7.420 76	−3.934 73	7.510 82	−2.7689	8.051 17	4.2262	9.041 81	17.0505
BANFF, Ma QE	0.025	7.709 58	−0.195 853	7.7198	−0.063 538	7.729 48	0.061 781	7.738 62	0.180 117
BANFF, Ma RES	0.157 895	5.557 65	−28.0536	7.121 25	−7.8121	8.227 51	6.509	9.004 84	16.5719
BANFF; Fermi Momentum	0.057 777 8	7.730 57	0.075 9039	7.727 76	0.039 489	7.721 22	−0.045 097	7.713 41	−0.146 216
BANFF; Shape; CC Oth	0.4	7.537 21	−2.427 24	7.648 13	−0.991 37	7.801 29	0.991 37	7.954 45	2.974 11
BANFF; Norm, CC Coh	0.3	7.589 06	−1.755 98	7.679 49	−0.585 33	7.769 92	0.585 33	7.860 35	1.755 98
BANFF; Norm, NC Oth	0.3	7.522 66	−2.615 63	7.657 36	−0.871 88	7.792 06	0.871 88	7.926 76	2.615 63
BANFF; Norm, ν_e To ν_μ	0.028 284 3	7.118 46	−7.848 14	7.522 63	−2.616	7.926 79	2.616	8.330 95	7.848 14
BANFF; Norm; NC 1 γ	1	7.666 15	−0.758 101	7.666 15	−0.7581	7.783 27	0.7581	7.900 39	2.2743
BANFF; Norm, $\bar{\nu}_e$ To $\bar{\nu}_\mu$	0.028 284 3	7.7235	−0.015 681 4	7.7243	−0.005 227 1	7.725 11	0.005 227 1	7.725 92	0.015 681 4
BANFF; Norm; 2p2hBar	1	7.723 12	−0.020 606 6	7.723 12	−0.020 607	7.7263	0.020 607	7.729 48	0.061 819 8
BANFF; BeRPA A	0.118	7.6872	−0.485 527	7.712 21	−0.161 84	7.737 21	0.161 84	7.762 21	0.485 527
BANFF; BeRPA B	0.21	7.663 89	−0.787 311	7.704 43	−0.262 44	7.744 98	0.262 44	7.785 52	0.787 31
BANFF; BeRPA D	0.1695	7.686 58	−0.493 568	7.712	−0.164 52	7.737 42	0.164 52	7.762 83	0.493 568
BANFF; BeRPA E	0.352	7.722 55	−0.027 970 1	7.723 85	−0.011 147	7.725 57	0.011 148	7.727 29	0.033 442 8
BANFF; Shape; 2p2h	3	7.6796	−0.583 888	7.702 16	−0.291 94	7.742 43	0.229 37	7.760 14	0.458 739
BANFF; Norm; 2p2h C to O	0.2	7.662 12	−0.810 272	7.703 84	−0.270 09	7.745 57	0.270 09	7.7873	0.810 272
SKDet + FSI/SI 0; E_{reco}(0.30 - 0.80)GeV; Osc. ν_e CC ($MultiR_e$); MultiRing	0.248 951	4.603 12	−40.4104	6.684 18	−13.47	8.765 24	13.47	10.8463	40.4104
SKDet + FSI/SI 1; E_{reco}(0.80 - 1.25)GeV; Osc. ν_e CC ($MultiR_e$); MultiRing	0.214 994	6.460 97	−16.3597	7.303 46	−5.4532	8.145 95	5.4532	8.988 45	16.3597
SKDet + FSI/SI 2; E_{reco}(0.30 - 0.80)GeV; $\nu_\mu/\bar{\nu}_\mu$ CC ($MultiR_e$); MultiRing	0.507 403	7.569 86	−2.004 53	7.673 09	−0.668 18	7.776 32	0.668 18	7.879 55	2.004 53
SKDet + FSI/SI 3; E_{reco}(0.80 - 1.25)GeV; $\nu_\mu/\bar{\nu}_\mu$ CC ($MultiR_e$); MultiRing	0.236 118	7.687 85	−0.477 076	7.712 42	−0.159 03	7.736 99	0.159 03	7.761 56	0.477 076
SKDet + FSI/SI 4; E_{reco}(0.30 - 0.80)GeV; $\nu_e/\bar{\nu}_e$ CC ($MultiR_e$); MultiRing	0.244 289	7.426 86	−3.855 77	7.625 42	−1.2853	7.823 99	1.2853	8.022 55	3.855 77
SKDet + FSI/SI 5; E_{reco}(0.80 - 1.25)GeV; $\nu_e/\bar{\nu}_e$ CC ($MultiR_e$); MultiRing	0.234 55	7.292 99	−5.588 77	7.5808	−1.8629	7.868 61	1.8629	8.156 42	5.588 77
SKDet + FSI/SI 6; E_{reco}(0.30 - 0.80)GeV; all NC ($MultiR_e$); MultiRing	0.983 178	6.846 41	−11.37	7.431 94	−3.79	8.017 47	3.79	8.603 01	11.37
SKDet + FSI/SI 7; E_{reco}(0.80 - 1.25)GeV; all NC ($MultiR_e$); MultiRing	0.523 609	7.545 27	−2.3229	7.664 89	−0.7743	7.784 52	0.7743	7.904 14	2.3229
SK energy scale	0.024	7.724 71	0	7.724 71	2.2996×10^{-14}	7.724 71	0	7.724 71	$1.149\,79 \times 10^{-14}$

Table 37: $\pm1\sigma$ and $\pm3\sigma$ effect of the systematic parameters on the total number of e-like Super-K events with one decay electron with Run 1-8 POT for neutrino mode pre-BANFF Asimov data set A. Only the systematic parameters which affect this sample are shown.

Parameter	1 σ value	BANFF tuned -3σ		BANFF tuned -1σ		BANFF tuned +1σ		BANFF tuned +3σ	
All syst		107.89	−58.8929(%)	200.066	−23.773 (%)	337.7	28.666(%)	550.51	109.75 (%)
BANFF 0, SK numu flux, 0.0 - 0.4 GeV	0.057 496 7	257.64	−1.836 31	260.853	−0.6121	264.066	0.6121	267.28	1.836 31
BANFF 1, SK numu flux, 0.4 - 0.5 GeV	0.051 936 3	258.407	−1.544 03	261.109	−0.514 68	263.811	0.514 68	266.512	1.544 03
BANFF 2, SK numu flux, 0.5 - 0.6 GeV	0.044 568 2	261.573	−0.337 759	262.164	−0.112 59	262.755	0.112 59	263.346	0.337 759
BANFF 3, SK numu flux, 0.6 - 0.7 GeV	0.041 886 1	261.743	−0.273 275	262.221	−0.091 092	262.699	0.091 092	263.177	0.273 275
BANFF 4, SK numu flux, 0.7 - 1.0 GeV	0.054 16	253.774	−3.309 58	259.565	−1.1032	265.355	1.1032	271.146	3.309 58
BANFF 5, SK numu flux, 1.0 - 1.5 GeV	0.049 026 1	255.414	−2.684 71	260.111	−0.8949	264.809	0.8949	269.506	2.684 71
BANFF 6, SK numu flux, 1.5 - 2.5 GeV	0.041 778 9	257.572	−1.862 32	260.831	−0.620 77	264.089	0.620 77	267.348	1.862 32
BANFF 7, SK numu flux, 2.5 - 3.5 GeV	0.043 606 9	259.942	−0.959 321	261.621	−0.319 77	263.299	0.319 77	264.978	0.959 321
BANFF 8, SK numu flux, 3.5 - 5.0 GeV	0.040 772	260.44	−0.769 586	261.787	−0.256 53	263.133	0.256 53	264.48	0.769 586
BANFF 9, SK numu flux, 5.0 - 7.0 GeV	0.040 095 8	261.934	−0.200 252	262.285	−0.066 751	262.635	0.066 751	262.986	0.200 252
BANFF 10, SK numu flux, 7.0 - 30.0 GeV	0.048 214	262.415	−0.017 173 8	262.445	−0.005 724 6	262.475	0.005 724 6	262.505	0.017 173 8
BANFF 11, SK numubar flux, 0.0 - 0.7 GeV	0.073 717	262.238	−0.084 566 9	262.386	−0.028 189	262.534	0.028 189	262.682	0.084 566 9
BANFF 12, SK numubar flux, 0.7 - 1.0 GeV	0.0477	262.299	−0.061 424 1	262.406	−0.020 475	262.514	0.020 475	262.621	0.061 424 1
BANFF 13, SK numubar flux, 1.0 - 1.5 GeV	0.057 472 1	261.806	−0.249 264	262.242	−0.083 088	262.678	0.083 088	263.114	0.249 264
BANFF 14, SK numubar flux, 1.5 - 2.5 GeV	0.061 890 7	261.514	−0.360 463	262.145	−0.120 15	262.775	0.120 15	263.406	0.360 463
BANFF 15, SK numubar flux, 2.5 - 30.0 GeV	0.064 676 3	261.784	−0.257 58	262.235	−0.085 86	262.685	0.085 86	263.136	0.257 58
BANFF 16, SK nue flux, 0.0 - 0.5 GeV	0.046 471 9	262.459	−0.000 248 152	262.46	-8.2717×10^{-5}	262.46	8.2717×10^{-5}	262.461	0.000 248 152
BANFF 17, SK nue flux, 0.5 - 0.7 GeV	0.042 400 3	262.459	−0.000 425 298	262.46	−0.000 141 77	262.46	0.000 141 77	262.461	0.000 425 298
BANFF 18, SK nue flux, 0.7 - 0.8 GeV	0.041 175	262.459	−0.000 313 178	262.46	−0.000 104 39	262.46	0.000 104 39	262.461	0.000 313 178
BANFF 19, SK nue flux, 0.8 - 1.5 GeV	0.039 056 6	262.454	−0.002 205 89	262.458	−0.000 735 3	262.462	0.000 735 3	262.466	0.002 205 89
BANFF 20, SK nue flux, 1.5 - 2.5 GeV	0.040 055 2	262.453	−0.002 770 52	262.458	−0.000 923 51	262.462	0.000 923 51	262.467	0.002 770 52
BANFF 21, SK nue flux, 2.5 - 4.0 GeV	0.041 468 5	262.455	−0.001 941 69	262.458	−0.000 647 23	262.462	0.000 647 23	262.465	0.001 941 69
BANFF 22, SK nue flux, 4.0 - 30.0 GeV	0.058 681 4	262.457	−0.001 271 22	262.459	−0.000 423 74	262.461	0.000 423 74	262.463	0.001 271 22
BANFF 23, SK nuebar flux, 0.0 - 2.5 GeV	0.053 829 3	262.459	−0.000 493 865	262.46	−0.000 164 62	262.46	0.000 164 62	262.461	0.000 493 865
BANFF 24, SK nuebar flux, 2.5 - 30.0 GeV	0.114 234	262.456	−0.001 434 86	262.459	−0.000 478 29	262.461	0.000 478 29	262.464	0.001 434 86
BANFF; Norm; 2p2h	0.182 443	249.661	−4.876 37	258.194	−1.6255	266.726	1.6255	275.258	4.876 37
BANFF; CA5 RES	0.059 470 4	255.895	−2.501 22	260.15	−0.880 07	264.891	0.9264	270.119	2.918 18
BANFF; Norm; BgRES Isospin 1/2	0.189 77	258.529	−1.497 55	260.792	−0.635 64	264.486	0.7721	269.614	2.725 68
BANFF, Ma QE	0.056 648 3	234.686	−10.5823	253.493	−3.4164	271.097	3.2909	287.323	9.472 98
BANFF, Ma RES	0.045 292 1	256.99	−2.084 16	260.608	−0.705 68	264.33	0.712 44	268.095	2.146 89
BANFF; Fermi Momentum	0.071 668 1	260.153	−0.878 886	263.803	0.511 59	258.576	−1.4798	250.147	−4.691 34
BANFF; Shape; CC Oth	0.186 63	261.304	−0.440 596	262.074	−0.146 87	262.845	0.146 87	263.616	0.440 596
BANFF; Norm, CC Coh	0.274 896	262.097	−0.138 277	262.339	−0.046 092	262.581	0.046 092	262.823	0.138 277
BANFF; Norm, NC Coh	0.297 426	262.46	−0.000 172 198	262.46	-5.7399×10^{-5}	262.46	5.7399×10^{-5}	262.46	0.000 172 198
BANFF; Norm, NC Oth	0.3	260.395	−0.786 683	261.772	−0.262 23	263.148	0.262 23	264.525	0.786 683
BANFF; Norm, ν_e To ν_μ	0.028 284 3	262.453	−0.002 545 52	262.458	−0.000 848 51	262.462	0.000 848 51	262.467	0.002 545 52
BANFF; Norm, $\bar{\nu}_e$ To $\bar{\nu}_\mu$	0.028 284 3	262.46	-2.093×10^{-5}	262.46	-6.9767×10^{-6}	262.46	6.9767×10^{-6}	262.46	2.093×10^{-5}
BANFF; Norm; 2p2hBar	0.201 069	261.848	−0.233 015	262.237	−0.084 844	262.683	0.084 844	263.128	0.254 533
BANFF; BeRPA A	0.055 826 7	244.308	−6.916 12	256.409	−2.3054	268.511	2.3054	280.612	6.916 13
BANFF; BeRPA B	0.117 434	244.85	−6.709 68	256.59	−2.2366	268.33	2.2365	280.07	6.709 66
BANFF; BeRPA D	0.127 806	241.706	−7.907 32	255.542	−2.6358	269.378	2.6358	283.214	7.907 35
BANFF; BeRPA E	0.354 515	261.309	−0.438 695	261.985	−0.180 78	262.939	0.182 51	263.907	0.551 209
BANFF; Shape; 2p2h	0.338 124	264.48	0.769 509	263.47	0.384 76	262.061	−0.151 97	264.505	0.779 109
BANFF; Norm; 2p2h C to O	0.161 427	244.958	−6.668 47	256.626	−2.2228	268.294	2.2228	279.962	6.668 47
SKDet + FSI/SI 0; E_{reco}(0.00 - 0.40)GeV; $\nu_\mu/\bar{\nu}_\mu$ CCQE ($1R_\mu$)	0.008 785 93	261.76	−0.266 69	262.227	−0.088 897	262.693	0.088 897	263.16	0.266 69
SKDet + FSI/SI 1; E_{reco}(0.40 - 1.10)GeV; $\nu_\mu/\bar{\nu}_\mu$ CCQE ($1R_\mu$)	0.007 403 28	260.255	−0.840 095	261.725	−0.280 03	263.195	0.280 03	264.665	0.840 095
SKDet + FSI/SI 2; E_{reco}(1.10 - 30.00)GeV; $\nu_\mu/\bar{\nu}_\mu$ CCQE ($1R_\mu$)	0.007 204 4	260.47	−0.758 196	261.797	−0.252 73	263.123	0.252 73	264.45	0.758 196
SKDet + FSI/SI 3; E_{reco}(0.00 - 30.00)GeV; $\nu_\mu/\bar{\nu}_\mu$ CCnQE ($1R_\mu$)	0.169 035	243.977	−7.042 31	256.299	−2.3474	268.621	2.3474	280.943	7.042 31
SKDet + FSI/SI 4; E_{reco}(0.00 - 30.00)GeV; $\nu_e/\bar{\nu}_e$/sigν_e CC ($1R_\mu$)	1.005 48	262.22	−0.091 235 1	262.38	−0.030 412	262.54	0.030 412	262.699	0.091 235 1
SKDet + FSI/SI 5; E_{reco}(0.00 - 30.00)GeV; all NC ($1R_\mu$)	0.658 101	246.61	−6.039 01	257.177	−2.013	267.743	2.013	278.31	6.039 01
SK energy scale	0.024	262.46	$-4.331\,59 \times 10^{-14}$	262.46	-2.1658×10^{-14}	262.46	-4.3316×10^{-14}	262.46	$-2.165\,79 \times 10^{-14}$

Table 38: $\pm1\sigma$ and $\pm3\sigma$ effect of the systematic parameters on the total number of μ-like Super-K events with Run 1-8 POT for neutrino mode post-BANFF Asimov data set A. Only the systematic parameters which affect this sample are shown.

Parameter	1 σ value	BANFF tuned -3σ		BANFF tuned -1σ		BANFF tuned +1σ		BANFF tuned +3σ	
All syst		30.5141	−50.7005(%)	49.2477	−20.434 (%)	77.269	24.839(%)	121.097	95.6478(%)
BANFF 0, SK numu flux, 0.0 - 0.7 GeV; RHC	0.065 643 2	61.5824	−0.505 456	61.791	−0.168 49	61.9995	0.168 49	62.2081	0.505 456
BANFF 1, SK numu flux, 0.7 - 1.0 GeV; RHC	0.048 611 9	61.5733	−0.520 155	61.7879	−0.173 38	62.0026	0.173 38	62.2172	0.520 155
BANFF 2, SK numu flux, 1.0 - 1.5 GeV; RHC	0.045 404 8	61.0082	−1.4332	61.5996	−0.477 73	62.1909	0.477 73	62.7823	1.4332
BANFF 3, SK numu flux, 1.5 - 2.5 GeV; RHC	0.048 620 6	60.7269	−1.887 59	61.5058	−0.6292	62.2847	0.6292	63.0636	1.887 59
BANFF 4, SK numu flux, 2.5 - 30.0 GeV; RHC	0.043 062 7	61.1628	−1.183 34	61.6511	−0.394 45	62.1394	0.394 45	62.6277	1.183 34
BANFF 5, SK numubar flux, 0.0 - 0.4 GeV; RHC	0.064 779 5	60.8622	−1.669 06	61.5509	−0.556 35	62.2396	0.556 35	62.9283	1.669 06
BANFF 6, SK numubar flux, 0.4 - 0.5 GeV; RHC	0.052 486 4	61.34	−0.897 129	61.7102	−0.299 04	62.0803	0.299 04	62.4505	0.897 129
BANFF 7, SK numubar flux, 0.5 - 0.6 GeV; RHC	0.044 828 8	61.7886	−0.172 346	61.8597	−0.057 449	61.9308	0.057 449	62.0019	0.172 346
BANFF 8, SK numubar flux, 0.6 - 0.7 GeV; RHC	0.040 954 3	61.8144	−0.130 581	61.8683	−0.043 527	61.9222	0.043 527	61.9761	0.130 581
BANFF 9, SK numubar flux, 0.7 - 1.0 GeV; RHC	0.052 165 1	60.8576	−1.676 54	61.5494	−0.558 85	62.2411	0.558 85	62.9329	1.676 54
BANFF 10, SK numubar flux, 1.0 - 1.5 GeV; RHC	0.047 032 3	60.8991	−1.609 36	61.5632	−0.536 45	62.2273	0.536 45	62.8914	1.609 36
BANFF 11, SK numubar flux, 1.5 - 2.5 GeV; RHC	0.042 404 3	61.0423	−1.377 99	61.6109	−0.459 33	62.1796	0.459 33	62.7482	1.377 99
BANFF 12, SK numubar flux, 2.5 - 3.5 GeV; RHC	0.046 242 6	61.4714	−0.684 739	61.754	−0.228 25	62.0365	0.228 25	62.3191	0.684 739
BANFF 13, SK numubar flux, 3.5 - 5.0 GeV; RHC	0.062 880 7	61.4938	−0.648 57	61.7614	−0.216 19	62.0291	0.216 19	62.2967	0.648 57
BANFF 14, SK numubar flux, 5.0 - 7.0 GeV; RHC	0.055 519 7	61.8265	−0.111 012	61.8723	−0.037 004	61.9182	0.037 004	61.964	0.111 012
BANFF 15, SK numubar flux, 7.0 - 30.0 GeV; RHC	0.092 647 5	61.8878	−0.011 987 5	61.8928	−0.003 995 8	61.8977	0.003 995 8	61.9027	0.011 987 5
BANFF 16, SK nue flux, 0.0 - 2.5 GeV; RHC	0.046 545 2	61.893	−0.003 660 41	61.8945	−0.001 220 1	61.896	0.001 220 1	61.8975	0.003 660 41
BANFF 17, SK nue flux, 2.5 - 30.0 GeV; RHC	0.064 639	61.8919	−0.005 333 91	61.8941	−0.001 778	61.8963	0.001 778	61.8986	0.005 333 91
BANFF 18, SK nuebar flux, 0.0 - 0.5 GeV; RHC	0.051 375 5	61.8952	-9.4396×10^{-5}	61.8952	-3.1465×10^{-5}	61.8953	3.1465×10^{-5}	61.8953	9.4396×10^{-5}
BANFF 19, SK nuebar flux, 0.5 - 0.7 GeV; RHC	0.043 268	61.8951	−0.000 240 04	61.8952	-8.0013×10^{-5}	61.8953	8.0013×10^{-5}	61.8954	0.000 240 04
BANFF 20, SK nuebar flux, 0.7 - 0.8 GeV; RHC	0.043 836 5	61.8952	−0.000 123 246	61.8952	-4.1082×10^{-5}	61.8953	4.1082×10^{-5}	61.8953	0.000 123 246
BANFF 21, SK nuebar flux, 0.8 - 1.5 GeV; RHC	0.039 880 6	61.8946	−0.000 993 981	61.895	−0.000 331 33	61.8955	0.000 331 33	61.8959	0.000 993 981
BANFF 22, SK nuebar flux, 1.5 - 2.5 GeV; RHC	0.051 444 5	61.8944	−0.001 328 37	61.895	−0.000 442 79	61.8955	0.000 442 79	61.8961	0.001 328 37
BANFF 23, SK nuebar flux, 2.5 - 4.0 GeV; RHC	0.064 480 7	61.894	−0.002 020 42	61.8948	−0.000 673 47	61.8957	0.000 673 47	61.8965	0.002 020 42
BANFF 24, SK nuebar flux, 4.0 - 30.0 GeV; RHC	0.132 852	61.894	−0.001 945 78	61.8948	−0.000 648 59	61.8957	0.000 648 59	61.8965	0.001 945 78
BANFF; Norm; 2p2h	0.182 443	60.2874	−2.597 71	61.3593	−0.8659	62.4312	0.8659	63.5031	2.597 71
BANFF; CA5 RES	0.059 470 4	60.1443	−2.828 81	61.2586	−1.0287	62.585	1.1144	64.1236	3.600 24
BANFF; Norm; BgRES Isospin 1/2	0.189 77	60.7952	−1.777 35	61.4283	−0.754 41	62.4624	0.916 36	63.8975	3.234 95
BANFF, Ma QE	0.056 648 3	57.5502	−7.019 95	60.4125	−2.3956	63.3914	2.4172	66.3733	7.234 88
BANFF, Ma RES	0.045 292 1	60.824	−1.730 71	61.5157	−0.613 17	62.2944	0.6448	63.1423	2.014 83
BANFF; Fermi Momentum	0.071 668 1	61.3847	−0.824 911	62.1854	0.468 83	61.0781	−1.3202	59.4091	−4.016 69
BANFF; Shape; CC Oth	0.186 63	61.5816	−0.506 818	61.7907	−0.168 94	61.9998	0.168 94	62.2089	0.506 818
BANFF; Norm, CC Coh	0.274 896	61.683	−0.342 957	61.8245	−0.114 32	61.966	0.114 32	62.1075	0.342 957
BANFF; Norm, NC Coh	0.297 426	61.8933	−0.003 141 37	61.8946	−0.001 047 1	61.8959	0.001 047 1	61.8972	0.003 141 37
BANFF; Norm, NC Oth	0.3	61.4044	−0.792 965	61.7316	−0.264 32	62.0589	0.264 32	62.3861	0.792 965
BANFF; Norm, ν_e To ν_μ	0.028 284 3	61.895	−0.000 474 143	61.8952	−0.000 158 05	61.8953	0.000 158 05	61.8955	0.000 474 143
BANFF; Norm, $\bar{\nu}_e$ To $\bar{\nu}_\mu$	0.028 284 3	61.895	−0.000 475 571	61.8952	−0.000 158 52	61.8953	0.000 158 52	61.8955	0.000 475 571
BANFF; Norm; 2p2hBar	0.201 069	60.6651	−1.987 46	61.4473	−0.723 67	62.3432	0.723 67	63.239	2.171
BANFF; BeRPA A	0.055 826 7	58.3023	−5.804 84	60.6976	−1.935	63.0929	1.9349	65.4882	5.804 85
BANFF; BeRPA B	0.117 434	58.8486	−4.922 25	60.8797	−1.6407	62.9108	1.6408	64.9419	4.922 26
BANFF; BeRPA D	0.127 806	58.4343	−5.591 62	60.7416	−1.8639	63.0489	1.8639	65.3562	5.591 61
BANFF; BeRPA E	0.354 515	61.7148	−0.291 494	61.8205	−0.1208	61.9711	0.122 62	62.1253	0.371 737
BANFF; Shape; 2p2h	0.338 124	61.957	0.099 688 3	61.9261	0.049 839	62.0044	0.176 38	62.5317	1.028 21
BANFF; Norm; 2p2h C to O	0.161 427	59.1021	−4.512 71	60.9642	−1.5042	62.8263	1.5042	64.6884	4.512 71
SKDet + FSI/SI 0; E_{reco}(0.00 - 0.40)GeV; $\nu_\mu/\bar{\nu}_\mu$ CCQE ($1R_\mu$); RHC	0.008 512 49	61.7509	−0.233 153	61.8471	−0.077 718	61.9434	0.077 718	62.0396	0.233 153
SKDet + FSI/SI 1; E_{reco}(0.40 - 1.10)GeV; $\nu_\mu/\bar{\nu}_\mu$ CCQE ($1R_\mu$); RHC	0.007 050 99	61.5329	−0.585 454	61.7745	−0.195 15	62.016	0.195 15	62.2576	0.585 454
SKDet + FSI/SI 2; E_{reco}(1.10 - 30.00)GeV; $\nu_\mu/\bar{\nu}_\mu$ CCQE ($1R_\mu$); RHC	0.006 784	61.3387	−0.899 18	61.7097	−0.299 73	62.0808	0.299 73	62.4518	0.899 18
SKDet + FSI/SI 3; E_{reco}(0.00 - 30.00)GeV; $\nu_\mu/\bar{\nu}_\mu$ CCnQE ($1R_\mu$); RHC	0.129 646	57.9032	−6.449 68	60.5646	−2.1499	63.2259	2.1499	65.8873	6.449 68
SKDet + FSI/SI 4; E_{reco}(0.00 - 30.00)GeV; $\nu_e/\bar{\nu}_e$/sigν_e CC ($1R_\mu$); RHC	1.004 54	61.8744	−0.033 729 9	61.8883	−0.011 243	61.9022	0.011 243	61.9161	0.033 729 9
SKDet + FSI/SI 5; E_{reco}(0.00 - 30.00)GeV; all NC ($1R_\mu$); RHC	0.657 251	58.9447	−4.766 96	60.9117	−1.589	62.8788	1.589	64.8458	4.766 96
SK energy scale	0.024	61.8952	$1.147\,98 \times 10^{-14}$	61.8952	-5.7399×10^{-14}	61.8952	0	61.8952	-4.5919×10^{-14}

Table 39: $\pm1\sigma$ and $\pm3\sigma$ effect of the systematic parameters on the total number of μ-like Super-K events with Run 1-8 POT for antineutrino mode post-BANFF Asimov data set A. Only the systematic parameters which affect this sample are shown.

Parameter	1 σ value	BANFF tuned -3σ		BANFF tuned -1σ		BANFF tuned +1σ		BANFF tuned +3σ	
All syst		24.922	−65.9618(%)	51.4515	−29.728 (%)	102.19	39.569(%)	190.164	159.724(%)
BANFF 0, SK numu flux, 0.0 - 0.4 GeV	0.057 496 7	72.9223	−0.403 375	73.1192	−0.134 46	73.3161	0.134 46	73.513	0.403 375
BANFF 1, SK numu flux, 0.4 - 0.5 GeV	0.051 936 3	72.1503	−1.457 78	72.8619	−0.485 93	73.5734	0.485 93	74.285	1.457 78
BANFF 2, SK numu flux, 0.5 - 0.6 GeV	0.044 568 2	71.1561	−2.815 69	72.5305	−0.938 56	73.9049	0.938 56	75.2792	2.815 69
BANFF 3, SK numu flux, 0.6 - 0.7 GeV	0.041 886 1	70.9918	−3.040 09	72.4757	−1.0134	73.9596	1.0134	75.4435	3.040 09
BANFF 4, SK numu flux, 0.7 - 1.0 GeV	0.054 16	70.287	−4.002 65	72.2408	−1.3342	74.1945	1.3342	76.1483	4.002 65
BANFF 5, SK numu flux, 1.0 - 1.5 GeV	0.049 026 1	72.8314	−0.527 495	73.0889	−0.175 83	73.3464	0.175 83	73.6039	0.527 495
BANFF 6, SK numu flux, 1.5 - 2.5 GeV	0.041 778 9	73.1531	−0.088 139 6	73.1962	−0.029 38	73.2392	0.029 38	73.2822	0.088 139 6
BANFF 7, SK numu flux, 2.5 - 3.5 GeV	0.043 606 9	73.1782	−0.053 842 2	73.2045	−0.017 947	73.2308	0.017 947	73.2571	0.053 842 2
BANFF 8, SK numu flux, 3.5 - 5.0 GeV	0.040 772	73.1759	−0.056 987 9	73.2038	−0.018 996	73.2316	0.018 996	73.2594	0.056 987 9
BANFF 9, SK numu flux, 5.0 - 7.0 GeV	0.040 095 8	73.2002	−0.023 894 1	73.2118	−0.007 964 7	73.2235	0.007 964 7	73.2352	0.023 894 1
BANFF 10, SK numu flux, 7.0 - 30.0 GeV	0.048 214	73.208	−0.013 257 5	73.2144	−0.004 419 2	73.2209	0.004 419 2	73.2274	0.013 257 5
BANFF 11, SK numubar flux, 0.0 - 0.7 GeV	0.073 717	73.1847	−0.044 983 2	73.2067	−0.014 994	73.2286	0.014 994	73.2506	0.044 983 2
BANFF 12, SK numubar flux, 0.7 - 1.0 GeV	0.0477	73.1928	−0.033 892 6	73.2094	−0.011 298	73.2259	0.011 298	73.2425	0.033 892 6
BANFF 13, SK numubar flux, 1.0 - 1.5 GeV	0.057 472 1	73.1952	−0.030 620 8	73.2102	−0.010 207	73.2251	0.010 207	73.2401	0.030 620 8
BANFF 14, SK numubar flux, 1.5 - 2.5 GeV	0.061 890 7	73.2042	−0.018 418 1	73.2132	−0.006 139 4	73.2222	0.006 139 4	73.2311	0.018 418 1
BANFF 15, SK numubar flux, 2.5 - 30.0 GeV	0.064 676 3	73.2046	−0.017 792 4	73.2133	−0.005 930 8	73.222	0.005 930 8	73.2307	0.017 792 4
BANFF 16, SK nue flux, 0.0 - 0.5 GeV	0.046 471 9	72.9814	−0.322 666	73.1389	−0.107 56	73.2964	0.107 56	73.4539	0.322 666
BANFF 17, SK nue flux, 0.5 - 0.7 GeV	0.042 400 3	72.9412	−0.377 611	73.1255	−0.125 87	73.3098	0.125 87	73.4941	0.377 611
BANFF 18, SK nue flux, 0.7 - 0.8 GeV	0.041 175	73.0844	−0.182 013	73.1732	−0.060 671	73.2621	0.060 671	73.3509	0.182 013
BANFF 19, SK nue flux, 0.8 - 1.5 GeV	0.039 056 6	72.8167	−0.547 63	73.084	−0.182 54	73.3513	0.182 54	73.6186	0.547 63
BANFF 20, SK nue flux, 1.5 - 2.5 GeV	0.040 055 2	73.1876	−0.041 083	73.2076	−0.013 694	73.2277	0.013 694	73.2477	0.041 083
BANFF 21, SK nue flux, 2.5 - 4.0 GeV	0.041 468 5	73.2139	−0.005 123 28	73.2164	−0.001 707 8	73.2189	0.001 707 8	73.2214	0.005 123 28
BANFF 22, SK nue flux, 4.0 - 30.0 GeV	0.058 681 4	73.216	−0.002 314 32	73.2171	−0.000 771 44	73.2182	0.000 771 44	73.2194	0.002 314 32
BANFF 23, SK nuebar flux, 0.0 - 2.5 GeV	0.053 829 3	73.1626	−0.075 266 1	73.1993	−0.025 089	73.236	0.025 089	73.2728	0.075 266 1
BANFF 24, SK nuebar flux, 2.5 - 30.0 GeV	0.114 234	73.2155	−0.003 008 59	73.2169	−0.001 002 9	73.2184	0.001 002 9	73.2199	0.003 008 59
BANFF; Norm; 2p2h	0.182 443	69.5006	−5.076 71	71.9786	−1.6922	74.4567	1.6922	76.9347	5.076 71
BANFF; CA5 RES	0.059 470 4	71.8118	−1.920 13	72.7247	−0.673 29	73.735	0.706 54	74.8426	2.219 38
BANFF; Norm; BgRES Isospin 1/2	0.189 77	72.627	−0.806 784	72.9669	−0.342 44	73.5222	0.415 96	74.2928	1.468 42
BANFF, Ma QE	0.056 648 3	64.696	−11.6388	70.5376	−3.6604	75.731	3.4327	80.2664	9.6271
BANFF, Ma RES	0.045 292 1	72.1583	−1.4469	72.8705	−0.474 12	73.5578	0.464 58	74.2156	1.362 97
BANFF; Fermi Momentum	0.071 668 1	72.6124	−0.826 631	73.5743	0.487 05	72.1652	−1.4375	69.8285	−4.628 86
BANFF; Shape; CC Oth	0.186 63	73.137	−0.110 225	73.1908	−0.036 741	73.2446	0.036 741	73.2984	0.110 222
BANFF; Norm, CC Coh	0.274 896	73.1696	−0.065 688	73.2016	−0.021 896	73.2337	0.021 896	73.2658	0.065 688
BANFF; Norm, NC Coh	0.297 426	72.6524	−0.772 032	73.0292	−0.257 34	73.4061	0.257 34	73.7829	0.772 032
BANFF; Norm, NC Oth	0.3	72.8927	−0.443 771	73.1094	−0.147 92	73.326	0.147 92	73.5426	0.443 771
BANFF; Norm, ν_e To ν_μ	0.028 284 3	67.5071	−7.799 41	71.3141	−2.5998	75.1212	2.5998	78.9282	7.799 41
BANFF; Norm; NC 1 γ	1	71.1672	−2.800 47	71.1672	−2.8005	75.2681	2.8005	79.369	8.401 41
BANFF; Norm, $\bar{\nu}_e$ To $\bar{\nu}_\mu$	0.028 284 3	73.1582	−0.081 196 6	73.1978	−0.027 066	73.2375	0.027 066	73.2771	0.081 196 6
BANFF; Norm; 2p2hBar	0.201 069	73.1818	−0.048 996 4	73.2046	−0.017 84	73.2307	0.017 84	73.2568	0.053 521 1
BANFF; BeRPA A	0.055 826 7	68.1306	−6.9479	71.522	−2.316	74.9134	2.316	78.3048	6.9479
BANFF; BeRPA B	0.117 434	67.4635	−7.858 98	71.2996	−2.6197	75.1357	2.6197	78.9718	7.858 98
BANFF; BeRPA D	0.127 806	67.8054	−7.392 04	71.4136	−2.464	75.0218	2.464	78.6299	7.392 04
BANFF; BeRPA E	0.354 515	72.8409	−0.514 573	73.064	−0.209 88	73.3713	0.209 88	73.6787	0.629 665
BANFF; Shape; 2p2h	0.338 124	72.6822	−0.731 326	72.9499	−0.365 66	73.9761	1.0359	75.739	3.443 59
BANFF; Norm; 2p2h C to O	0.161 427	68.2076	−6.842 65	71.5477	−2.2809	74.8877	2.2809	78.2277	6.842 65
SKDet + FSI/SI 6; E_{reco}(0.00 - 0.35)GeV; Osc. ν_e CC ($1R_e$)	0.181 271	68.7493	−6.1029	71.7282	−2.0343	74.7071	2.0343	77.6861	6.1029
SKDet + FSI/SI 7; E_{reco}(0.35 - 0.80)GeV; Osc. ν_e CC ($1R_e$)	0.036 083	68.6468	−6.242 86	71.694	−2.081	74.7413	2.081	77.7885	6.242 86
SKDet + FSI/SI 8; E_{reco}(0.80 - 1.25)GeV; Osc. ν_e CC ($1R_e$)	0.041 663 6	72.1642	−1.438 86	72.8665	−0.479 62	73.5688	0.479 62	74.2712	1.438 86
SKDet + FSI/SI 9; E_{reco}(0.00 - 0.35)GeV; $\nu_\mu/\bar{\nu}_\mu$ CC ($1R_e$)	0.300 431	73.1543	−0.086 531 7	73.1965	−0.028 844	73.2388	0.028 844	73.281	0.086 531 7
SKDet + FSI/SI 10; E_{reco}(0.35 - 0.80)GeV; $\nu_\mu/\bar{\nu}_\mu$ CC ($1R_e$)	0.321 346	73.0587	−0.217 089	73.1647	−0.072 363	73.2706	0.072 363	73.3766	0.217 089
SKDet + FSI/SI 11; E_{reco}(0.80 - 1.25)GeV; $\nu_\mu/\bar{\nu}_\mu$ CC ($1R_e$)	0.392 969	73.1904	−0.037 232 2	73.2086	−0.012 411	73.2267	0.012 411	73.2449	0.037 232 2
SKDet + FSI/SI 12; E_{reco}(0.00 - 0.35)GeV; $\nu_e/\bar{\nu}_e$ CC ($1R_e$)	0.135 399	72.7003	−0.706 59	73.0452	−0.235 53	73.3901	0.235 53	73.735	0.706 59
SKDet + FSI/SI 13; E_{reco}(0.35 - 0.80)GeV; $\nu_e/\bar{\nu}_e$ CC ($1R_e$)	0.069 892 8	72.1921	−1.400 77	72.8758	−0.466 92	73.5595	0.466 92	74.2433	1.400 77
SKDet + FSI/SI 14; E_{reco}(0.80 - 1.25)GeV; $\nu_e/\bar{\nu}_e$ CC ($1R_e$)	0.077 136 1	72.5317	−0.936 824	72.989	−0.312 27	73.4463	0.312 27	73.9036	0.936 824
SKDet + FSI/SI 15; E_{reco}(0.00 - 0.35)GeV; all NC ($1R_e$)	0.194 453	72.0874	−1.543 71	72.8409	−0.514 57	73.5944	0.514 57	74.3479	1.543 71
SKDet + FSI/SI 16; E_{reco}(0.35 - 0.80)GeV; all NC ($1R_e$)	0.181 015	71.849	−1.869 26	72.7615	−0.623 09	73.6739	0.623 09	74.5863	1.869 26
SKDet + FSI/SI 17; E_{reco}(0.80 - 1.25)GeV; all NC ($1R_e$)	0.472 078	72.5075	−0.969 968	72.9809	−0.323 32	73.4544	0.323 32	73.9279	0.969 968
SK energy scale	0.024	73.2177	$1.940\,91 \times 10^{-14}$	73.2177	1.9409×10^{-14}	73.2177	1.9409×10^{-14}	73.2177	0

Table 40: $\pm1\sigma$ and $\pm3\sigma$ effect of the systematic parameters on the total number of e-like Super-K events with Run 1-8 POT for neutrino mode post-BANFF Asimov data set A. Only the systematic parameters which affect this sample are shown.

Parameter	1 σ value	BANFF tuned -3σ		BANFF tuned -1σ		BANFF tuned +1σ		BANFF tuned +3σ	
All syst		2.963 71	−62.2766(%)	5.462 14	−30.475 (%)	11.115	41.472(%)	21.274	170.785(%)
BANFF 0, SK numu flux, 0.0 - 0.7 GeV; RHC	0.065 643 2	7.723 62	−1.690 35	7.812 16	−0.563 45	7.900 69	0.563 45	7.989 22	1.690 35
BANFF 1, SK numu flux, 0.7 - 1.0 GeV; RHC	0.048 611 9	7.776 33	−1.019 45	7.829 73	−0.339 82	7.883 12	0.339 82	7.936 52	1.019 45
BANFF 2, SK numu flux, 1.0 - 1.5 GeV; RHC	0.045 404 8	7.812 29	−0.561 803	7.841 71	−0.187 27	7.871 14	0.187 27	7.900 56	0.561 803
BANFF 3, SK numu flux, 1.5 - 2.5 GeV; RHC	0.048 620 6	7.837 35	−0.242 765	7.850 07	−0.080 922	7.862 78	0.080 922	7.8755	0.242 765
BANFF 4, SK numu flux, 2.5 - 30.0 GeV; RHC	0.043 062 7	7.838 32	−0.230 369	7.850 39	−0.076 79	7.862 46	0.076 79	7.874 52	0.230 369
BANFF 5, SK numubar flux, 0.0 - 0.4 GeV; RHC	0.064 779 5	7.84	−0.209 058	7.850 95	−0.069 686	7.8619	0.069 686	7.872 85	0.209 058
BANFF 6, SK numubar flux, 0.4 - 0.5 GeV; RHC	0.052 486 4	7.807 95	−0.616 928	7.840 27	−0.205 64	7.872 58	0.205 64	7.904 89	0.616 928
BANFF 7, SK numubar flux, 0.5 - 0.6 GeV; RHC	0.044 828 8	7.746 07	−1.404 56	7.819 64	−0.468 19	7.893 21	0.468 19	7.966 77	1.404 56
BANFF 8, SK numubar flux, 0.6 - 0.7 GeV; RHC	0.040 954 3	7.720 03	−1.736 07	7.810 96	−0.578 69	7.901 89	0.578 69	7.992 82	1.736 07
BANFF 9, SK numubar flux, 0.7 - 1.0 GeV; RHC	0.052 165 1	7.640 76	−2.745	7.784 54	−0.915	7.928 31	0.915	8.072 08	2.745
BANFF 10, SK numubar flux, 1.0 - 1.5 GeV; RHC	0.047 032 3	7.816 42	−0.509 156	7.843 09	−0.169 72	7.869 76	0.169 72	7.896 42	0.509 156
BANFF 11, SK numubar flux, 1.5 - 2.5 GeV; RHC	0.042 404 3	7.844 95	−0.146 092	7.8526	−0.048 697	7.860 25	0.048 697	7.8679	0.146 092
BANFF 12, SK numubar flux, 2.5 - 3.5 GeV; RHC	0.046 242 6	7.850 33	−0.077 579 9	7.854 39	−0.025 86	7.858 45	0.025 86	7.862 52	0.077 579 9
BANFF 13, SK numubar flux, 3.5 - 5.0 GeV; RHC	0.062 880 7	7.849 05	−0.093 860 5	7.853 96	−0.031 287	7.858 88	0.031 287	7.8638	0.093 860 5
BANFF 14, SK numubar flux, 5.0 - 7.0 GeV; RHC	0.055 519 7	7.854 31	−0.026 909 5	7.855 72	−0.008 969 8	7.857 13	0.008 969 8	7.858 54	0.026 909 5
BANFF 15, SK numubar flux, 7.0 - 30.0 GeV; RHC	0.092 647 5	7.8547	−0.021 895 7	7.855 85	−0.007 298 6	7.857	0.007 298 6	7.858 14	0.021 895 7
BANFF 16, SK nue flux, 0.0 - 2.5 GeV; RHC	0.046 545 2	7.754 35	−1.299 28	7.8224	−0.433 09	7.890 45	0.433 09	7.9585	1.299 28
BANFF 17, SK nue flux, 2.5 - 30.0 GeV; RHC	0.064 639	7.854 84	−0.020 132 8	7.8559	−0.006 710 9	7.856 95	0.006 710 9	7.858	0.020 132 8
BANFF 18, SK nuebar flux, 0.0 - 0.5 GeV; RHC	0.051 375 5	7.824 84	−0.401 976	7.8459	−0.133 99	7.866 95	0.133 99	7.888	0.401 976
BANFF 19, SK nuebar flux, 0.5 - 0.7 GeV; RHC	0.043 268	7.8255	−0.393 588	7.846 12	−0.1312	7.866 73	0.1312	7.887 34	0.393 588
BANFF 20, SK nuebar flux, 0.7 - 0.8 GeV; RHC	0.043 836 5	7.839 41	−0.216 534	7.850 75	−0.072 178	7.862 09	0.072 178	7.873 43	0.216 534
BANFF 21, SK nuebar flux, 0.8 - 1.5 GeV; RHC	0.039 880 6	7.802 53	−0.685 932	7.838 46	−0.228 64	7.874 39	0.228 64	7.910 31	0.685 932
BANFF 22, SK nuebar flux, 1.5 - 2.5 GeV; RHC	0.051 444 5	7.852 38	−0.051 433 9	7.855 08	−0.017 145	7.857 77	0.017 145	7.860 46	0.051 433 9
BANFF 23, SK nuebar flux, 2.5 - 4.0 GeV; RHC	0.064 480 7	7.855 54	−0.011 252 1	7.856 13	−0.003 750 7	7.856 72	0.003 750 7	7.857 31	0.011 252 1
BANFF 24, SK nuebar flux, 4.0 - 30.0 GeV; RHC	0.132 852	7.855 62	−0.010 160 3	7.856 16	−0.003 386 8	7.856 69	0.003 386 8	7.857 22	0.010 160 3
BANFF; Norm; 2p2h	0.182 443	7.720 15	−1.7346	7.811	−0.5782	7.901 85	0.5782	7.9927	1.7346
BANFF; CA5 RES	0.059 470 4	7.633 37	−2.8391	7.770 72	−1.0909	7.953 48	1.2354	8.181 66	4.139 76
BANFF; Norm; BgRES Isospin 1/2	0.189 77	7.740 41	−1.476 65	7.807 18	−0.626 77	7.916 24	0.761 32	8.067 57	2.687 63
BANFF, Ma QE	0.056 648 3	7.380 72	−6.054 95	7.693 53	−2.0734	8.019 55	2.0764	8.338 05	6.130 37
BANFF, Ma RES	0.045 292 1	7.748 05	−1.379 41	7.8167	−0.505 59	7.899 09	0.543 13	7.991 45	1.718 63
BANFF; Fermi Momentum	0.071 668 1	7.7857	−0.900 157	7.8963	0.507 53	7.743 12	−1.4422	7.506 25	−4.457 13
BANFF; Shape; CC Oth	0.186 63	7.839 38	−0.216 893	7.850 74	−0.072 298	7.8621	0.072 298	7.873 46	0.216 893
BANFF; Norm, CC Coh	0.274 896	7.813 83	−0.542 185	7.842 22	−0.180 73	7.870 62	0.180 73	7.899 02	0.542 185
BANFF; Norm, NC Coh	0.297 426	7.6102	−3.134 05	7.774 35	−1.0447	7.9385	1.0447	8.102 65	3.134 05
BANFF; Norm, NC Oth	0.3	7.777 88	−0.999 766	7.830 24	−0.333 26	7.8826	0.333 26	7.934 97	0.999 766
BANFF; Norm, ν_e To ν_μ	0.028 284 3	7.6748	−2.311 78	7.795 88	−0.770 59	7.916 96	0.770 59	8.038 05	2.311 78
BANFF; Norm; NC 1 γ	1	7.325 03	−6.763 74	7.325 03	−6.7637	8.387 81	6.7637	9.450 59	20.2912
BANFF; Norm, $\bar{\nu}_e$ To $\bar{\nu}_\mu$	0.028 284 3	7.481 99	−4.765 99	7.731 61	−1.5887	7.981 24	1.5887	8.230 86	4.765 99
BANFF; Norm; 2p2hBar	0.201 069	7.662 82	−2.464 32	7.785 93	−0.8973	7.926 92	0.8973	8.067 91	2.6919
BANFF; BeRPA A	0.055 826 7	7.376 28	−6.111 45	7.696 38	−2.0371	8.016 47	2.0371	8.336 56	6.111 44
BANFF; BeRPA B	0.117 434	7.471 56	−4.898 73	7.728 13	−1.6329	7.984 71	1.6329	8.241 29	4.898 73
BANFF; BeRPA D	0.127 806	7.584 43	−3.461 99	7.765 76	−1.154	7.947 09	1.154	8.128 41	3.461 99
BANFF; BeRPA E	0.354 515	7.837 36	−0.242 58	7.848 65	−0.098 949	7.8642	0.098 955	7.879 75	0.296 901
BANFF; Shape; 2p2h	0.338 124	7.810 59	−0.583 371	7.833 51	−0.291 69	7.916 07	0.759 24	8.046	2.413 06
BANFF; Norm; 2p2h C to O	0.161 427	7.573 64	−3.599 42	7.762 16	−1.1998	7.950 68	1.1998	8.139 21	3.599 42
SKDet + FSI/SI 6; E_{reco}(0.00 - 0.35)GeV; Osc. ν_e CC ($1R_e$); RHC	0.108 165	7.719 43	−1.743 77	7.810 76	−0.581 26	7.902 09	0.581 26	7.993 42	1.743 77
SKDet + FSI/SI 7; E_{reco}(0.35 - 0.80)GeV; Osc. ν_e CC ($1R_e$); RHC	0.036 303 9	7.494 43	−4.607 64	7.735 76	−1.5359	7.977 09	1.5359	8.218 42	4.607 64
SKDet + FSI/SI 8; E_{reco}(0.80 - 1.25)GeV; Osc. ν_e CC ($1R_e$); RHC	0.056 731 5	7.687 41	−2.151 25	7.800 09	−0.717 08	7.912 76	0.717 08	8.025 43	2.151 25
SKDet + FSI/SI 9; E_{reco}(0.00 - 0.35)GeV; $\nu_\mu/\bar{\nu}_\mu$ CC ($1R_e$); RHC	0.348 644	7.8459	−0.133 966	7.852 91	−0.044 655	7.859 93	0.044 655	7.866 95	0.133 966
SKDet + FSI/SI 10; E_{reco}(0.35 - 0.80)GeV; $\nu_\mu/\bar{\nu}_\mu$ CC ($1R_e$); RHC	0.341 647	7.8176	−0.494 211	7.843 48	−0.164 74	7.869 37	0.164 74	7.895 25	0.494 211
SKDet + FSI/SI 11; E_{reco}(0.80 - 1.25)GeV; $\nu_\mu/\bar{\nu}_\mu$ CC ($1R_e$); RHC	0.417 164	7.840 41	−0.203 851	7.851 08	−0.067 95	7.861 76	0.067 95	7.872 44	0.203 851
SKDet + FSI/SI 12; E_{reco}(0.00 - 0.35)GeV; $\nu_e/\bar{\nu}_e$ CC ($1R_e$); RHC	0.084 556 3	7.805 85	−0.643 709	7.839 57	−0.214 57	7.873 28	0.214 57	7.907	0.643 709
SKDet + FSI/SI 13; E_{reco}(0.35 - 0.80)GeV; $\nu_e/\bar{\nu}_e$ CC ($1R_e$); RHC	0.056 069 7	7.704 77	−1.930 28	7.805 87	−0.643 43	7.906 97	0.643 43	8.008 07	1.930 28
SKDet + FSI/SI 14; E_{reco}(0.80 - 1.25)GeV; $\nu_e/\bar{\nu}_e$ CC ($1R_e$); RHC	0.078 024 1	7.689 46	−2.125 19	7.800 77	−0.7084	7.912 08	0.7084	8.023 39	2.125 19
SKDet + FSI/SI 15; E_{reco}(0.00 - 0.35)GeV; all NC ($1R_e$); RHC	0.214 066	7.571 17	−3.630 77	7.761 34	−1.2103	7.951 51	1.2103	8.141 67	3.630 77
SKDet + FSI/SI 16; E_{reco}(0.35 - 0.80)GeV; all NC ($1R_e$); RHC	0.190 575	7.487 66	−4.693 75	7.7335	−1.5646	7.979 34	1.5646	8.225 18	4.693 75
SKDet + FSI/SI 17; E_{reco}(0.80 - 1.25)GeV; all NC ($1R_e$); RHC	0.464 86	7.642 65	−2.721	7.785 17	−0.907	7.927 68	0.907	8.0702	2.721
SK energy scale	0.024	7.856 42	$2.261\,02 \times 10^{-14}$	7.856 42	2.261×10^{-14}	7.856 42	3.3915×10^{-14}	7.856 42	0

Table 41: $\pm1\sigma$ and $\pm3\sigma$ effect of the systematic parameters on the total number of e-like Super-K events with Run 1-8 POT for antineutrino mode post-BANFF Asimov data set A. Only the systematic parameters which affect this sample are shown.

Parameter	1 σ value	BANFF tuned -3σ		BANFF tuned -1σ		BANFF tuned +1σ		BANFF tuned +3σ	
All syst		0.804 711	−88.4314(%)	4.008 33	−42.376 (%)	11.207	61.106(%)	25.0389	259.961(%)
BANFF 0, SK numu flux, 0.0 - 0.4 GeV	0.057 496 7	6.954 24	−0.025 206 3	6.955 41	−0.008 402 1	6.956 58	0.008 402 1	6.957 75	0.025 206 3
BANFF 1, SK numu flux, 0.4 - 0.5 GeV	0.051 936 3	6.910 04	−0.660 661	6.940 68	−0.220 22	6.971 31	0.220 22	7.001 95	0.660 661
BANFF 2, SK numu flux, 0.5 - 0.6 GeV	0.044 568 2	6.798 99	−2.257 11	6.903 66	−0.752 37	7.008 33	0.752 37	7.113	2.257 11
BANFF 3, SK numu flux, 0.6 - 0.7 GeV	0.041 886 1	6.727 93	−3.278 66	6.879 97	−1.0929	7.032 01	1.0929	7.184 06	3.278 66
BANFF 4, SK numu flux, 0.7 - 1.0 GeV	0.054 16	6.610 26	−4.970 27	6.840 75	−1.6568	7.071 24	1.6568	7.301 73	4.970 27
BANFF 5, SK numu flux, 1.0 - 1.5 GeV	0.049 026 1	6.906 32	−0.714 124	6.939 44	−0.238 04	6.972 55	0.238 04	7.005 67	0.714 124
BANFF 6, SK numu flux, 1.5 - 2.5 GeV	0.041 778 9	6.9436	−0.178 111	6.951 86	−0.059 37	6.960 12	0.059 37	6.968 38	0.178 111
BANFF 7, SK numu flux, 2.5 - 3.5 GeV	0.043 606 9	6.948 64	−0.105 658	6.953 54	−0.035 219	6.958 44	0.035 219	6.963 34	0.105 658
BANFF 8, SK numu flux, 3.5 - 5.0 GeV	0.040 772	6.944 79	−0.161 07	6.952 26	−0.053 69	6.959 73	0.053 69	6.9672	0.161 07
BANFF 9, SK numu flux, 5.0 - 7.0 GeV	0.040 095 8	6.951 21	−0.068 745 4	6.9544	−0.022 915	6.957 59	0.022 915	6.960 78	0.068 745 4
BANFF 10, SK numu flux, 7.0 - 30.0 GeV	0.048 214	6.9554	−0.008 540 26	6.9558	−0.002 846 8	6.956 19	0.002 846 8	6.956 59	0.008 540 26
BANFF 11, SK numubar flux, 0.0 - 0.7 GeV	0.073 717	6.955 31	−0.009 815 25	6.955 77	−0.003 271 8	6.956 22	0.003 271 8	6.956 68	0.009 815 25
BANFF 12, SK numubar flux, 0.7 - 1.0 GeV	0.0477	6.955 39	−0.008 743 09	6.955 79	−0.002 914 4	6.9562	0.002 914 4	6.9566	0.008 743 09
BANFF 13, SK numubar flux, 1.0 - 1.5 GeV	0.057 472 1	6.954 97	−0.014 683 7	6.955 65	−0.004 894 6	6.956 33	0.004 894 6	6.957 02	0.014 683 7
BANFF 14, SK numubar flux, 1.5 - 2.5 GeV	0.061 890 7	6.954 56	−0.020 658 8	6.955 51	−0.006 886 3	6.956 47	0.006 886 3	6.957 43	0.020 658 8
BANFF 15, SK numubar flux, 2.5 - 30.0 GeV	0.064 676 3	6.9542	−0.025 784 6	6.9554	−0.008 594 9	6.956 59	0.008 594 9	6.957 79	0.025 784 6
BANFF 16, SK nue flux, 0.0 - 0.5 GeV	0.046 471 9	6.950 48	−0.079 277 3	6.954 16	−0.026 426	6.957 83	0.026 426	6.961 51	0.079 277 3
BANFF 17, SK nue flux, 0.5 - 0.7 GeV	0.042 400 3	6.931 68	−0.349 548	6.947 89	−0.116 52	6.9641	0.116 52	6.980 31	0.349 548
BANFF 18, SK nue flux, 0.7 - 0.8 GeV	0.041 175	6.939 98	−0.230 278	6.950 65	−0.076 759	6.961 33	0.076 759	6.972 01	0.230 278
BANFF 19, SK nue flux, 0.8 - 1.5 GeV	0.039 056 6	6.896 41	−0.856 582	6.936 13	−0.285 53	6.975 86	0.285 53	7.015 58	0.856 582
BANFF 20, SK nue flux, 1.5 - 2.5 GeV	0.040 055 2	6.949 68	−0.090 741 5	6.953 89	−0.030 247	6.9581	0.030 247	6.962 31	0.090 741 5
BANFF 21, SK nue flux, 2.5 - 4.0 GeV	0.041 468 5	6.955 29	−0.010 118 7	6.955 76	−0.003 372 9	6.956 23	0.003 372 9	6.9567	0.010 118 7
BANFF 22, SK nue flux, 4.0 - 30.0 GeV	0.058 681 4	6.955 63	−0.005 302 11	6.955 87	−0.001 767 4	6.956 12	0.001 767 4	6.956 36	0.005 302 11
BANFF 23, SK nuebar flux, 0.0 - 2.5 GeV	0.053 829 3	6.954 56	−0.020 660 1	6.955 51	−0.006 886 7	6.956 47	0.006 886 7	6.957 43	0.020 660 1
BANFF 24, SK nuebar flux, 2.5 - 30.0 GeV	0.114 234	6.955 64	−0.005 042 17	6.955 88	−0.001 680 7	6.956 11	0.001 680 7	6.956 34	0.005 042 17
BANFF; Norm; 2p2h	0.182 443	6.886 67	−0.996 676	6.932 88	−0.332 23	6.9791	0.332 23	7.025 32	0.996 676
BANFF; CA5 RES	0.059 470 4	5.897 54	−15.2164	6.587 37	−5.2994	7.340 43	5.5266	8.156 71	17.2616
BANFF; Norm; BgRES Isospin 1/2	0.189 77	6.570 82	−5.537 26	6.792 51	−2.3503	7.154 58	2.8549	7.657 04	10.0783
BANFF, Ma QE	0.056 648 3	6.9079	−0.691 353	6.940 74	−0.219 33	6.970 39	0.206 94	6.996 54	0.582 87
BANFF, Ma RES	0.045 292 1	6.208 12	−10.7516	6.717 13	−3.4339	7.184 06	3.2788	7.608 06	9.374 09
BANFF; Fermi Momentum	0.071 668 1	6.954 02	−0.028 404 5	6.9574	0.020 253	6.951 51	−0.064 517	6.941 42	−0.209 459
BANFF; Shape; CC Oth	0.186 63	6.850 23	−1.520 53	6.920 74	−0.506 84	6.991 25	0.506 84	7.061 76	1.520 53
BANFF; Norm, CC Coh	0.274 896	6.832 07	−1.781 54	6.914 69	−0.593 85	6.9973	0.593 85	7.079 92	1.781 54
BANFF; Norm, NC Oth	0.3	6.7486	−2.9815	6.886 86	−0.993 83	7.025 12	0.993 83	7.163 39	2.9815
BANFF; Norm, ν_e To ν_μ	0.028 284 3	6.412 77	−7.809 49	6.774 92	−2.6032	7.137 07	2.6032	7.499 22	7.809 49
BANFF; Norm; NC 1 γ	1	6.896 95	−0.848 755	6.896 95	−0.848 76	7.015 03	0.848 76	7.133 11	2.546 27
BANFF; Norm, $\bar{\nu}_e$ To $\bar{\nu}_\mu$	0.028 284 3	6.954 62	−0.019 721 7	6.955 54	−0.006 573 9	6.956 45	0.006 573 9	6.957 37	0.019 721 7
BANFF; Norm; 2p2hBar	0.201 069	6.954 91	−0.015 515	6.9556	−0.005 649 2	6.956 39	0.005 649 2	6.957 17	0.016 947 7
BANFF; BeRPA A	0.055 826 7	6.935 41	−0.295 95	6.949 13	−0.098 65	6.962 86	0.098 65	6.976 58	0.295 95
BANFF; BeRPA B	0.117 434	6.9248	−0.448 374	6.9456	−0.149 46	6.966 39	0.149 46	6.987 18	0.448 374
BANFF; BeRPA D	0.127 806	6.923 23	−0.470 98	6.945 07	−0.156 99	6.966 91	0.156 99	6.988 76	0.470 98
BANFF; BeRPA E	0.354 515	6.953 69	−0.033 073 6	6.955 06	−0.013 489	6.956 93	0.013 489	6.958 81	0.040 466 6
BANFF; Shape; 2p2h	0.338 124	6.909 88	−0.662 967	6.932 94	−0.331 47	6.982 02	0.374 12	7.014 03	0.834 343
BANFF; Norm; 2p2h C to O	0.161 427	6.862 34	−1.346 41	6.924 78	−0.4488	6.987 21	0.4488	7.049 65	1.346 41
SKDet + FSI/SI 0; E_{reco}(0.30 - 0.80)GeV; Osc. ν_e CC ($MultiR_e$); MultiRing	0.248 951	4.116 61	−40.8193	6.009 53	−13.606	7.902 46	13.606	9.795 38	40.8193
SKDet + FSI/SI 1; E_{reco}(0.80 - 1.25)GeV; Osc. ν_e CC ($MultiR_e$); MultiRing	0.214 994	5.8806	−15.46	6.597 53	−5.1533	7.314 46	5.1533	8.031 39	15.46
SKDet + FSI/SI 2; E_{reco}(0.30 - 0.80)GeV; $\nu_\mu/\bar{\nu}_\mu$ CC ($MultiR_e$); MultiRing	0.507 403	6.789 88	−2.388 12	6.900 62	−0.796 04	7.011 37	0.796 04	7.122 11	2.388 12
SKDet + FSI/SI 3; E_{reco}(0.80 - 1.25)GeV; $\nu_\mu/\bar{\nu}_\mu$ CC ($MultiR_e$); MultiRing	0.236 118	6.9201	−0.516 076	6.944 03	−0.172 03	6.967 96	0.172 03	6.991 89	0.516 076
SKDet + FSI/SI 4; E_{reco}(0.30 - 0.80)GeV; $\nu_e/\bar{\nu}_e$ CC ($MultiR_e$); MultiRing	0.244 289	6.6652	−4.180 41	6.859 06	−1.3935	7.052 92	1.3935	7.246 78	4.180 41
SKDet + FSI/SI 5; E_{reco}(0.80 - 1.25)GeV; $\nu_e/\bar{\nu}_e$ CC ($MultiR_e$); MultiRing	0.234 55	6.5674	−5.586 39	6.826 46	−1.8621	7.085 52	1.8621	7.344 58	5.586 39
SKDet + FSI/SI 6; E_{reco}(0.30 - 0.80)GeV; all NC ($MultiR_e$); MultiRing	0.983 178	6.165 29	−11.3673	6.692 42	−3.7891	7.219 56	3.7891	7.7467	11.3673
SKDet + FSI/SI 7; E_{reco}(0.80 - 1.25)GeV; all NC ($MultiR_e$); MultiRing	0.523 609	6.783 29	−2.482 74	6.898 43	−0.827 58	7.013 56	0.827 58	7.128 69	2.482 74
SK energy scale	0.024	6.955 99	$-1.276\,85 \times 10^{-14}$	6.955 99	1.2769×10^{-14}	6.955 99	0	6.955 99	$1.276\,85 \times 10^{-14}$

Table 42: $\pm1\sigma$ and $\pm3\sigma$ effect of the systematic parameters on the total number of e-like Super-K events with one decay electron with Run 1-8 POT for neutrino mode post-BANFF Asimov data set A. Only the systematic parameters which affect this sample are shown.

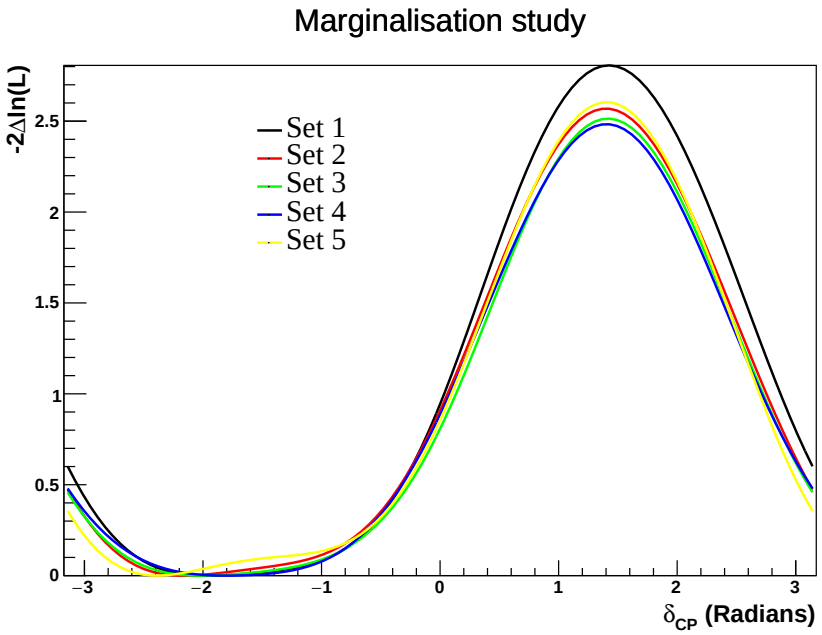


Figure 66: Asimov A δ_{CP} normal hierarchy contours without reactor constraint for independent sets of 10K marginalisation toys

D. Marginalisation studies

The increase in sensitivity to δ_{CP} provided by a doubling of FHC statistics in Run 1-8 relative to Run 1-7c has made sampling of the tails of oscillation parameters (particularly $\sin^2\theta_{13}$) more significant for 1D fits. In previous analyses all fits were performed with 10K marginalisation toys, but in the present analysis this is no longer sufficient to ensure that fits return consistent results for independent sets of marginalisation toys. One can see the nature of the problem in fig. 66, in which five independent sets of 10K marginalisation toys were fitted under the Asimov A oscillation parameter hypothesis without the reactor constraint; a large spread in $\Delta\chi^2$ contours is evident.

To determine a sufficient number of marginalisation toys to use checks of fit stability were carried out for different numbers of marginalisation toys. Figures 67 to 68 show the $\Delta\chi^2$ contours for the case of 40K and 80K marginalisation toys respectively. These studies indicate that good agreement among fits is achieved for 80K marginalisation toys 40k toys was found to be sufficient in the with reactor case due to the smaller marginalisation range for $\sin^2\theta_{13}$).

Further to the Asimov-based marginalisation studies, the stronger constraint seen in the data fits prompted a check of the data fit stability. Figures 69 to 69 show the result of fitting δ_{CP} with the reactor constraint for both normal and inverted hierarchy. Variations of $\Delta\chi^2 \sim 1$ are evident around the maximum in the region of $\delta_{CP} \sim \pi/2$, but that good stability is maintained in the preferred region and at the values of $\delta_{CP} = 0$ and π.

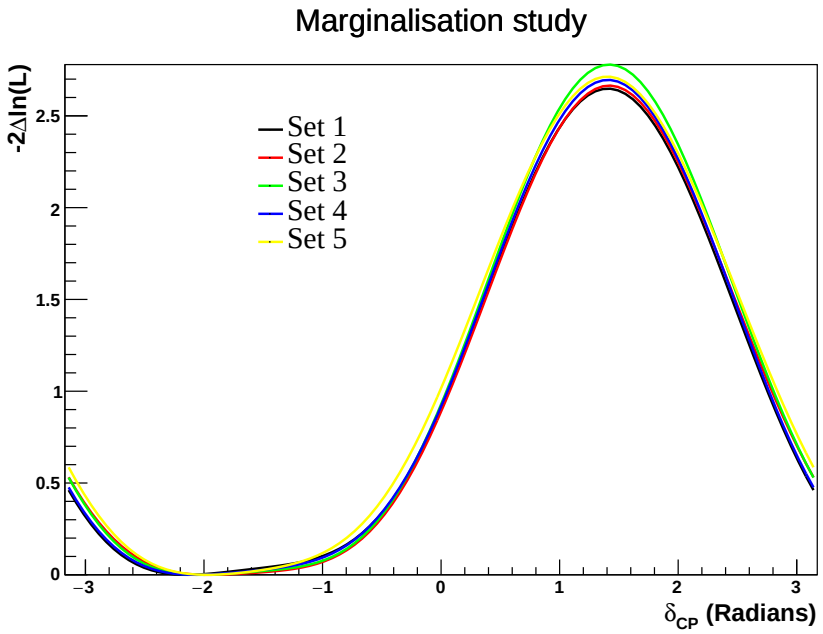


Figure 67: Asimov A δ_{CP} normal hierarchy contours without reactor constraint for independent sets of 40K marginalisation toys

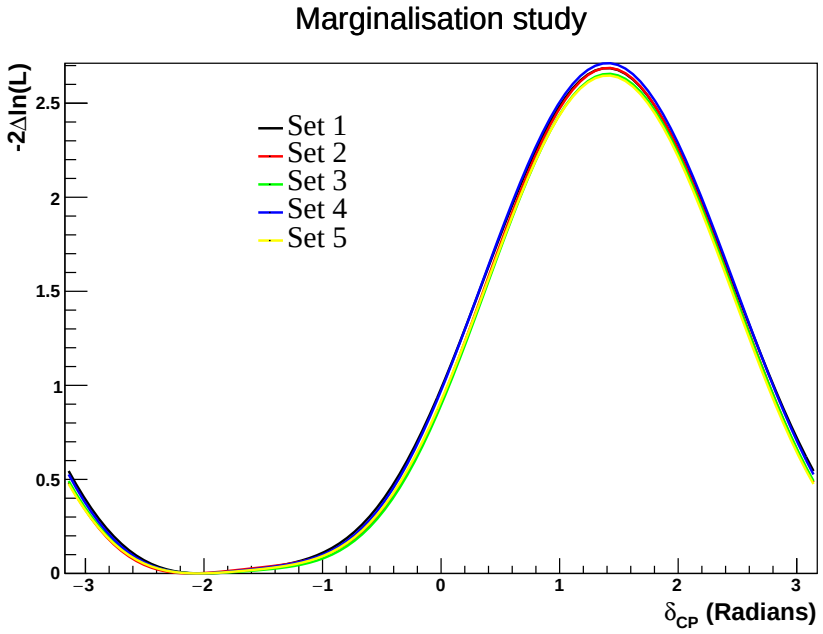


Figure 68: Asimov A δ_{CP} normal hierarchy contours without reactor constraint for independent sets of 80K marginalisation toys

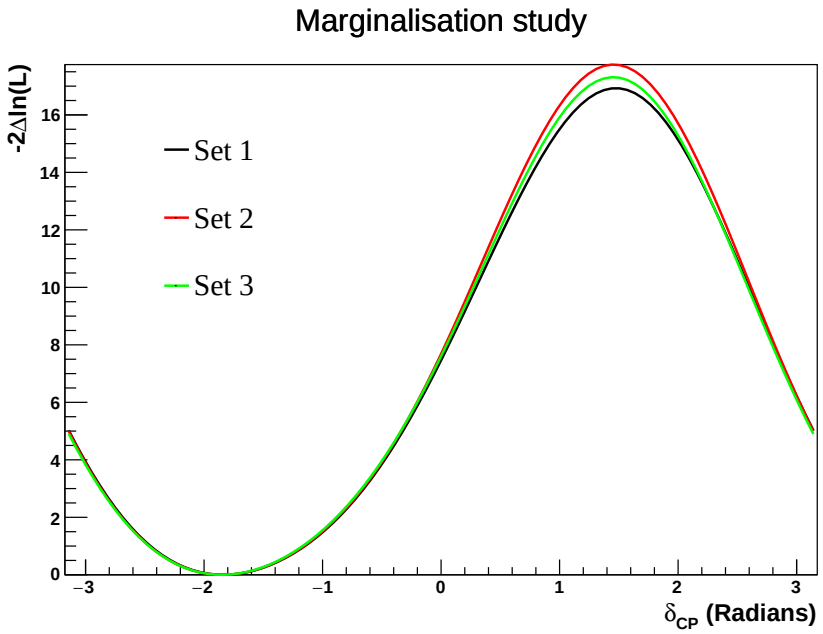


Figure 69: Data fit of δ_{CP} normal hierarchy contours with reactor constraint for independent sets of 40K marginalisation toys

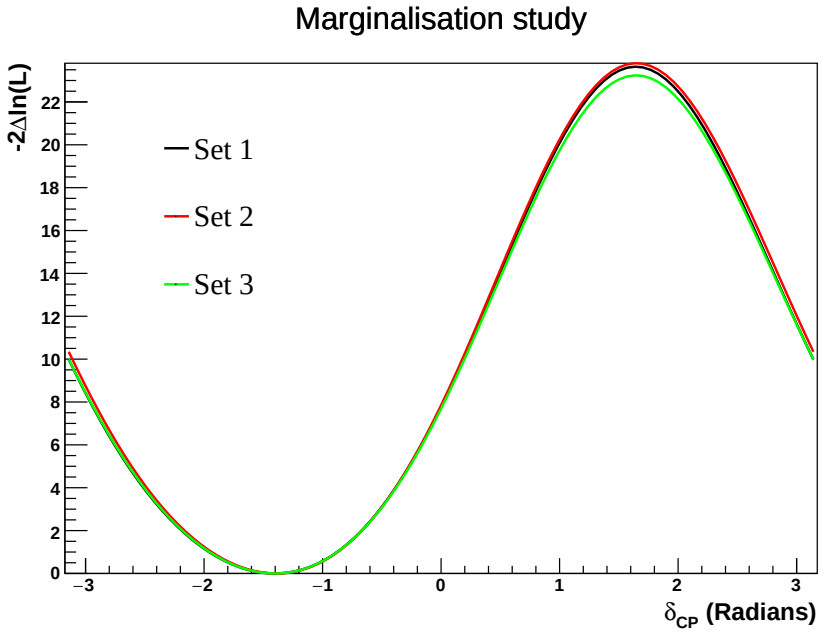


Figure 70: Data fit of δ_{CP} inverted hierarchy contours with reactor constraint for independent sets of 40K marginalisation toys

E. Effect of systematic errors on Super-K predictions

	Pre-BANFF			Post-BANFF		
Group	Mean	1σ	%	Mean	1σ	%
SK Detector	-	-	-	269.11	5.01	1.86
SK FSI+SI+PN	-	-	-	268.38	5.91	2.20
Flux+Xsec constrained	250.34	35.35	14.12	266.68	8.60	3.22
$\sigma(\nu_e)/\sigma(\bar{\nu}_e)$	-	-	-	268.42	0.00	0.00
NC1γ	-	-	-	268.42	0.00	0.00
NC Other	-	-	-	268.42	0.67	0.25
Osc	-	-	-	268.42	0.09	0.04
All	250.95	35.71	14.23	267.78	11.79	4.40
All with osc	250.95	35.71	14.23	267.78	11.79	4.40

Table 43: Average event rate and RMS error broken down by systematic parameter group for μ-like Super-K events with Run 1-8 POT for neutrino mode.

	Pre-BANFF			Post-BANFF		
Group	Mean	1σ	%	Mean	1σ	%
SK Detector	-	-	-	64.41	0.97	1.51
SK FSI+SI+PN	-	-	-	64.27	1.27	1.98
Flux+Xsec constrained	62.09	7.18	11.56	63.97	1.74	2.72
$\sigma(\nu_e)/\sigma(\bar{\nu}_e)$	-	-	-	64.28	0.00	0.00
NC1γ	-	-	-	64.28	0.00	0.00
NC Other	-	-	-	64.28	0.16	0.25
Osc	-	-	-	64.28	0.02	0.03
All	62.18	7.31	11.76	64.17	2.41	3.76
All with osc	62.18	7.31	11.76	64.17	2.41	3.76

Table 44: Average event rate and RMS error broken down by systematic parameter group for μ-like Super-K events with Run 1-8 POT for antineutrino mode.

	Pre-BANFF			Post-BANFF		
Group	Mean	1σ	%	Mean	1σ	%
SK Detector	-	-	-	73.64	2.23	3.03
SK FSI+SI+PN	-	-	-	73.46	2.21	3.01
Flux+Xsec constrained	65.47	9.80	14.97	72.57	2.34	3.22
$\sigma(\nu_e)/\sigma(\bar{\nu}_e)$	-	-	-	73.46	1.93	2.63
NC1γ	-	-	-	73.76	0.80	1.08
NC Other	-	-	-	73.48	0.11	0.14
Osc	-	-	-	73.49	2.84	3.86
All	65.91	10.26	15.56	73.12	4.46	6.10
All with osc	65.92	10.57	16.03	73.14	5.32	7.27

Table 45: Average event rate and RMS error broken down by systematic parameter group for e-like Super-K events with Run 1-8 POT for neutrino mode.

	Pre-BANFF			Post-BANFF		
Group	Mean	1σ	%	Mean	1σ	%
SK Detector	-	-	-	7.94	0.33	4.22
SK FSI+SI+PN	-	-	-	7.91	0.18	2.31
Flux+Xsec constrained	7.63	0.91	11.94	7.84	0.23	2.88
$\sigma(\nu_e)/\sigma(\bar{\nu}_e)$	-	-	-	7.91	0.12	1.46
NC1γ	-	-	-	7.98	0.21	2.59
NC Other	-	-	-	7.91	0.03	0.33
Osc	-	-	-	7.91	0.29	3.60
All	7.74	1.03	13.32	7.95	0.52	6.51
All with osc	7.74	1.07	13.79	7.96	0.59	7.44

Table 46: Average event rate and RMS error broken down by systematic parameter group for e-like Super-K events with Run 1-8 POT for antineutrino mode.

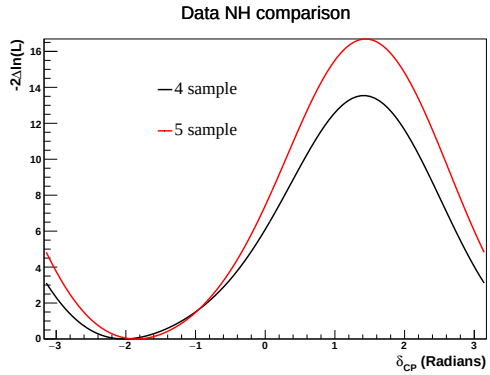
	Pre-BANFF			Post-BANFF		
Group	Mean	1σ	%	Mean	1σ	%
SK Detector	-	-	-	7.02	1.17	16.69
SK FSI+SI+PN	-	-	-	6.92	0.79	11.43
Flux+Xsec constrained	7.71	0.91	11.79	6.91	0.28	4.05
$\sigma(\nu_e)/\sigma(\bar{\nu}_e)$	-	-	-	6.92	0.18	2.62
NC1γ	-	-	-	6.93	0.02	0.33
NC Other	-	-	-	6.92	0.07	0.98
Osc	-	-	-	6.92	0.26	3.78
All	7.79	1.85	23.71	7.00	1.47	20.94
All with osc	7.79	1.87	24.05	7.00	1.49	21.24

Table 47: Average event rate and RMS error broken down by systematic parameter group for ν_e CC1π^+-like Super-K events with Run 1-8 POT for neutrino mode.

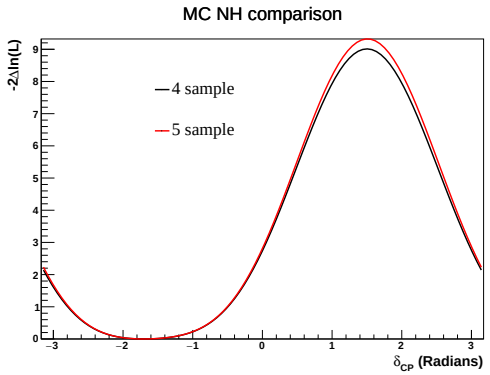
F. Effect of ν_e CC1π^+sample on constraint strength

Section 5 showed stronger than expected constraints from data for δ_{CP}, appearance and disappearance fits. To determine a potential explanation for this we chose to run an additional fit without the ν_e CC1π^+sample, hereafter referred to as the 4-sample fit, with the fit using all samples being referred to as the 5-sample fit. It should be noted that the full 5-sample marginalisation toys were used for all fits, rather than using 4-sample marginalisation toys for the 4-sample fit. This means that the correlations among samples for the 4-sample fit are not handled correctly, but the effect from this is expected to be small.

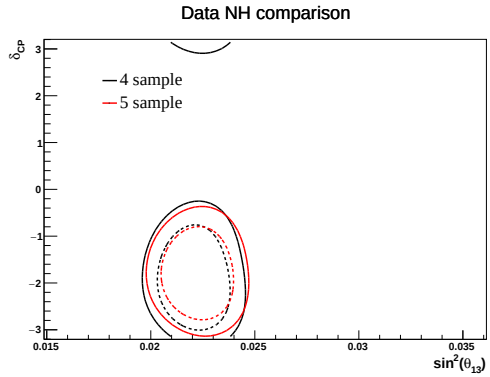
Figure 71 shows comparisons of the aforementioned fits for normal hierarchy. It can be seen that the constraint on the δ_{CP} and appearance fits is weakened in the absence of the ν_e CC1π^+sample, while the disappearance contour is unaffected.



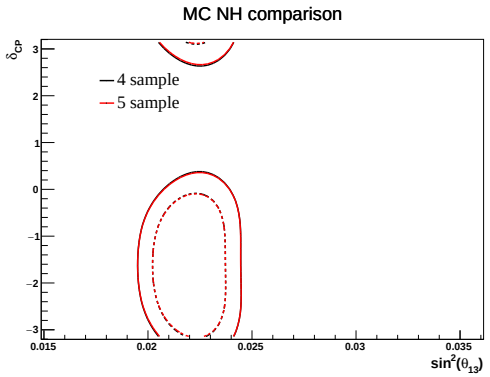
(a) δ_{CP} data fit



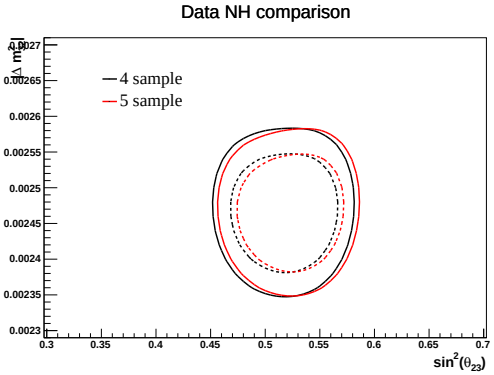
(b) δ_{CP} MC fit



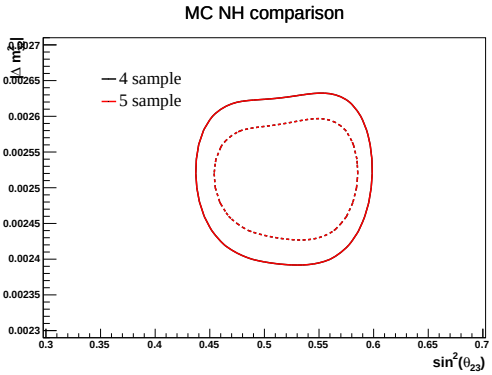
(c) Appearance data fit



(d) Appearance MC fit



(e) Disappearance data fit



(f) Disappearance MC fit

Figure 71: Fits of data and the Asimov A data set with four and five samples respectively.

G. Matter effects and crust density

To check the effect of the error on the average crust density [91] a number of fits of δ_{CP} were performed varying this value. Figures 72 to 73 show the effect of varying the average crust density between 2.5, 2.6 and 2.7 g cm^{-3} in normal and inverted hierarchy respectively. 1×10^4 marginalisation toys were generated with the average crust density fixed at the value of 2.6 g cm^{-3}, whilst fits were performed using the Asimov A data set where the crust density was varied using the aforementioned values. The effect on the fit of δ_{CP} can be seen to be small, supporting the use of a fixed value for the average crust density.

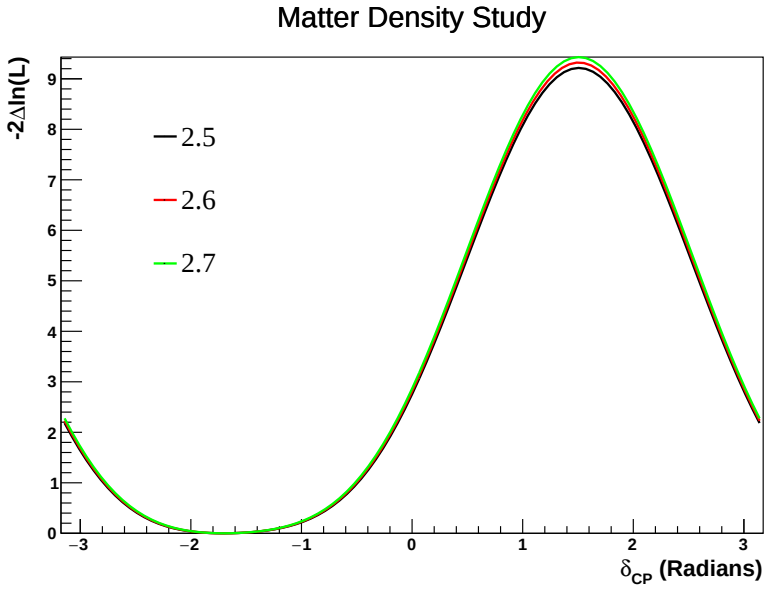


Figure 72: Fits of δ_{CP} using the Asimov A data set whilst varying the average crust density in normal hierarchy for 10K marginalisation toys.

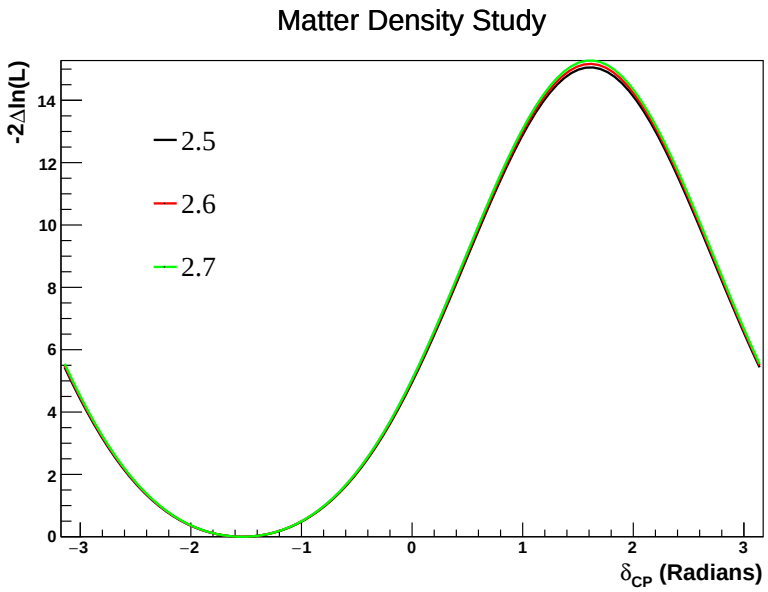
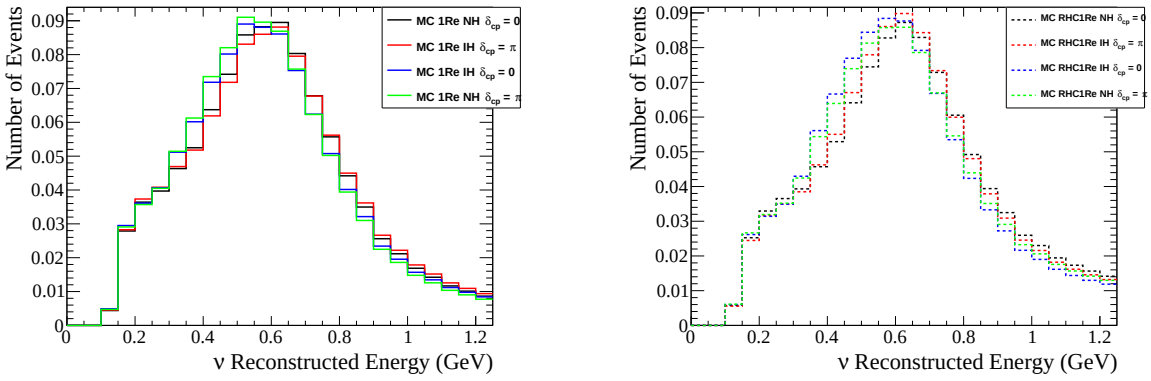


Figure 73: Fits of δ_{CP} using the Asimov A data set whilst varying the average crust density in inverted hierarchy for 10K marginalisation toys.

H. Asimov B δ_{CP}/mass-hierarchy degeneracy

In section 4.2 it was noted that Asimov B fits are affected by a lack of sensitivity to distinguish $\delta_{CP} = 0$ and $\delta_{CP} = \pi$, for example, the fit of δ_{CP} shown in fig. 22 shows a best-fit point shifted away from the true value. fig. 74 shows normalised one-ring e-like spectra for Asimov data set B with mass hierarchy and δ_{CP} varied to demonstrate the problem. It is clear from the plots that the spectrum for $\delta_{CP} = 0$ in normal hierarchy is similar in shape to the spectrum for $\delta_{CP} = \pi$ in inverted hierarchy and vice versa.



(a) MC one-ring FHC e-like spectra

(b) MC one-ring RHC e-like spectra

Figure 74: Normalised spectra for one-ring e-like samples showing the effect of changing the mass hierarchy and true value of δ_{CP} for the Asimov B data set.

I. Effect of systematic errors on Super-K predictions in updated analysis

Section 6 presents the updated fit results following the inclusion of the E_b parameter, this section presents the effect of systematic errors on Super-K predictions including variations of the E_b parameter. The procedure for determining the systematics is as described in section 3.5.

I.1. Systematic error on Super-K predictions

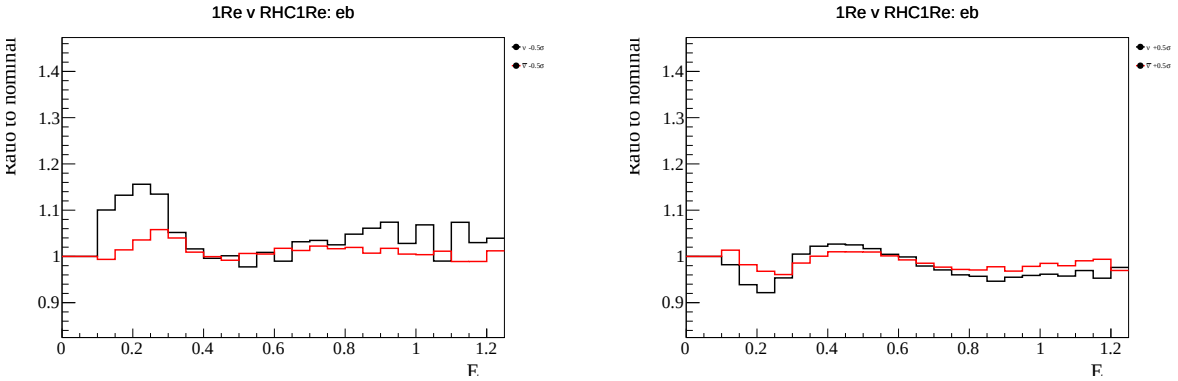
	1-Ring μ		**1-Ring e**			
Error source	FHC	RHC	FHC	RHC	FHC 1 d.e.	FHC/RHC
SK Detector	2.41	2.02	2.85	3.83	13.26	1.47
SK FSI+SI+PN	2.21	1.99	3.03	2.34	11.51	1.57
Flux + Xsec constrained	3.25	2.74	3.24	2.90	4.08	2.50
$\sigma(\nu_e)/\sigma(\bar{\nu}_e)$	0.00	0.00	2.64	1.45	2.63	3.04
NC1γ	0.00	0.00	1.08	2.60	0.33	1.50
NC Other	0.25	0.25	0.15	0.33	0.98	0.18
E_b	2.42	1.73	7.27	3.70	2.99	3.71
Osc	0.04	0.03	3.87	3.60	3.78	0.80
All Systematics	5.07	4.32	8.81	7.02	18.41	5.87
All with osc	5.07	4.32	9.61	7.85	18.74	5.93

Table 48: Percentage error on event rate by error source and sample. Final column is the percentage error on the ratio of FHC/RHC events in the one-ring e sample.

Looking at the ratio of events with respect to nominal for variations of $\pm 0.5\sigma$ and $\pm 1.5\sigma$ at the level of individual E_{reco} bins (see fig. 75) shows a pattern of larger variations due to E_b in the FHC electron-like sample relative to the corresponding RHC sample. This effect is also apparent in the muon-like samples at lower energies, which can be seen in table 49, which applies a cut of $E_{\text{reco}} < 1.25$ GeV to the muon-like samples.

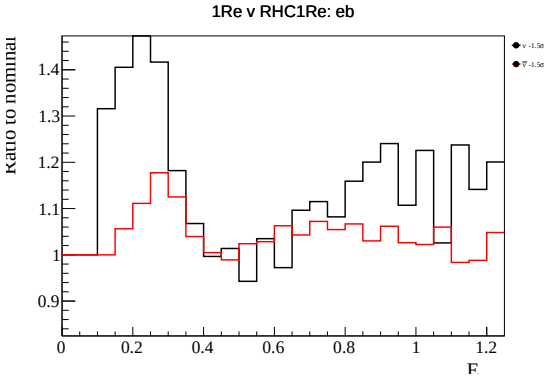
	1-Ring μ		**1-Ring e**			
Error source	FHC	RHC	FHC	RHC	FHC 1 d.e.	FHC/RHC
SK Detector	3.09	2.87	2.85	3.83	13.26	1.47
SK FSI+SI+PN	2.31	2.91	3.03	2.34	11.51	1.57
Flux + Xsec constrained	3.44	2.93	3.24	2.90	4.08	2.50
$\sigma(\nu_e)/\sigma(\bar{\nu}_e)$	0.00	0.00	2.64	1.45	2.63	3.04
NC1γ	0.00	0.00	1.08	2.60	0.33	1.50
NC Other	0.40	0.48	0.15	0.33	0.98	0.18
E_b	3.61	1.63	7.27	3.70	2.99	3.71
Osc	0.05	0.04	3.87	3.60	3.78	0.80
All Systematics	6.13	5.29	8.81	7.02	18.41	5.87
All with osc	6.13	5.29	9.61	7.85	18.74	5.93

Table 49: Percentage error on event rate by error source and sample with a cut of $E_{\text{reco}} < 1.25$ GeV applied to the muon samples.

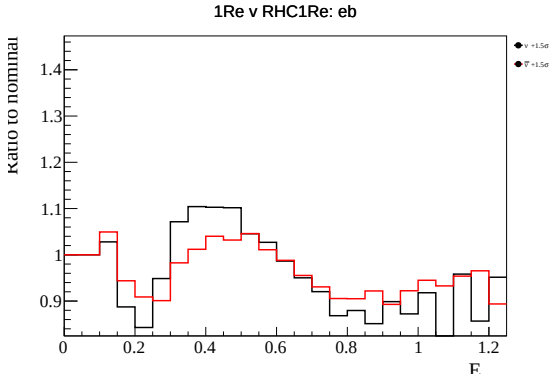


(a) $E_b = -0.5\sigma$

(b) $E_b = +0.5\sigma$



(c) $E_b = -1.5\sigma$

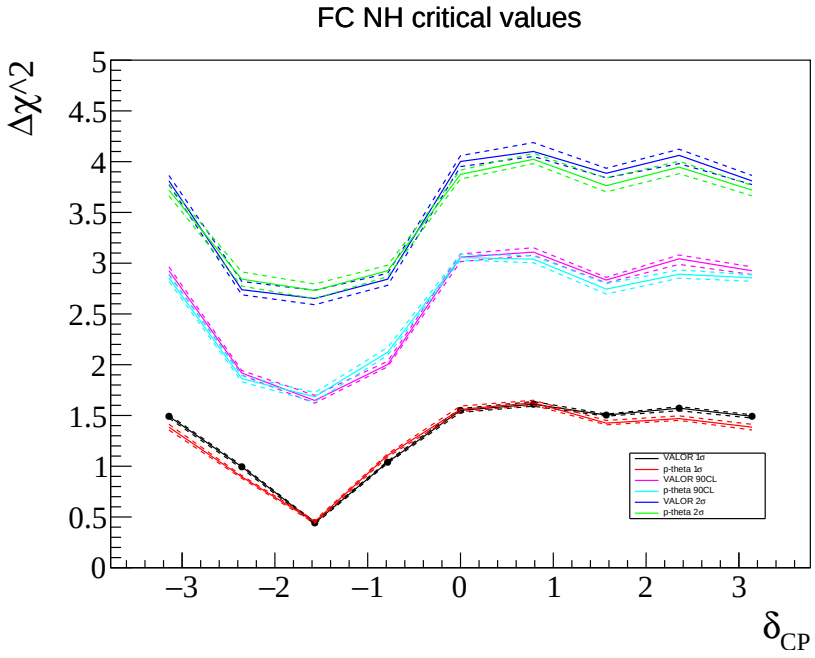


(d) $E_b = +1.5\sigma$

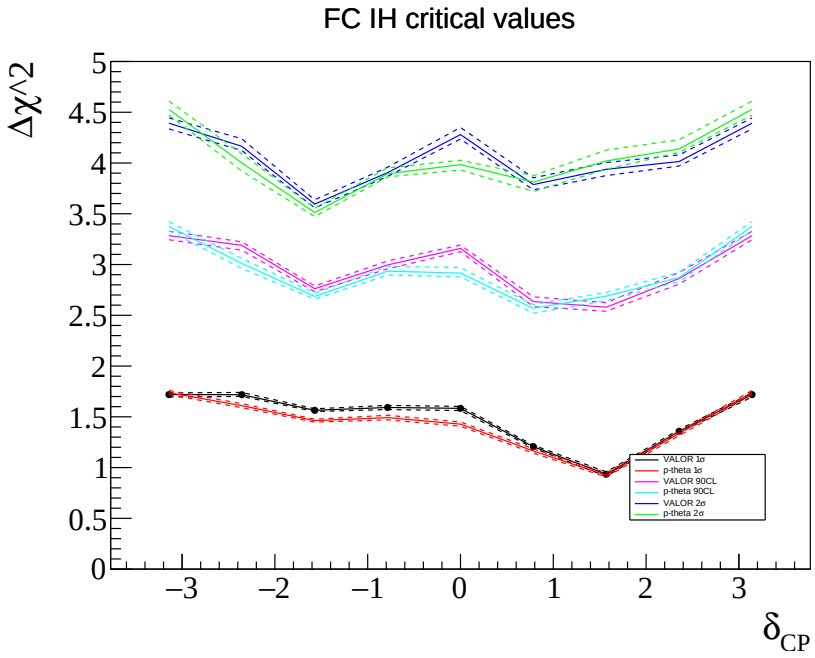
Figure 75: The ratio of events with respect to nominal for variations of E_b with all other systematics at nominal for Asimov data set A.

J. Feldman-Cousins comparison with p-theta

This section presents comparisons of VALOR and p-theta Feldman-Cousins critical values corresponding to the results in section 6 for reference.



(a) NH



(b) IH

Figure 76: Feldman-Cousins critical values.